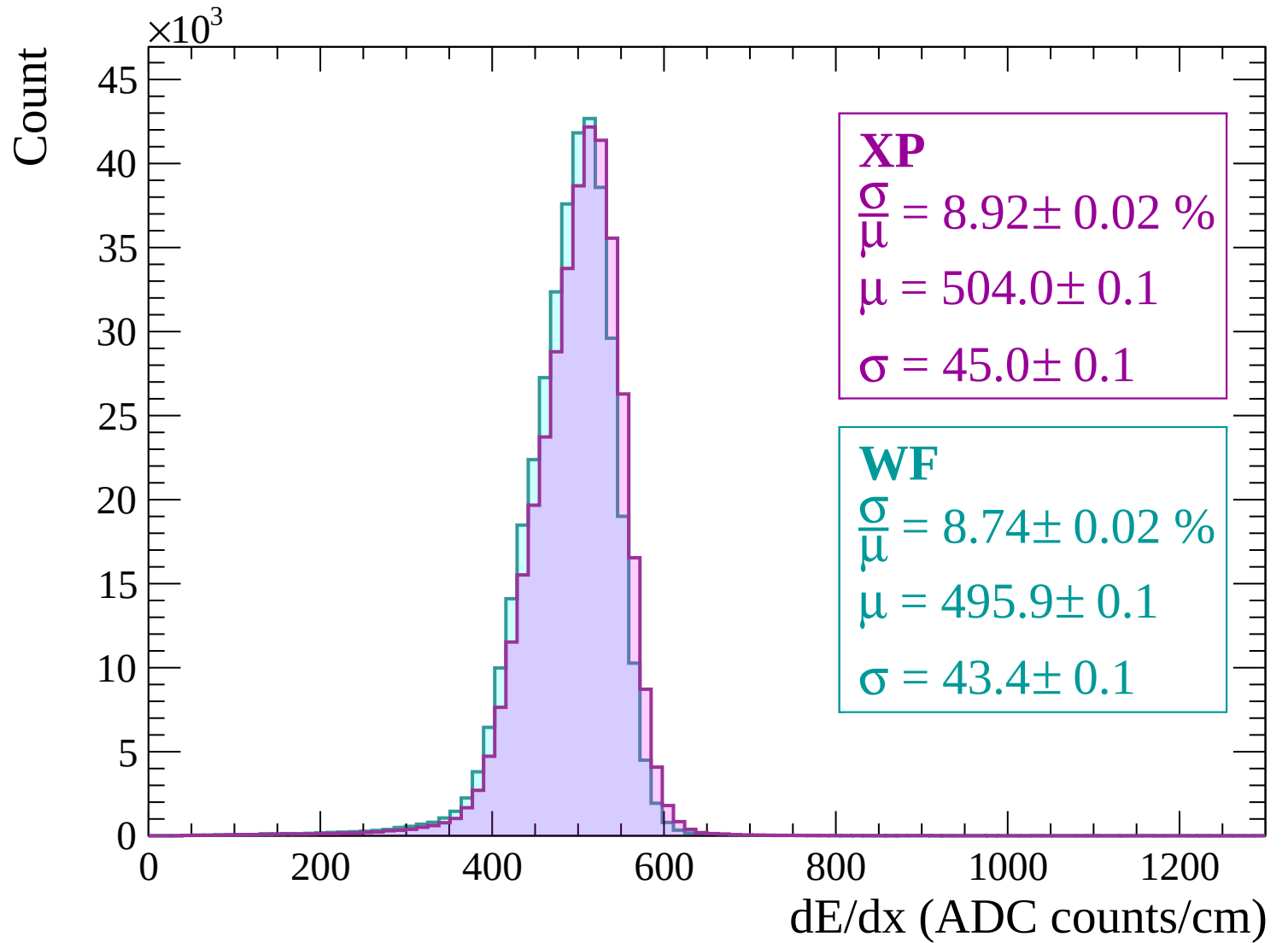
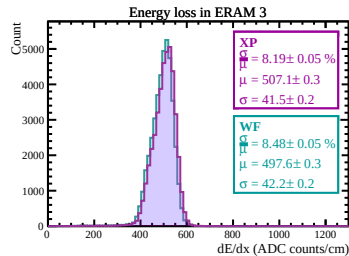
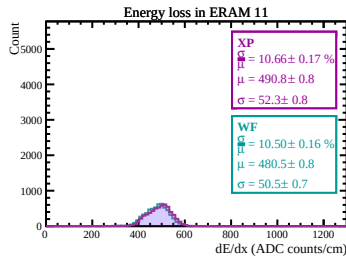
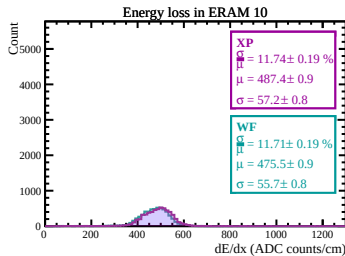
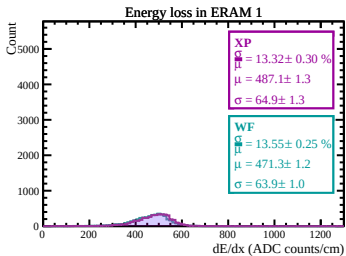
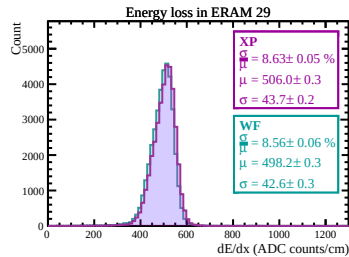
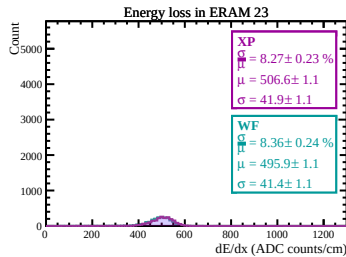
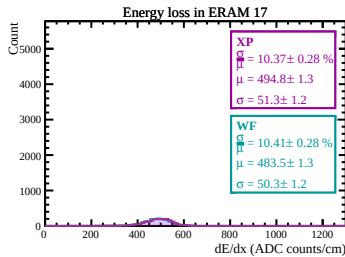
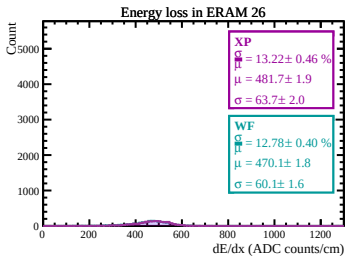
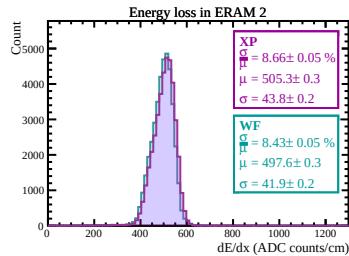
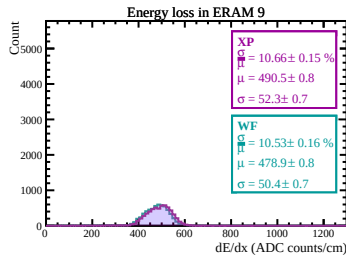
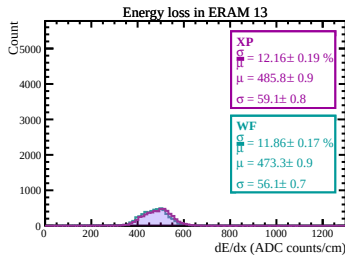
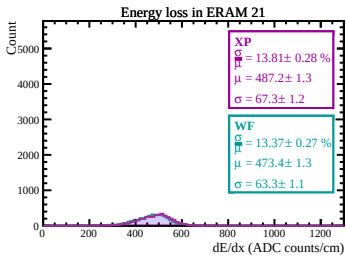
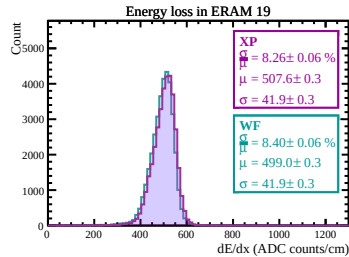
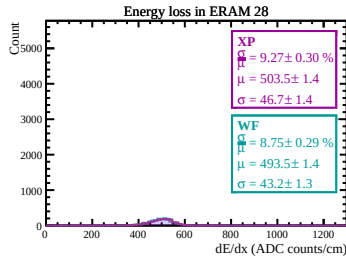
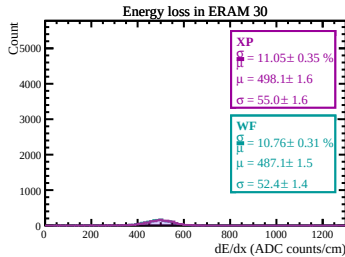
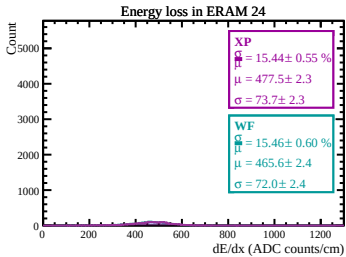
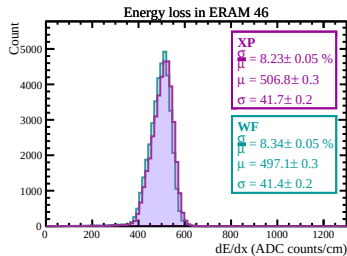
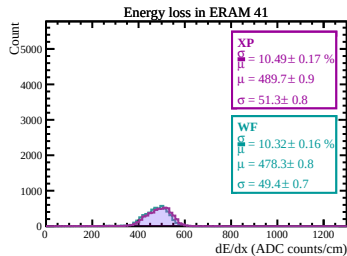
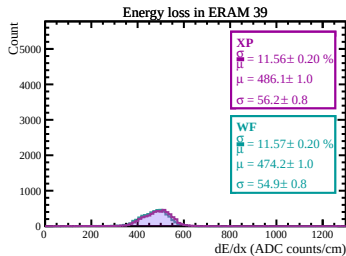
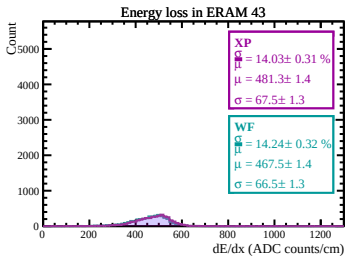
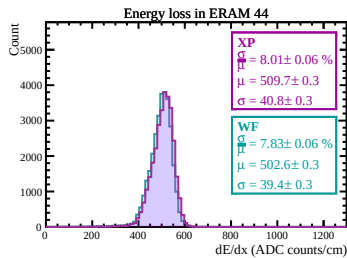
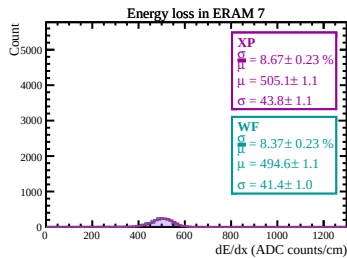
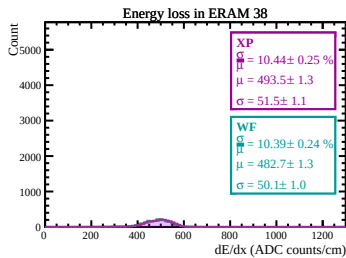
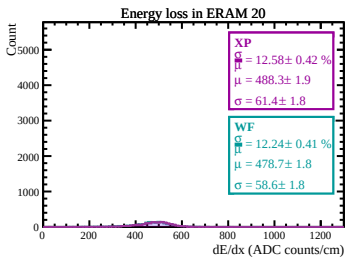
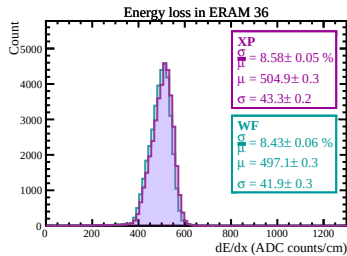
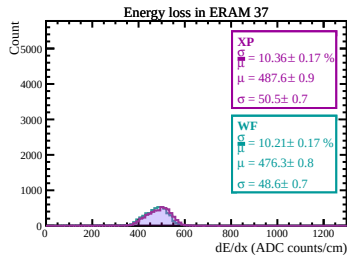
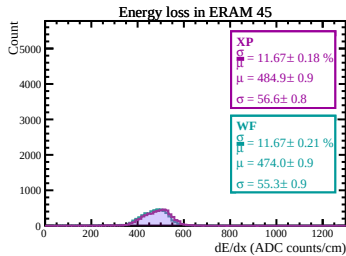
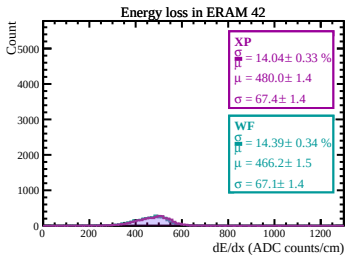
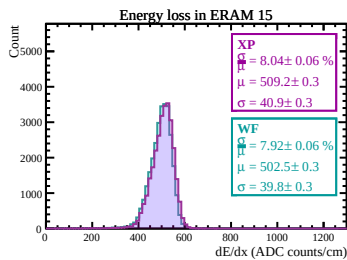
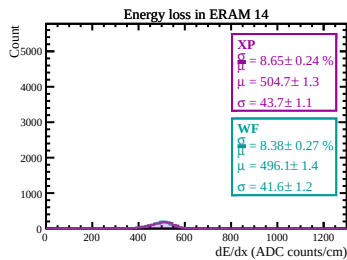
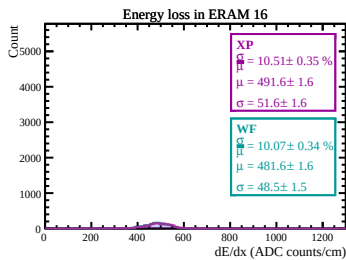
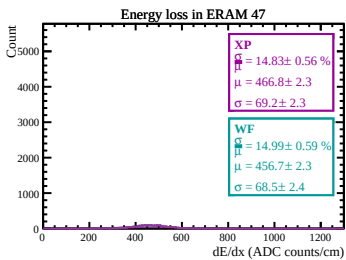
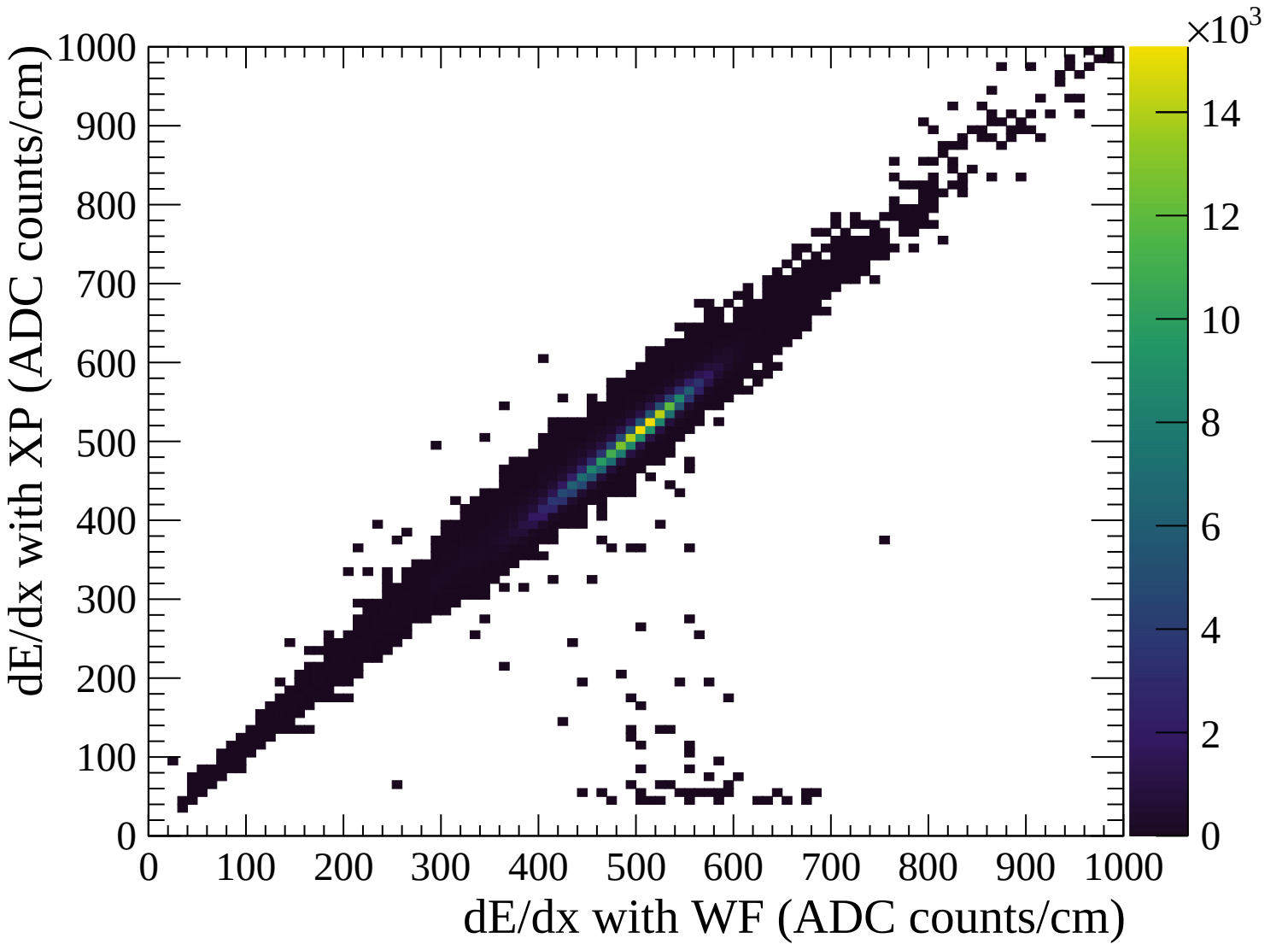
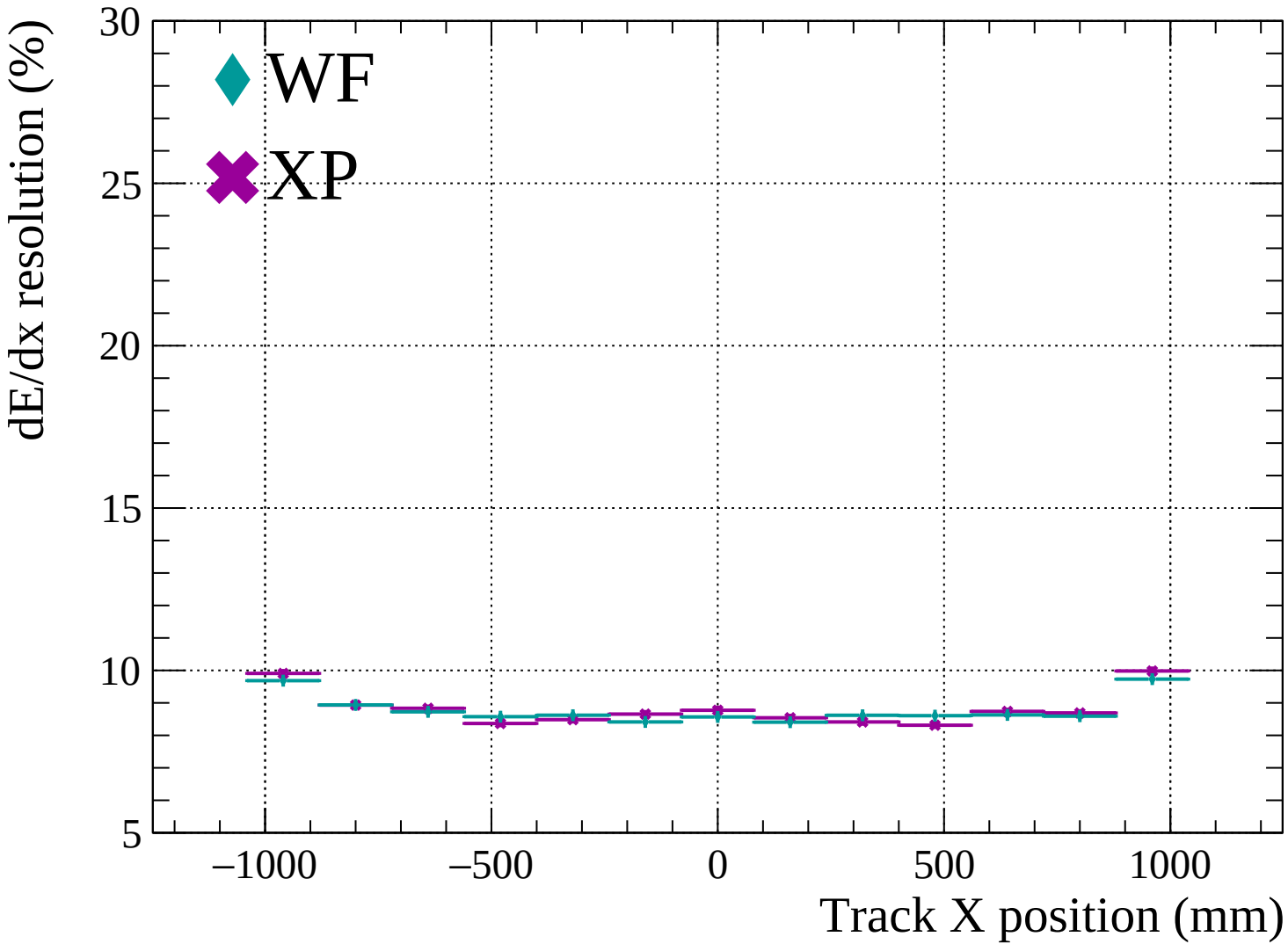


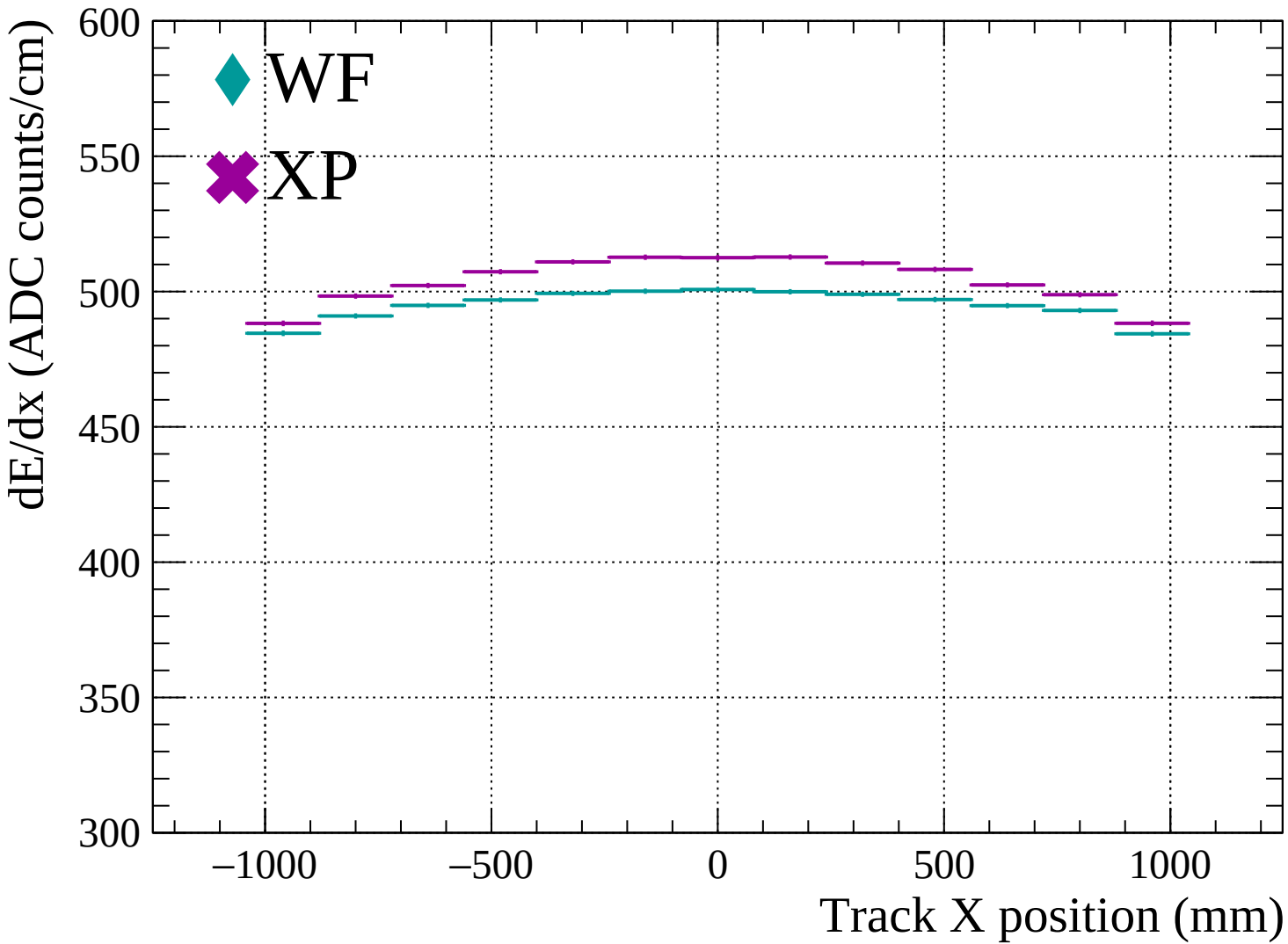


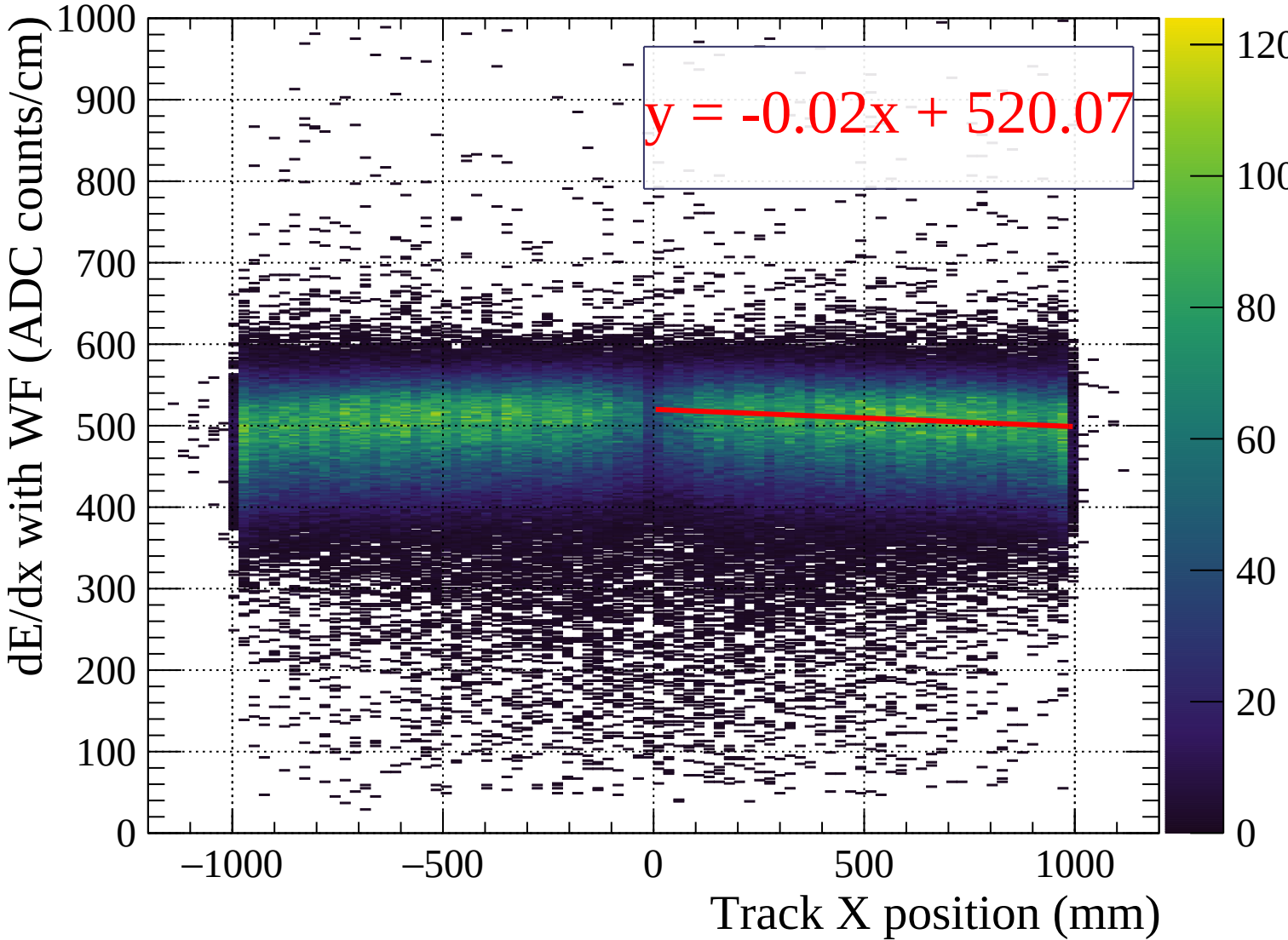


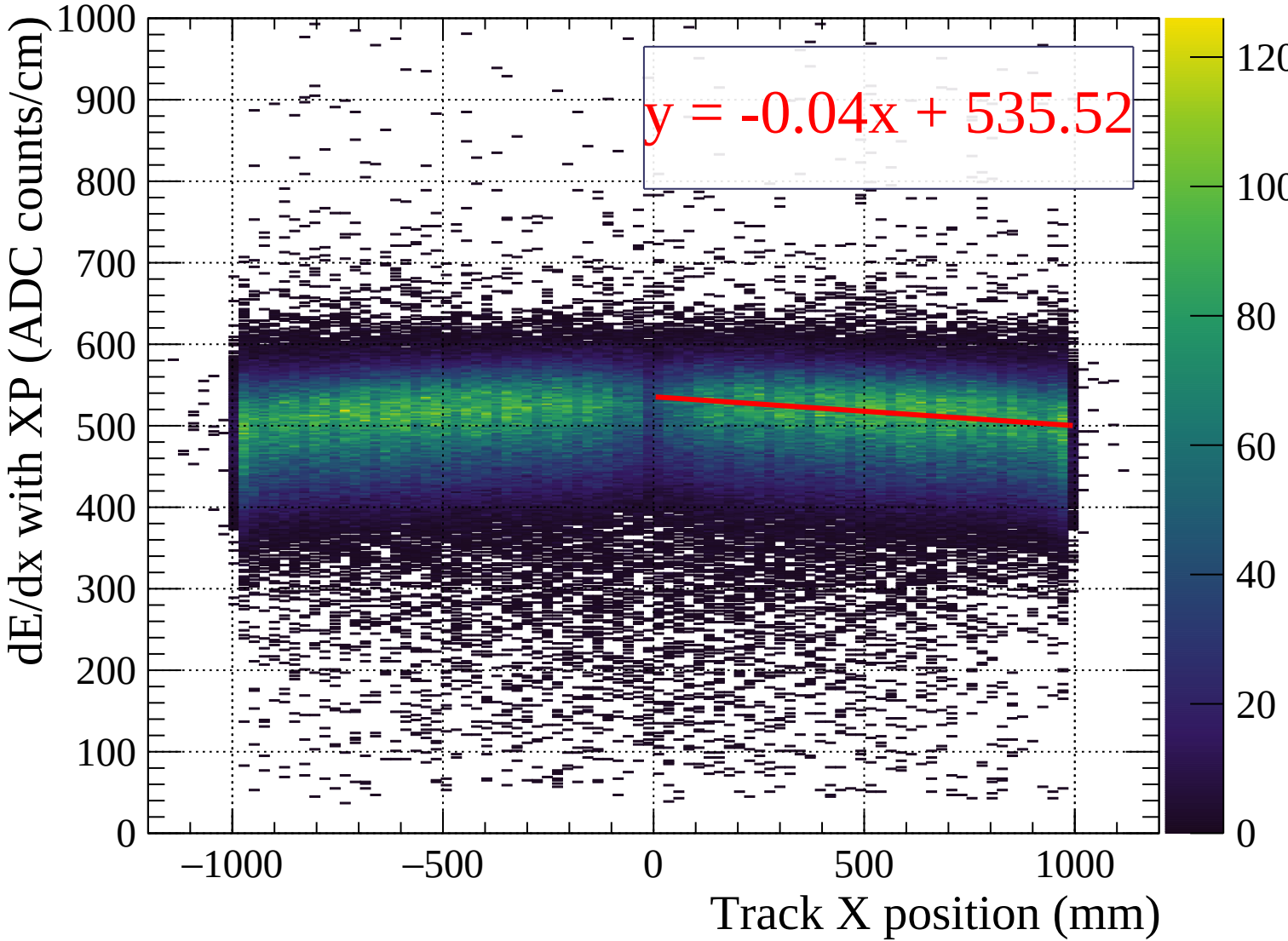


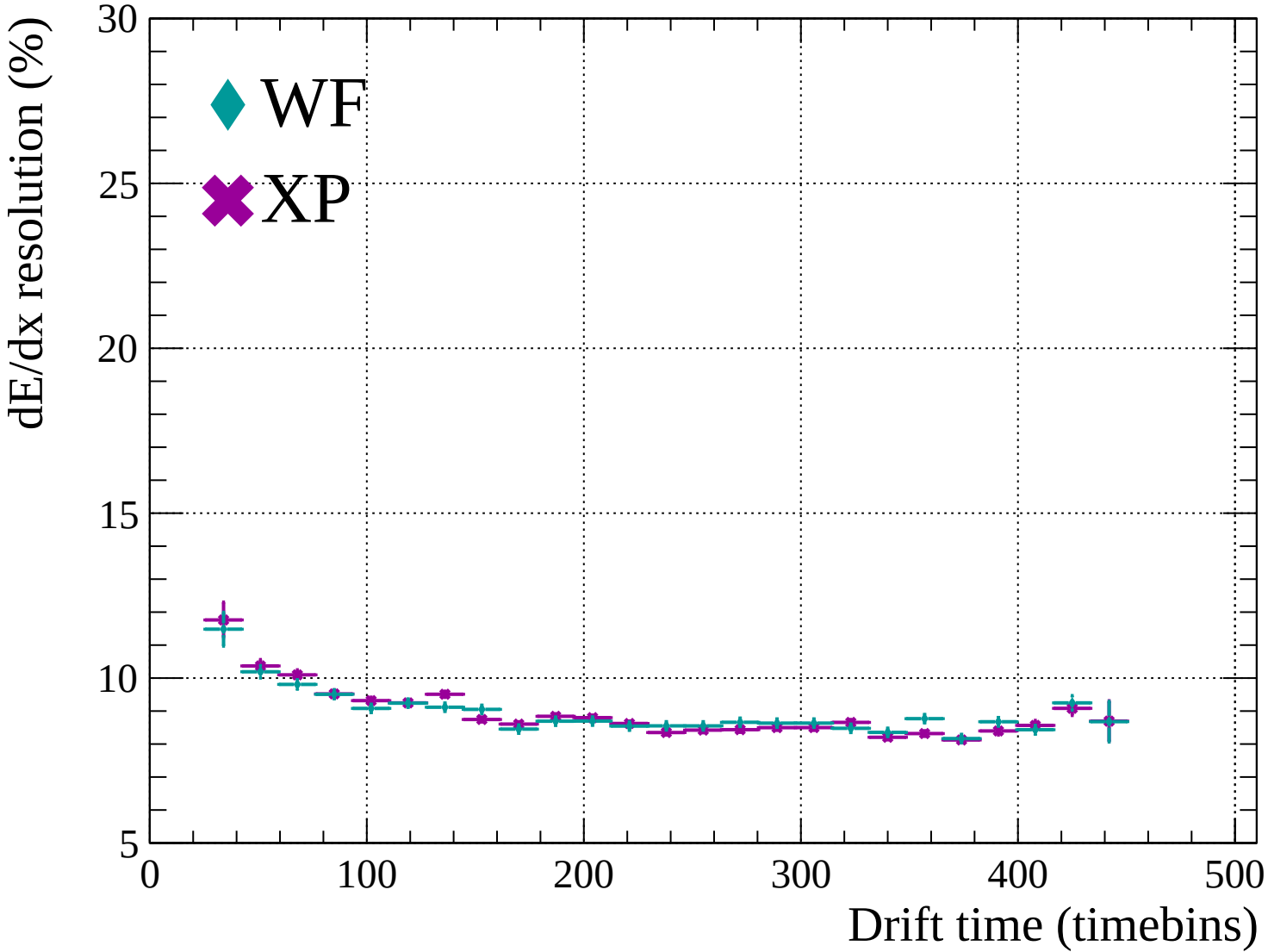


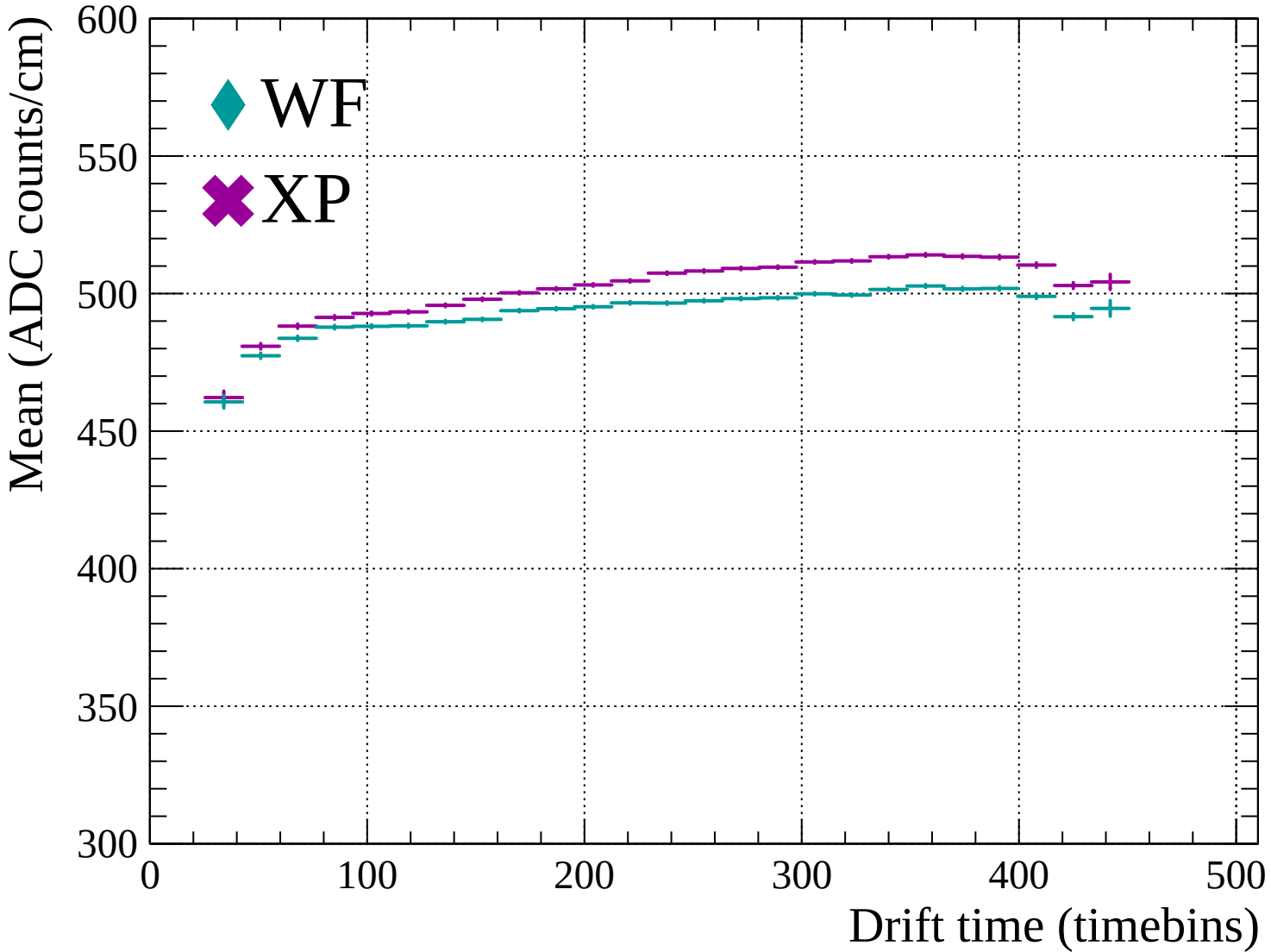


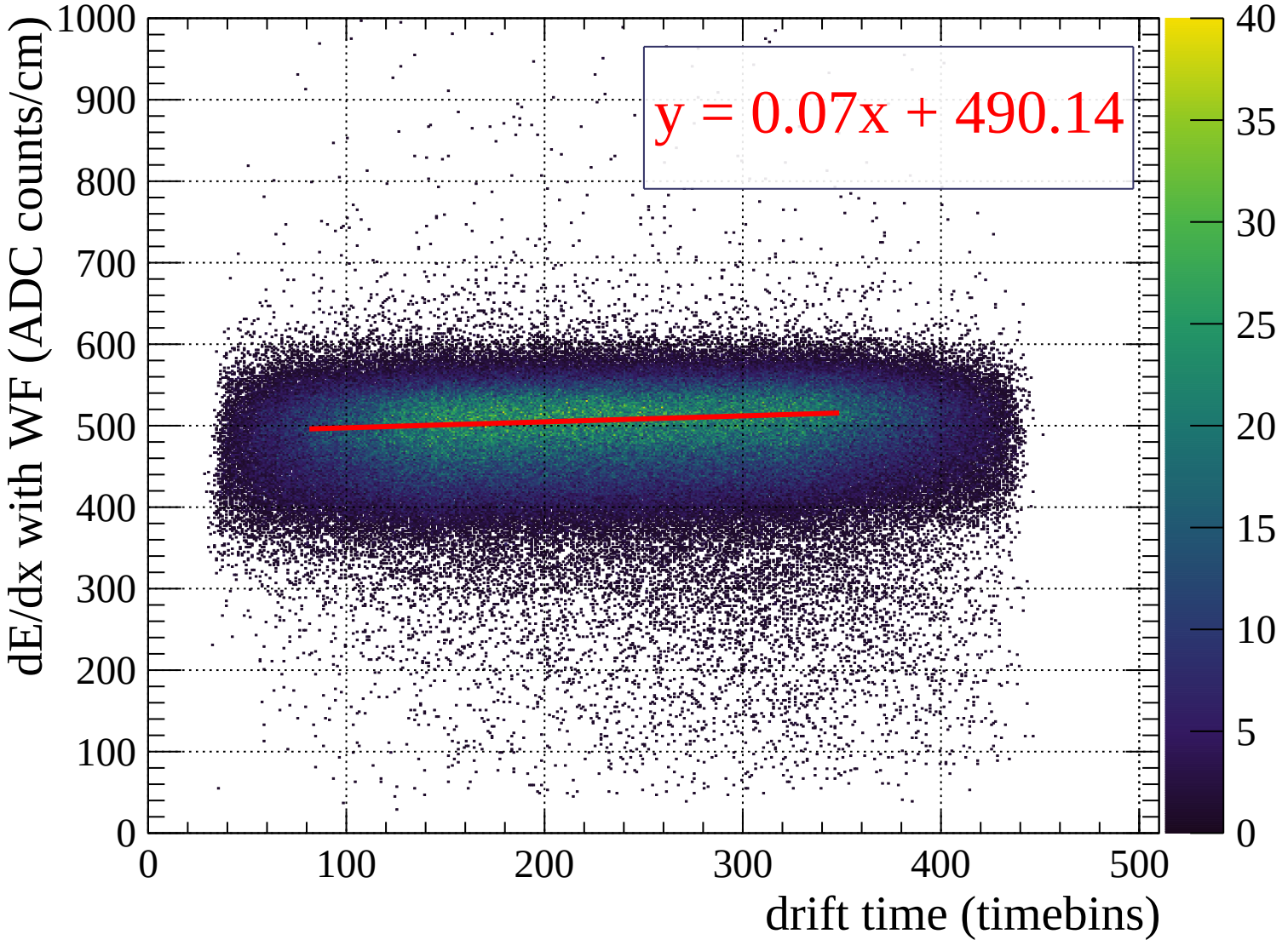


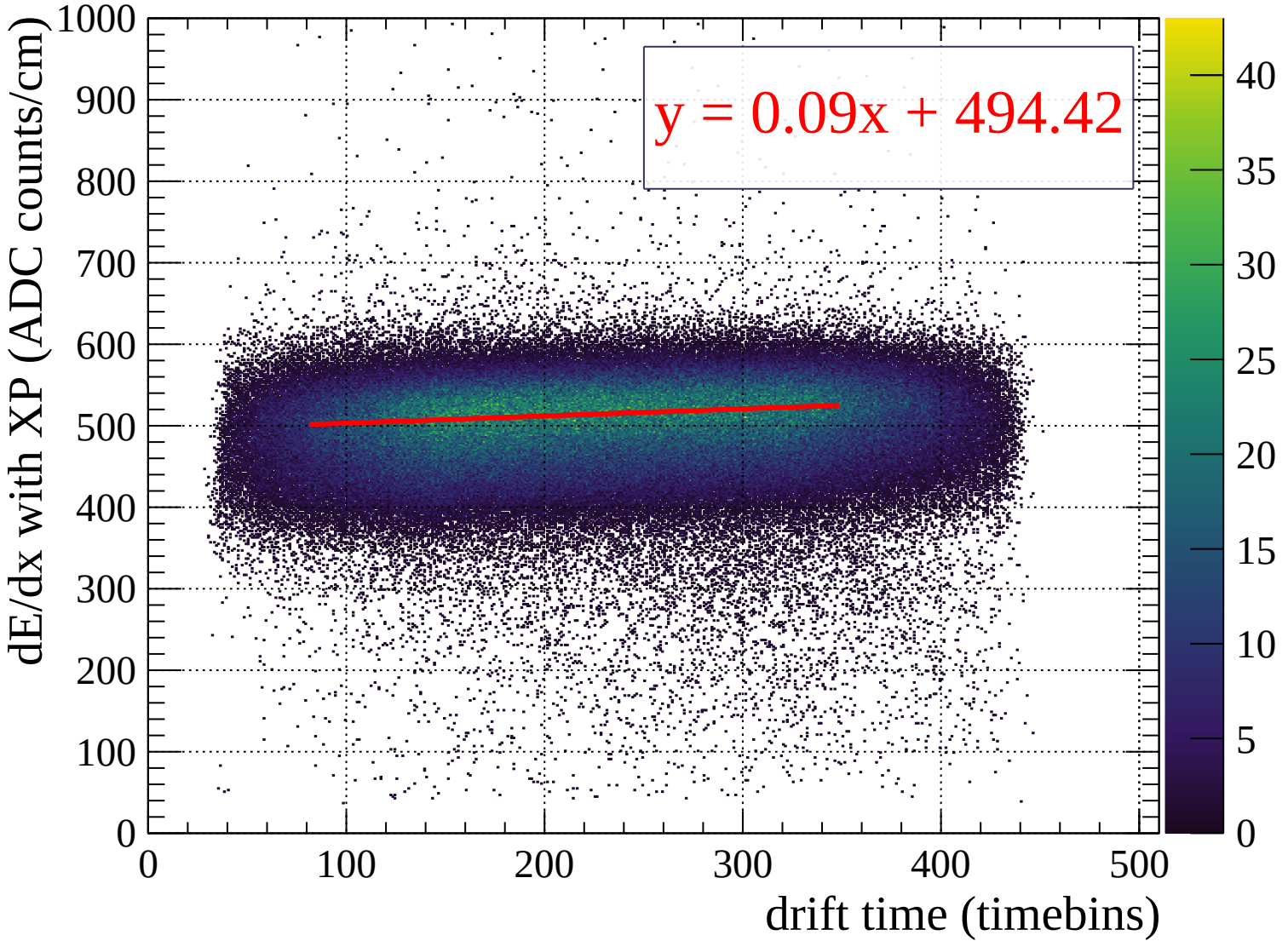


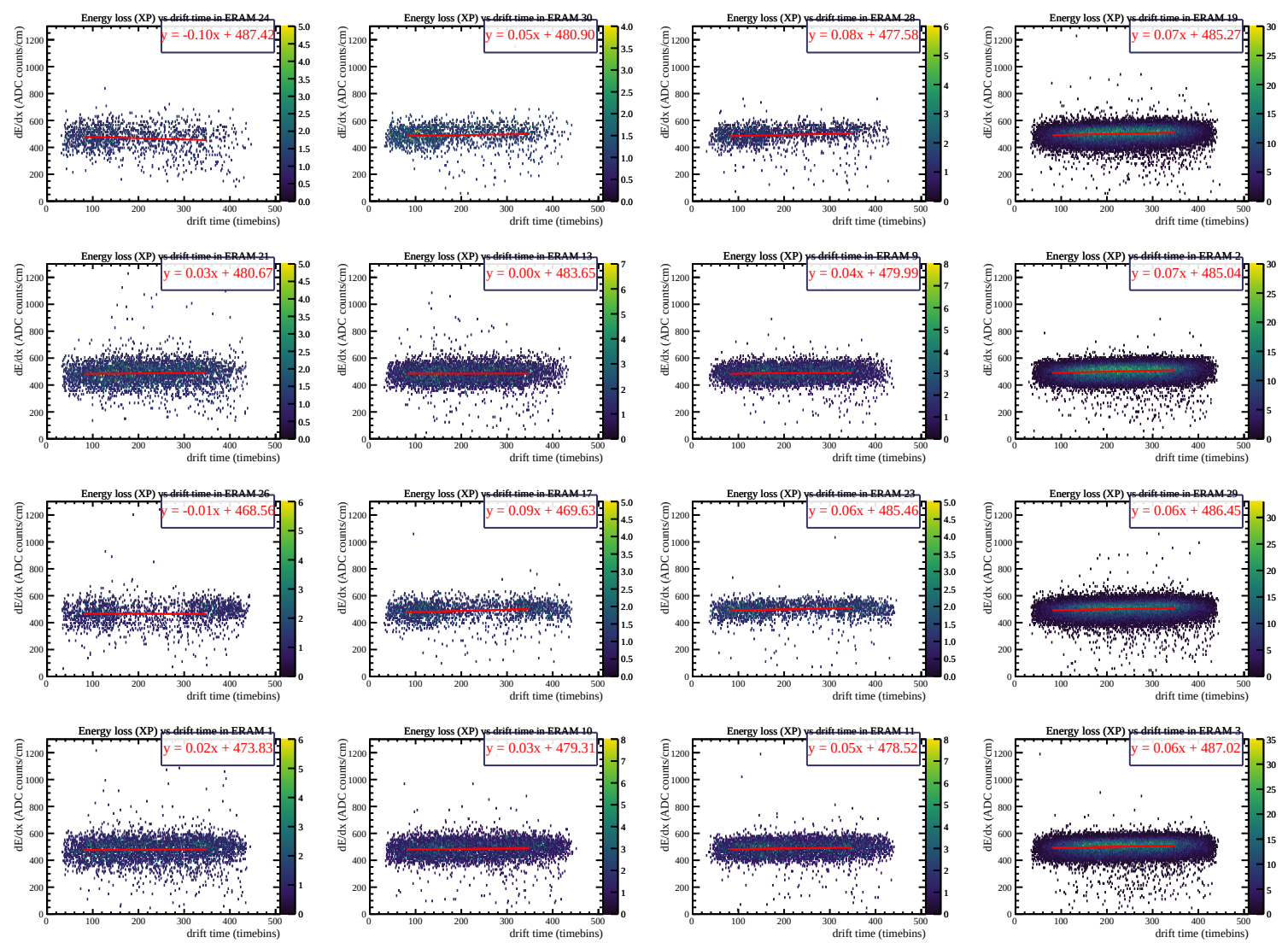


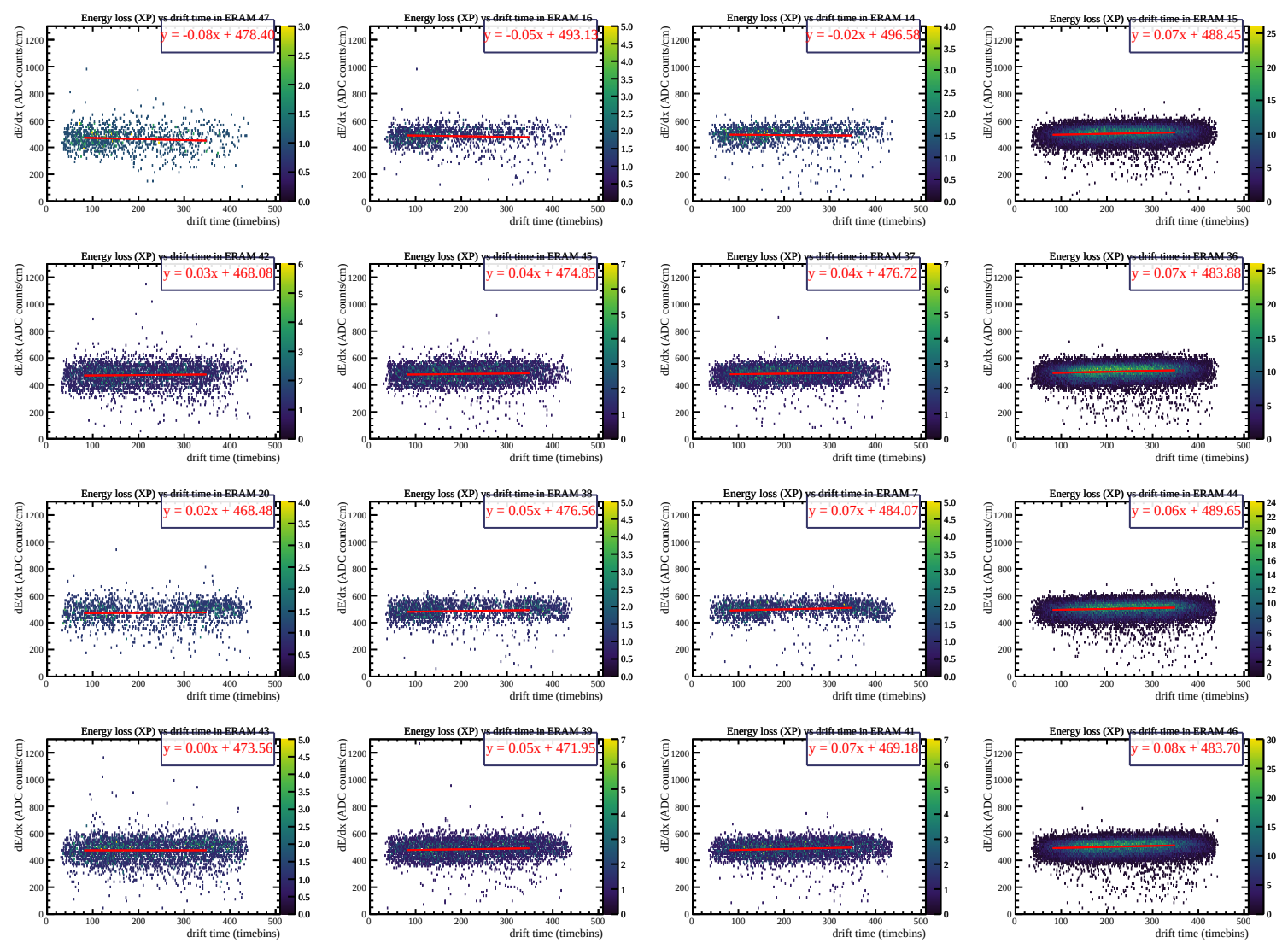


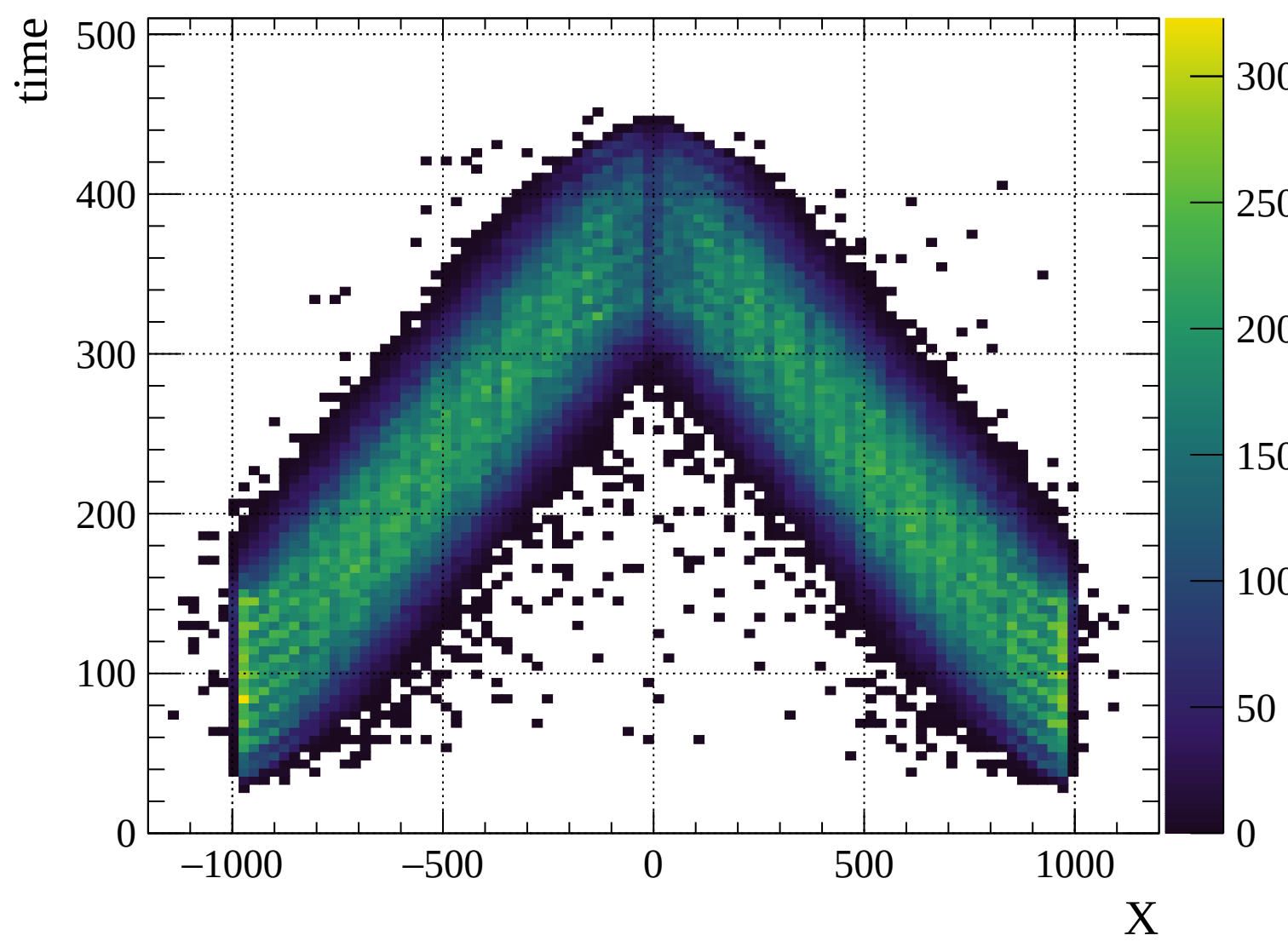


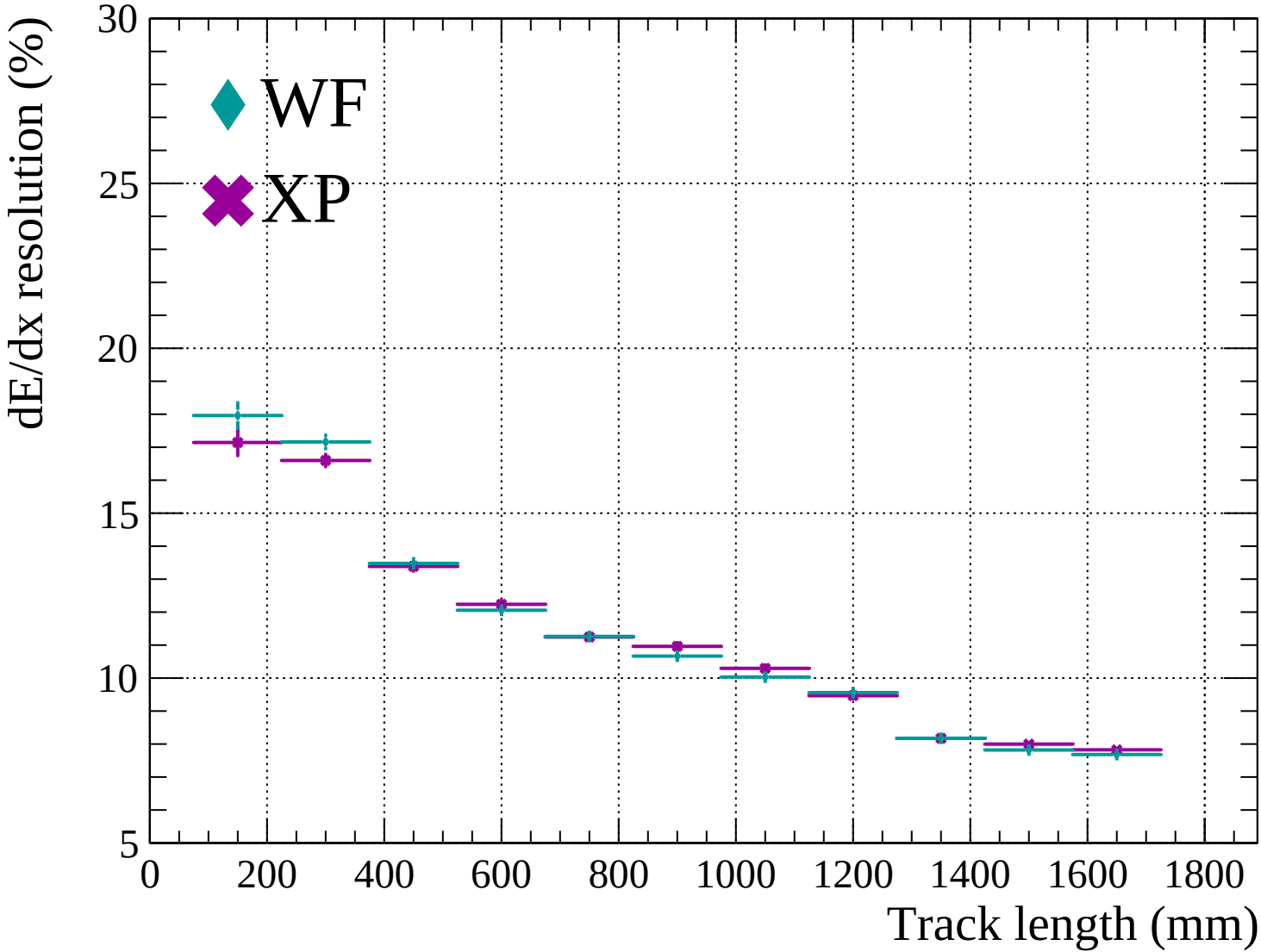


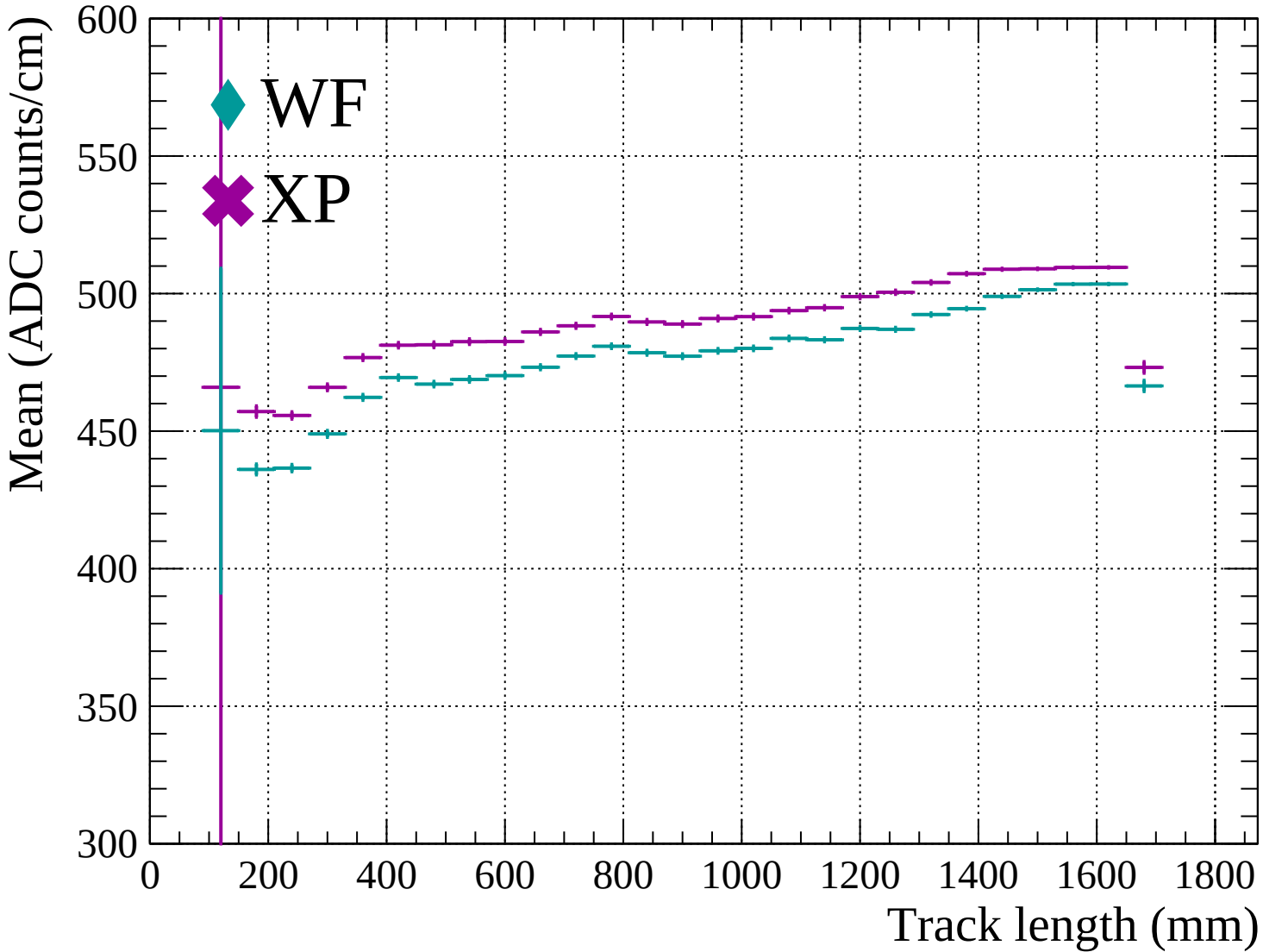


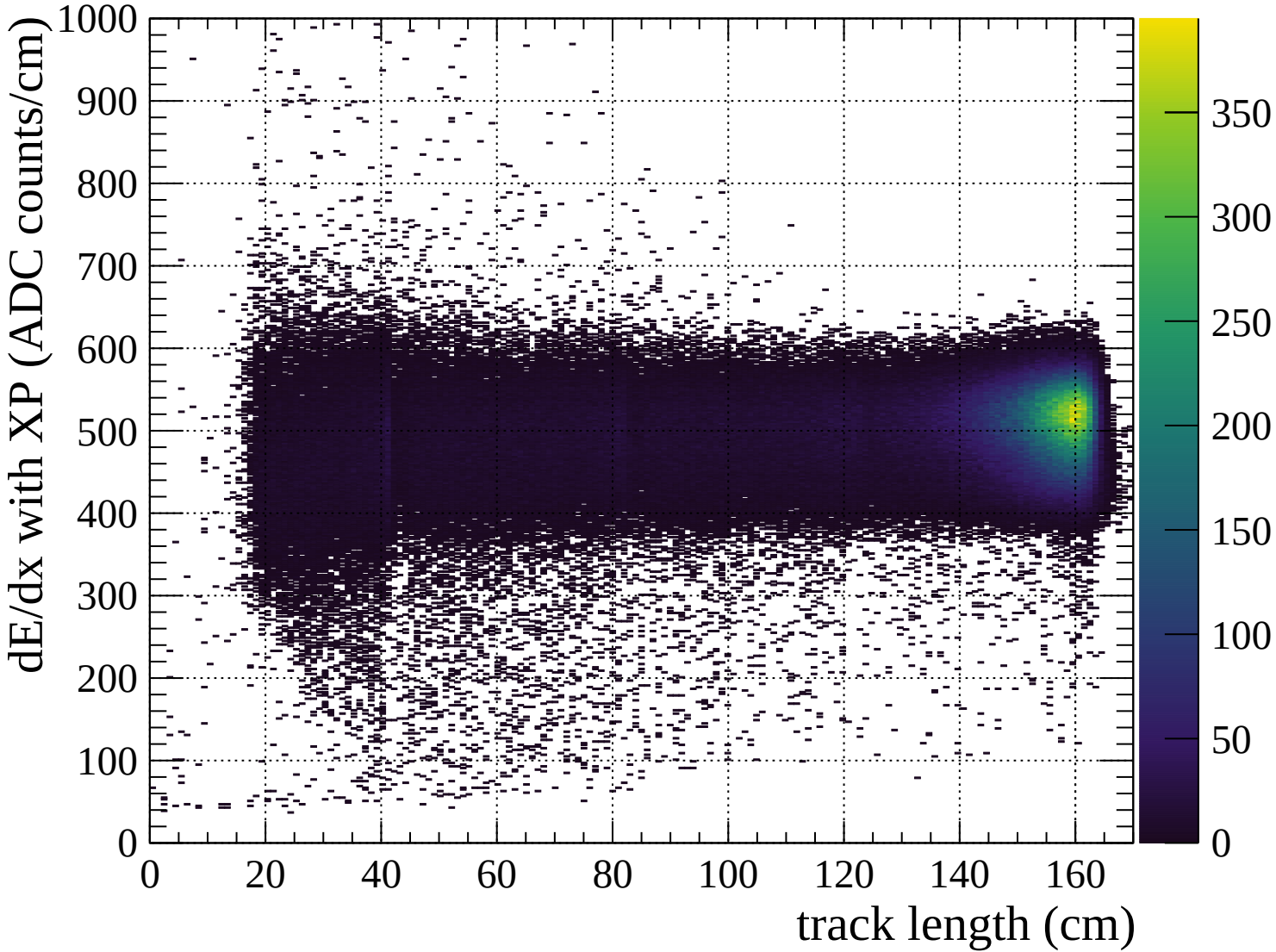


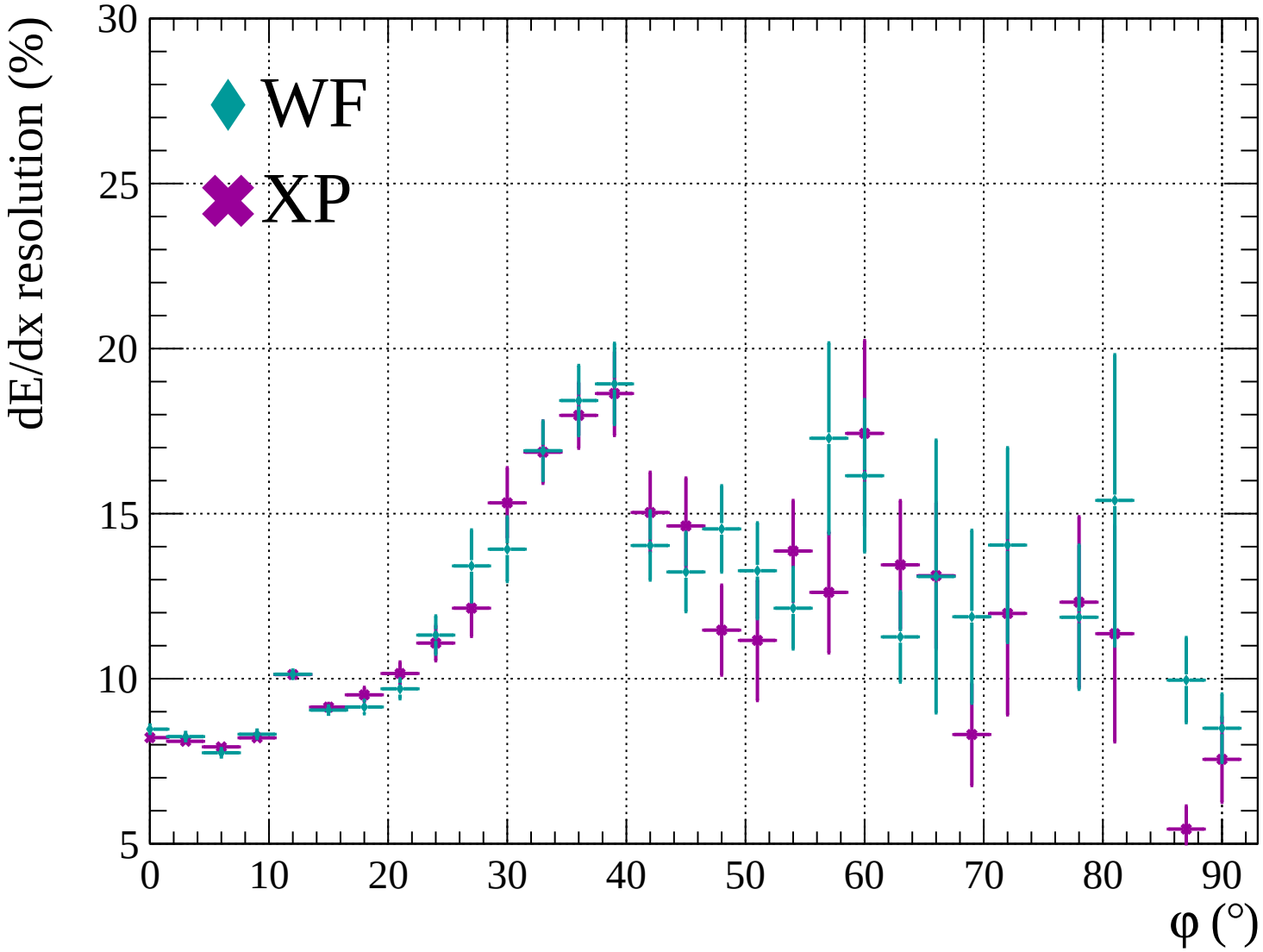


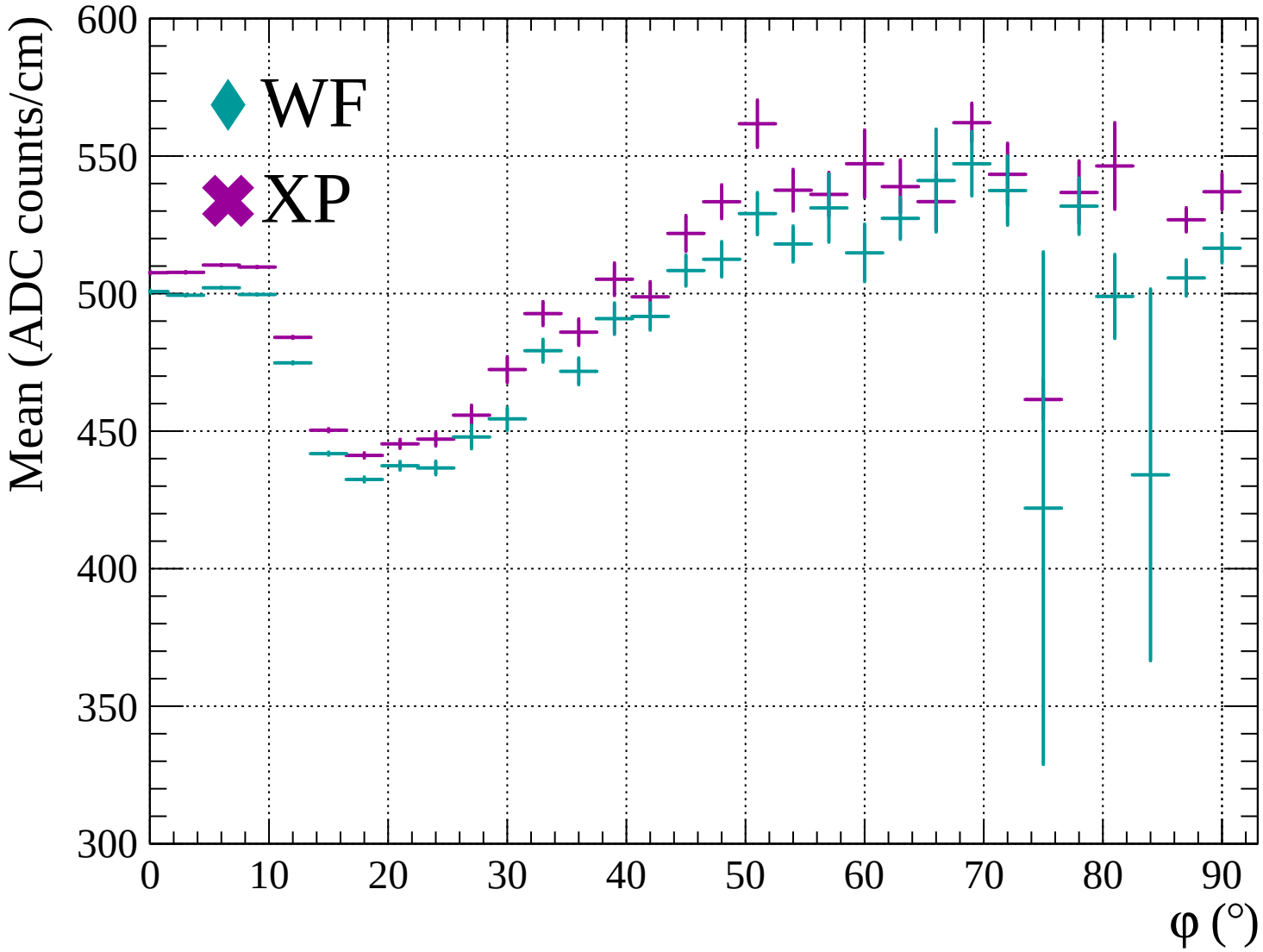


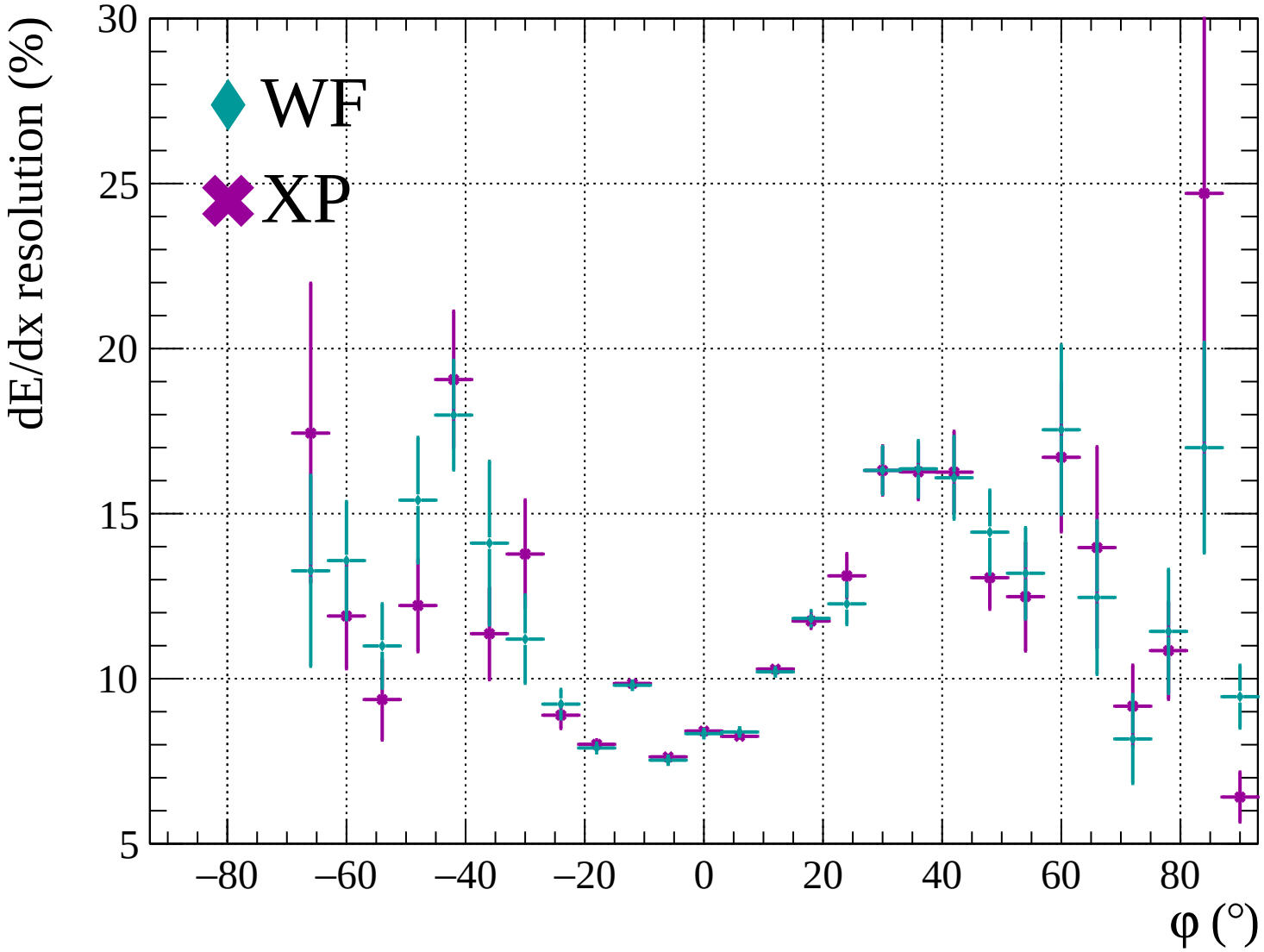


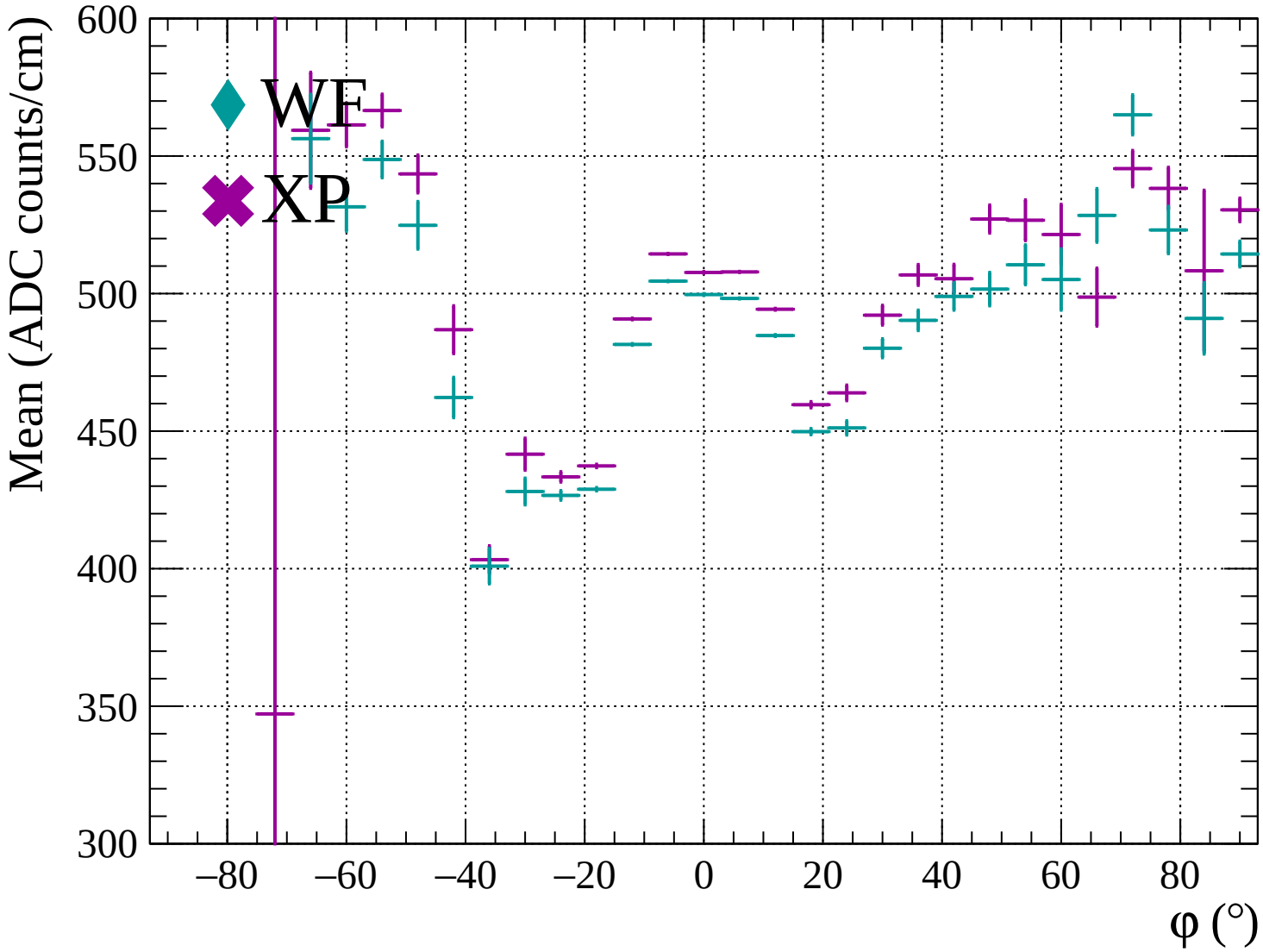


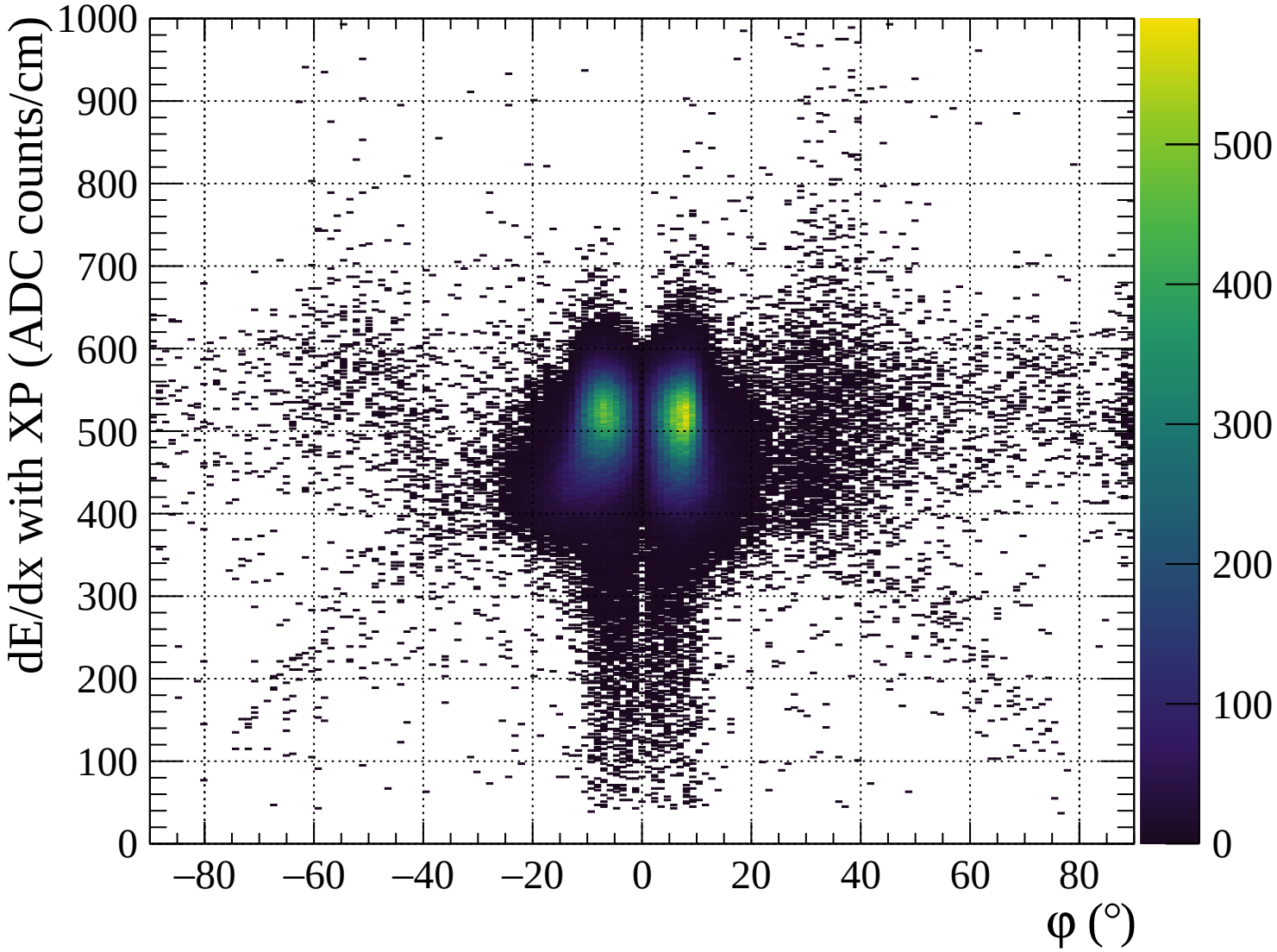


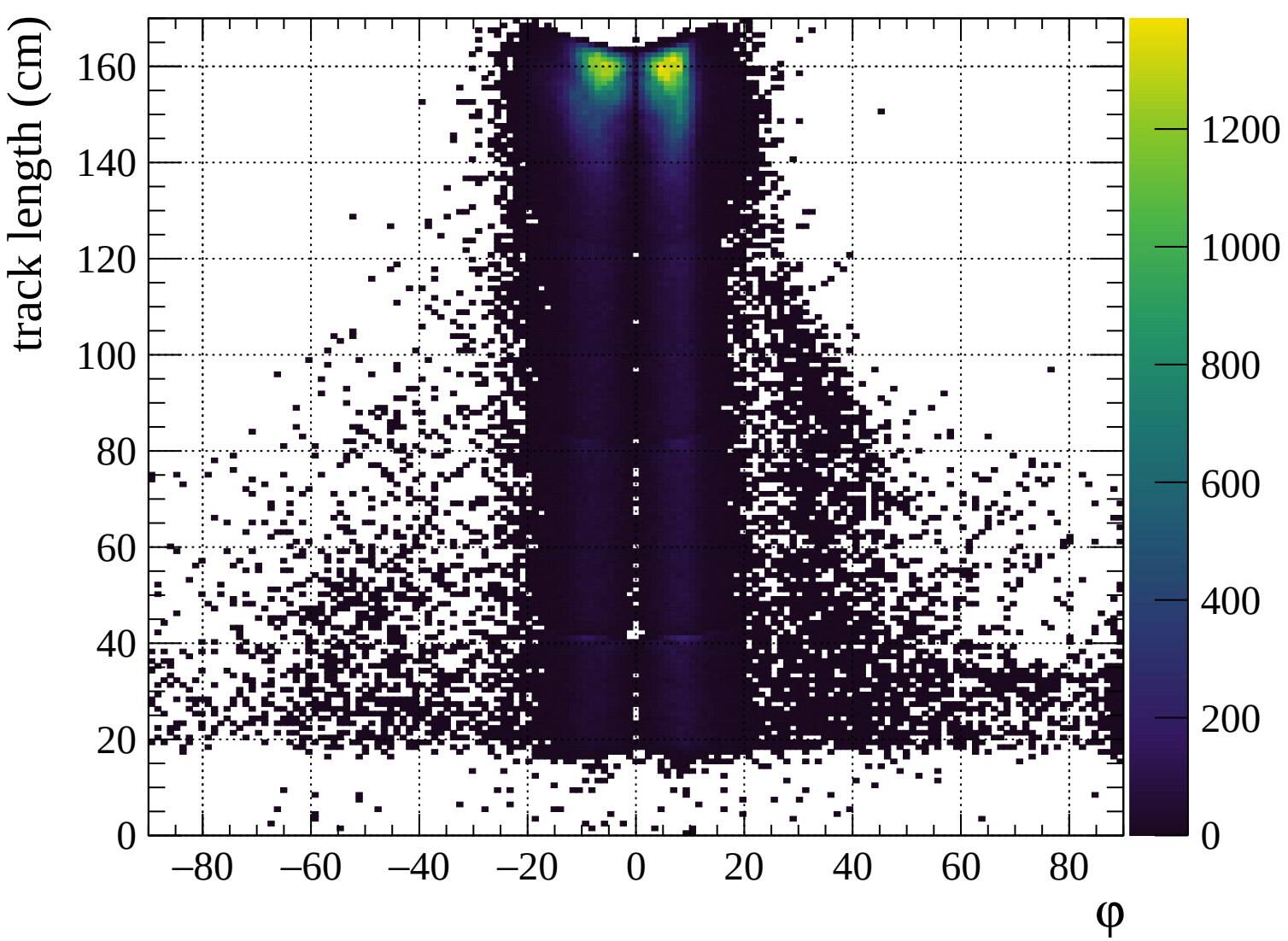


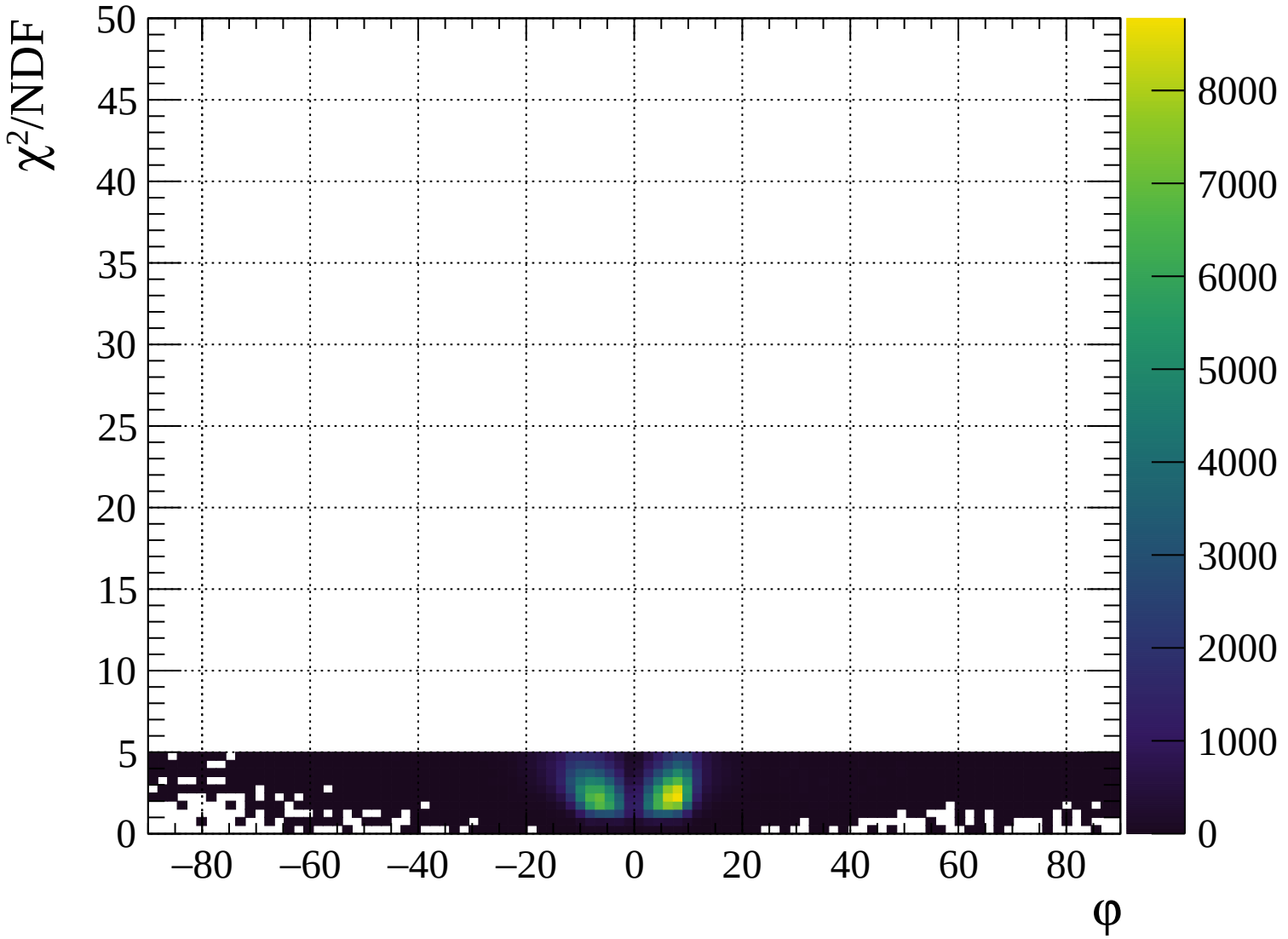


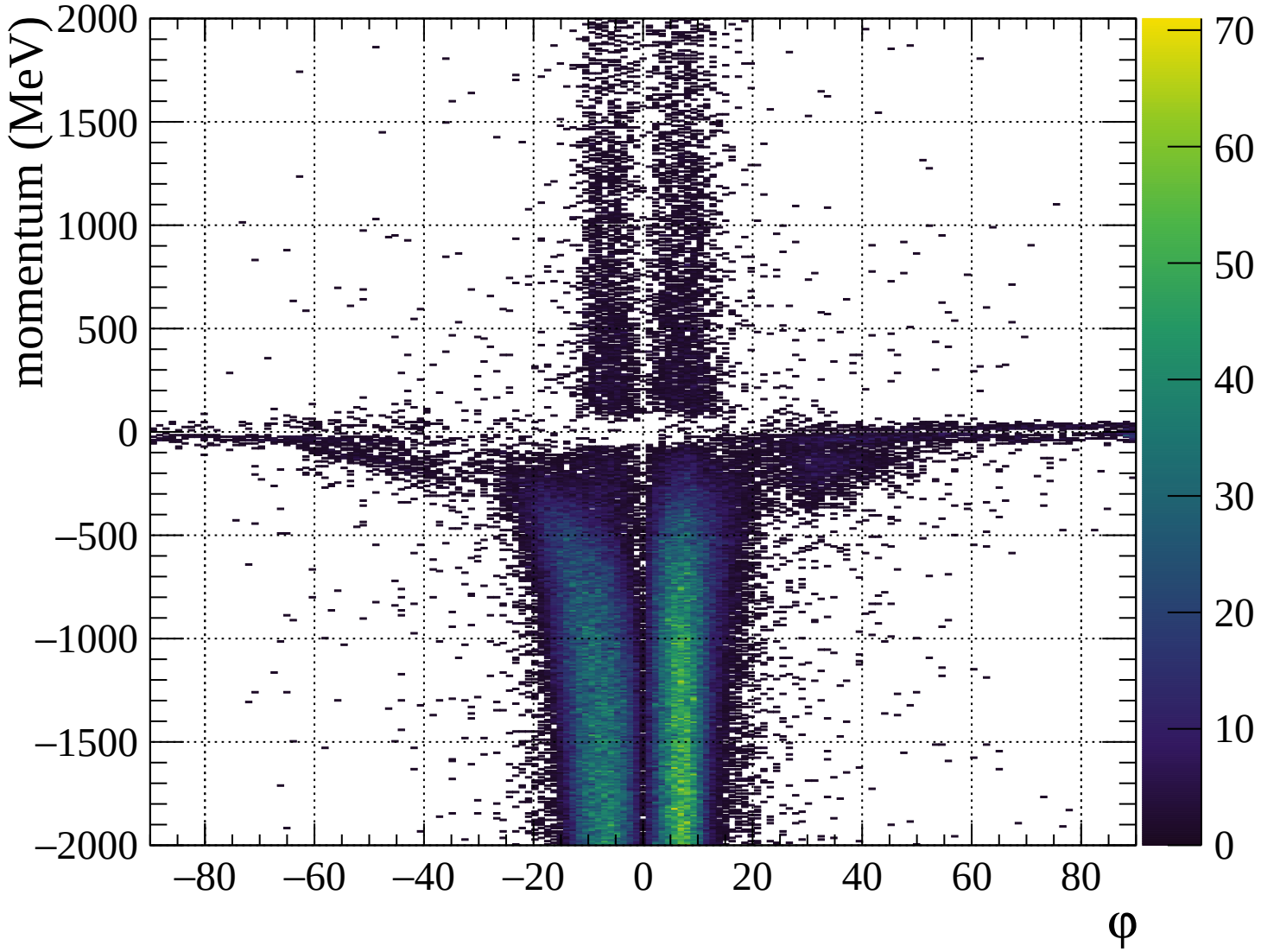


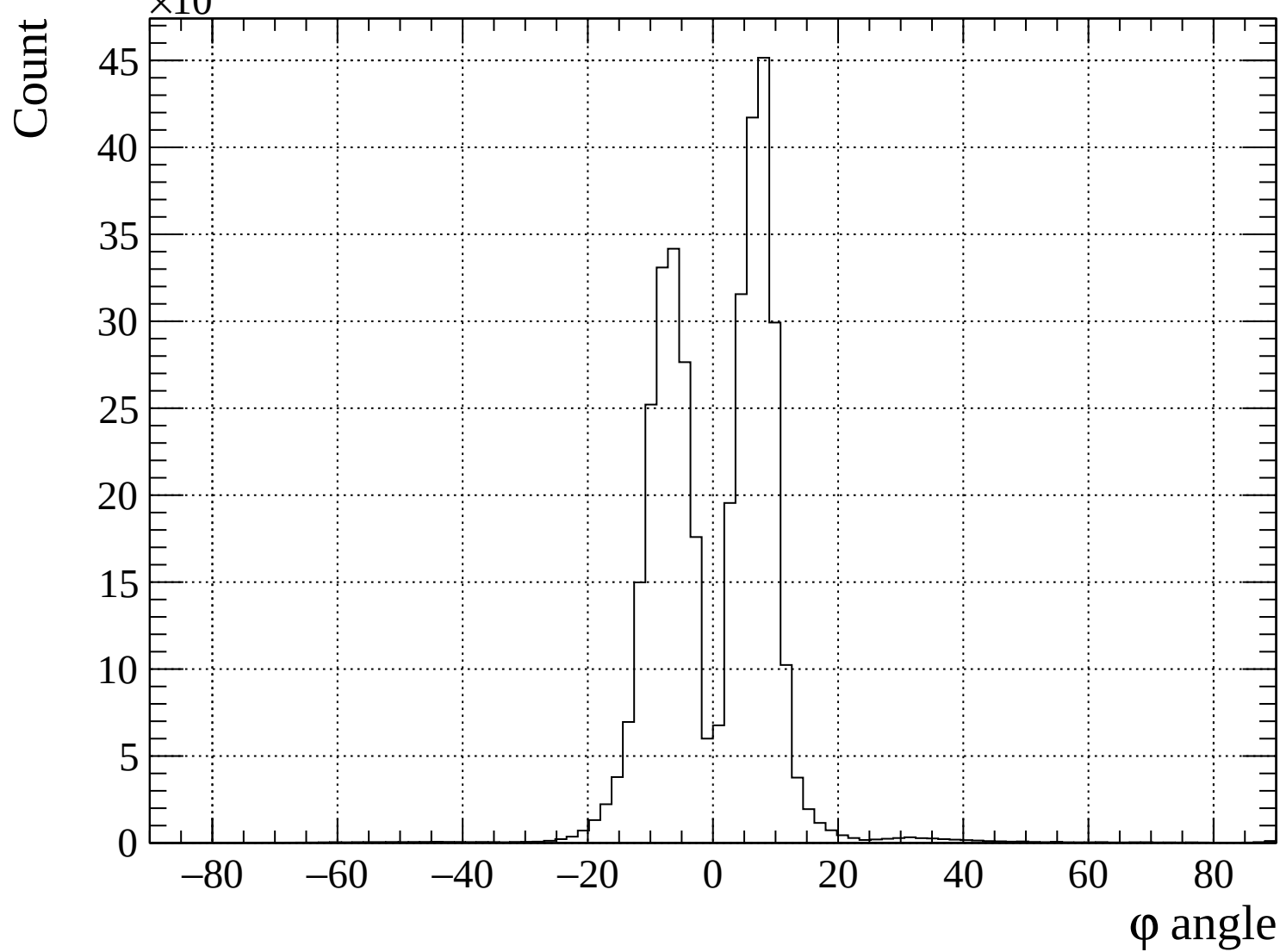


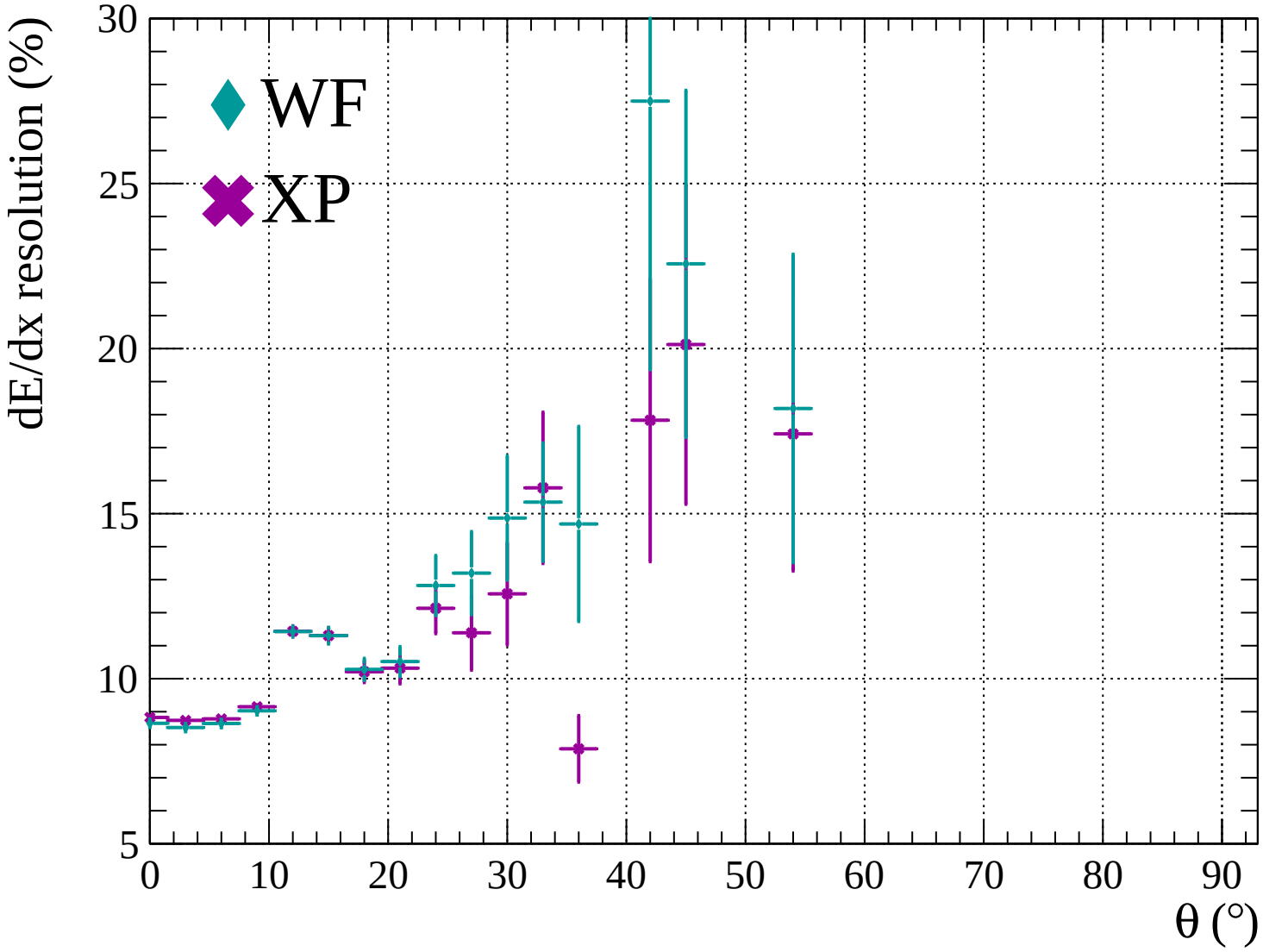


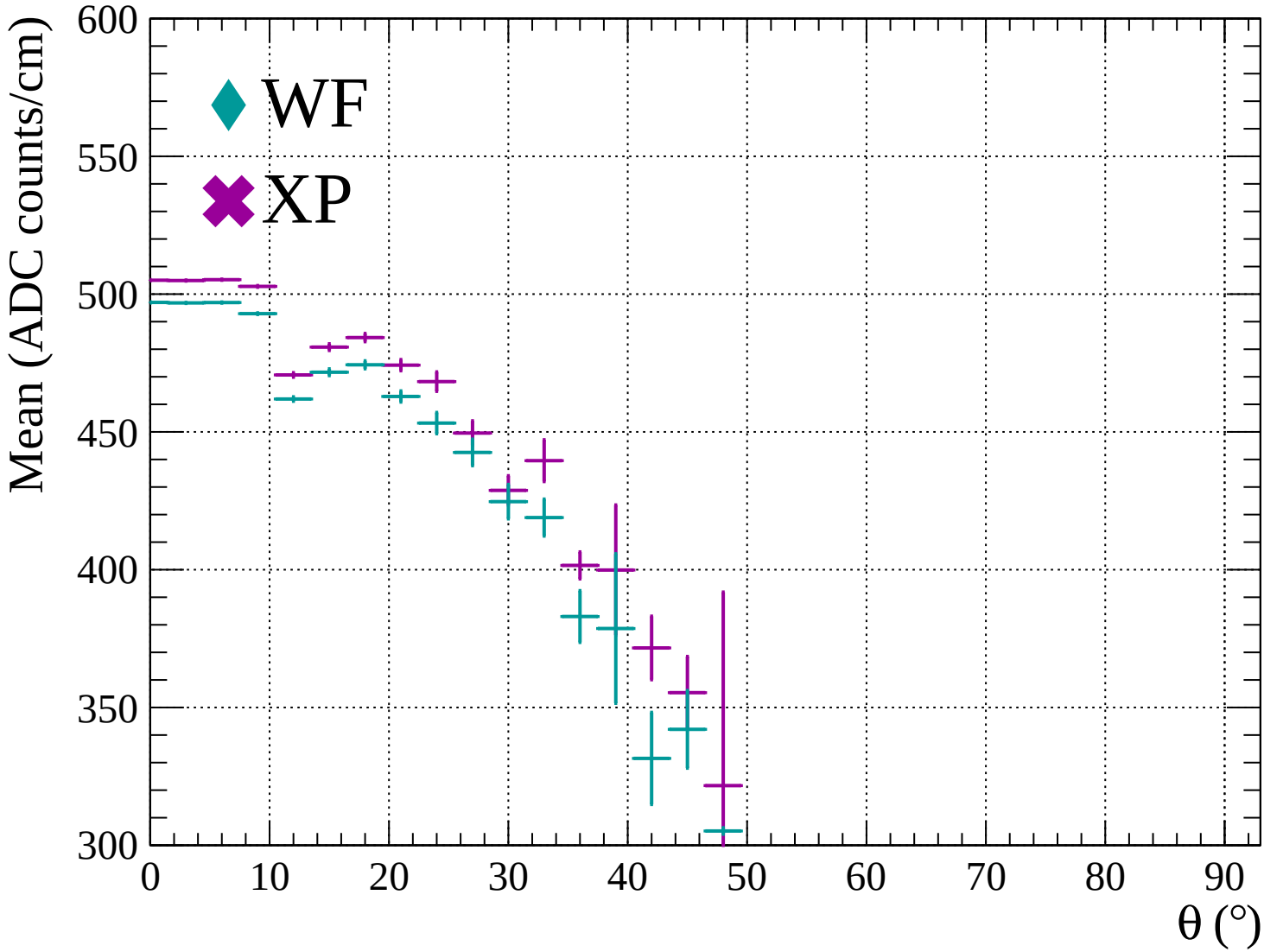


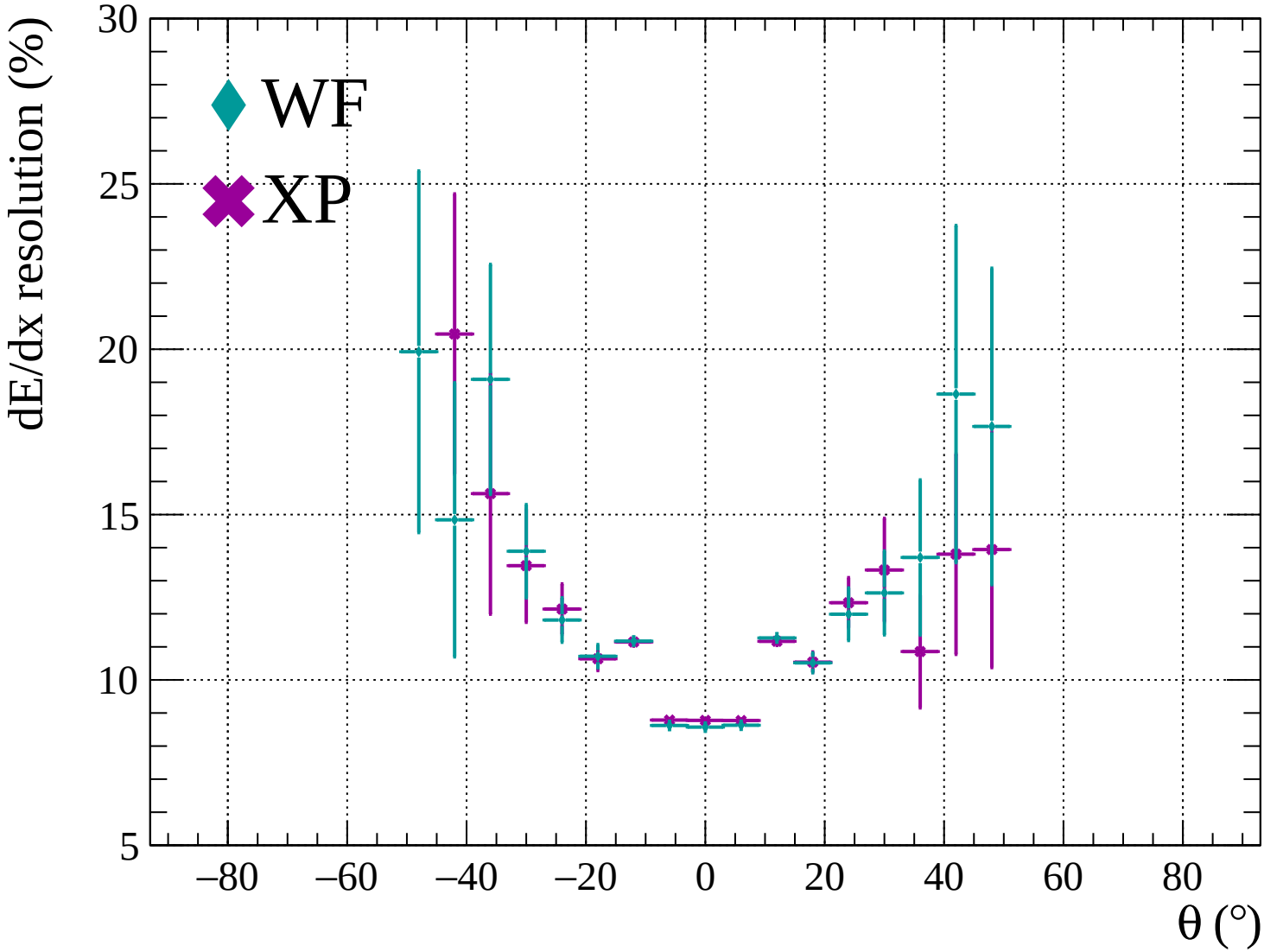


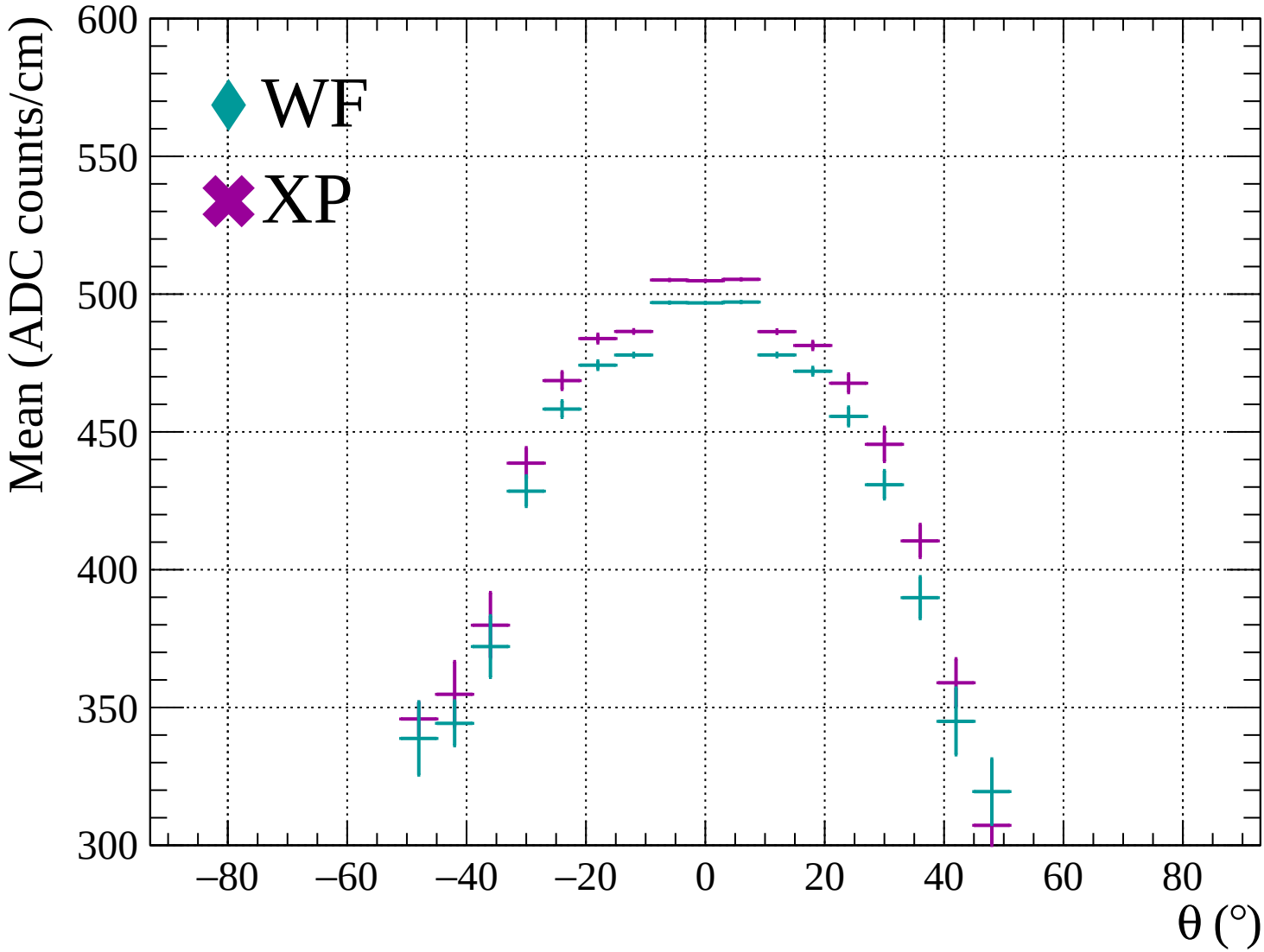


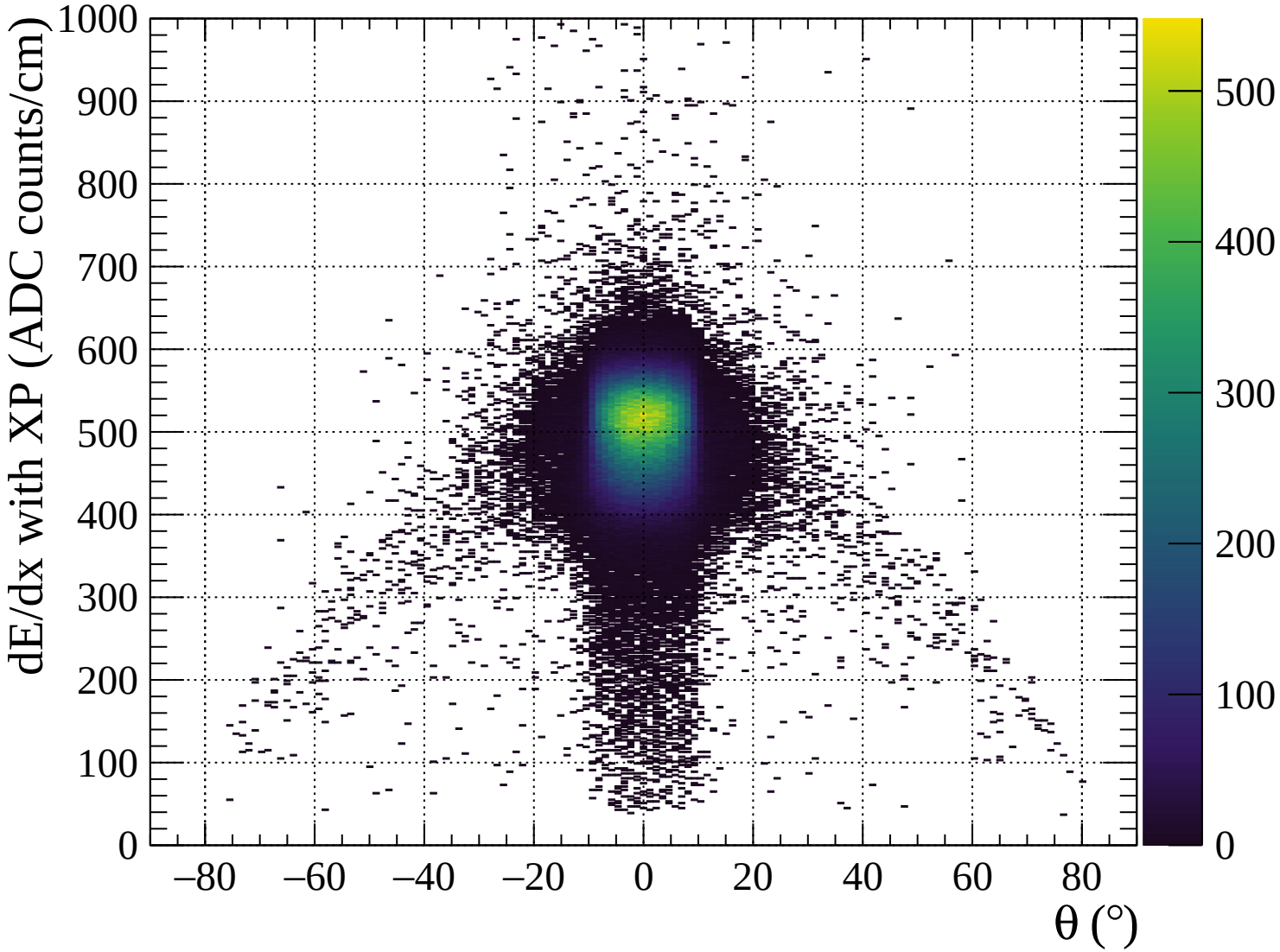


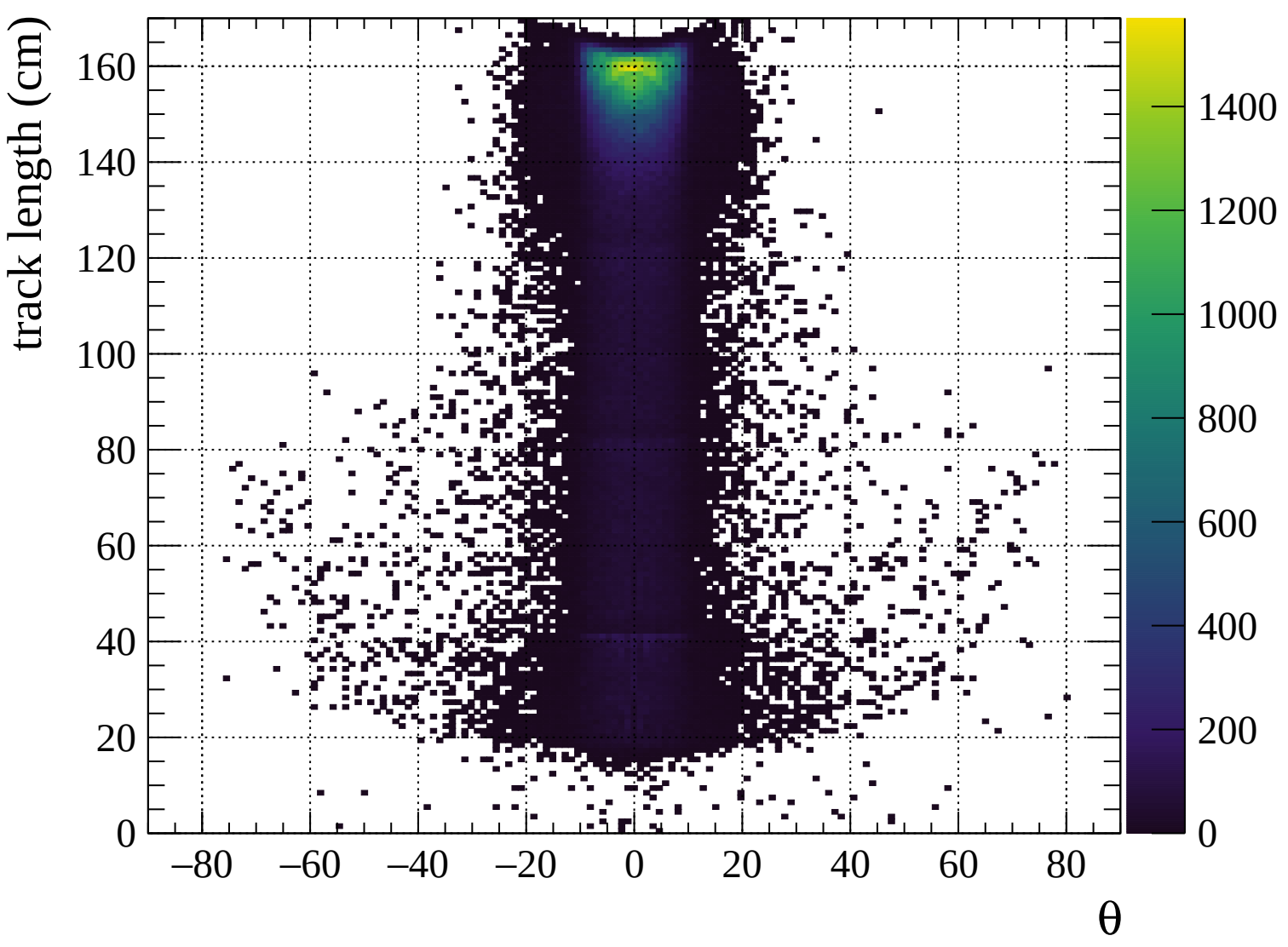


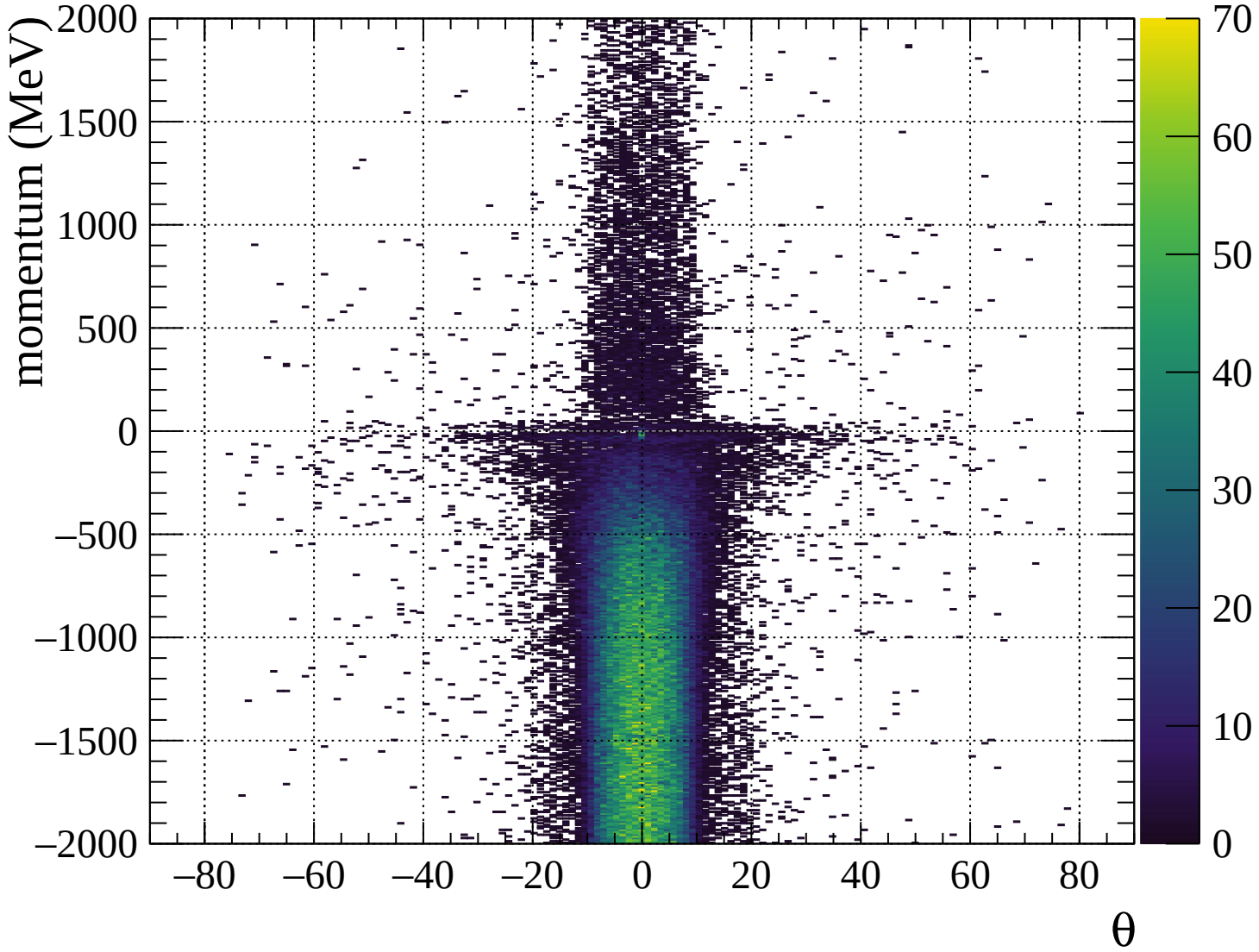


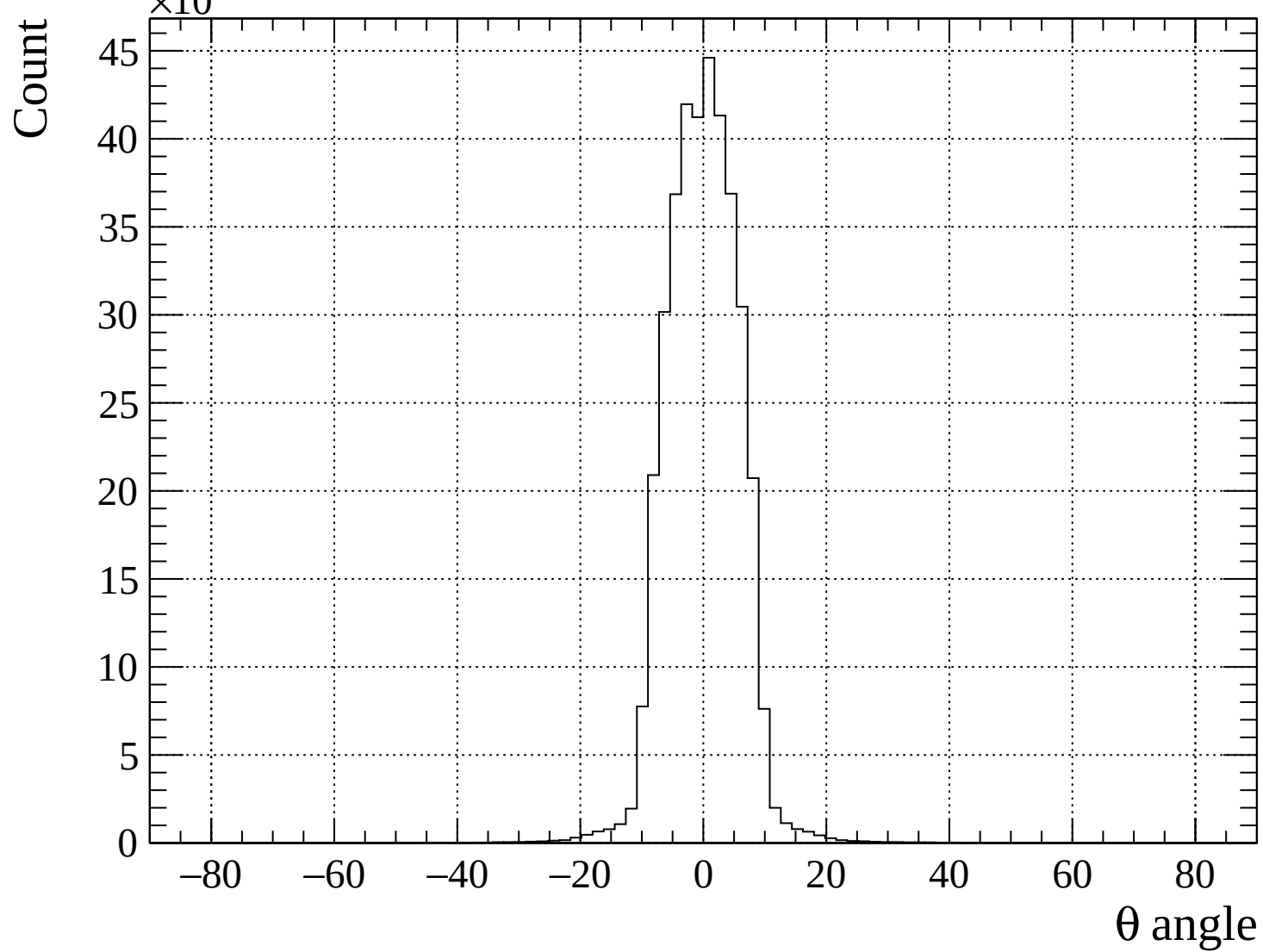


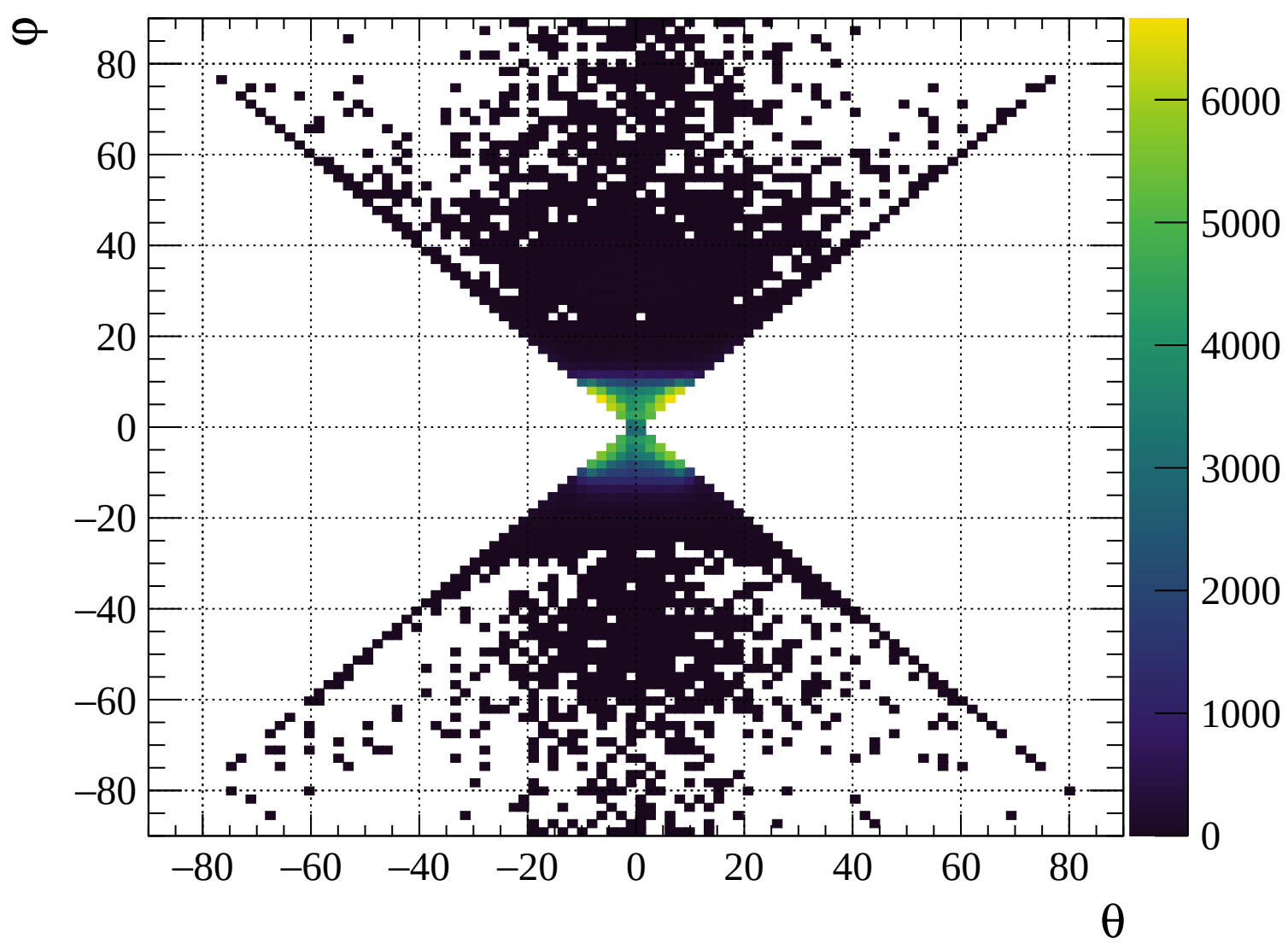


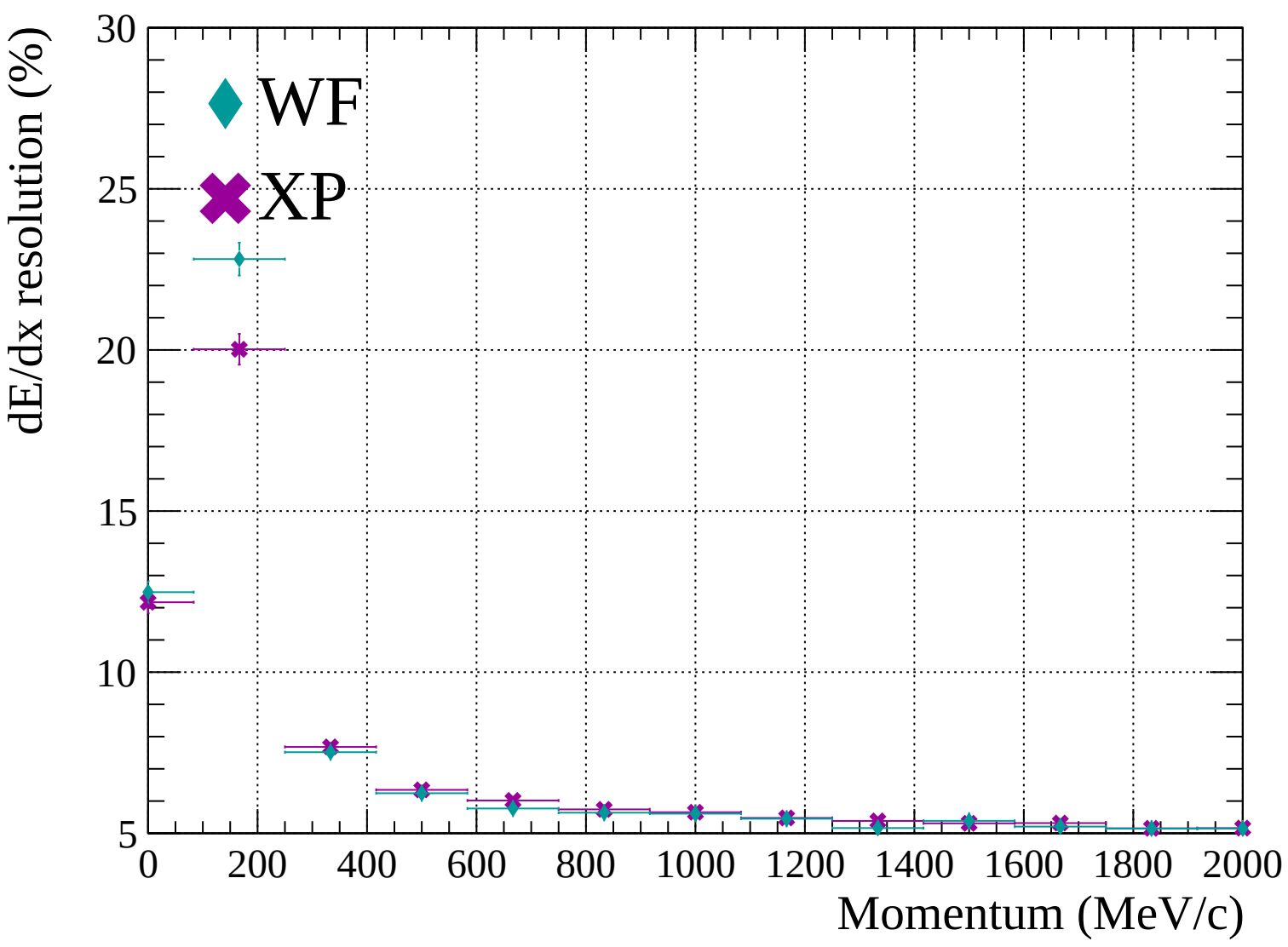


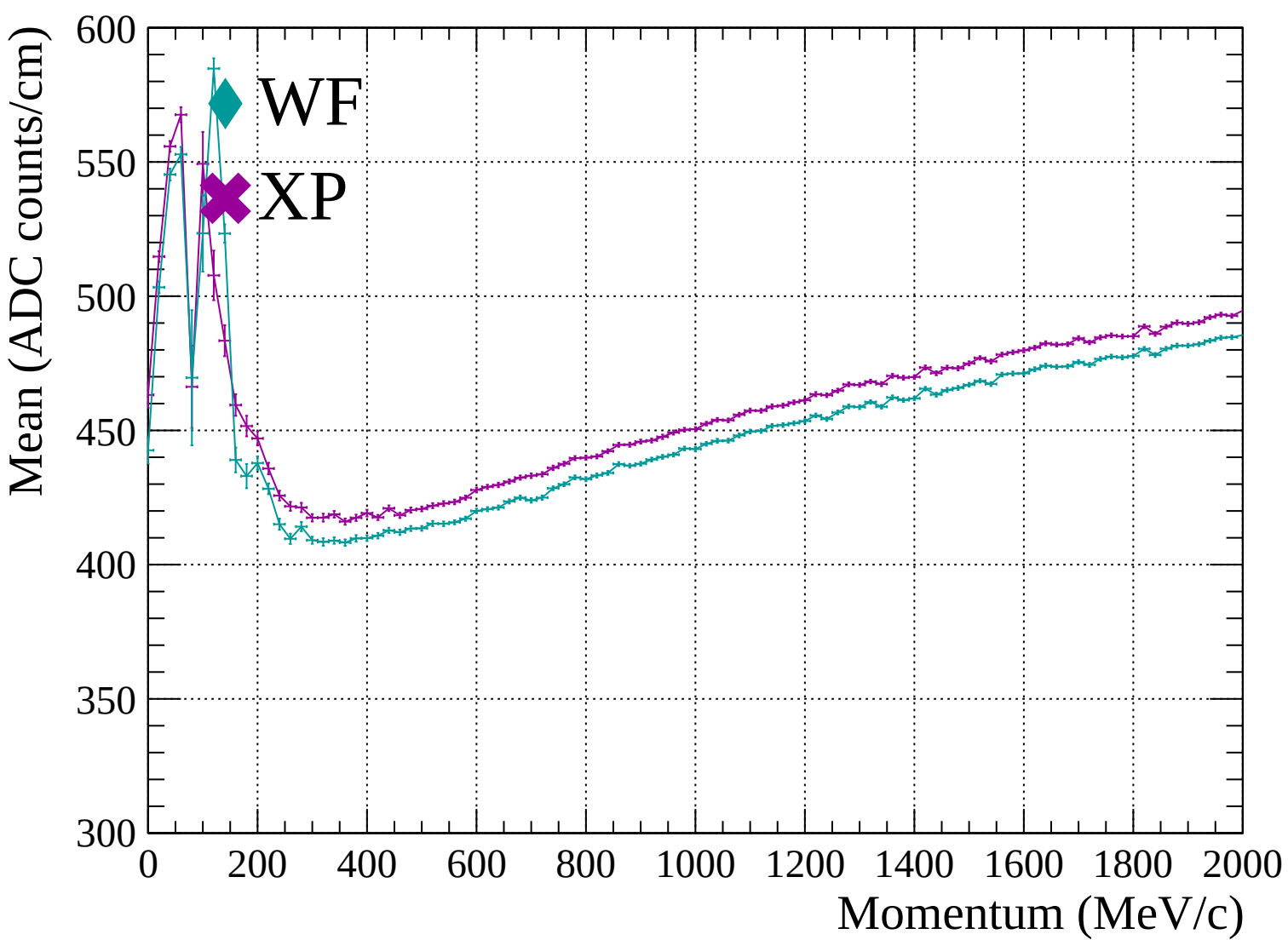


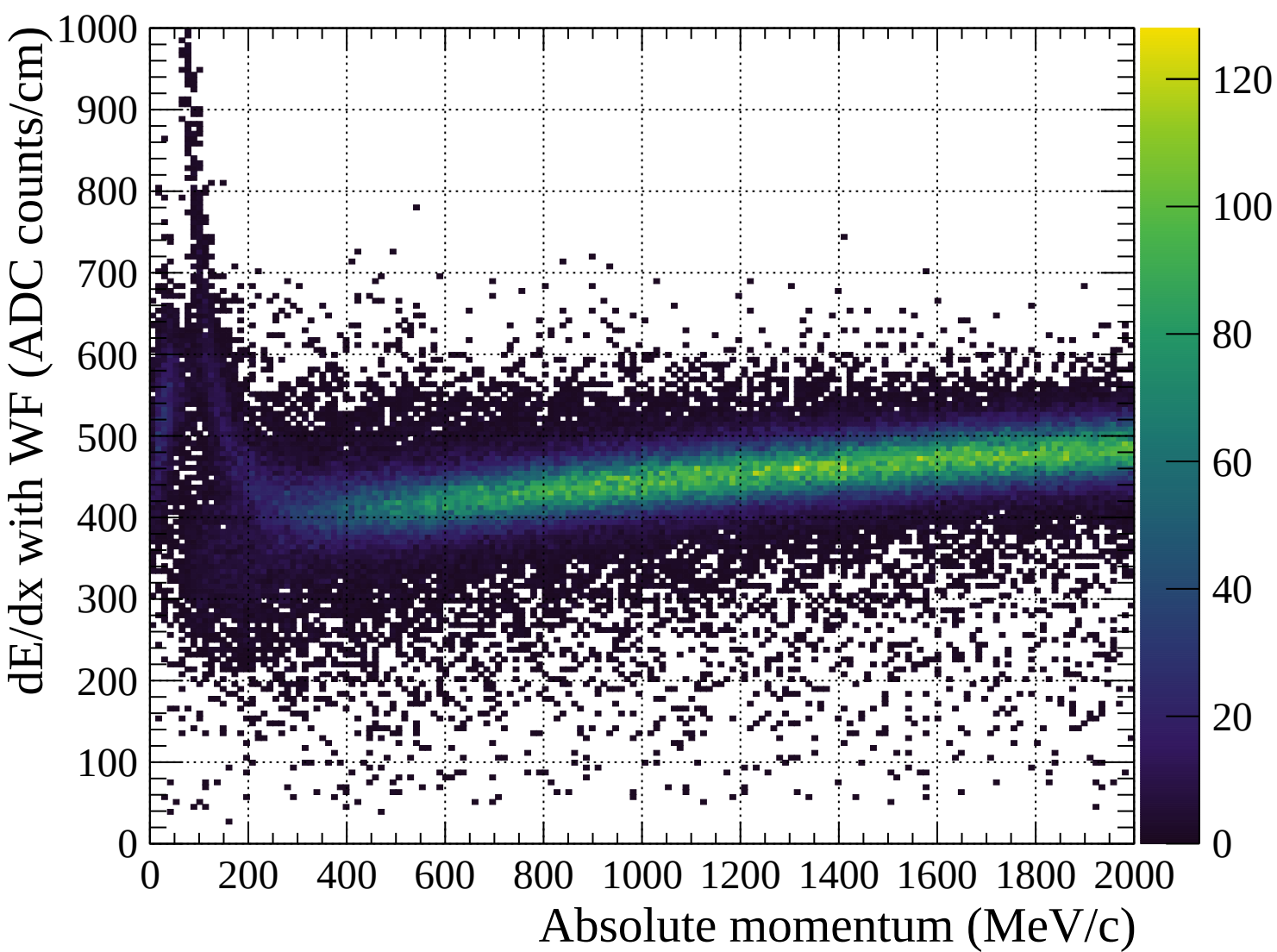


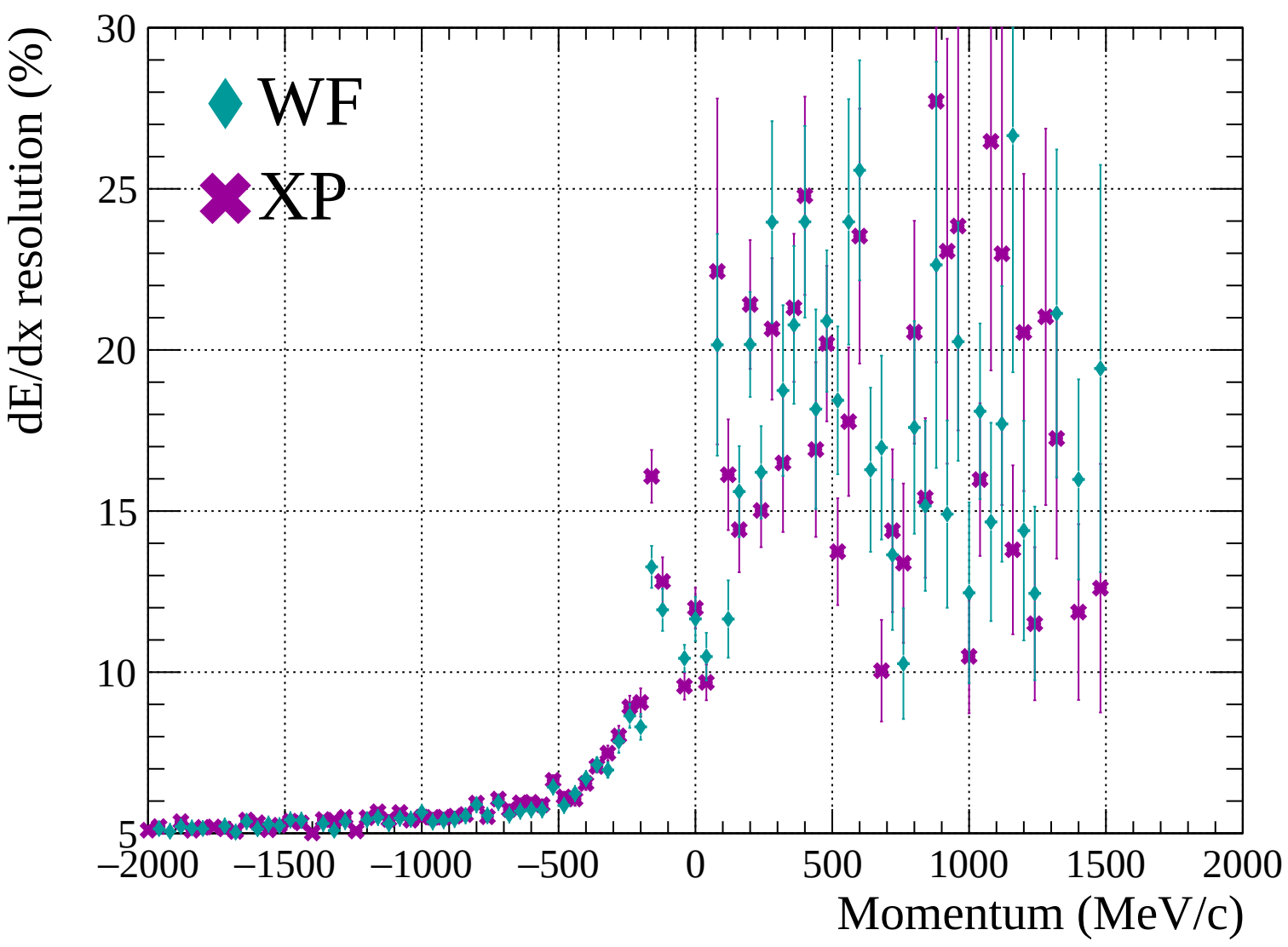


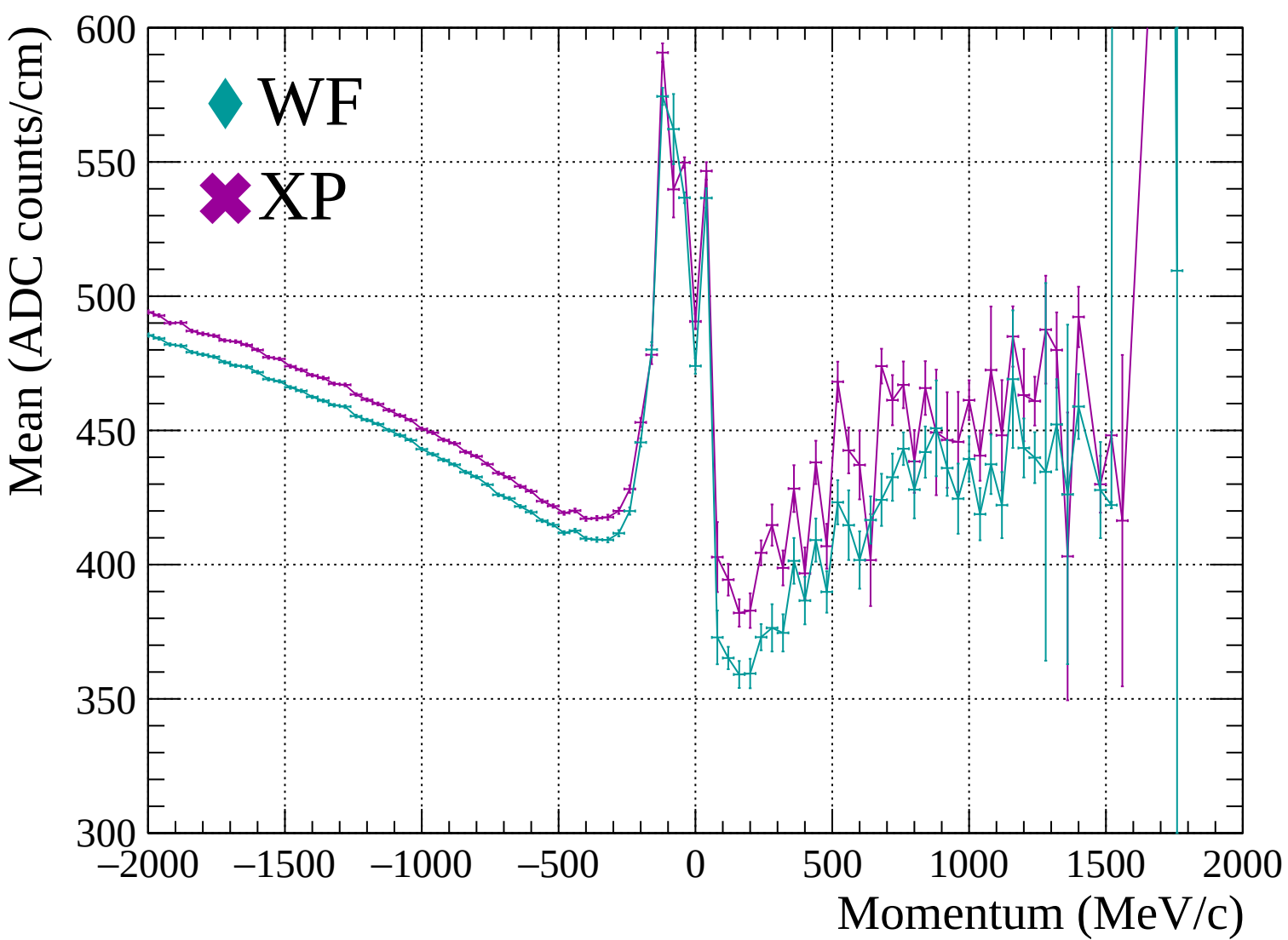


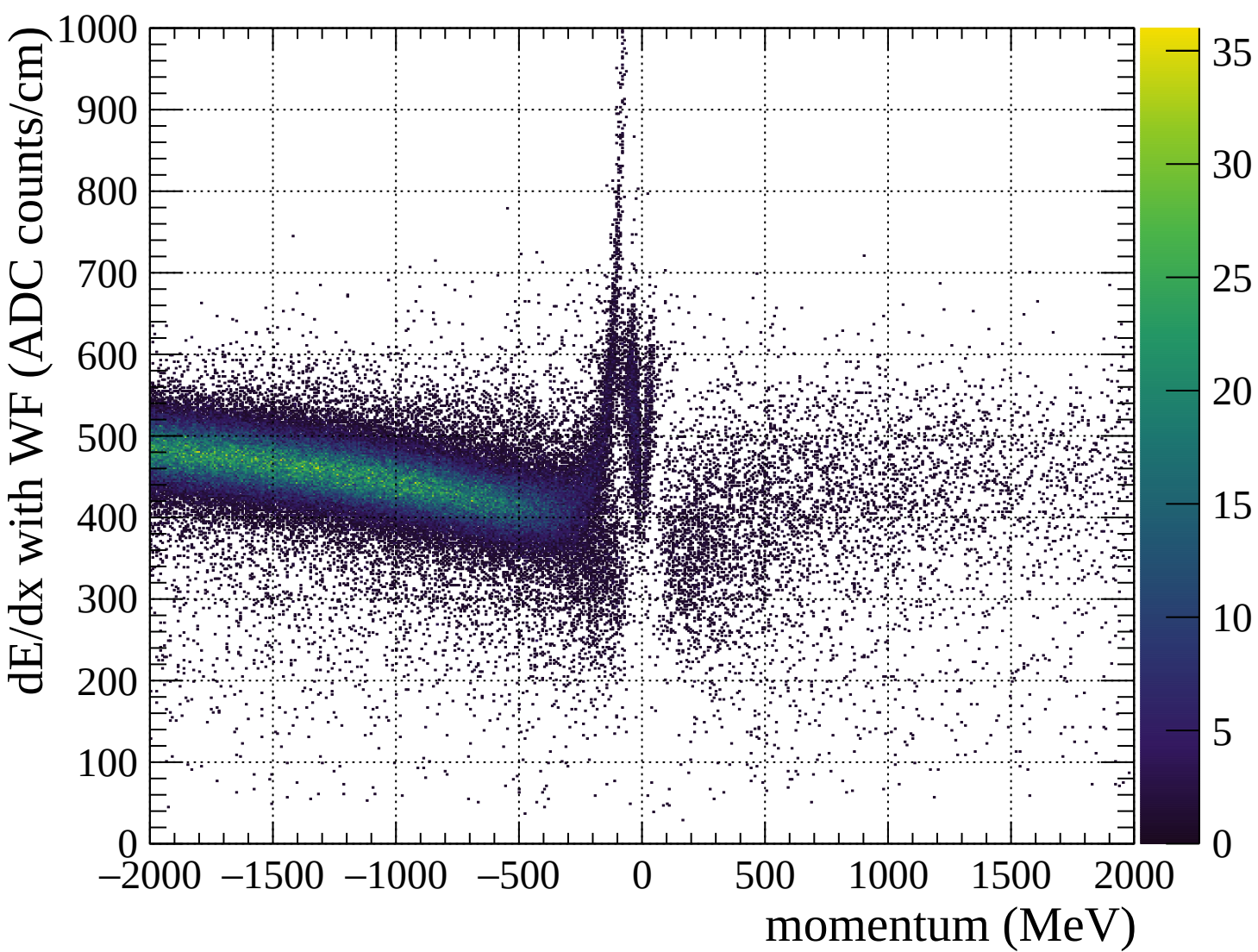


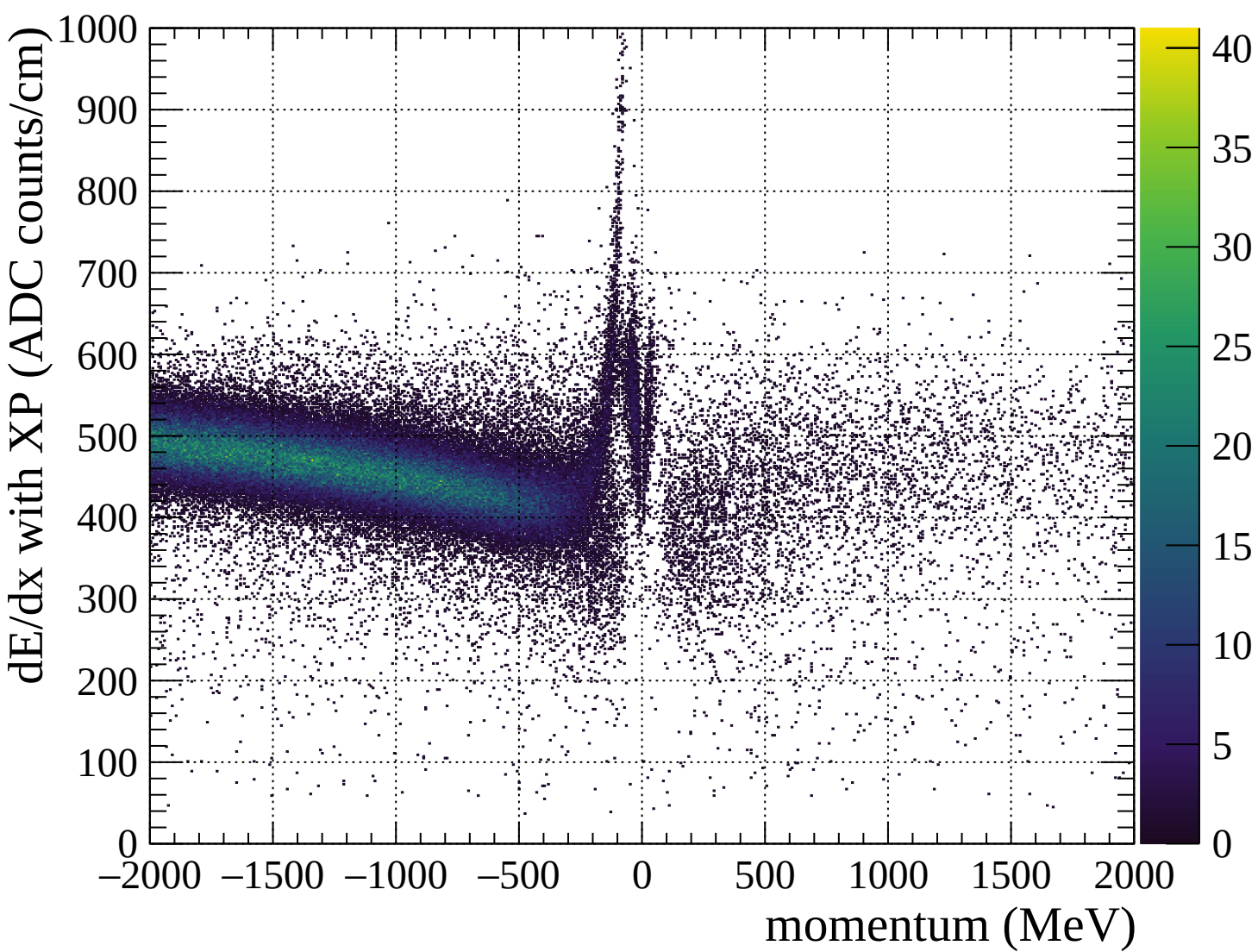


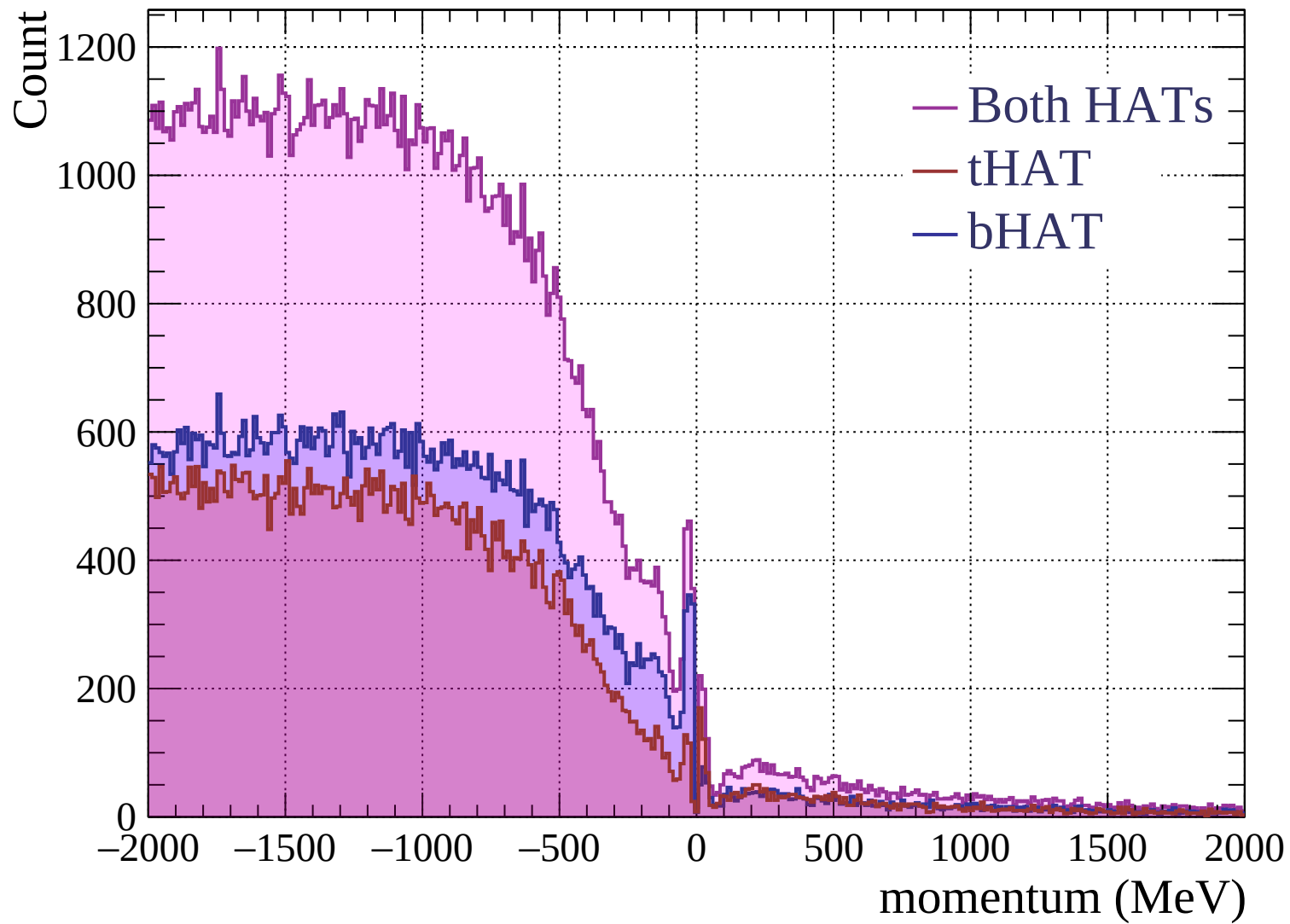


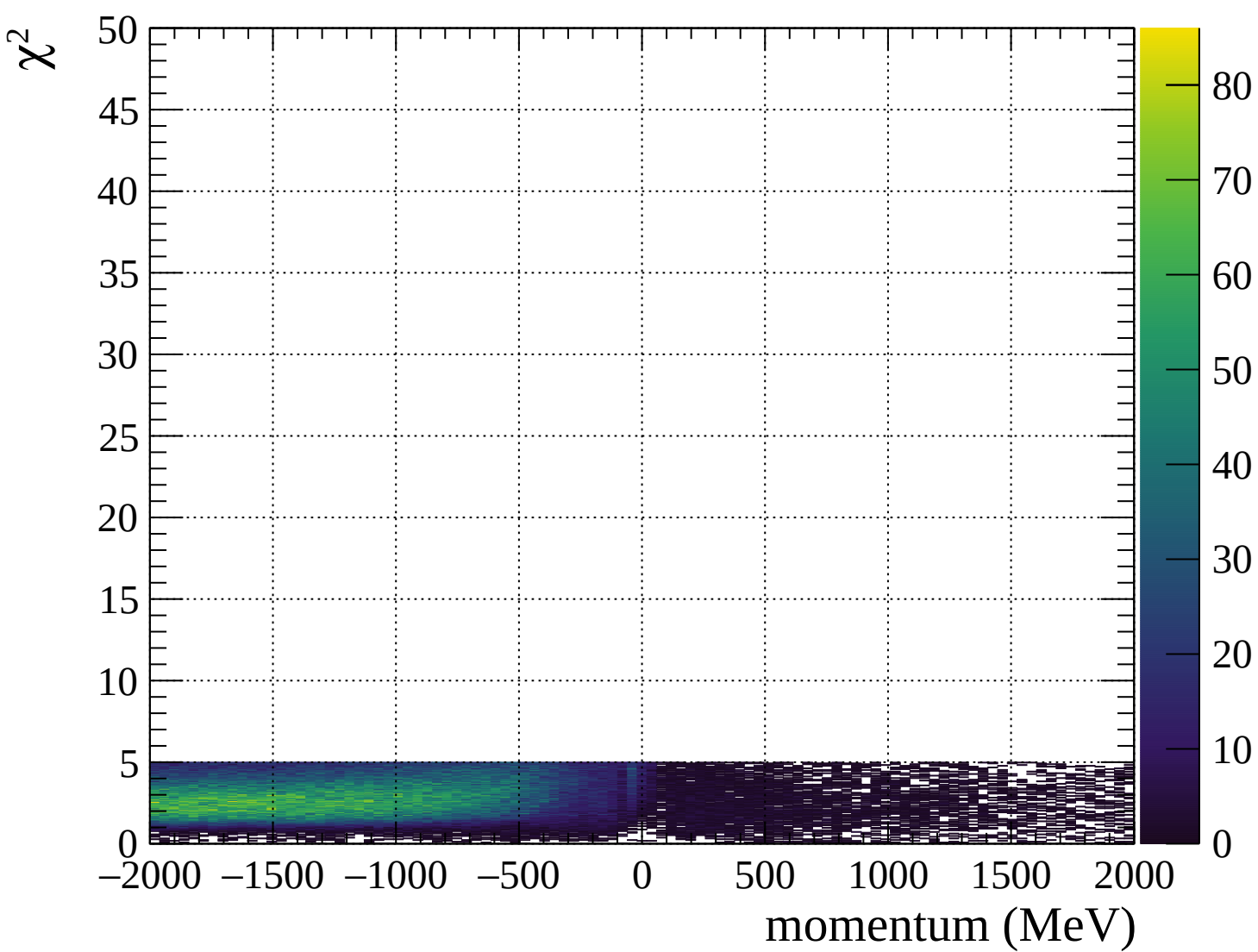




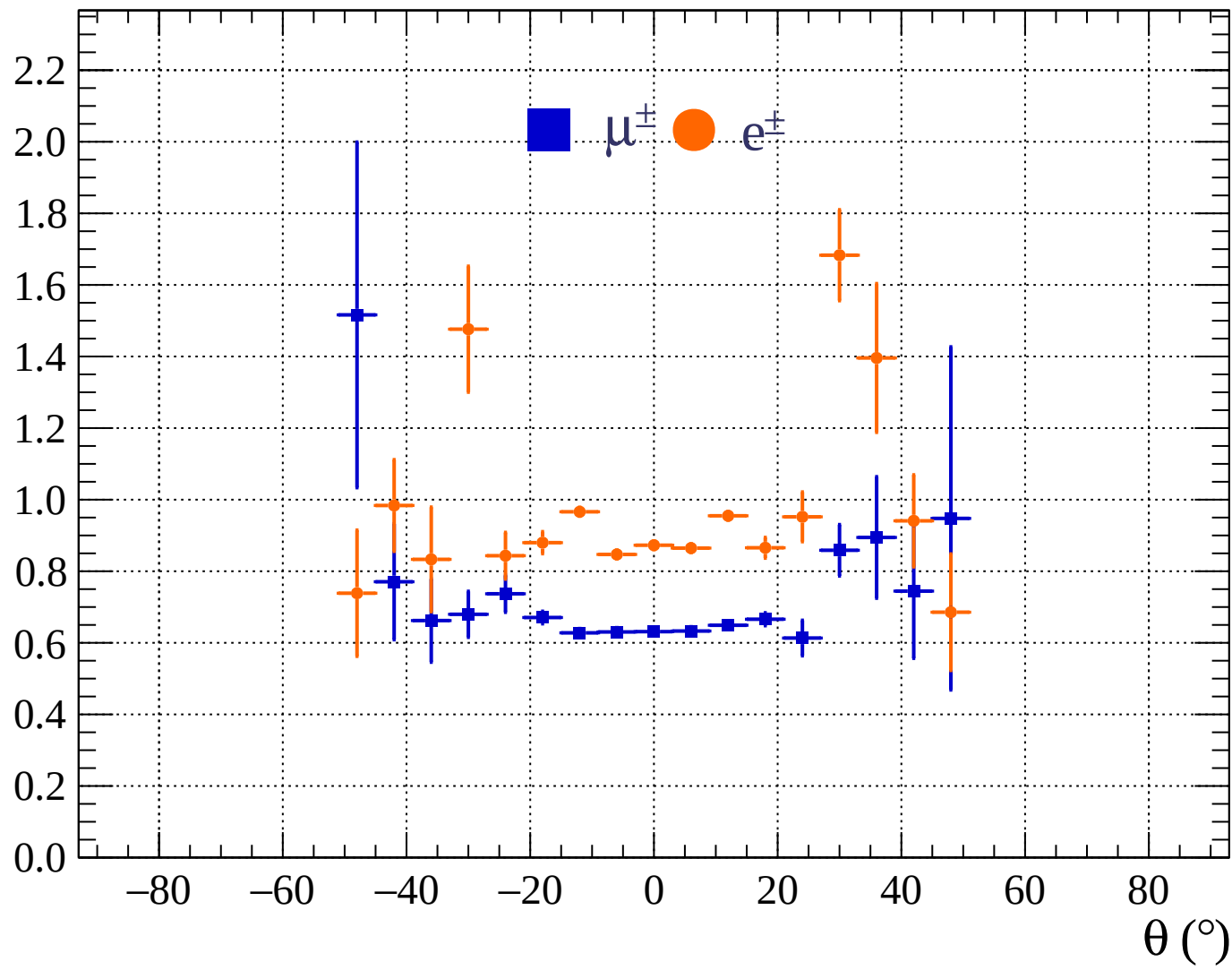


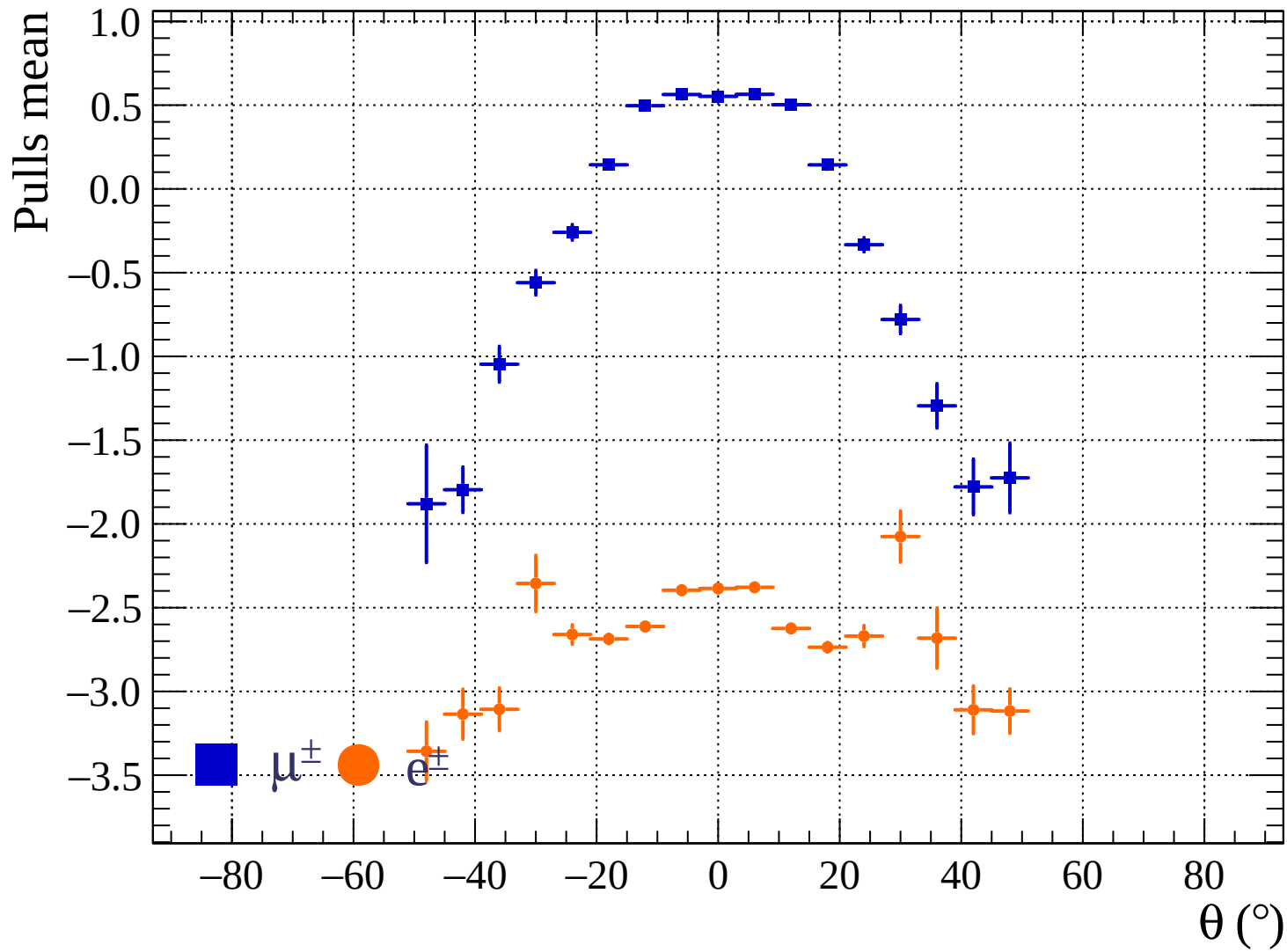




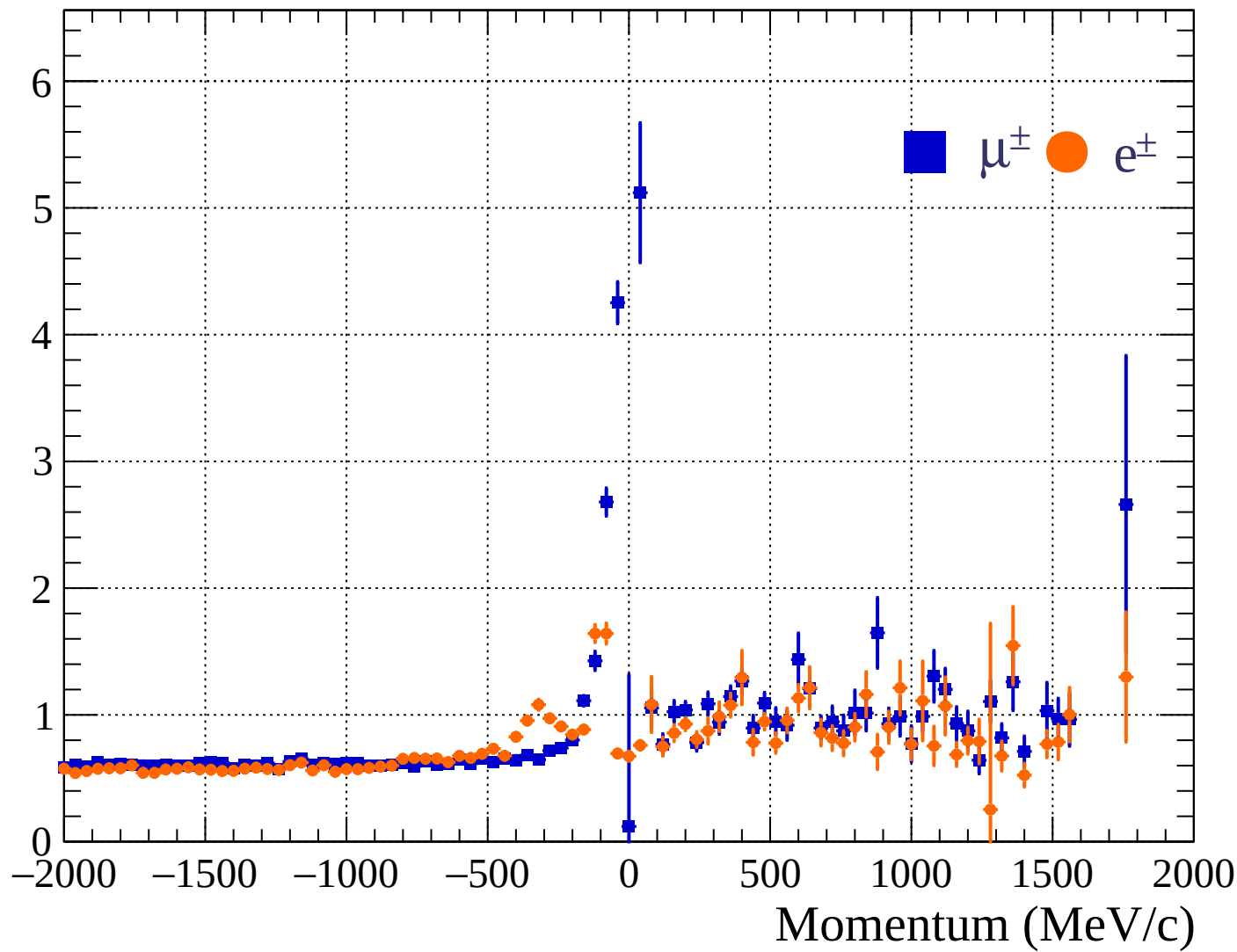


Pulls standard deviation

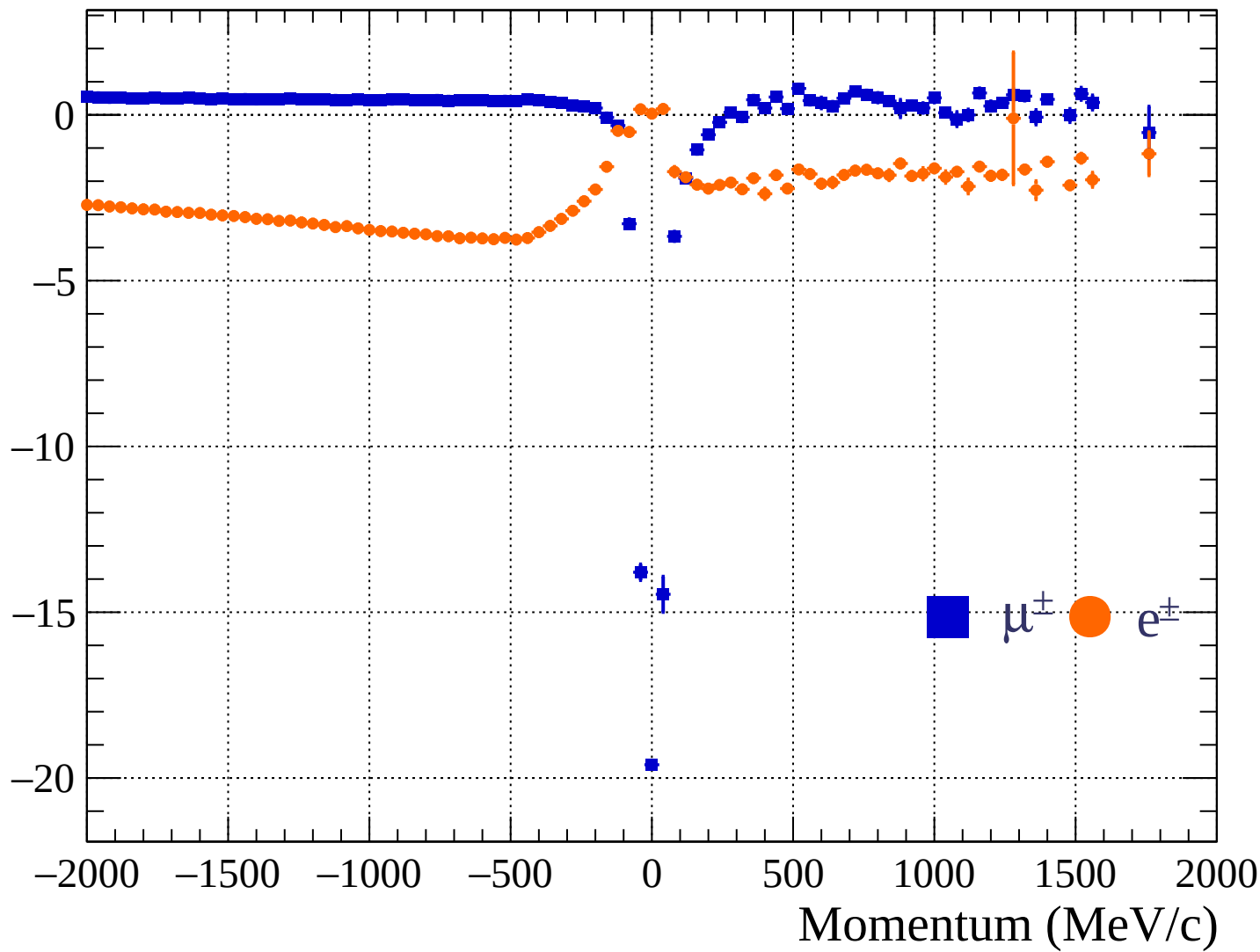


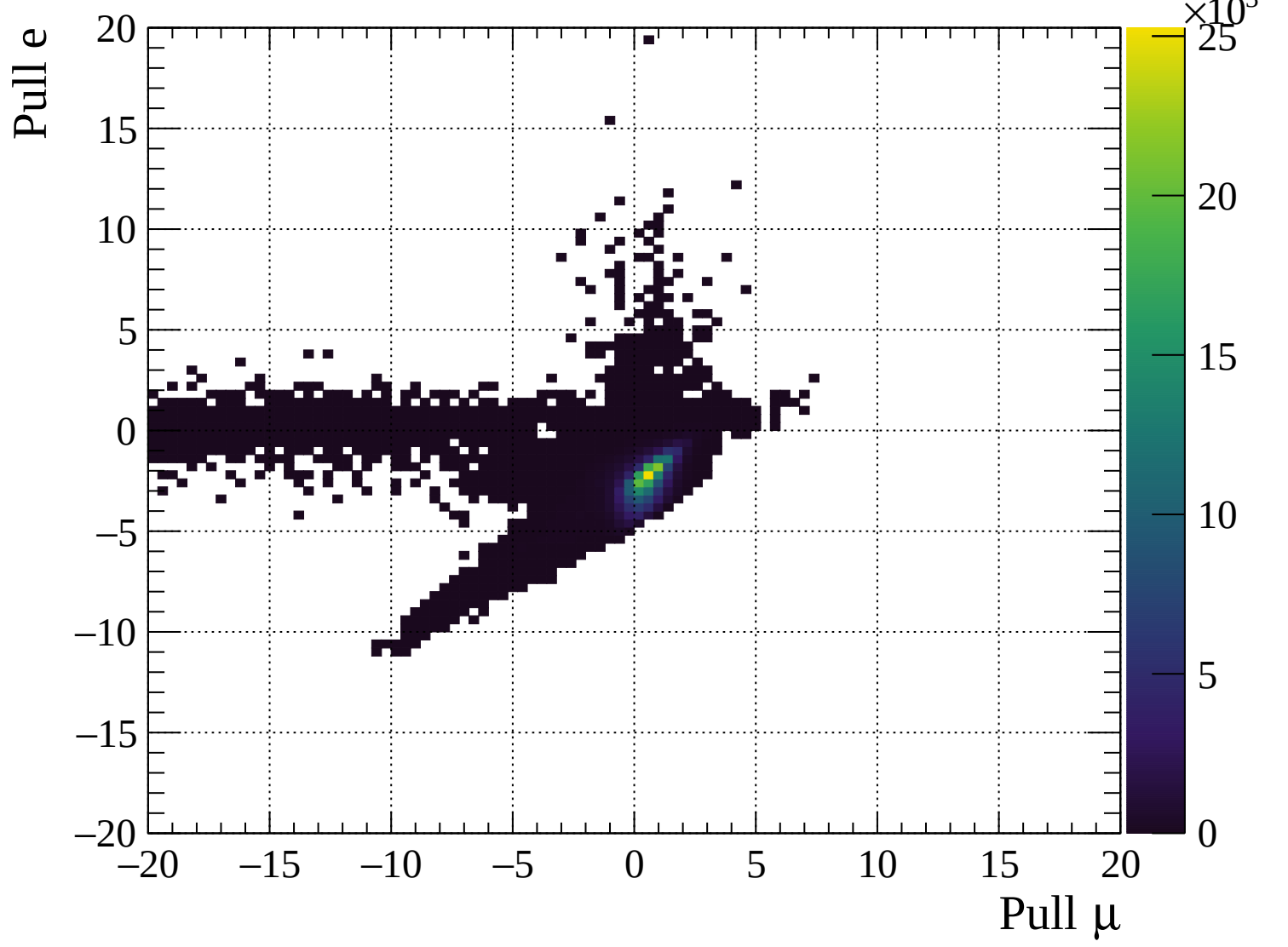


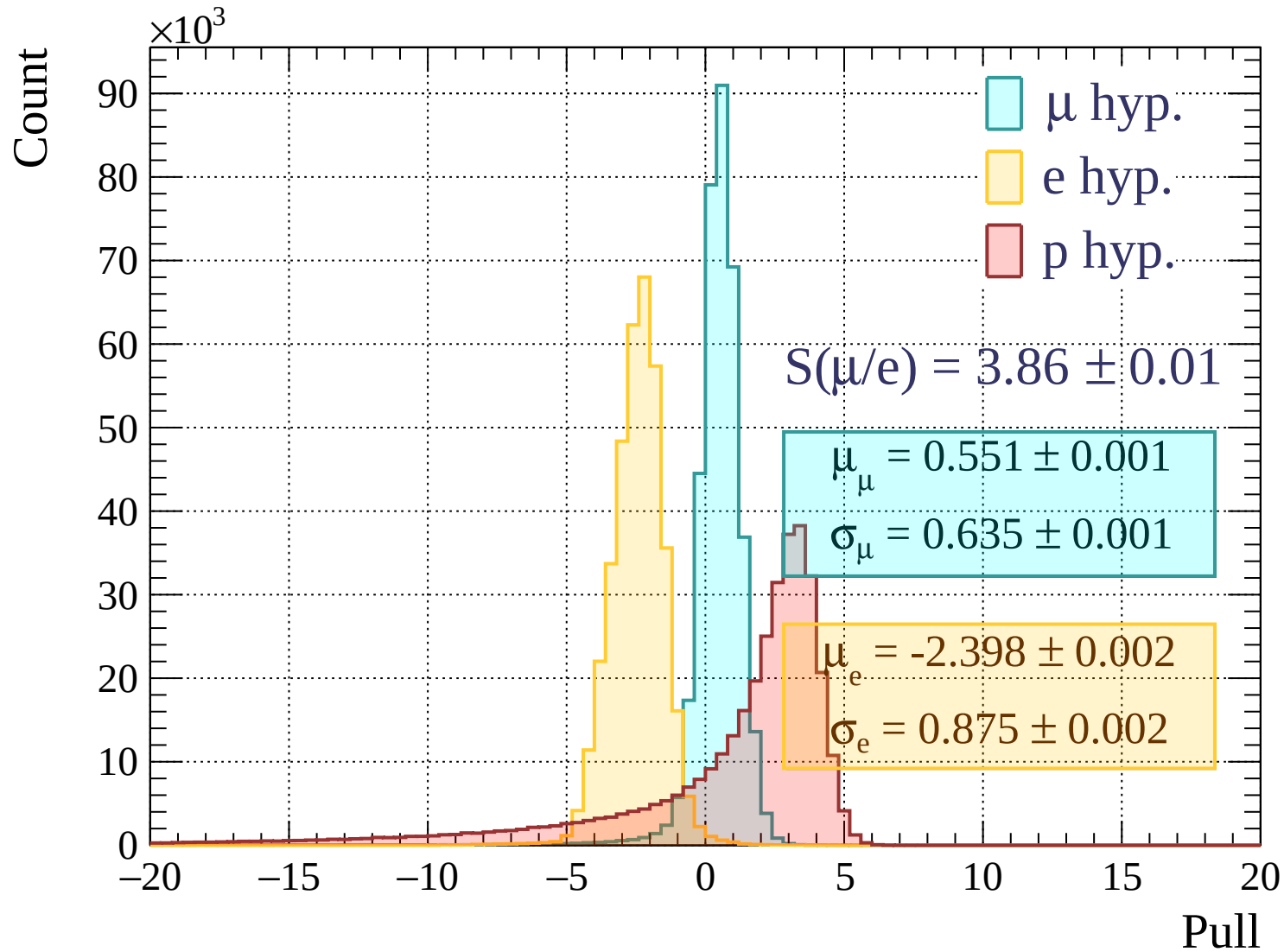
Pulls standard deviation

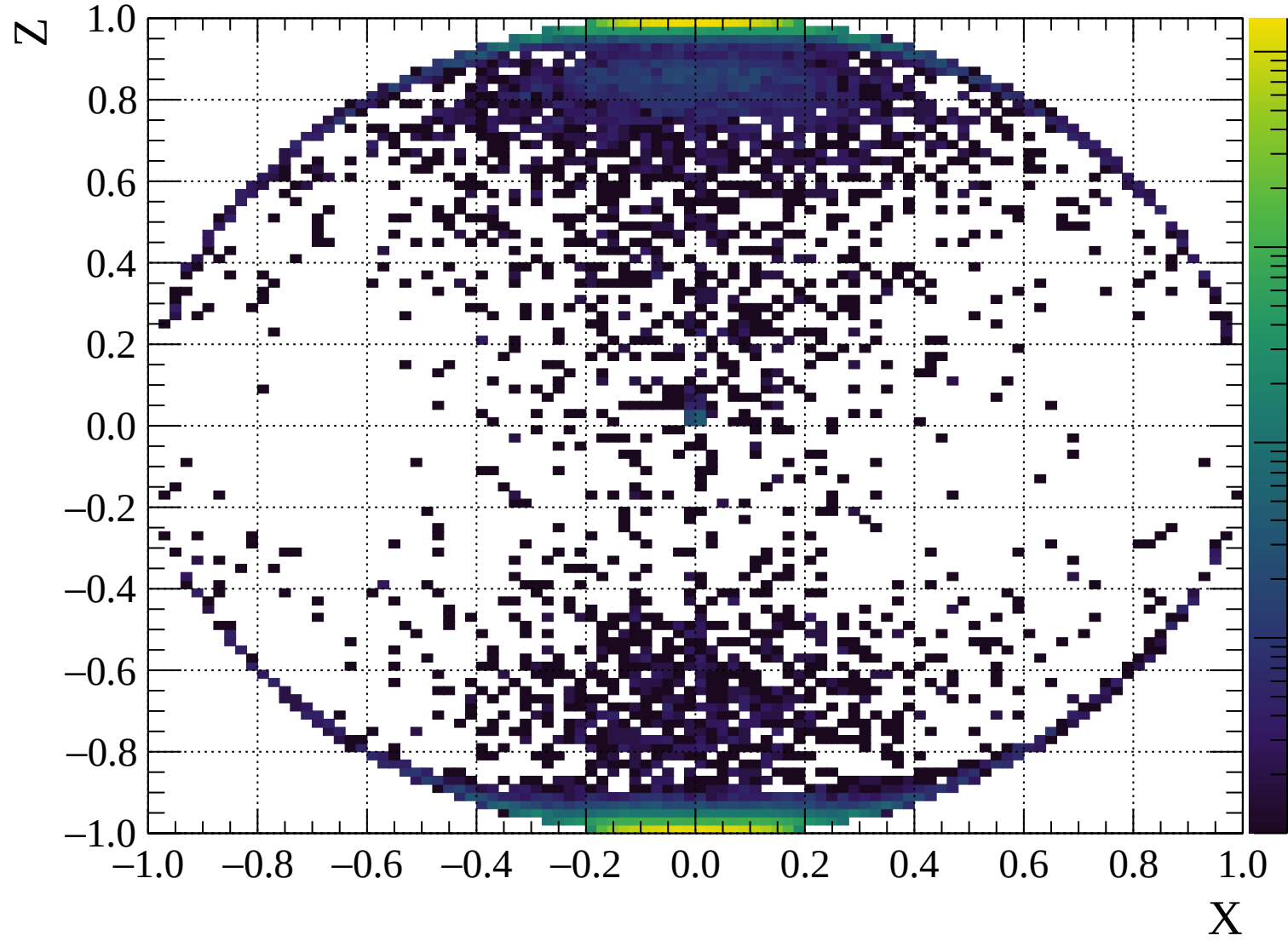


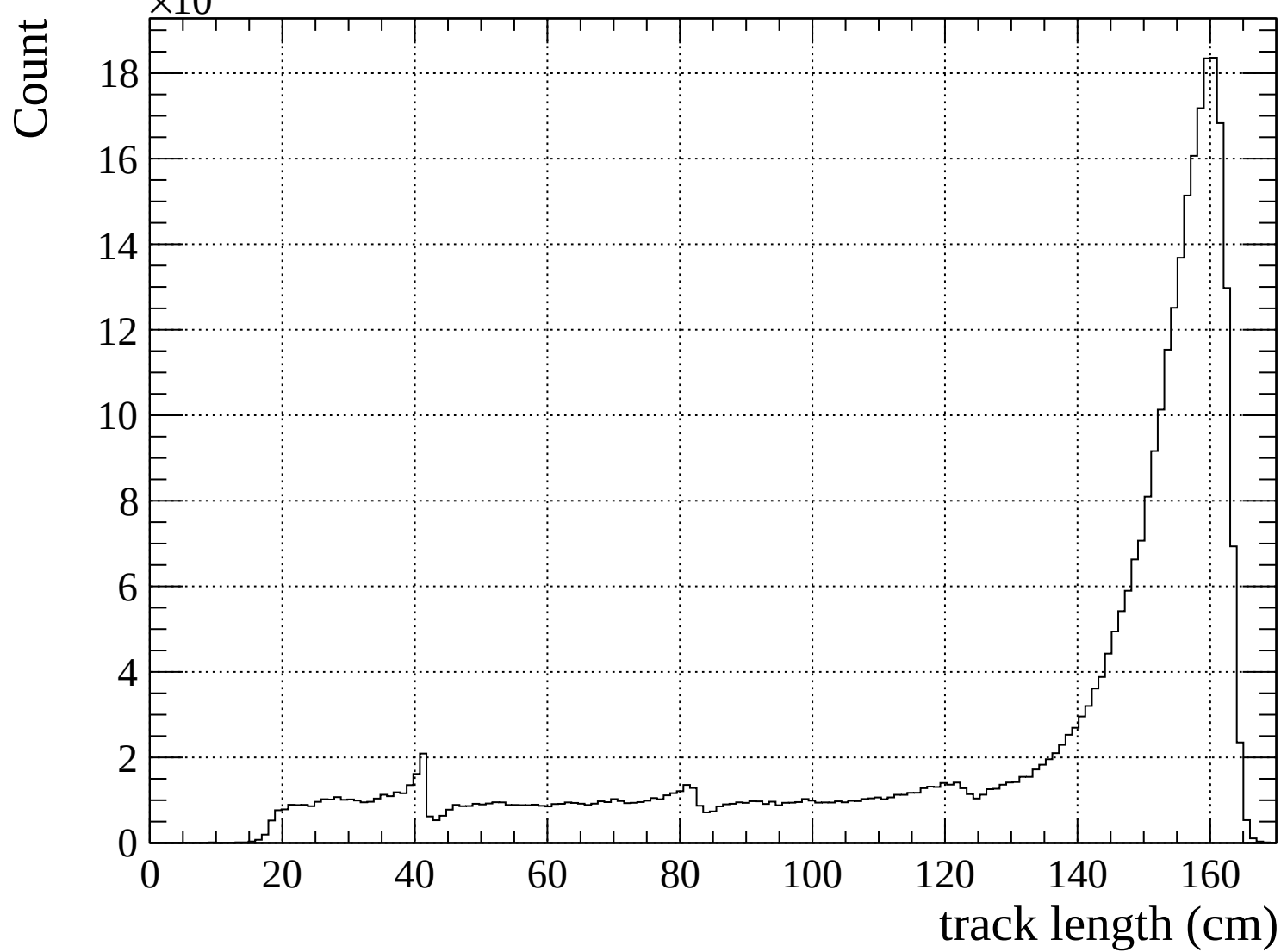
Pulls mean

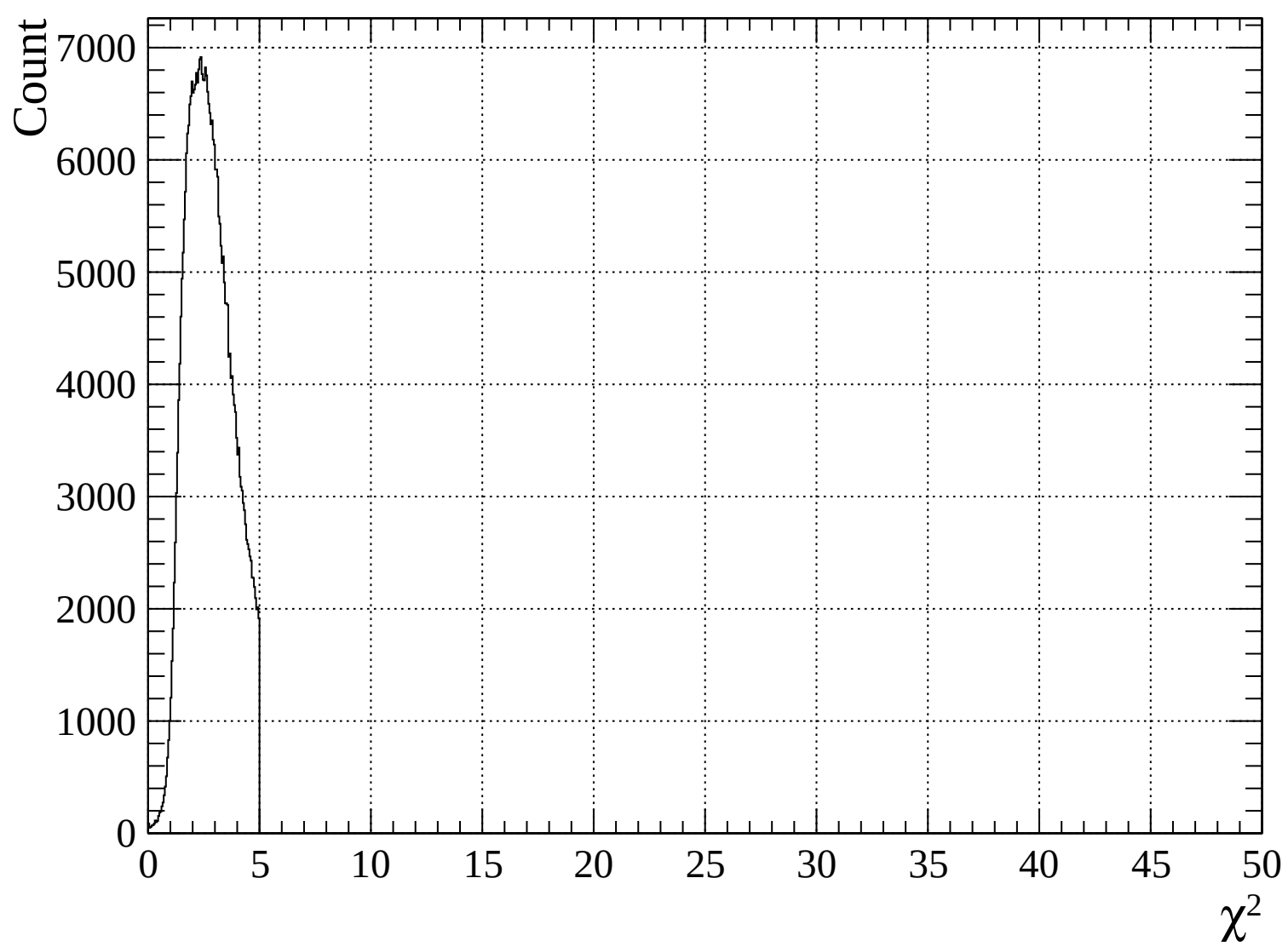


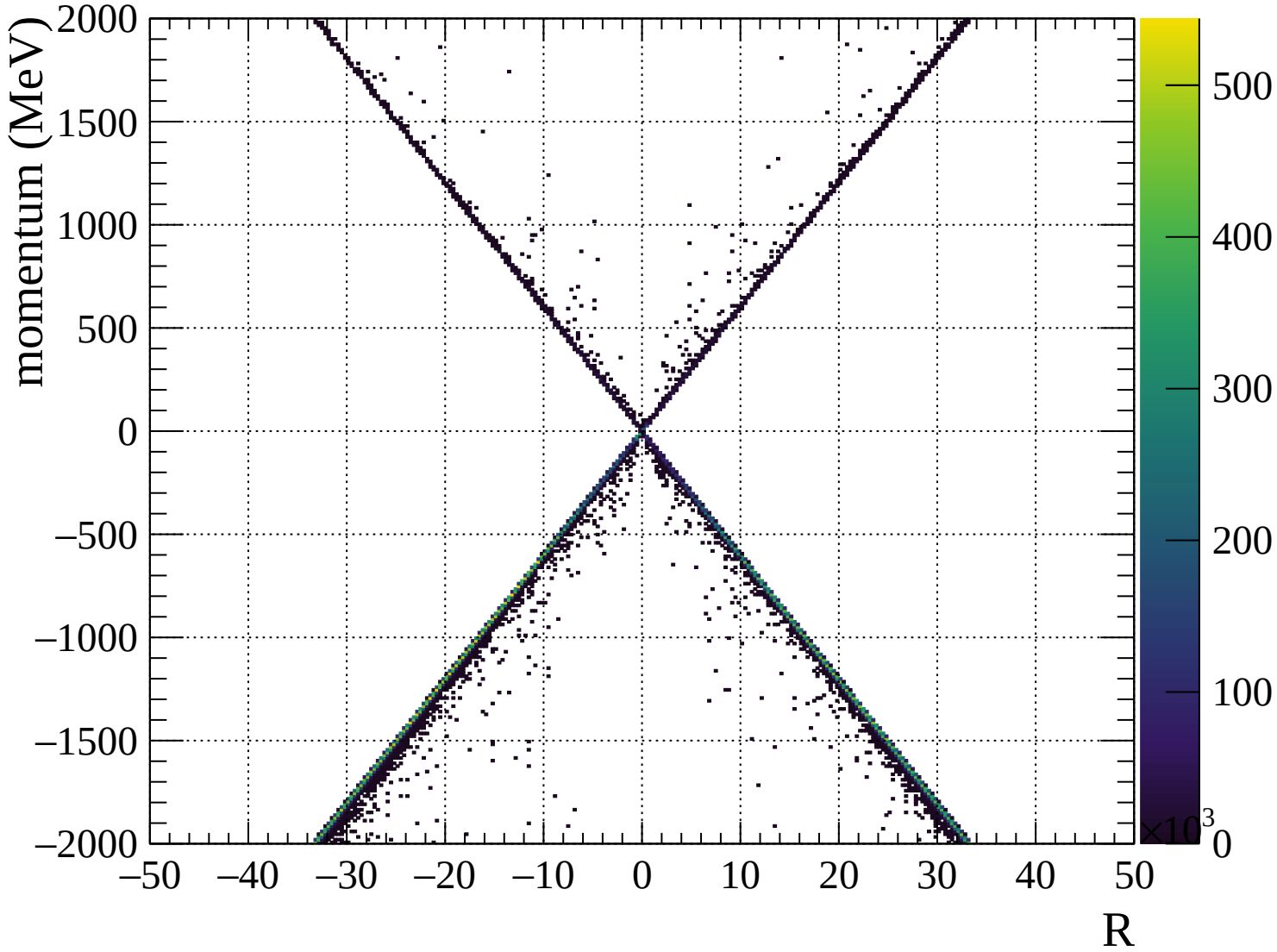


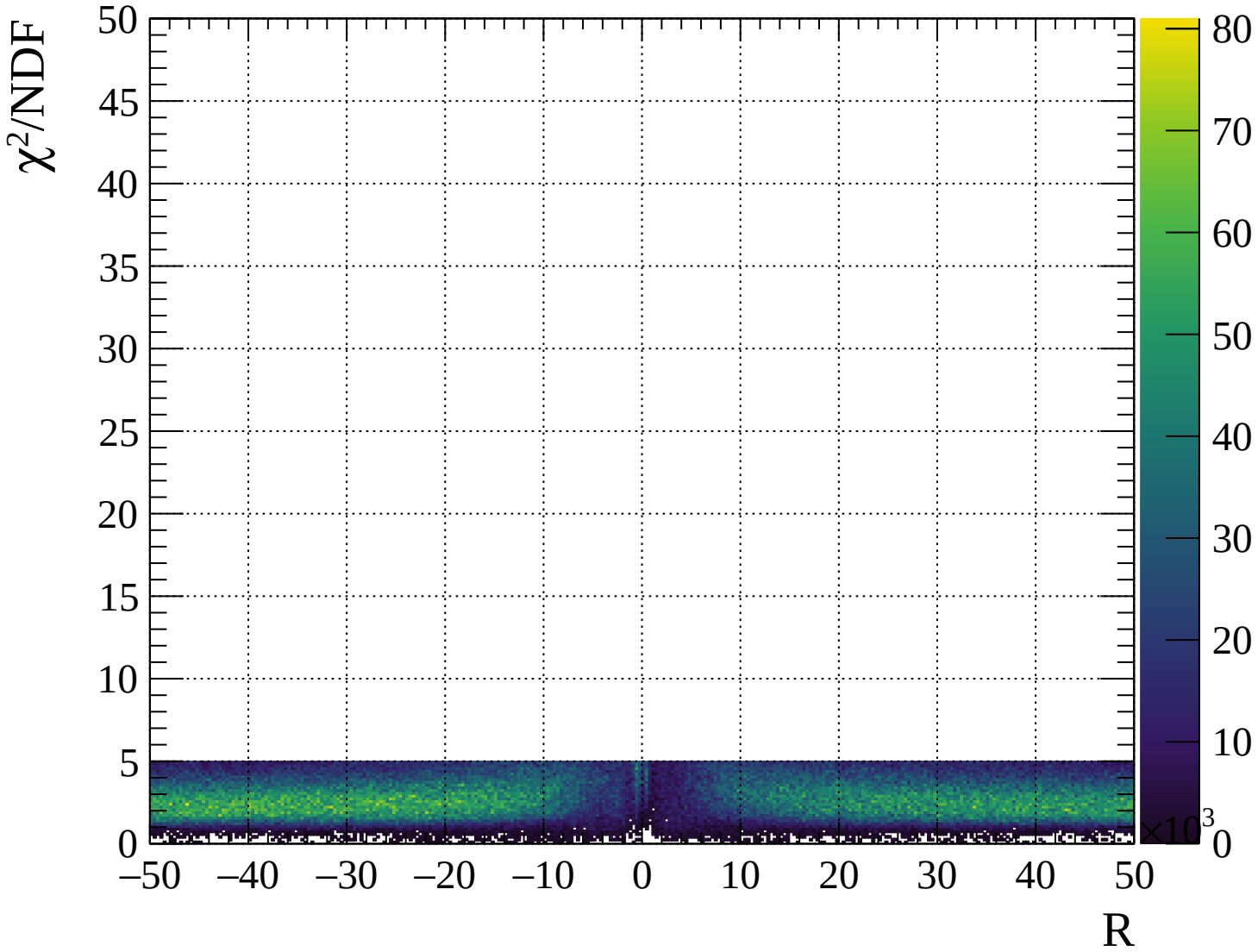




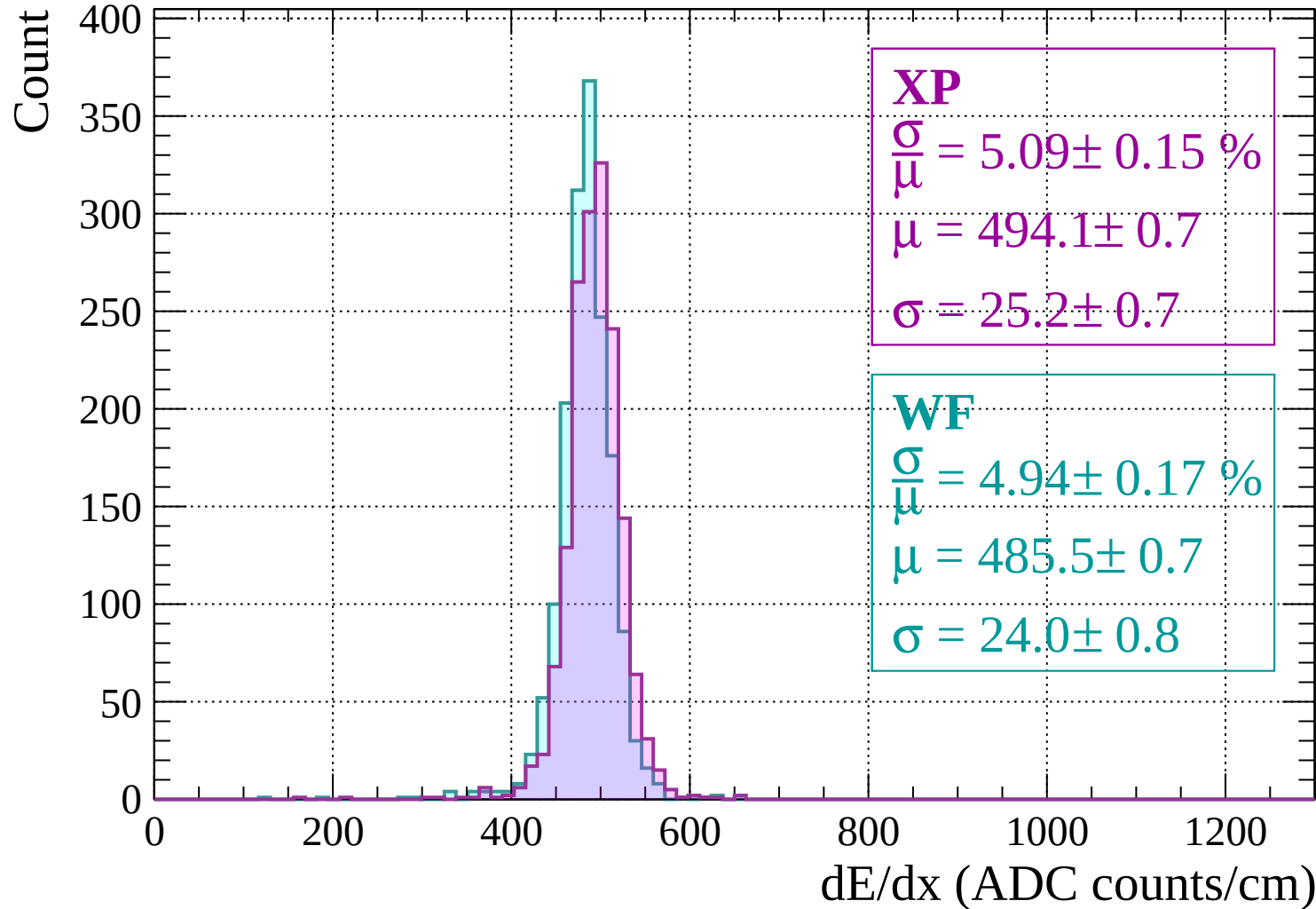






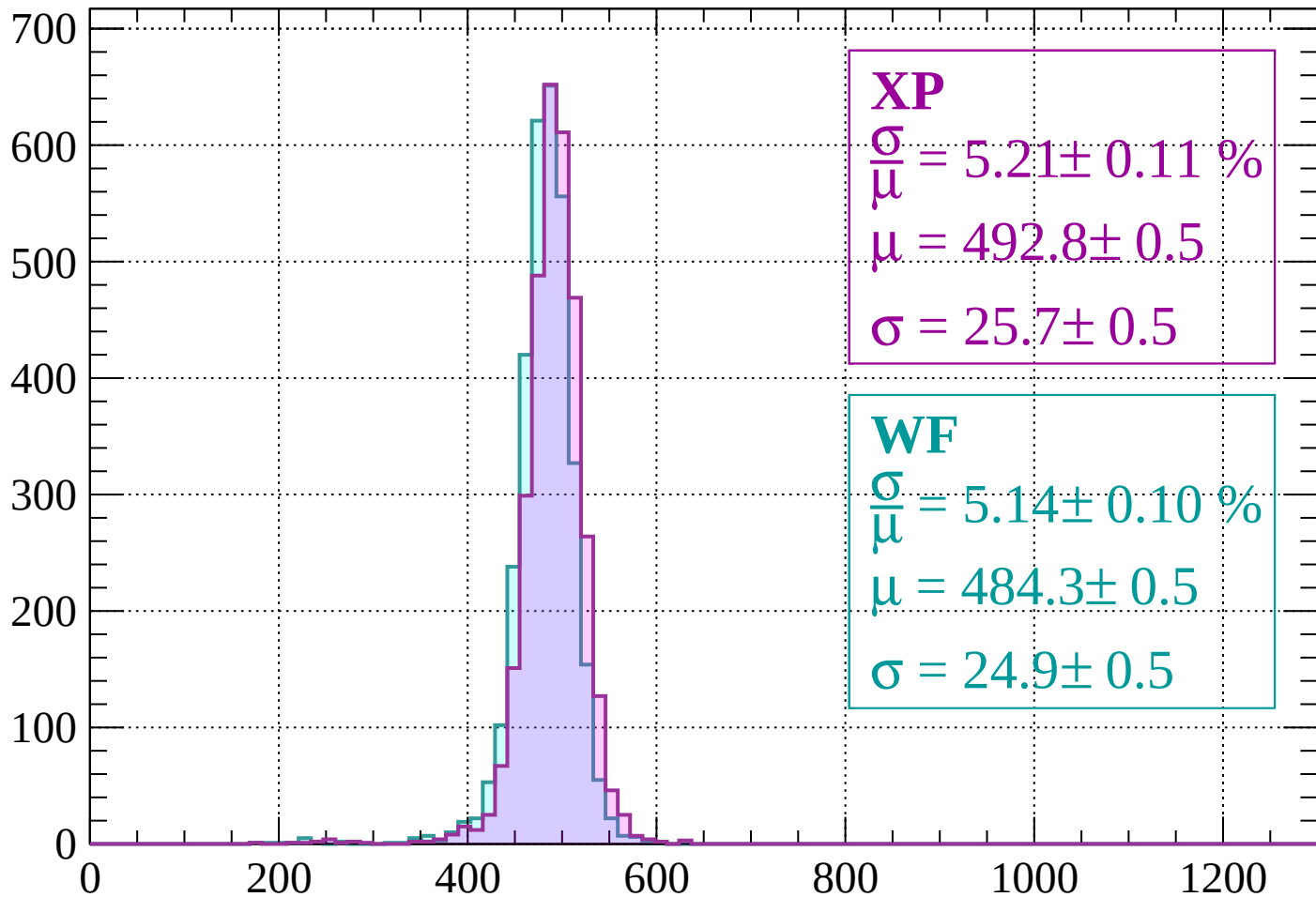


Energy loss | -2000 < p < -1960



Energy loss | -1960 < p < -1920

Count



XP

$$\frac{\sigma}{\mu} = 5.21 \pm 0.11 \%$$

$$\mu = 492.8 \pm 0.5$$

$$\sigma = 25.7 \pm 0.5$$

WF

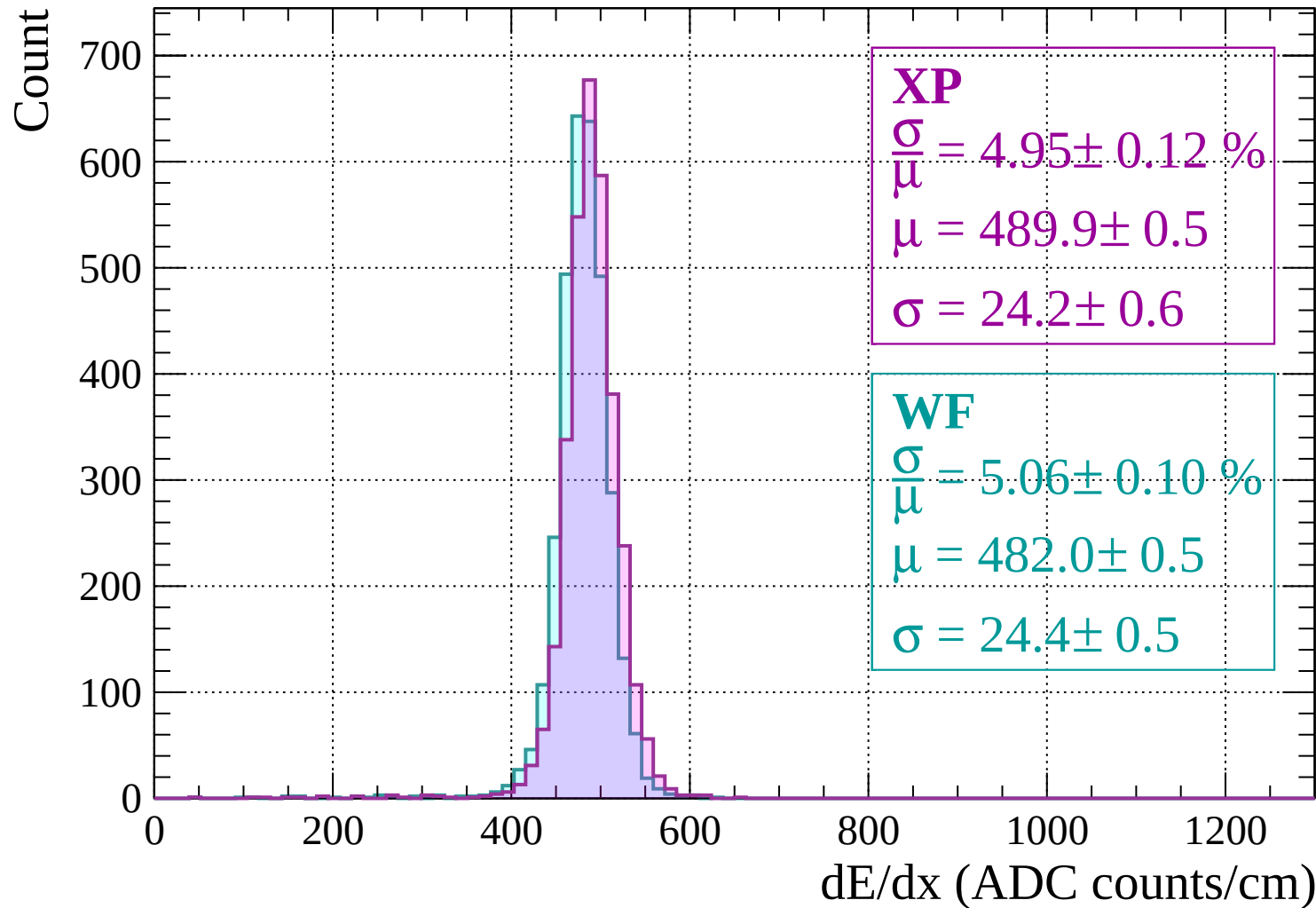
$$\frac{\sigma}{\mu} = 5.14 \pm 0.10 \%$$

$$\mu = 484.3 \pm 0.5$$

$$\sigma = 24.9 \pm 0.5$$

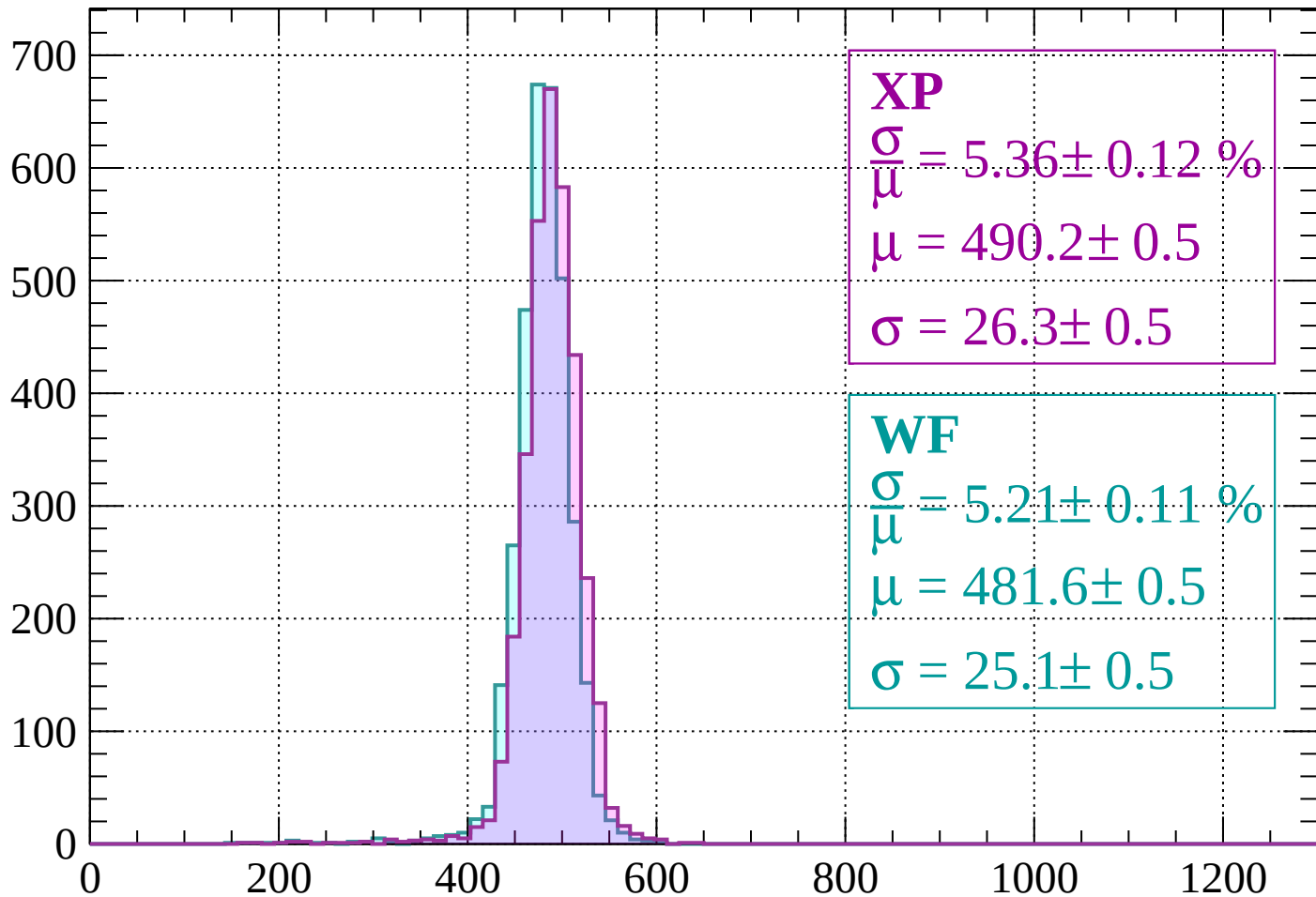
dE/dx (ADC counts/cm)

Energy loss | $-1920 < p < -1880$



Energy loss | $-1880 < p < -1840$

Count



XP

$$\frac{\sigma}{\mu} = 5.36 \pm 0.12 \%$$

$$\mu = 490.2 \pm 0.5$$

$$\sigma = 26.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.21 \pm 0.11 \%$$

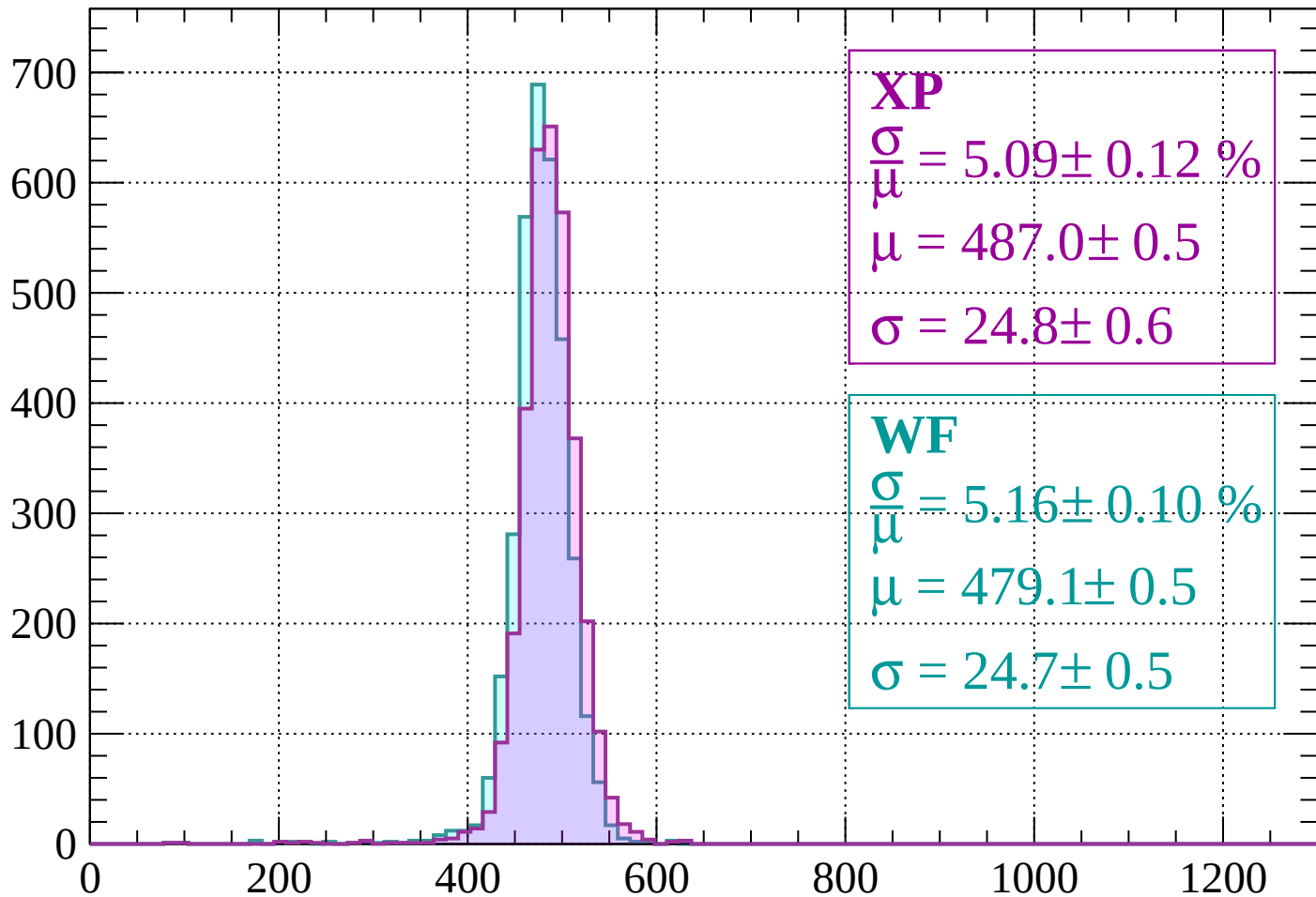
$$\mu = 481.6 \pm 0.5$$

$$\sigma = 25.1 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1840 < p < -1800$

Count



XP

$$\frac{\sigma}{\mu} = 5.09 \pm 0.12 \%$$

$$\mu = 487.0 \pm 0.5$$

$$\sigma = 24.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.16 \pm 0.10 \%$$

$$\mu = 479.1 \pm 0.5$$

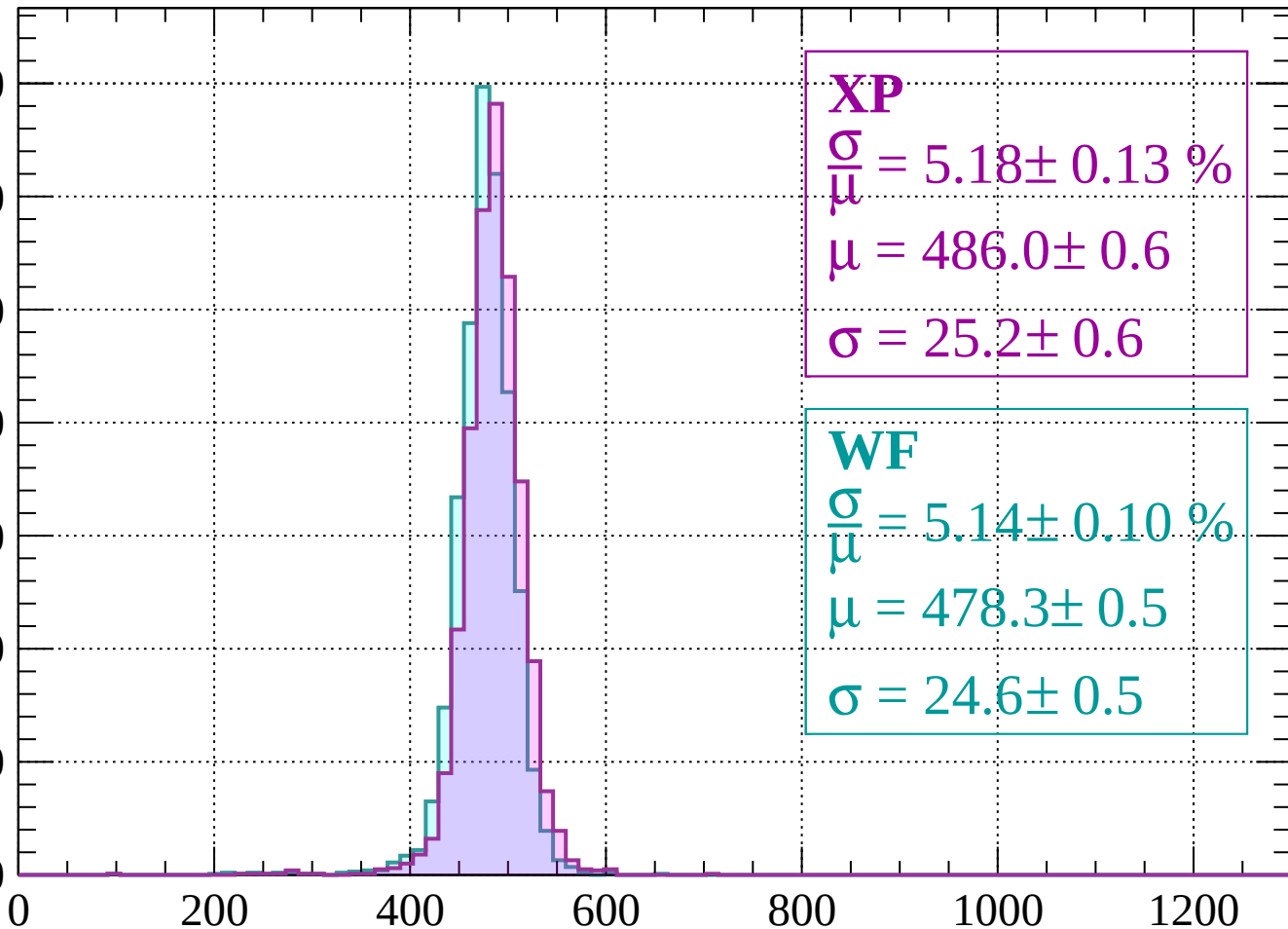
$$\sigma = 24.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1800 < p < -1760$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 5.18 \pm 0.13 \%$$

$$\mu = 486.0 \pm 0.6$$

$$\sigma = 25.2 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.14 \pm 0.10 \%$$

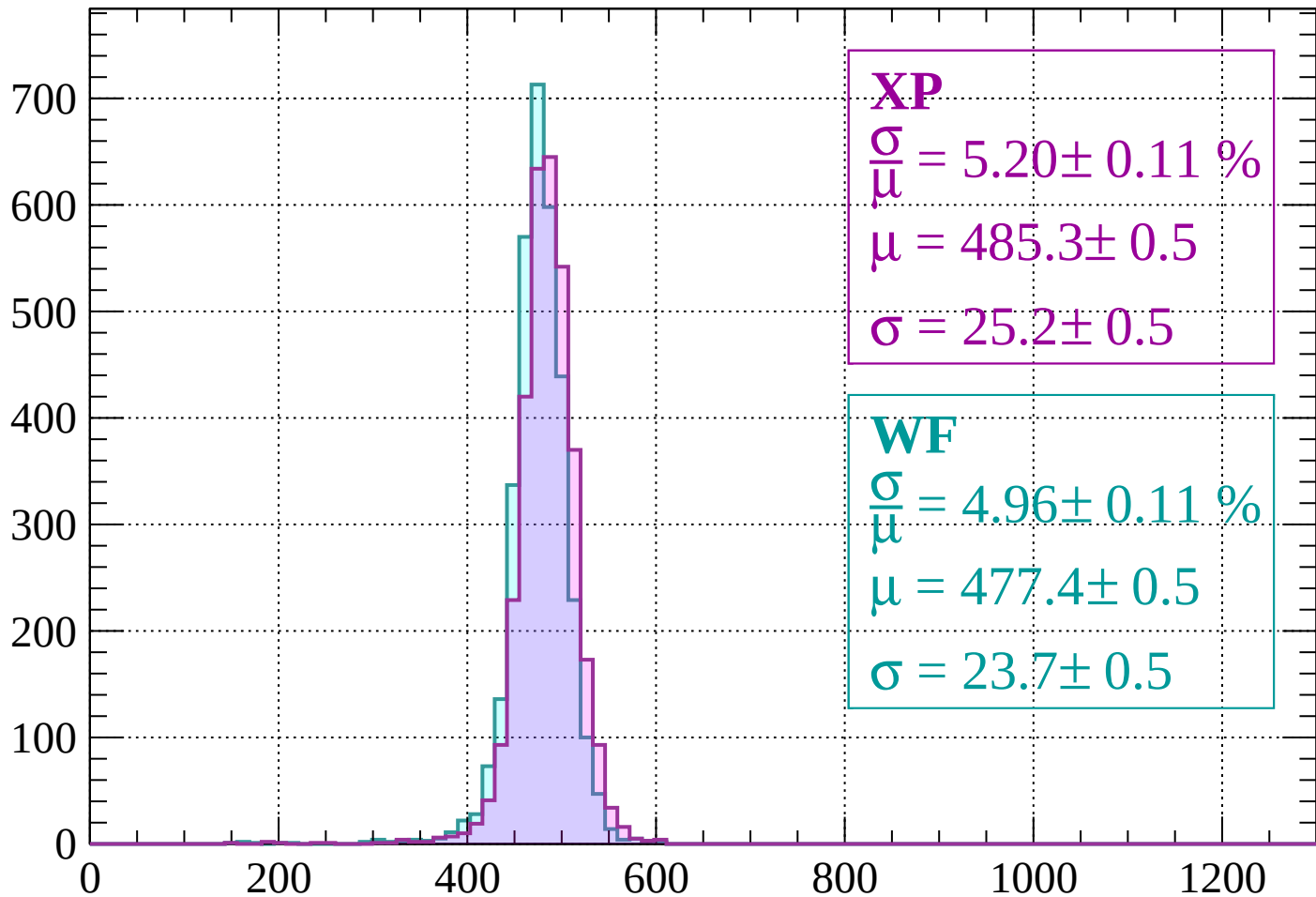
$$\mu = 478.3 \pm 0.5$$

$$\sigma = 24.6 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1760 < p < -1720$

Count



XP

$$\frac{\sigma}{\mu} = 5.20 \pm 0.11 \%$$

$$\mu = 485.3 \pm 0.5$$

$$\sigma = 25.2 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 4.96 \pm 0.11 \%$$

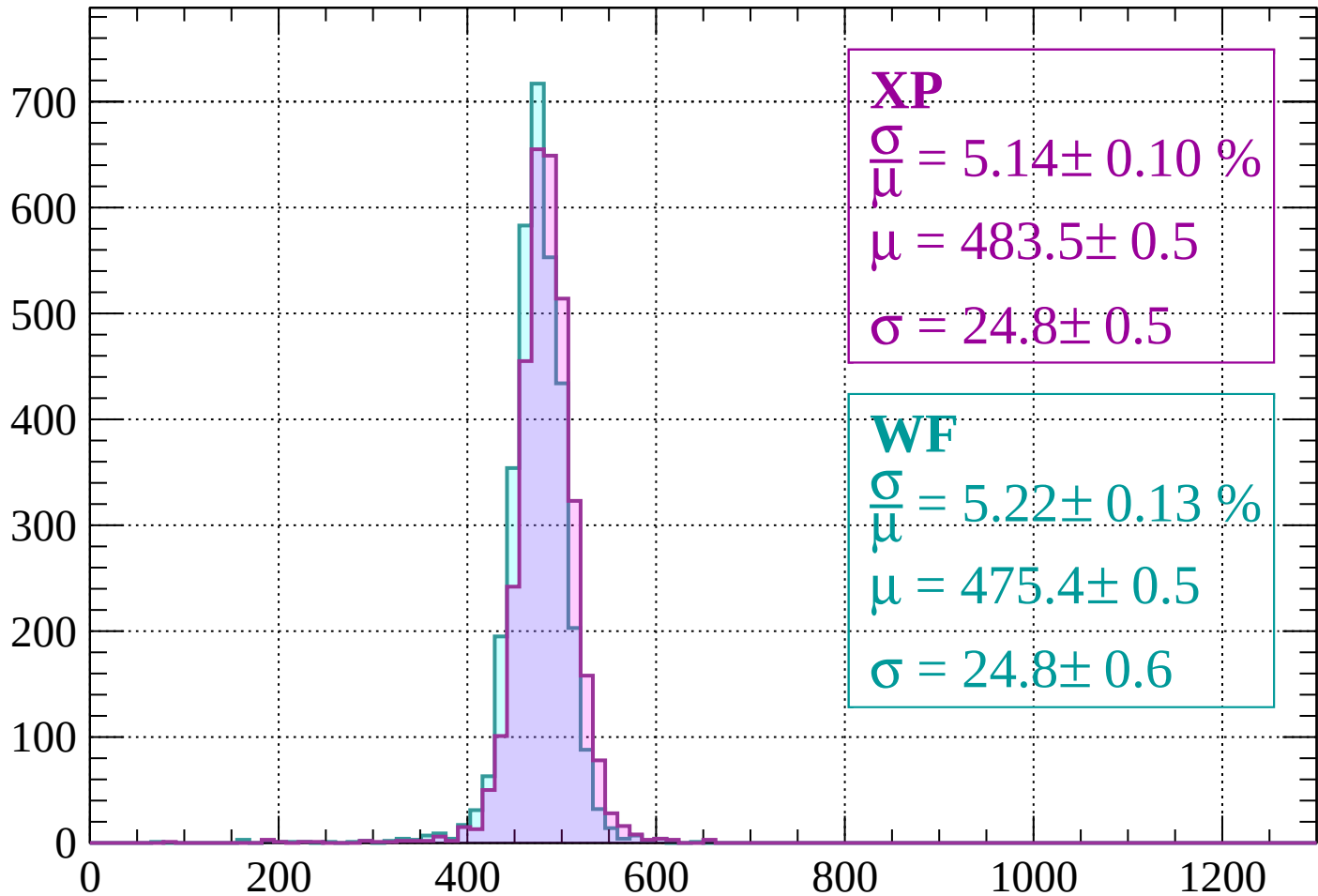
$$\mu = 477.4 \pm 0.5$$

$$\sigma = 23.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1720 < p < -1680$

Count



XP

$$\frac{\sigma}{\mu} = 5.14 \pm 0.10 \%$$

$$\mu = 483.5 \pm 0.5$$

$$\sigma = 24.8 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.22 \pm 0.13 \%$$

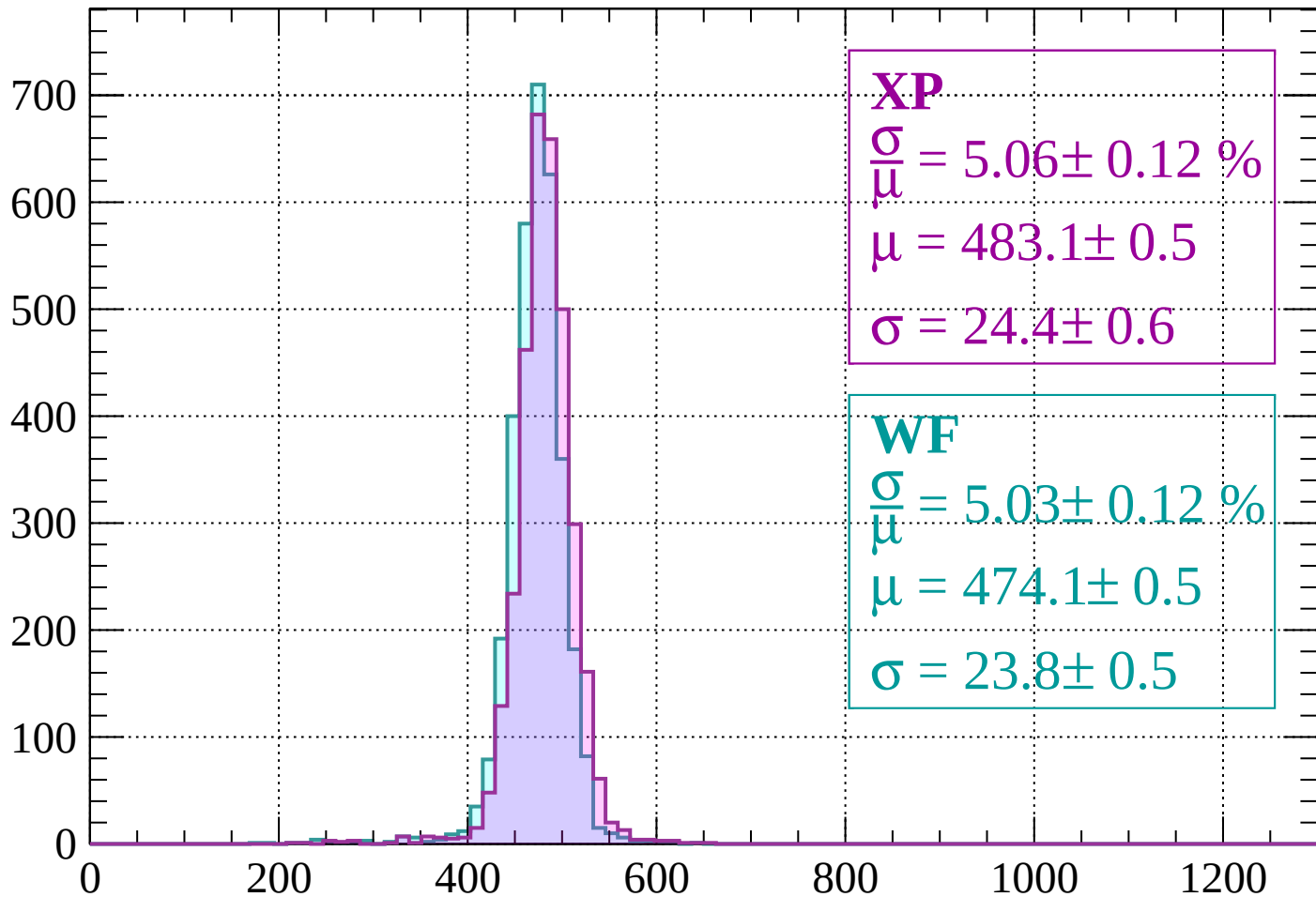
$$\mu = 475.4 \pm 0.5$$

$$\sigma = 24.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1680 < p < -1640$

Count



XP

$$\frac{\sigma}{\mu} = 5.06 \pm 0.12 \%$$

$$\mu = 483.1 \pm 0.5$$

$$\sigma = 24.4 \pm 0.6$$

WF

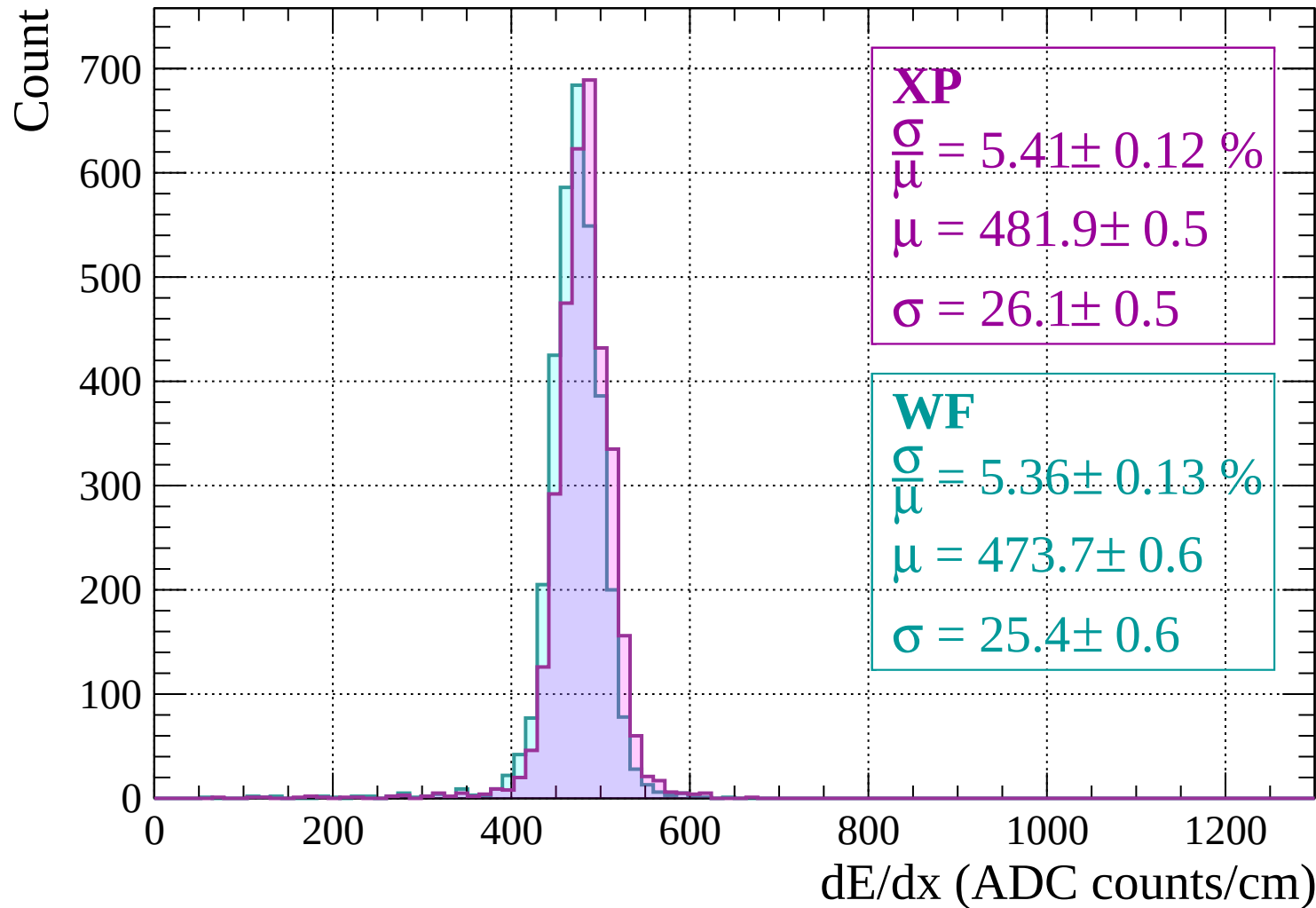
$$\frac{\sigma}{\mu} = 5.03 \pm 0.12 \%$$

$$\mu = 474.1 \pm 0.5$$

$$\sigma = 23.8 \pm 0.5$$

dE/dx (ADC counts/cm)

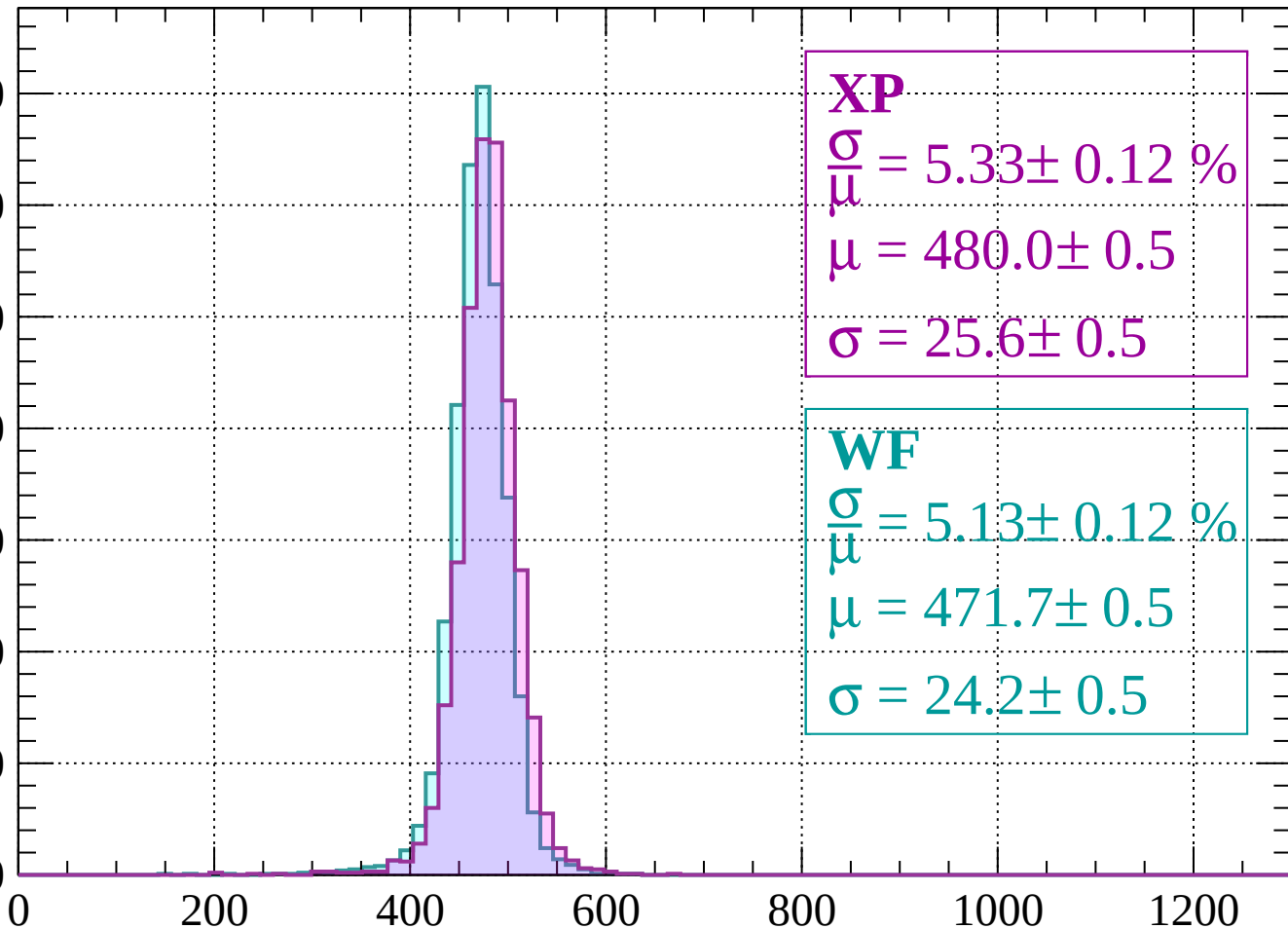
Energy loss | -1640 < p < -1600



Energy loss | $-1600 < p < -1560$

Count

700
600
500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 5.33 \pm 0.12 \%$$

$$\mu = 480.0 \pm 0.5$$

$$\sigma = 25.6 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.13 \pm 0.12 \%$$

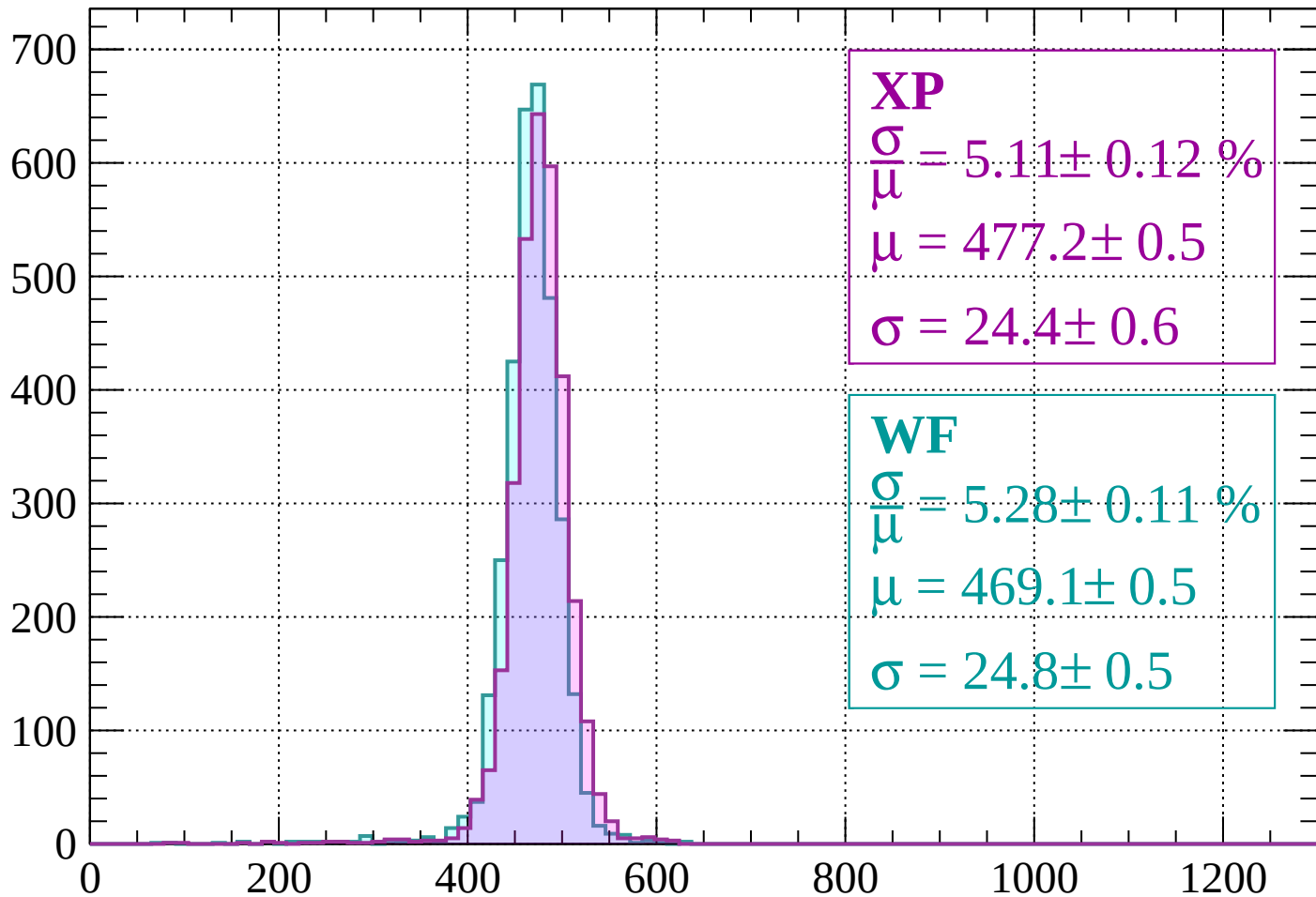
$$\mu = 471.7 \pm 0.5$$

$$\sigma = 24.2 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1560 < p < -1520$

Count



XP

$$\frac{\sigma}{\mu} = 5.11 \pm 0.12 \%$$

$$\mu = 477.2 \pm 0.5$$

$$\sigma = 24.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.28 \pm 0.11 \%$$

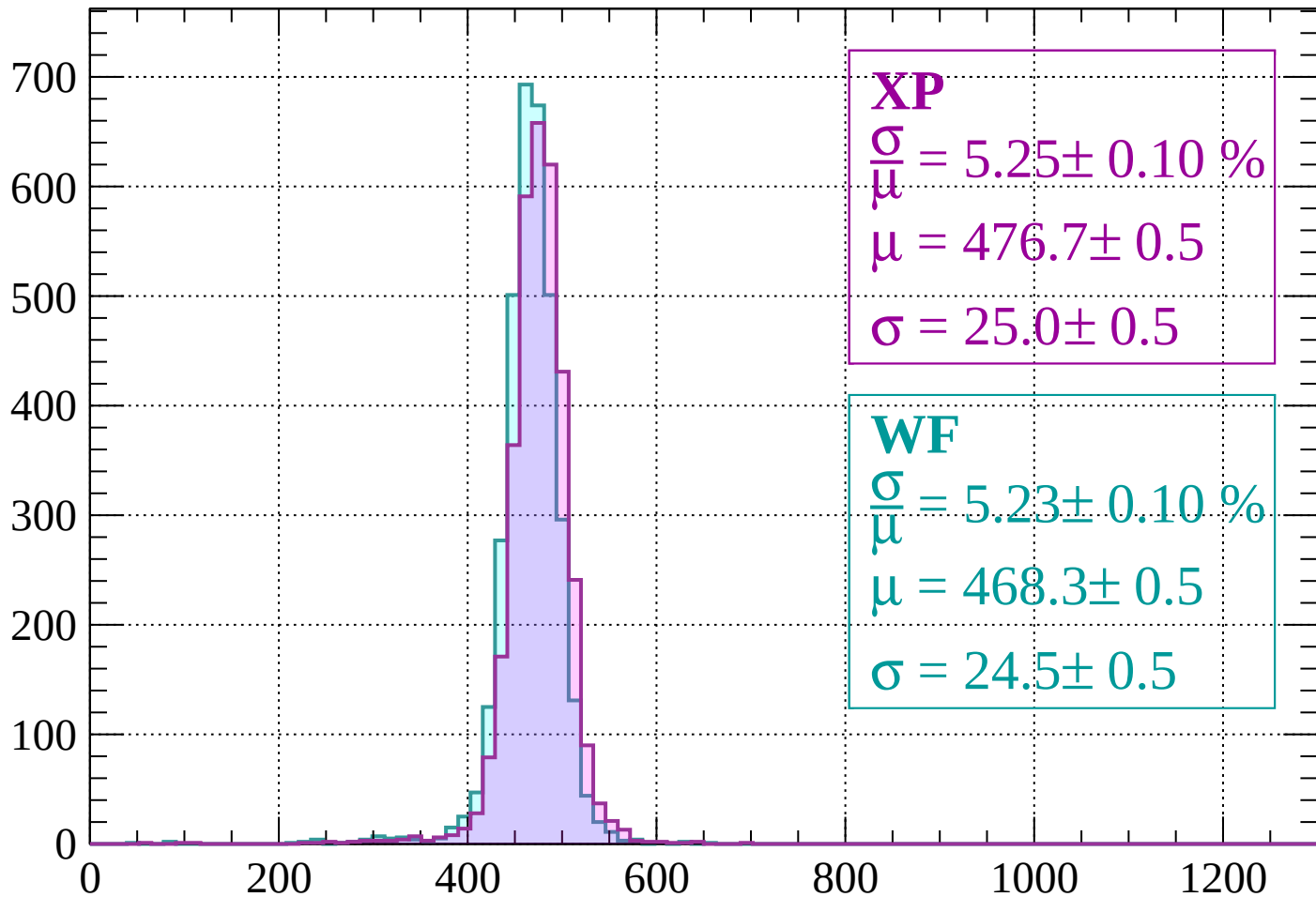
$$\mu = 469.1 \pm 0.5$$

$$\sigma = 24.8 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1520 < p < -1480$

Count



XP

$$\frac{\sigma}{\mu} = 5.25 \pm 0.10 \%$$

$$\mu = 476.7 \pm 0.5$$

$$\sigma = 25.0 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.23 \pm 0.10 \%$$

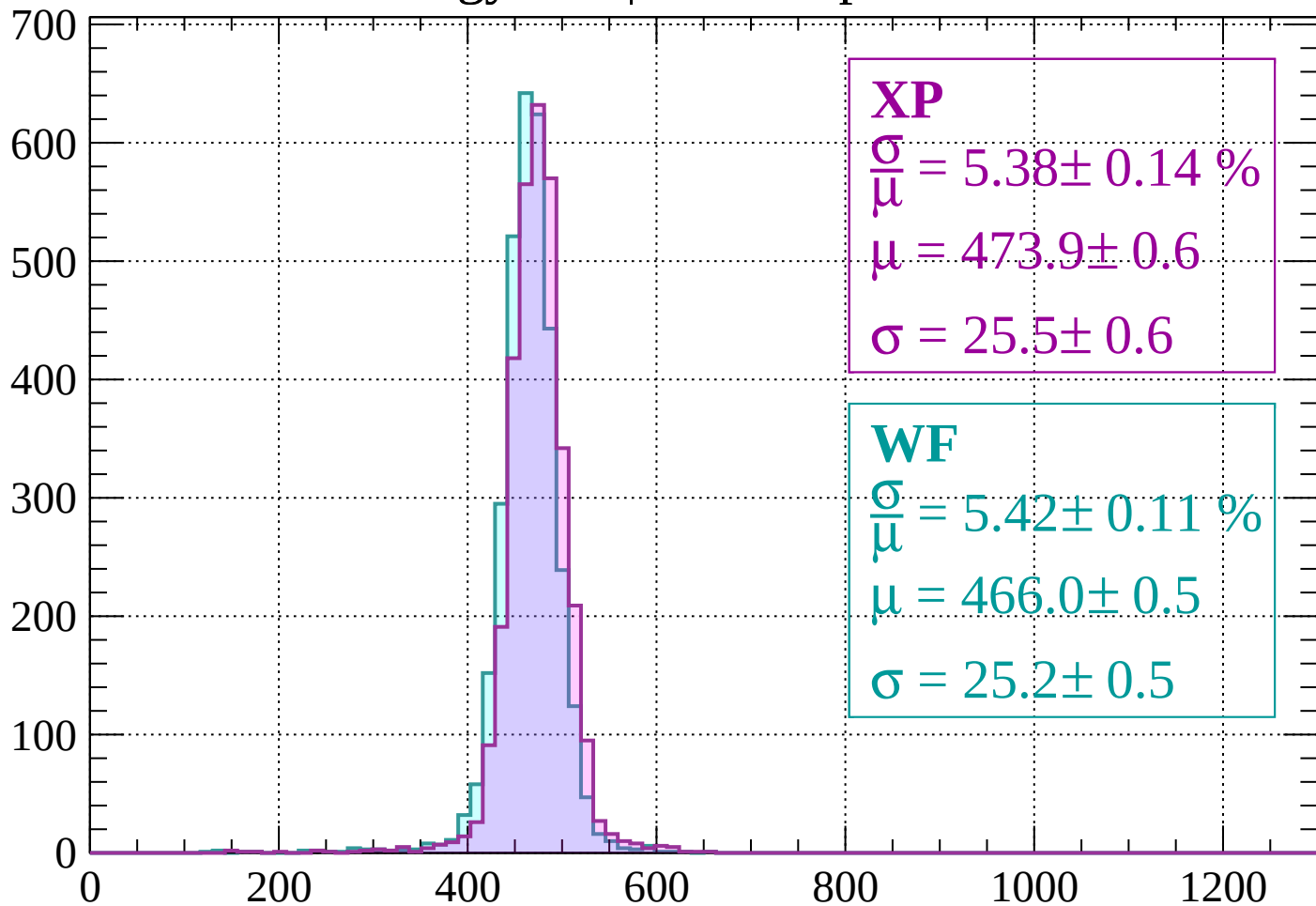
$$\mu = 468.3 \pm 0.5$$

$$\sigma = 24.5 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1480 < p < -1440$

Count



XP

$$\frac{\sigma}{\mu} = 5.38 \pm 0.14 \%$$

$$\mu = 473.9 \pm 0.6$$

$$\sigma = 25.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.42 \pm 0.11 \%$$

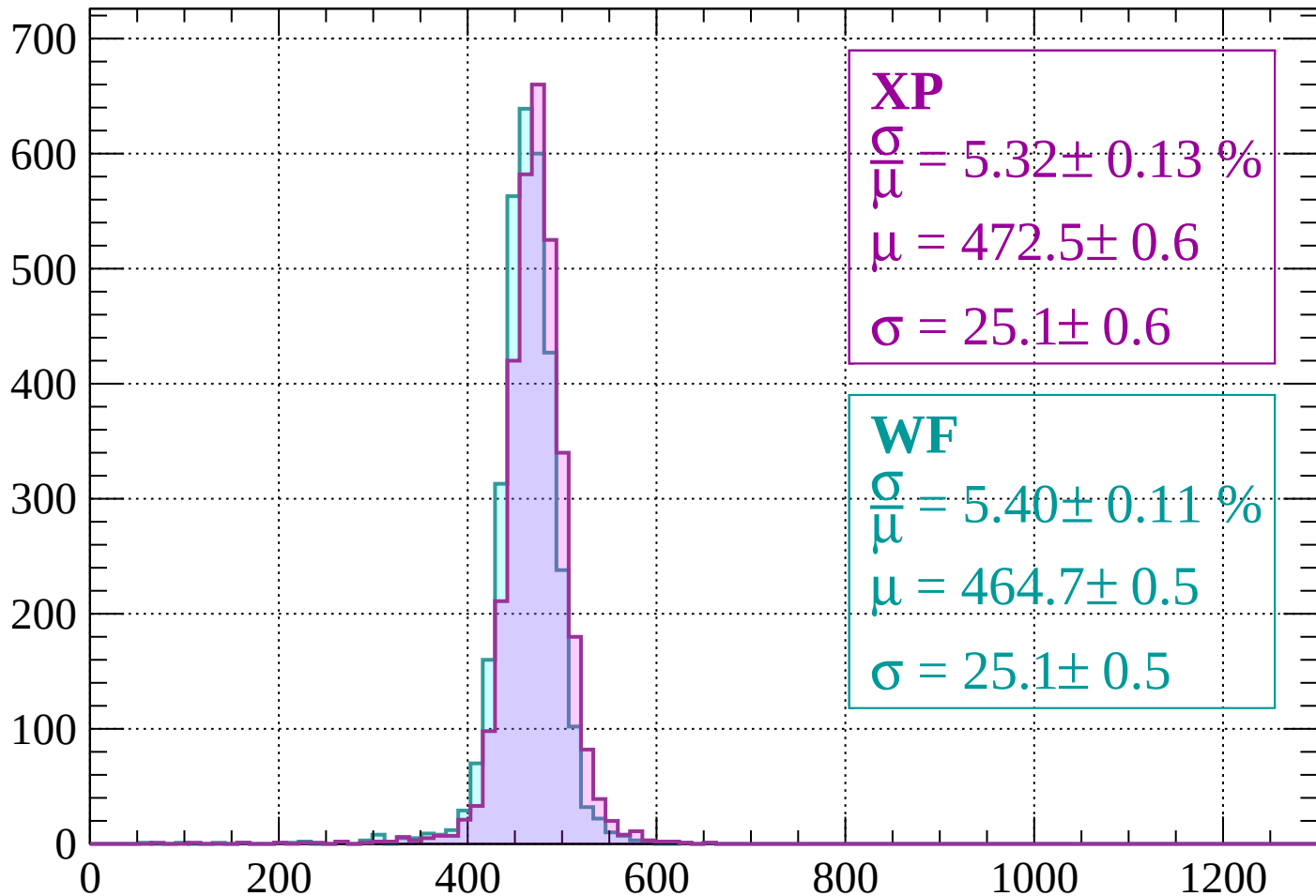
$$\mu = 466.0 \pm 0.5$$

$$\sigma = 25.2 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1440 < p < -1400$

Count



XP

$$\frac{\sigma}{\mu} = 5.32 \pm 0.13 \%$$

$$\mu = 472.5 \pm 0.6$$

$$\sigma = 25.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.40 \pm 0.11 \%$$

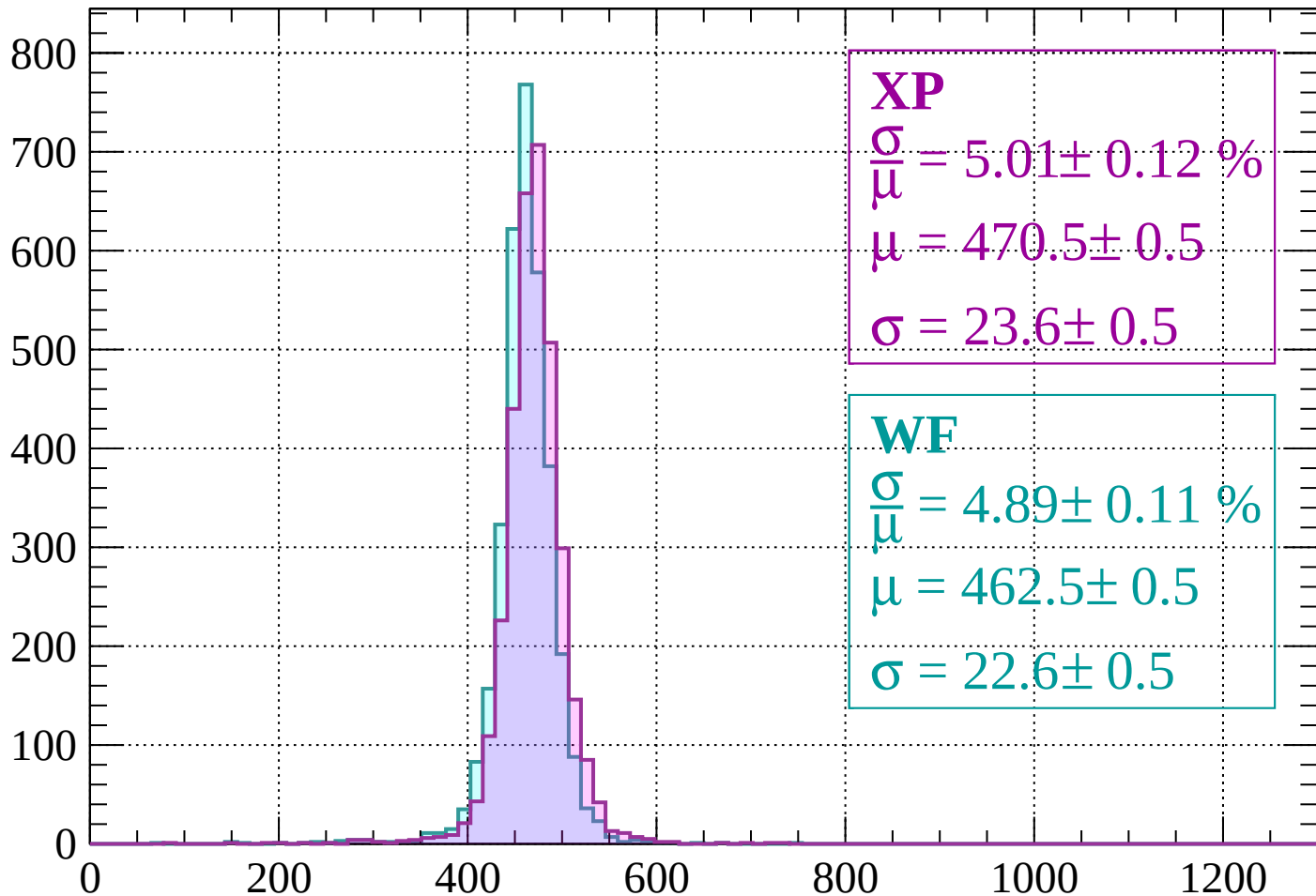
$$\mu = 464.7 \pm 0.5$$

$$\sigma = 25.1 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1400 < p < -1360$

Count



XP

$$\frac{\sigma}{\mu} = 5.01 \pm 0.12 \%$$

$$\mu = 470.5 \pm 0.5$$

$$\sigma = 23.6 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 4.89 \pm 0.11 \%$$

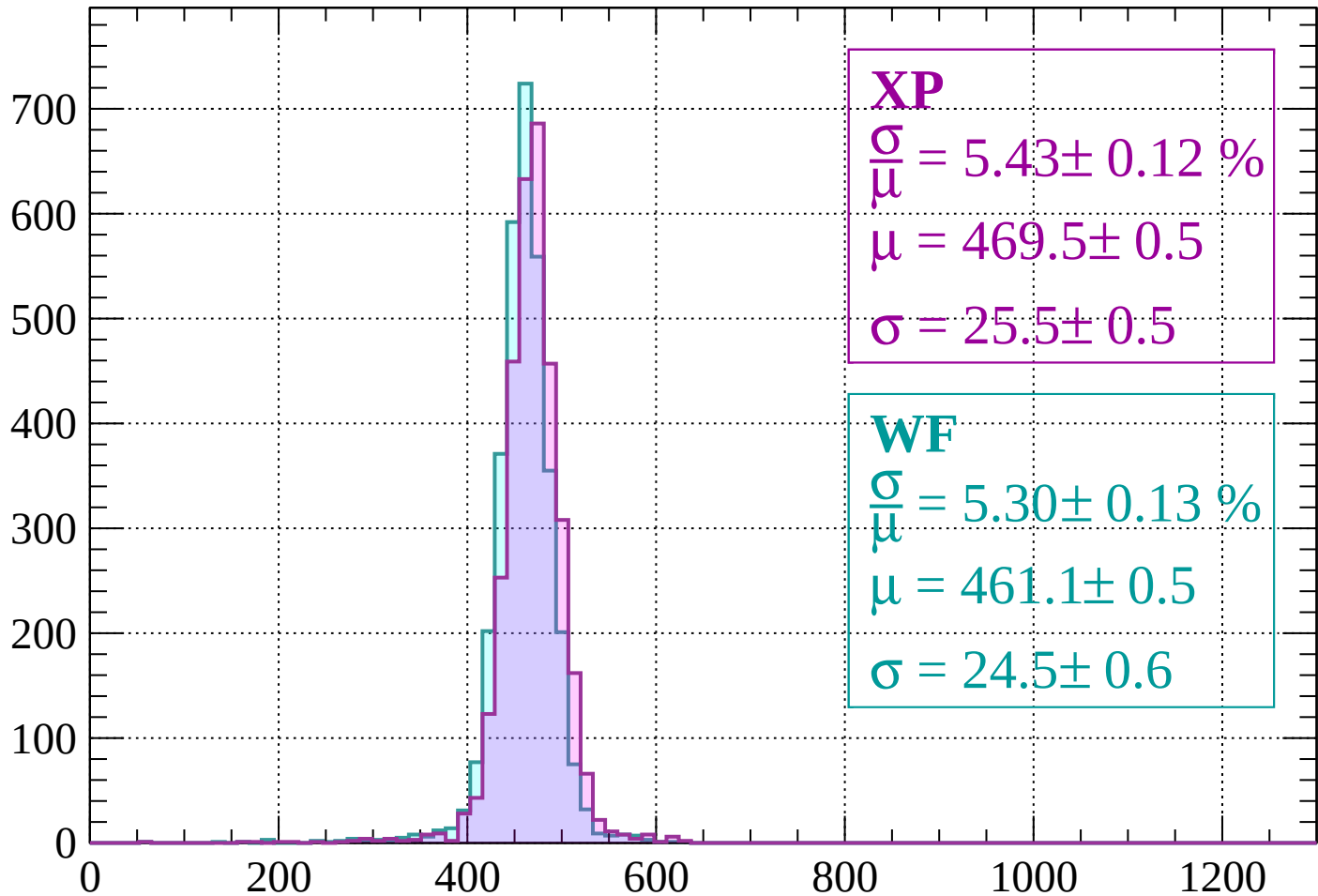
$$\mu = 462.5 \pm 0.5$$

$$\sigma = 22.6 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1360 < p < -1320$

Count



XP

$$\frac{\sigma}{\mu} = 5.43 \pm 0.12 \%$$

$$\mu = 469.5 \pm 0.5$$

$$\sigma = 25.5 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.30 \pm 0.13 \%$$

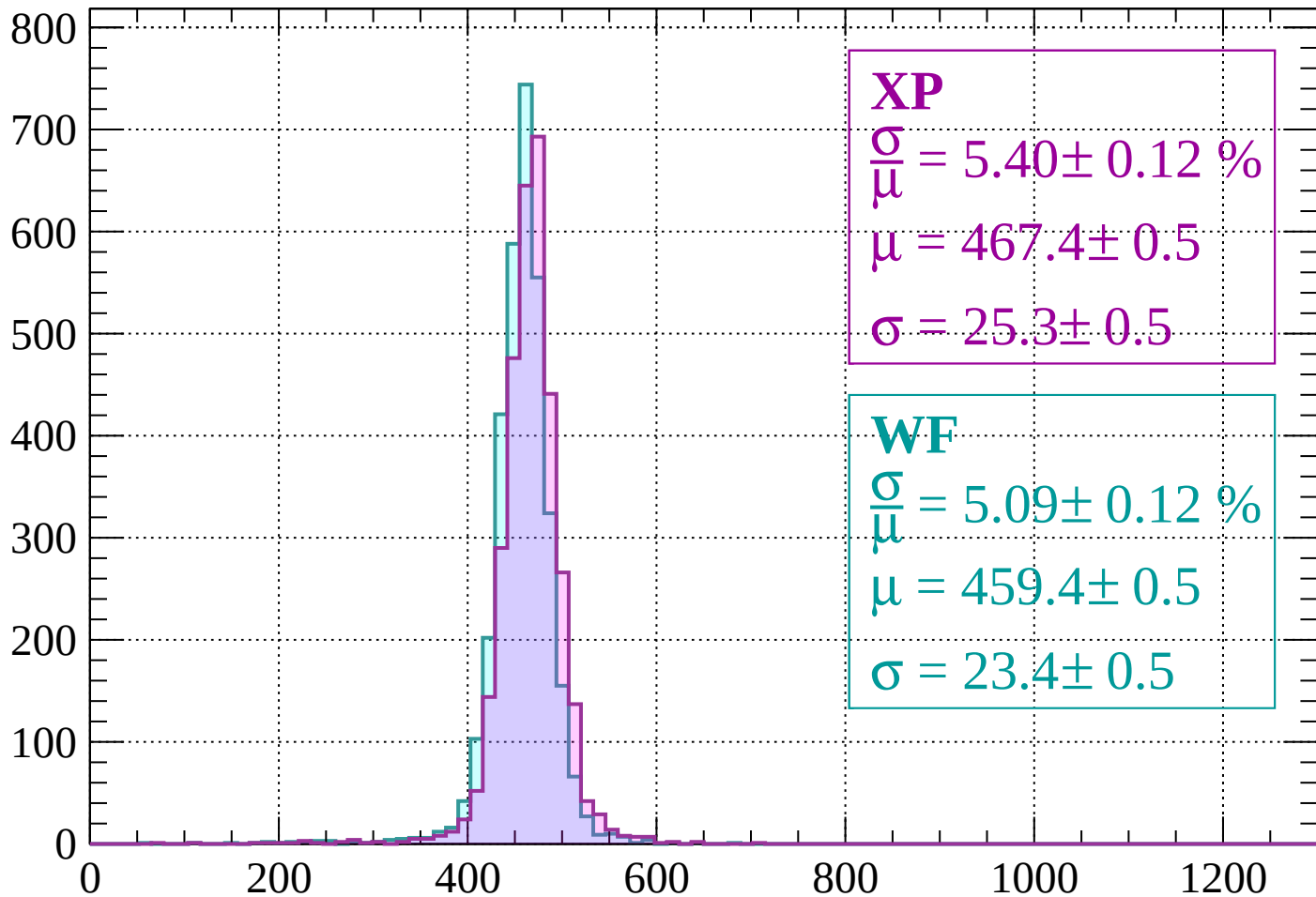
$$\mu = 461.1 \pm 0.5$$

$$\sigma = 24.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1320 < p < -1280$

Count



XP

$$\frac{\sigma}{\mu} = 5.40 \pm 0.12 \%$$

$$\mu = 467.4 \pm 0.5$$

$$\sigma = 25.3 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.09 \pm 0.12 \%$$

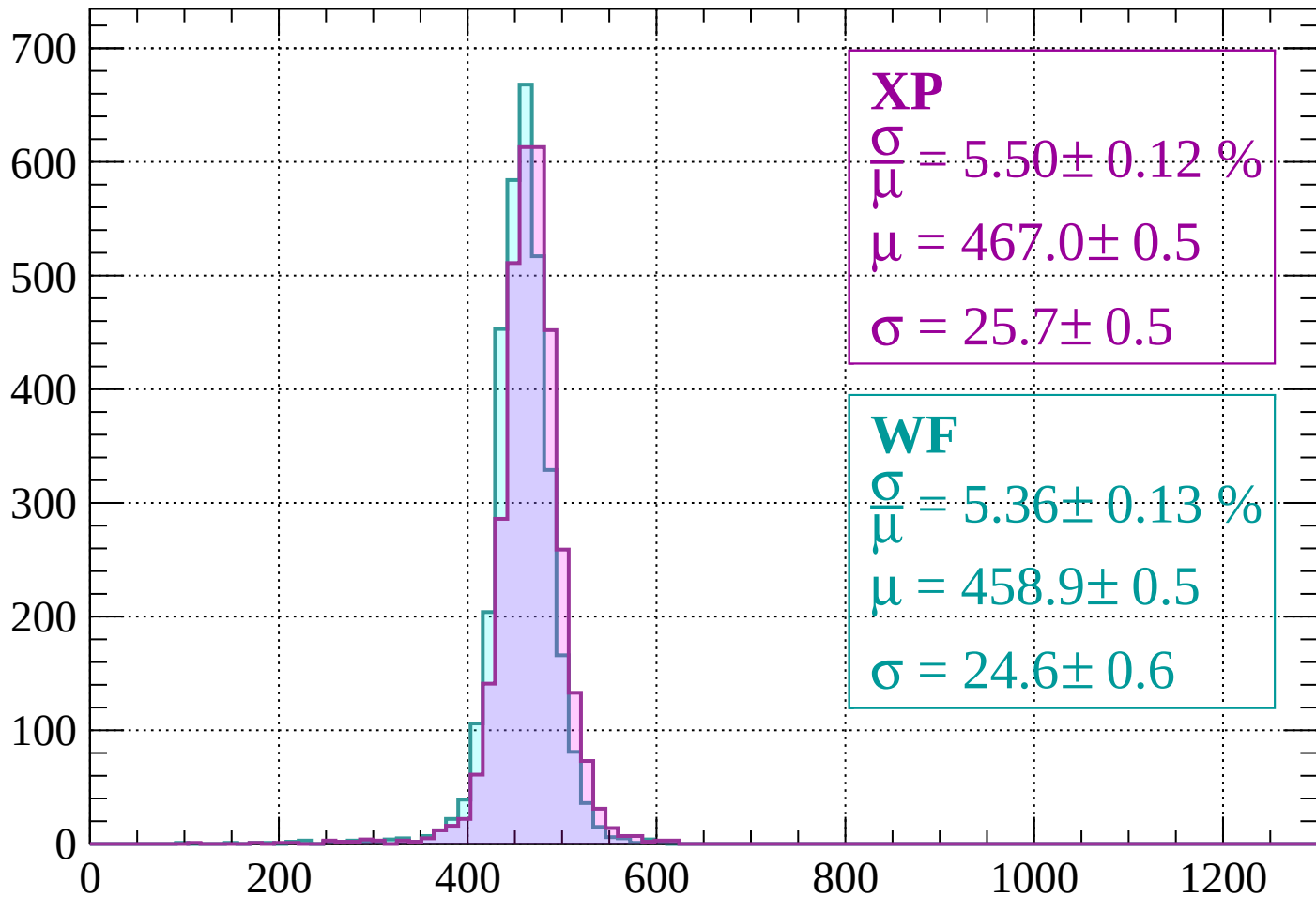
$$\mu = 459.4 \pm 0.5$$

$$\sigma = 23.4 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1280 < p < -1240$

Count



XP

$$\frac{\sigma}{\mu} = 5.50 \pm 0.12 \%$$

$$\mu = 467.0 \pm 0.5$$

$$\sigma = 25.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.36 \pm 0.13 \%$$

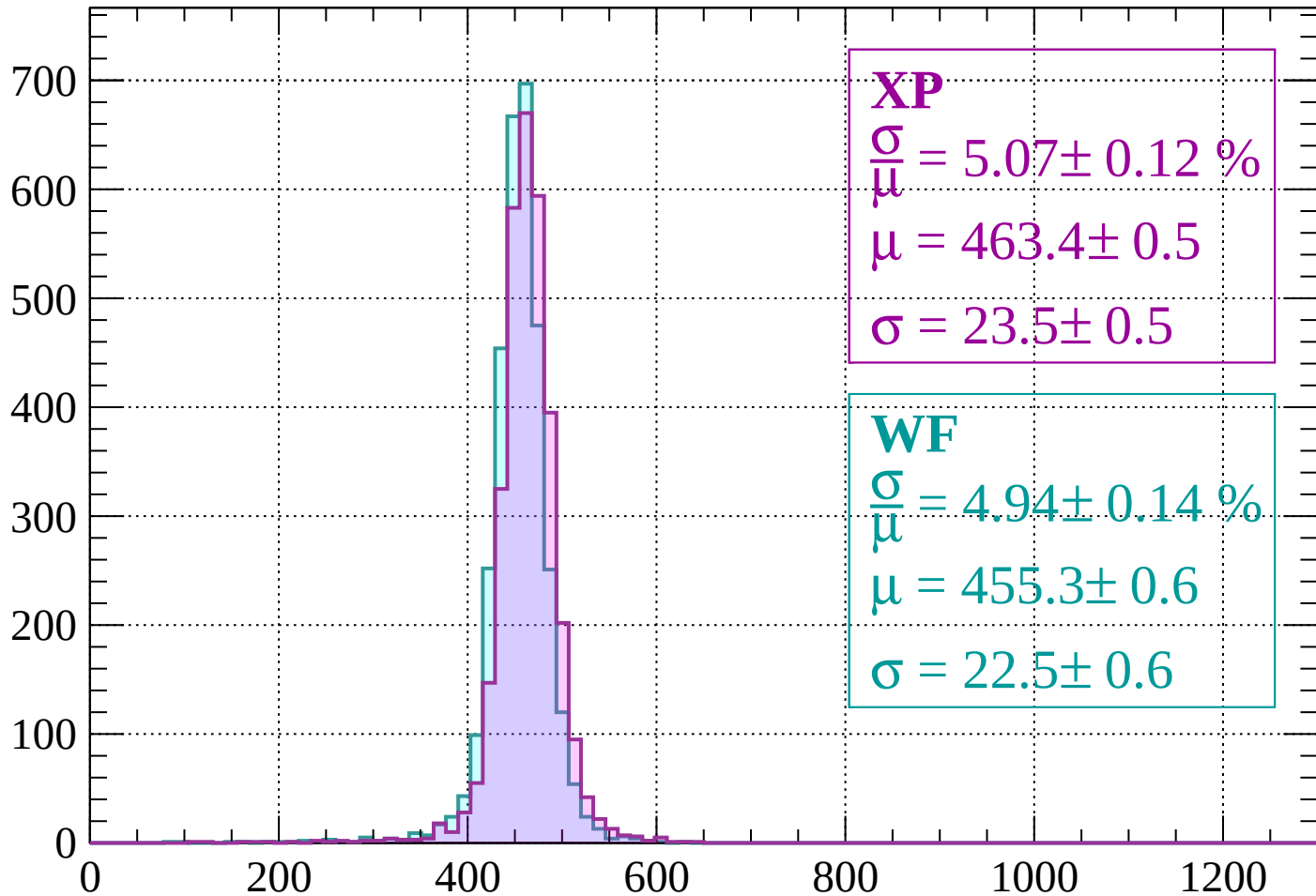
$$\mu = 458.9 \pm 0.5$$

$$\sigma = 24.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1240 < p < -1200$

Count



XP

$$\frac{\sigma}{\mu} = 5.07 \pm 0.12 \%$$

$$\mu = 463.4 \pm 0.5$$

$$\sigma = 23.5 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 4.94 \pm 0.14 \%$$

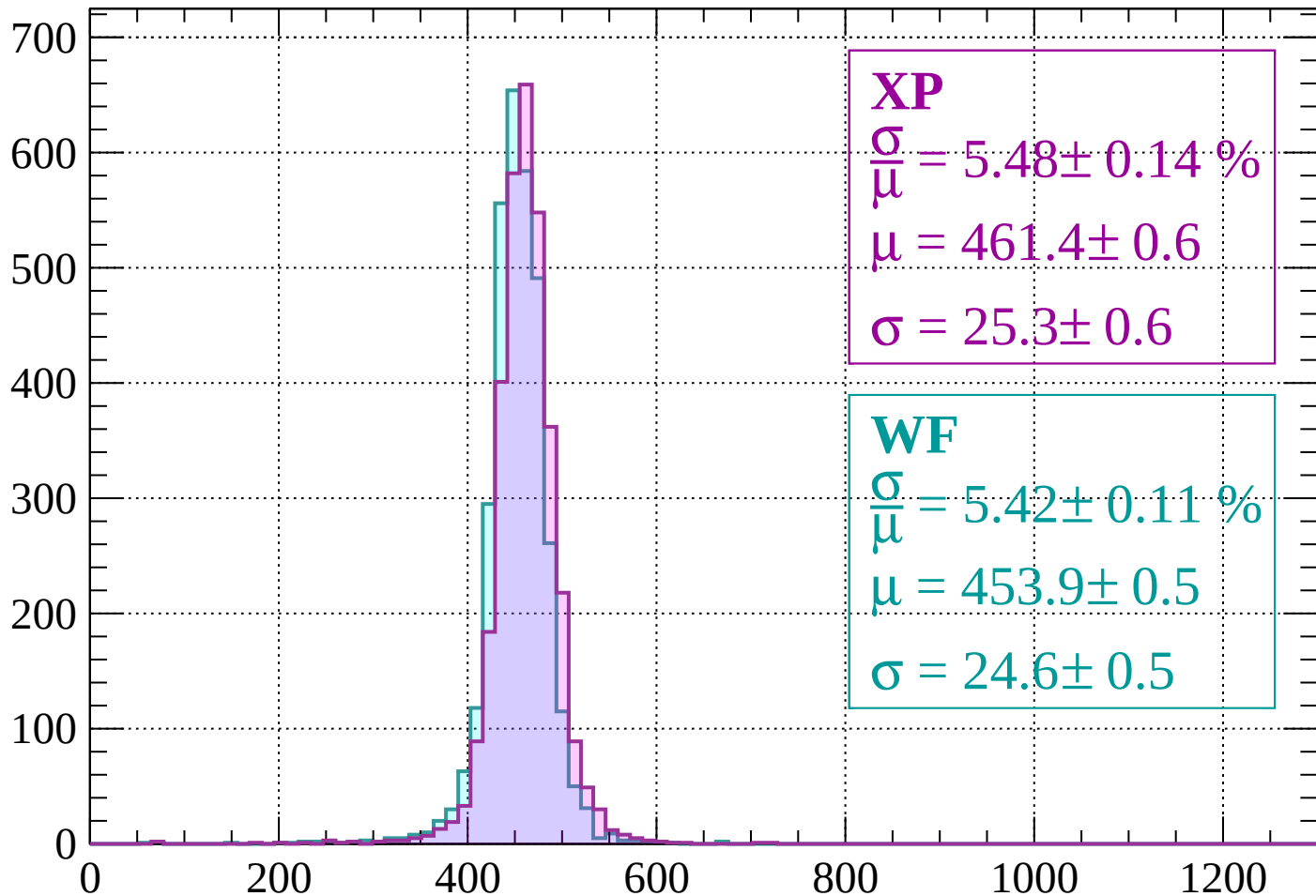
$$\mu = 455.3 \pm 0.6$$

$$\sigma = 22.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1200 < p < -1160$

Count



XP

$$\frac{\sigma}{\mu} = 5.48 \pm 0.14 \%$$

$$\mu = 461.4 \pm 0.6$$

$$\sigma = 25.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.42 \pm 0.11 \%$$

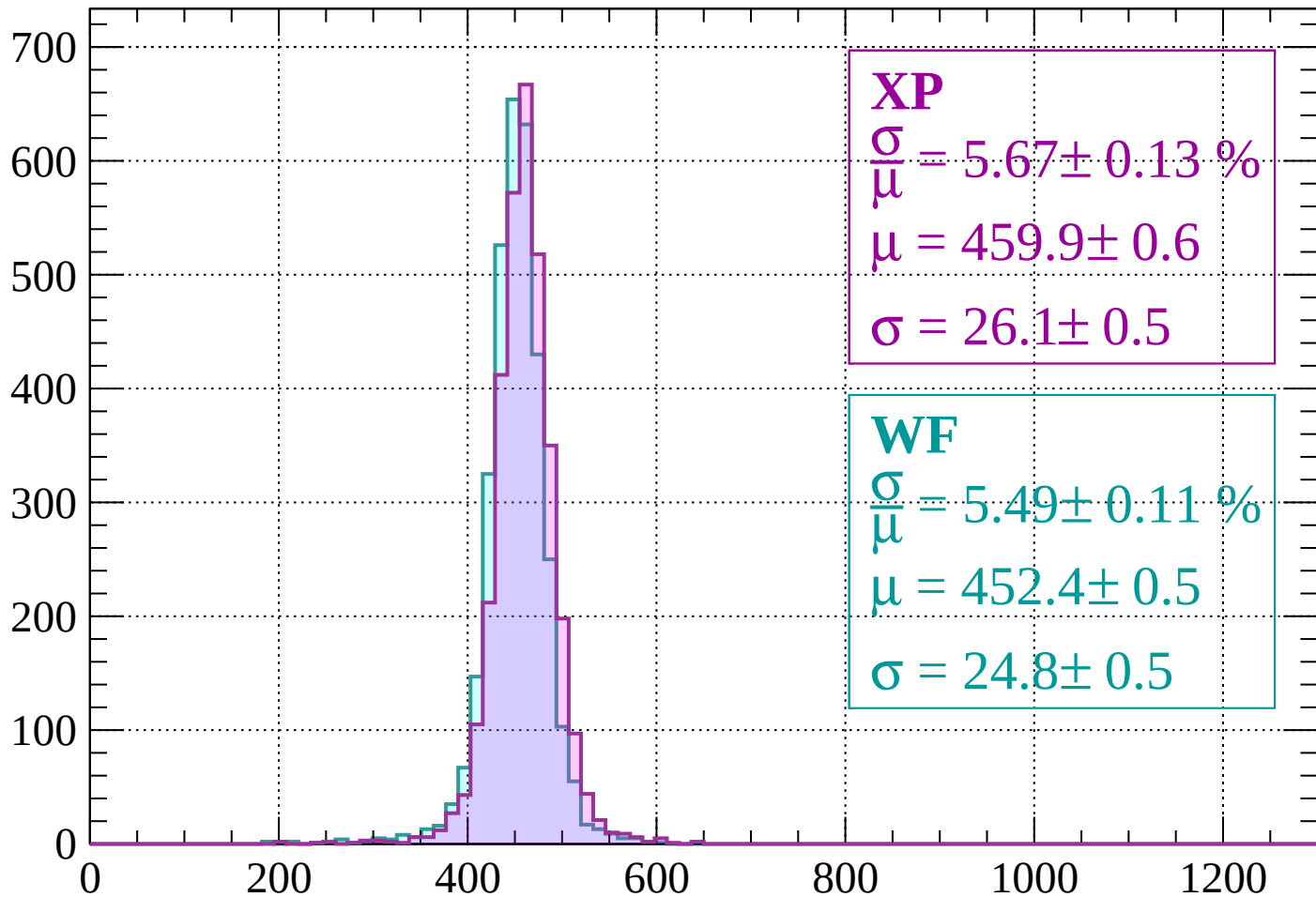
$$\mu = 453.9 \pm 0.5$$

$$\sigma = 24.6 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1160 < p < -1120$

Count



XP

$$\frac{\sigma}{\mu} = 5.67 \pm 0.13 \%$$

$$\mu = 459.9 \pm 0.6$$

$$\sigma = 26.1 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.49 \pm 0.11 \%$$

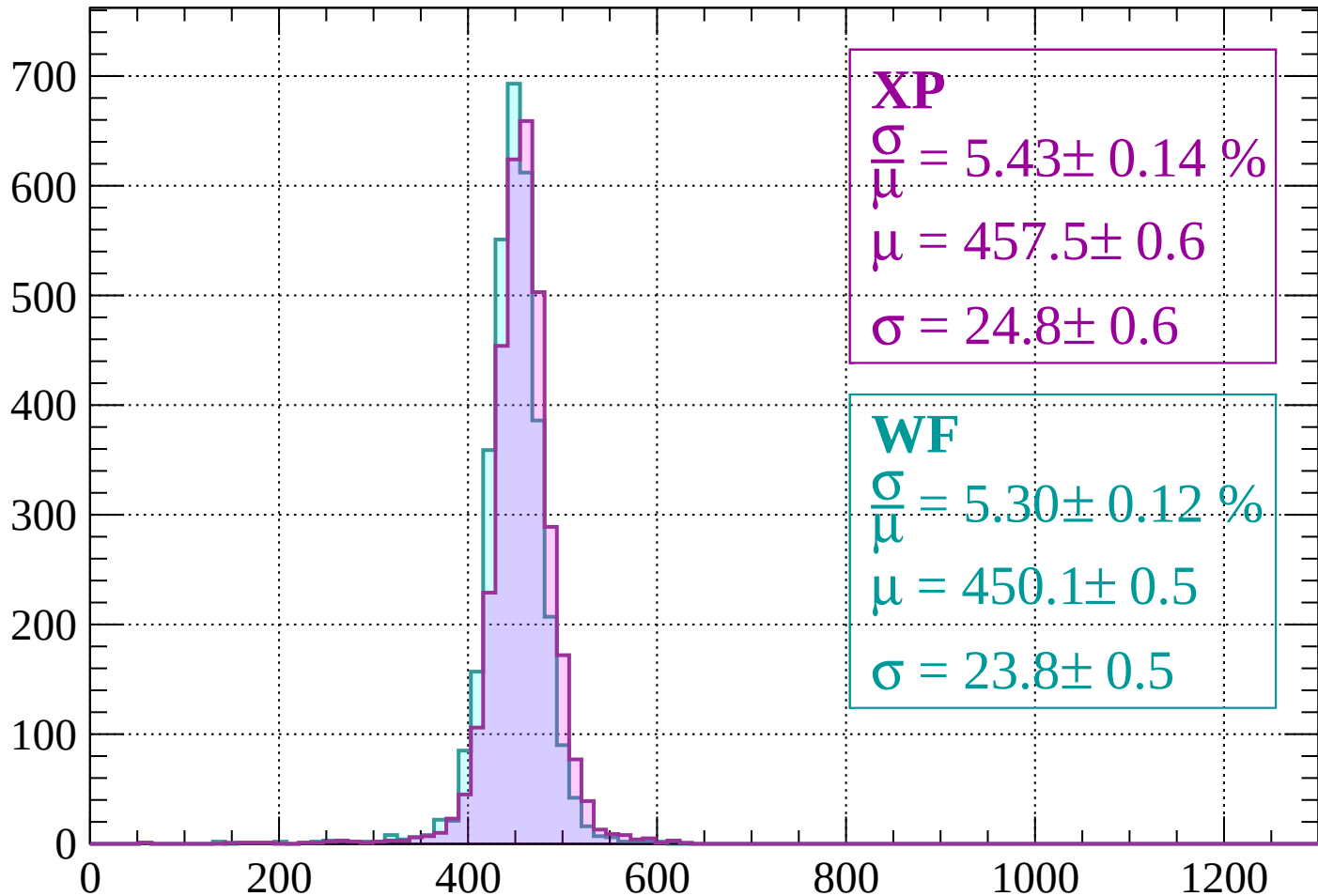
$$\mu = 452.4 \pm 0.5$$

$$\sigma = 24.8 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | -1120 < p < -1080

Count



XP

$$\frac{\sigma}{\mu} = 5.43 \pm 0.14 \%$$

$$\mu = 457.5 \pm 0.6$$

$$\sigma = 24.8 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.30 \pm 0.12 \%$$

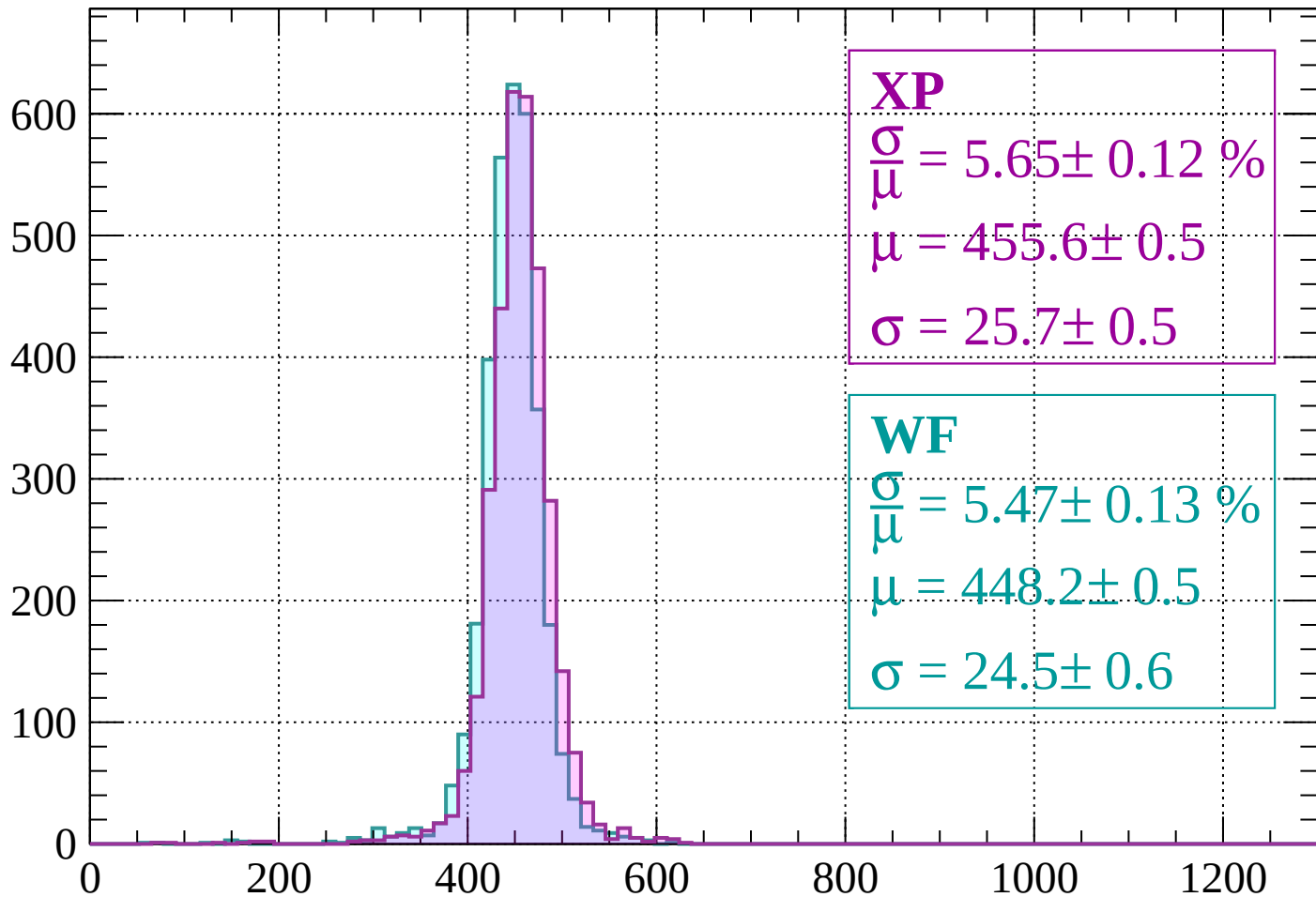
$$\mu = 450.1 \pm 0.5$$

$$\sigma = 23.8 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-1080 < p < -1040$

Count



XP

$$\frac{\sigma}{\mu} = 5.65 \pm 0.12 \%$$

$$\mu = 455.6 \pm 0.5$$

$$\sigma = 25.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.47 \pm 0.13 \%$$

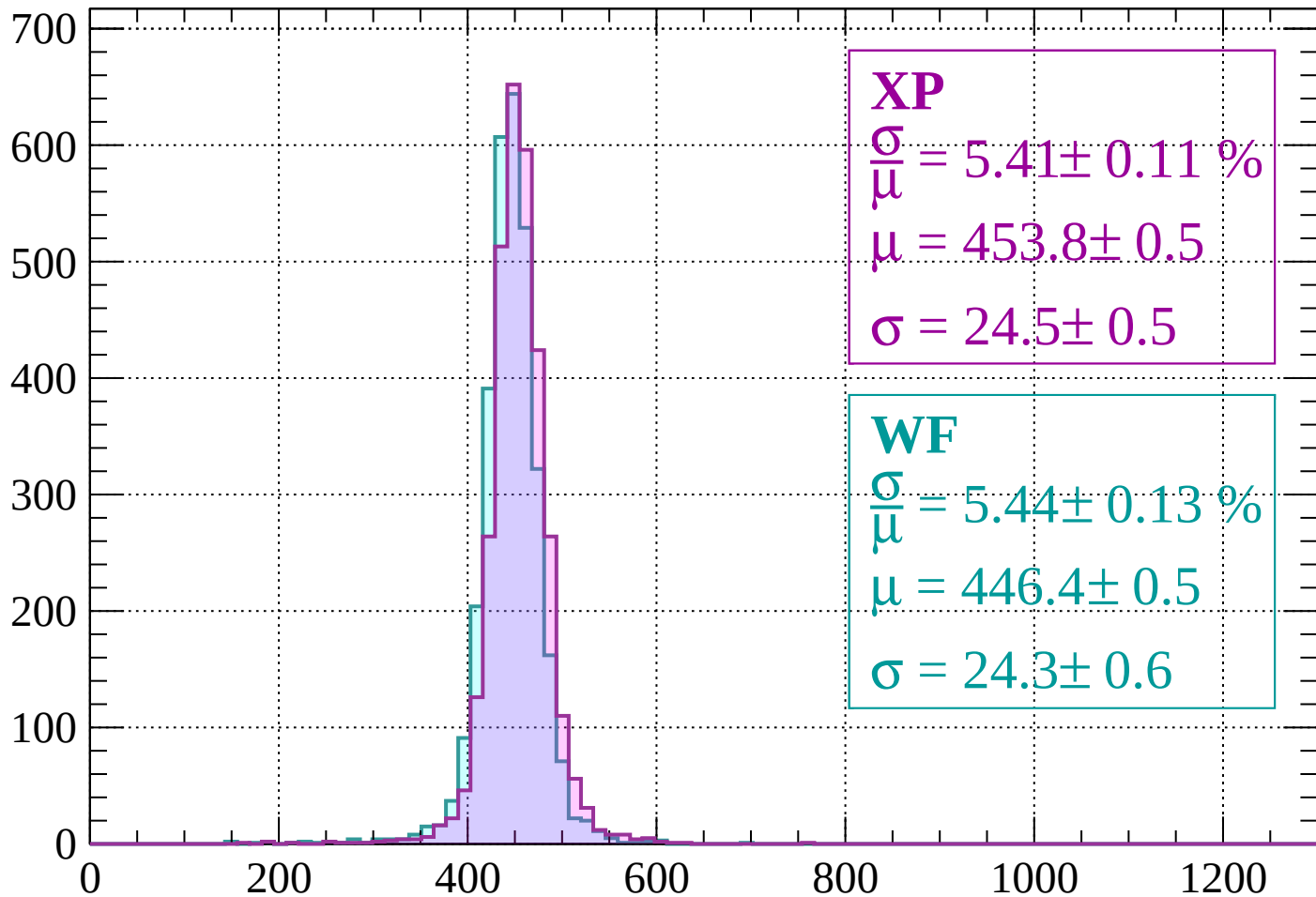
$$\mu = 448.2 \pm 0.5$$

$$\sigma = 24.5 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-1040 < p < -1000$

Count



XP

$$\frac{\sigma}{\mu} = 5.41 \pm 0.11 \%$$

$$\mu = 453.8 \pm 0.5$$

$$\sigma = 24.5 \pm 0.5$$

WF

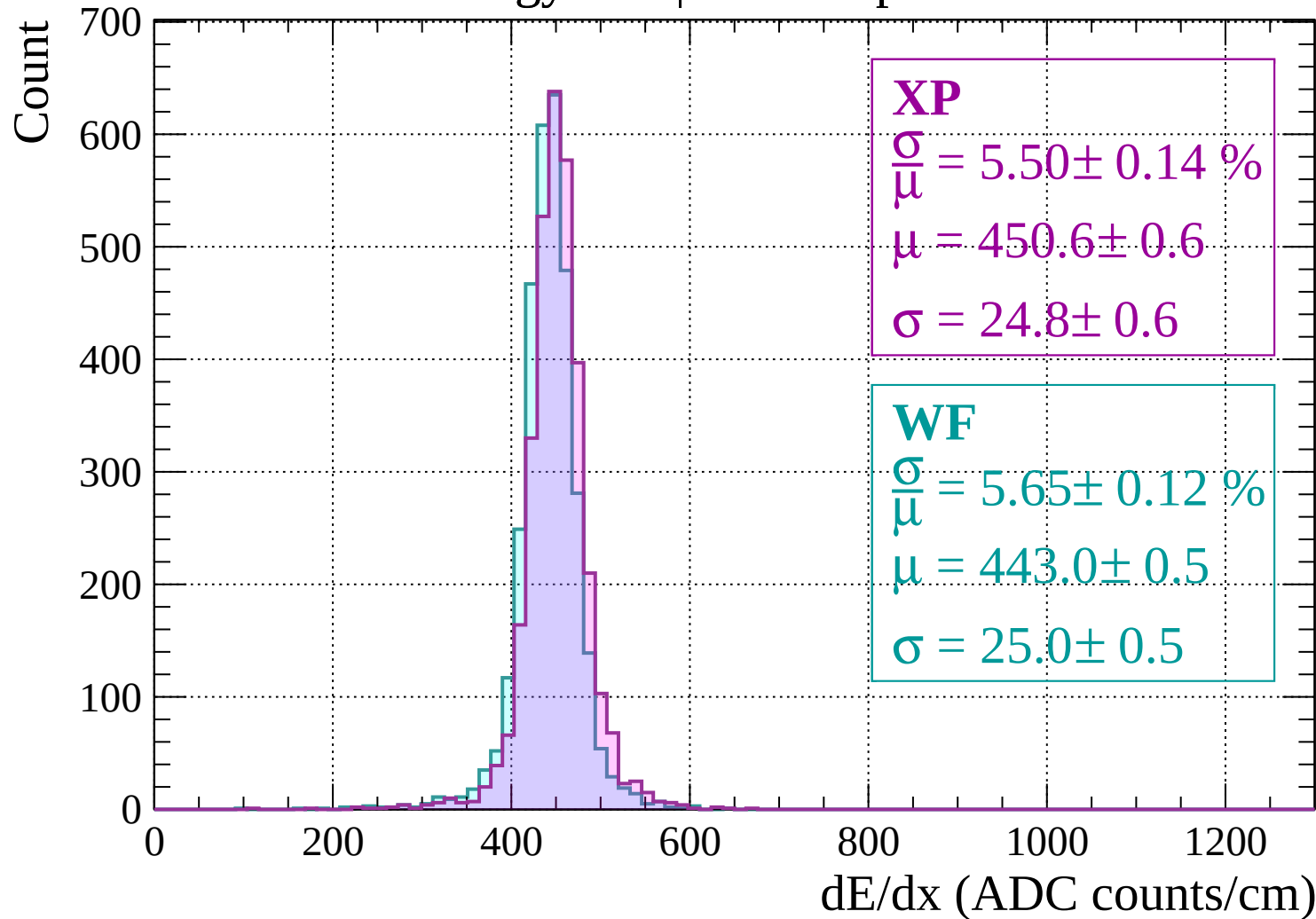
$$\frac{\sigma}{\mu} = 5.44 \pm 0.13 \%$$

$$\mu = 446.4 \pm 0.5$$

$$\sigma = 24.3 \pm 0.6$$

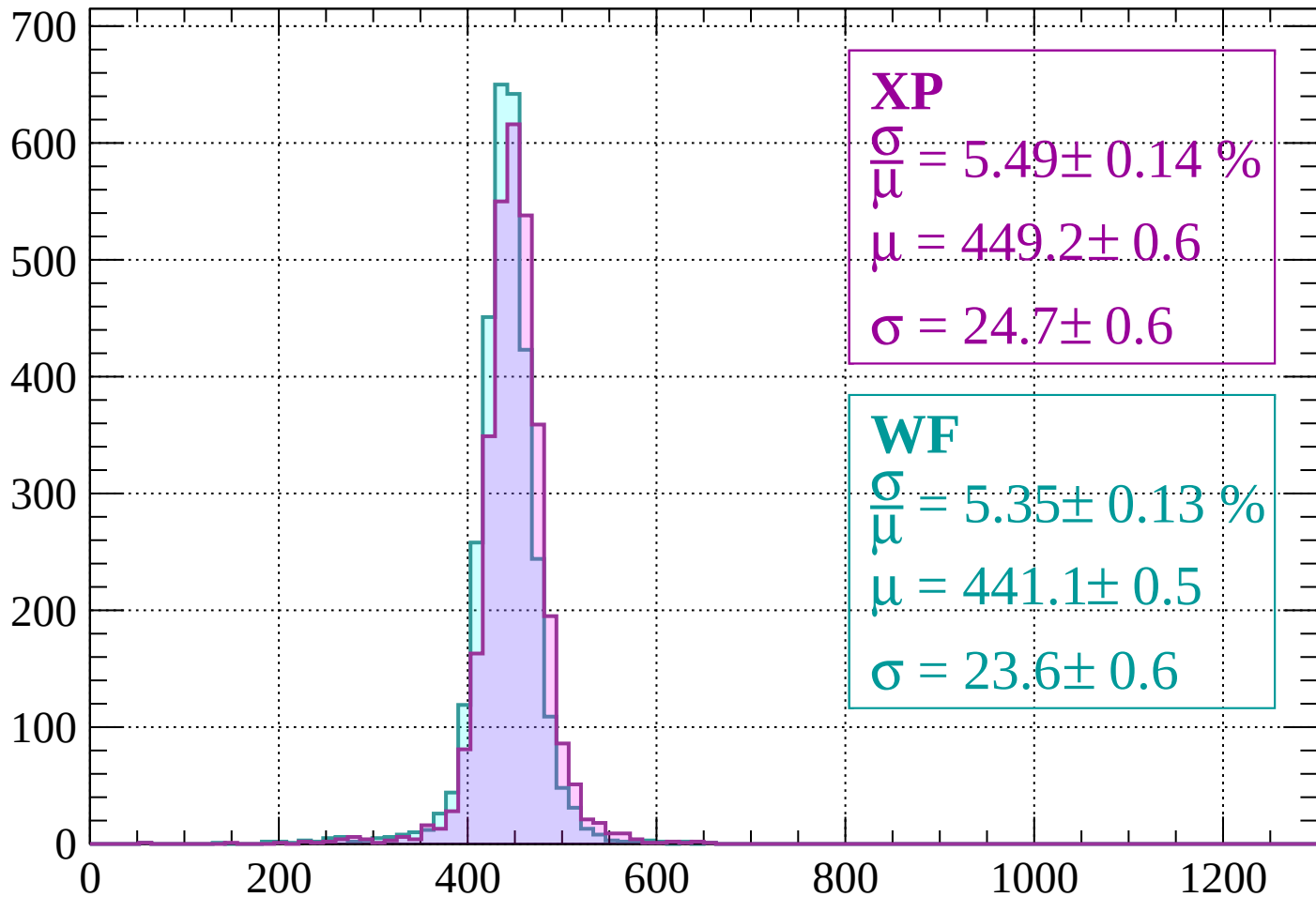
dE/dx (ADC counts/cm)

Energy loss | $-1000 < p < -960$



Energy loss | $-960 < p < -920$

Count



XP

$$\frac{\sigma}{\mu} = 5.49 \pm 0.14 \%$$

$$\mu = 449.2 \pm 0.6$$

$$\sigma = 24.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.35 \pm 0.13 \%$$

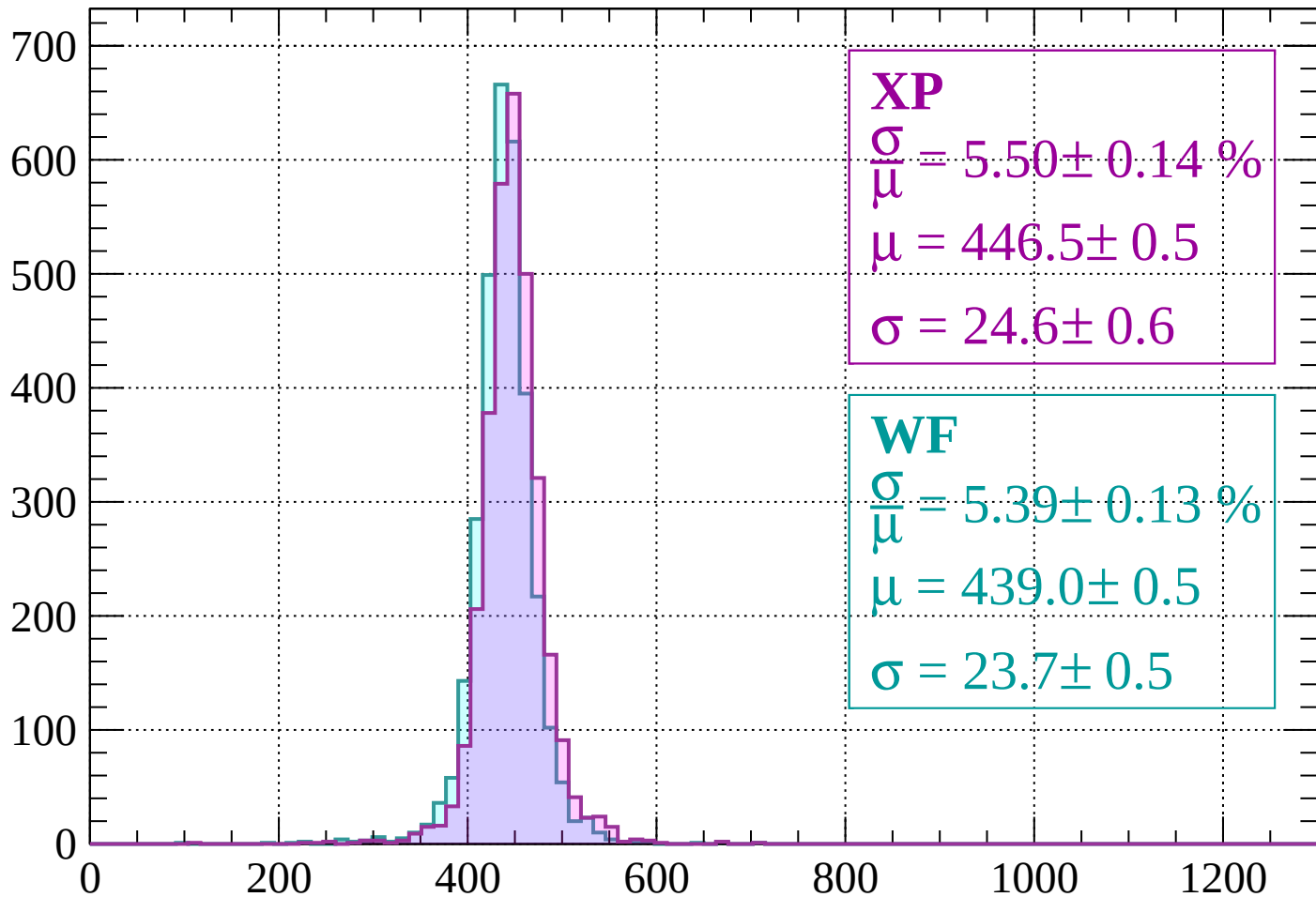
$$\mu = 441.1 \pm 0.5$$

$$\sigma = 23.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-920 < p < -880$

Count



XP

$$\frac{\sigma}{\mu} = 5.50 \pm 0.14 \%$$

$$\mu = 446.5 \pm 0.5$$

$$\sigma = 24.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.39 \pm 0.13 \%$$

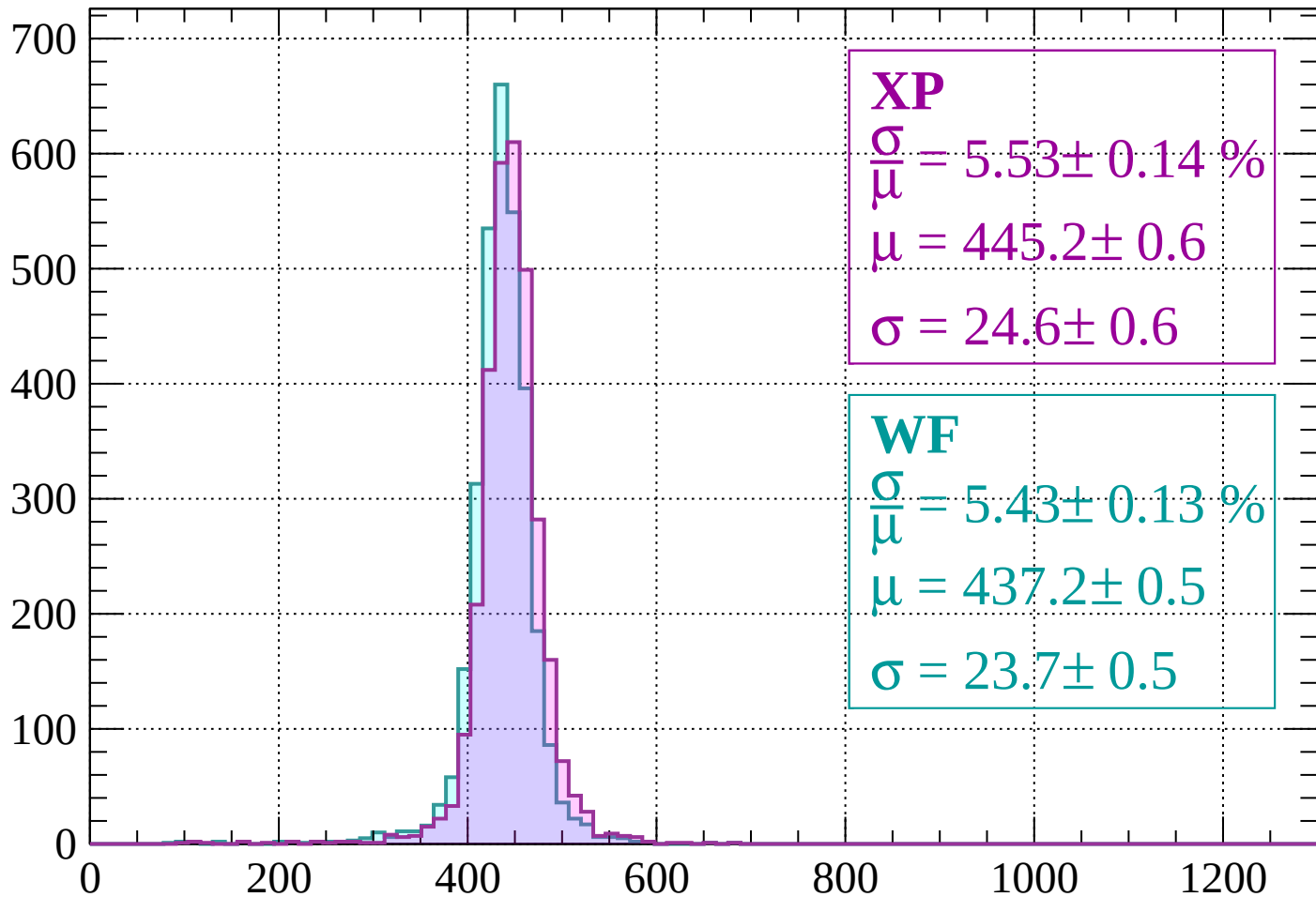
$$\mu = 439.0 \pm 0.5$$

$$\sigma = 23.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-880 < p < -840$

Count



XP

$$\frac{\sigma}{\mu} = 5.53 \pm 0.14 \%$$

$$\mu = 445.2 \pm 0.6$$

$$\sigma = 24.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.43 \pm 0.13 \%$$

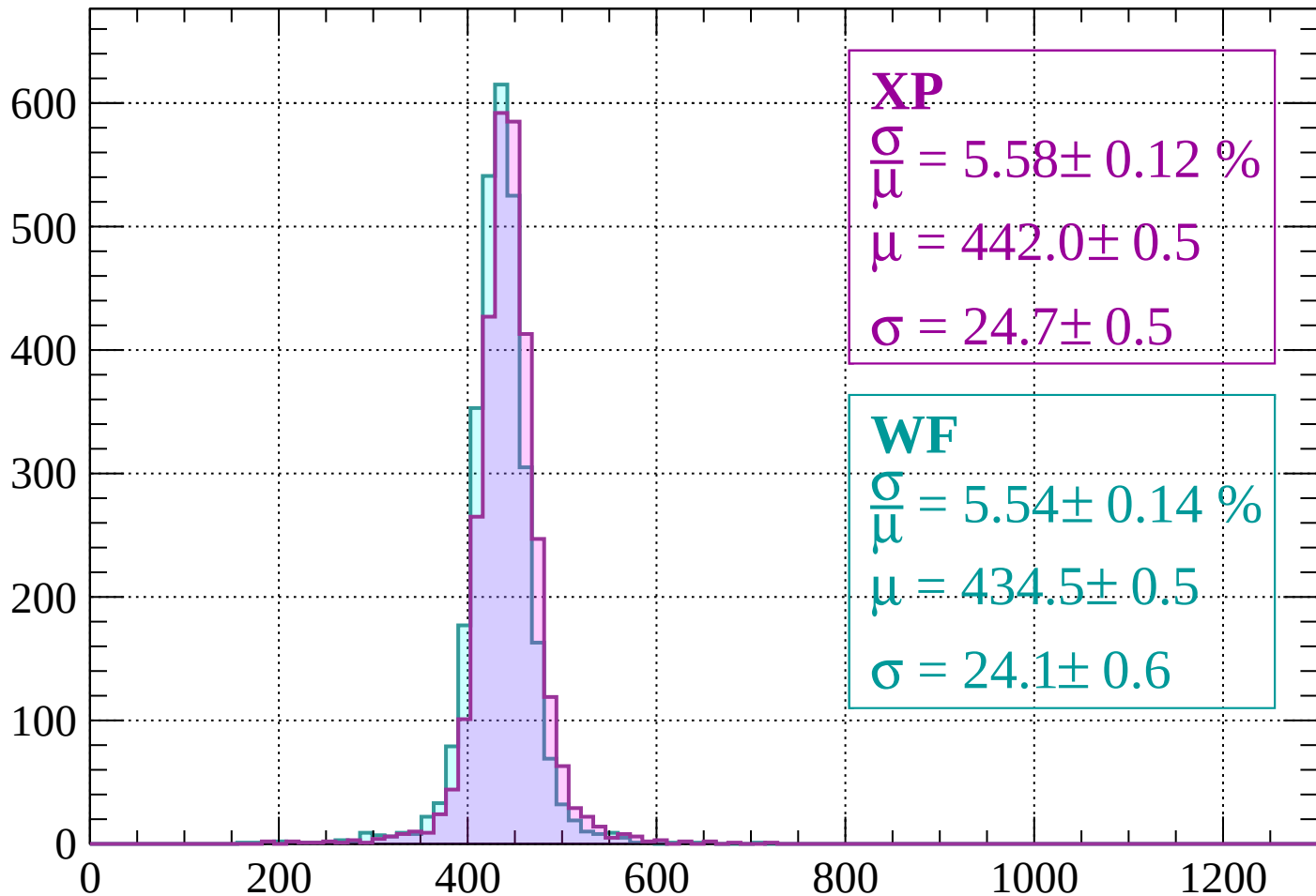
$$\mu = 437.2 \pm 0.5$$

$$\sigma = 23.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-840 < p < -800$

Count



XP

$$\frac{\sigma}{\mu} = 5.58 \pm 0.12 \%$$

$$\mu = 442.0 \pm 0.5$$

$$\sigma = 24.7 \pm 0.5$$

WF

$$\frac{\sigma}{\mu} = 5.54 \pm 0.14 \%$$

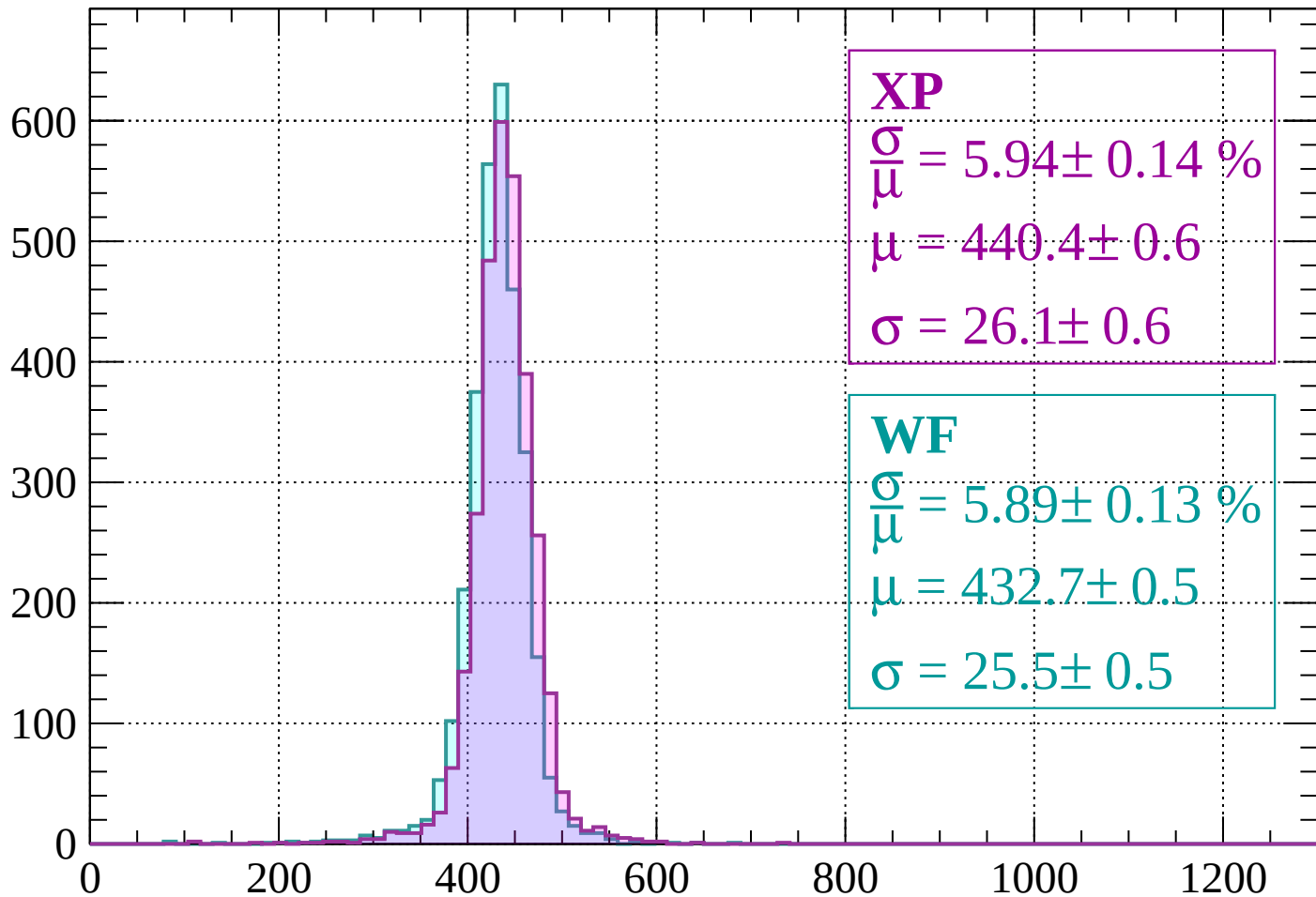
$$\mu = 434.5 \pm 0.5$$

$$\sigma = 24.1 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-800 < p < -760$

Count



XP

$$\frac{\sigma}{\mu} = 5.94 \pm 0.14 \%$$

$$\mu = 440.4 \pm 0.6$$

$$\sigma = 26.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.89 \pm 0.13 \%$$

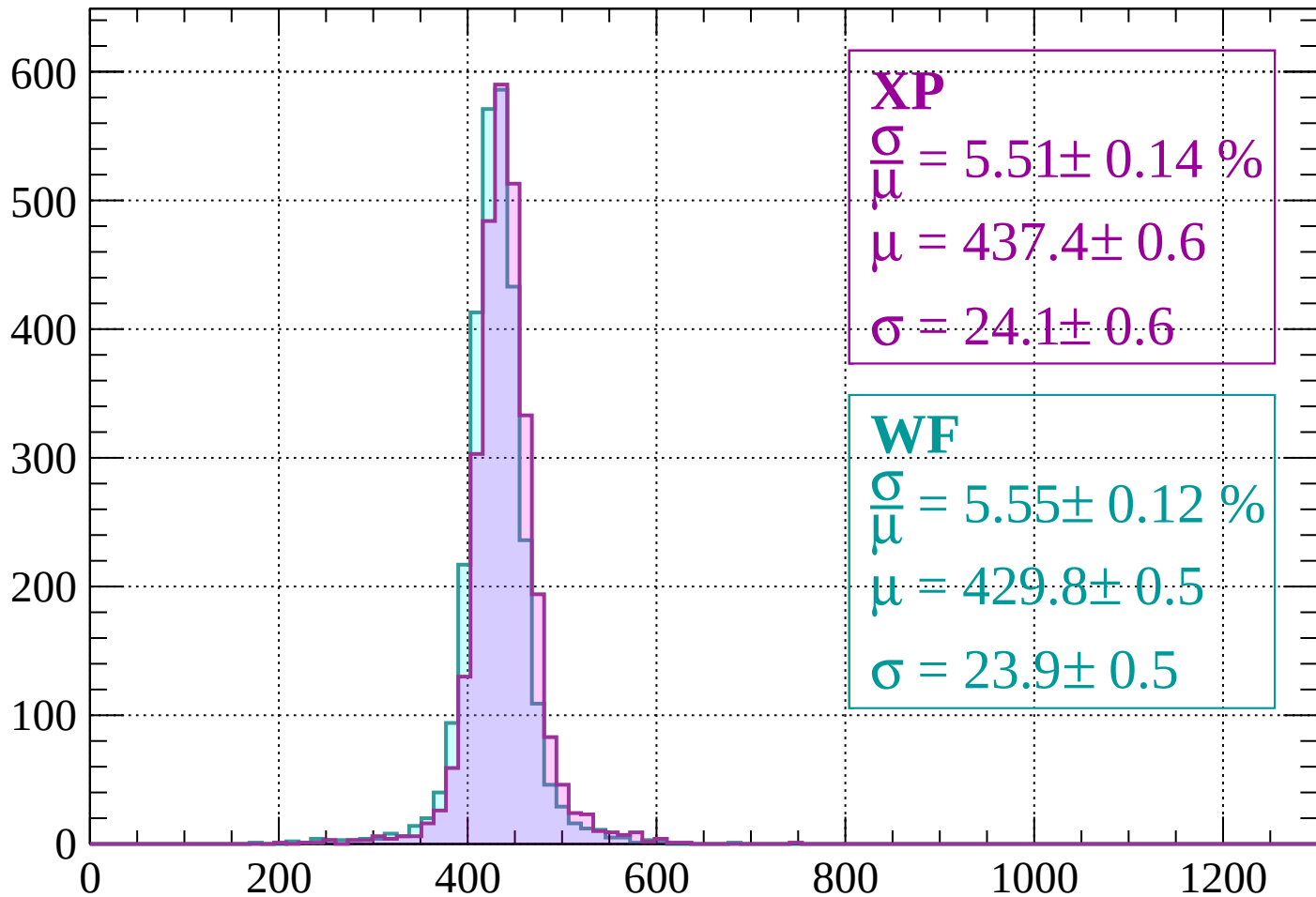
$$\mu = 432.7 \pm 0.5$$

$$\sigma = 25.5 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-760 < p < -720$

Count



XP

$$\frac{\sigma}{\mu} = 5.51 \pm 0.14 \%$$

$$\mu = 437.4 \pm 0.6$$

$$\sigma = 24.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.55 \pm 0.12 \%$$

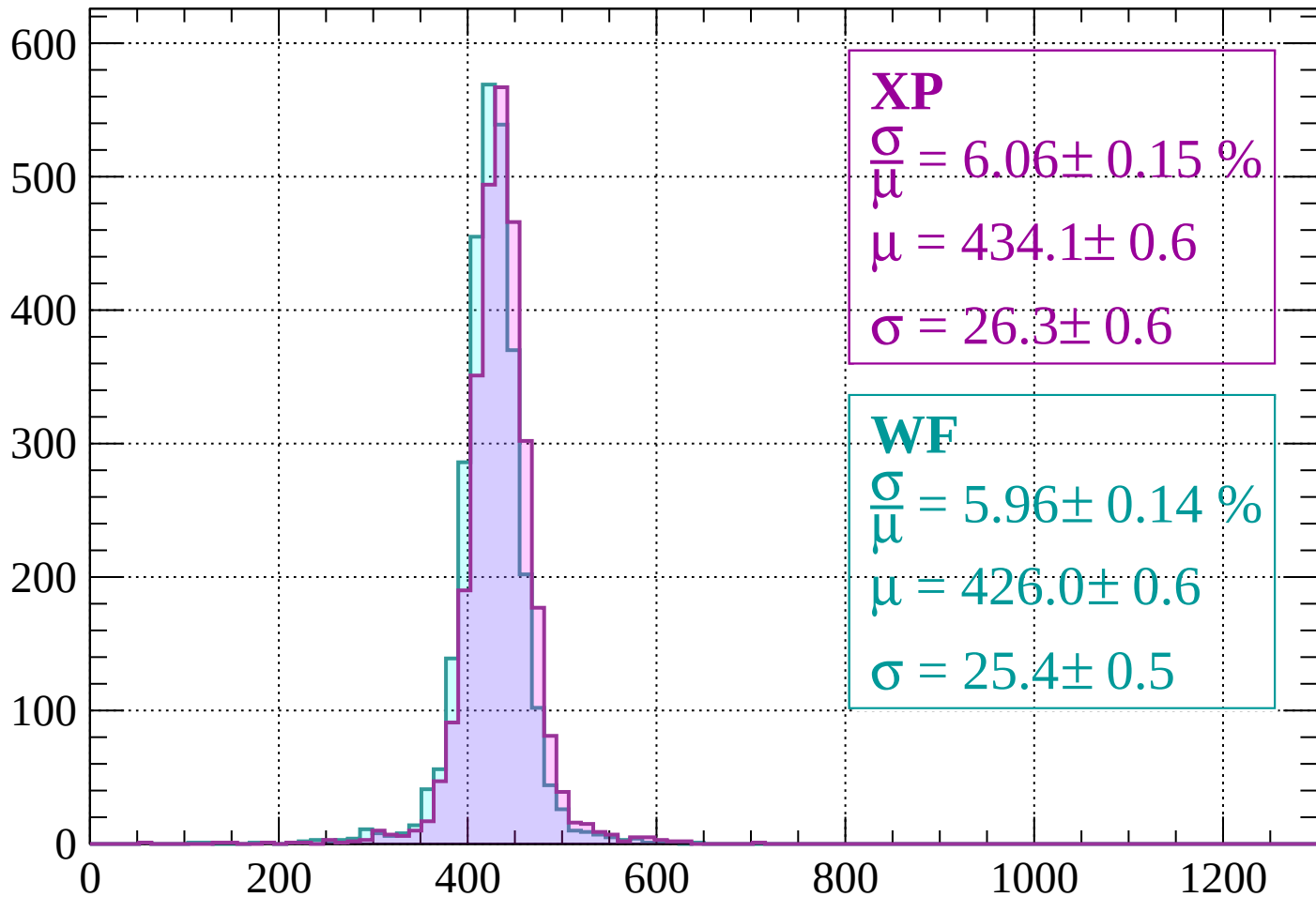
$$\mu = 429.8 \pm 0.5$$

$$\sigma = 23.9 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-720 < p < -680$

Count



XP

$$\frac{\sigma}{\mu} = 6.06 \pm 0.15 \%$$

$$\mu = 434.1 \pm 0.6$$

$$\sigma = 26.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.96 \pm 0.14 \%$$

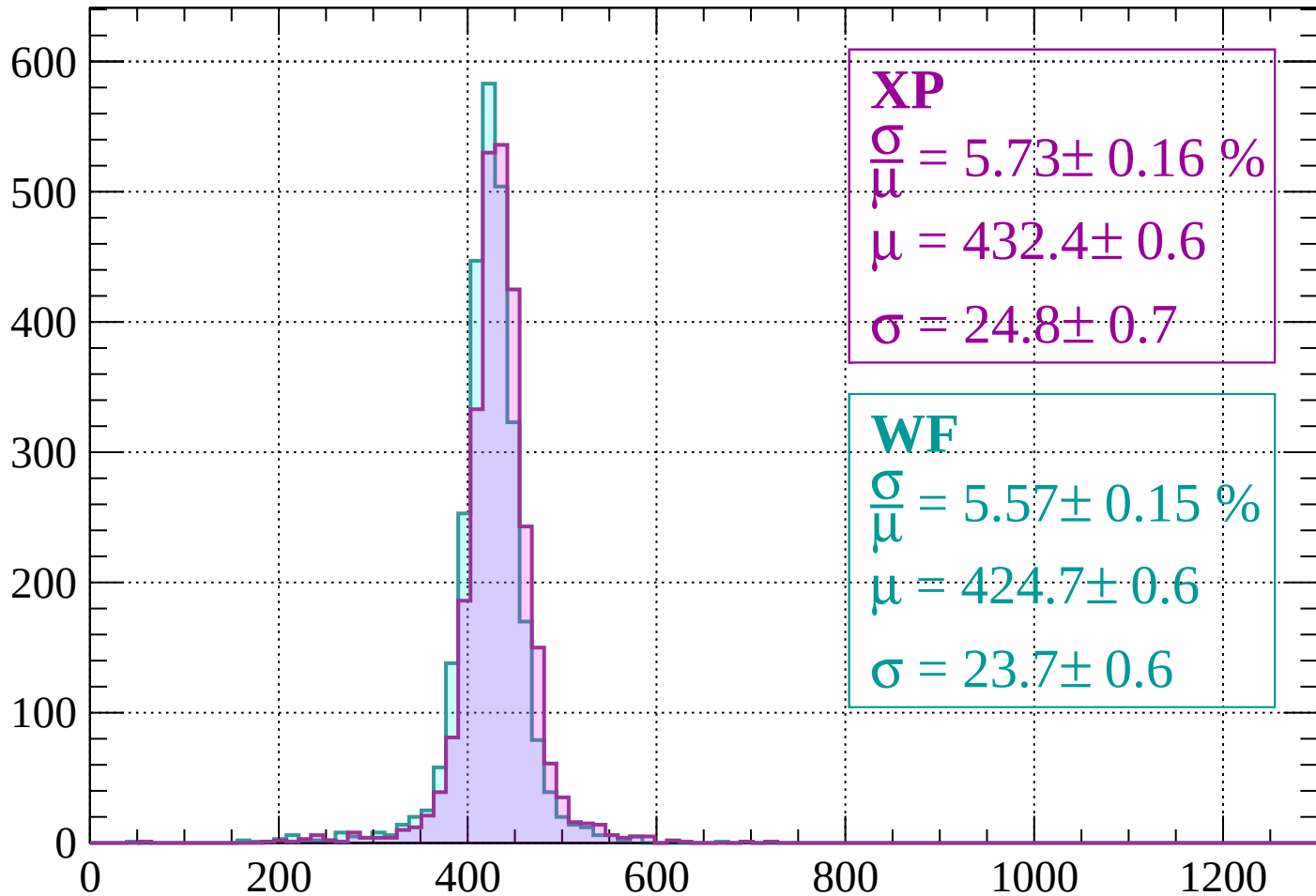
$$\mu = 426.0 \pm 0.6$$

$$\sigma = 25.4 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-680 < p < -640$

Count



XP

$$\frac{\sigma}{\mu} = 5.73 \pm 0.16 \%$$

$$\mu = 432.4 \pm 0.6$$

$$\sigma = 24.8 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.57 \pm 0.15 \%$$

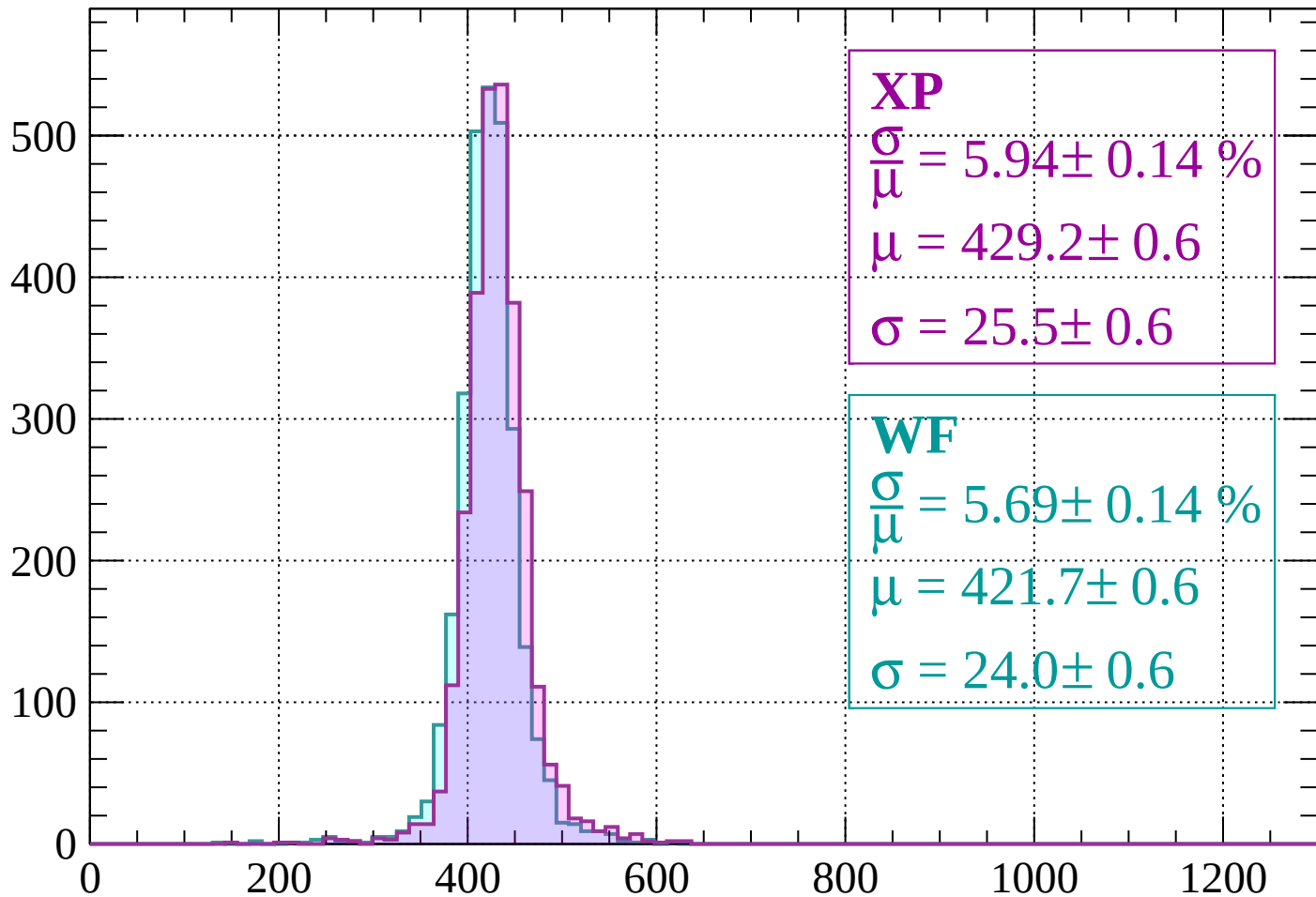
$$\mu = 424.7 \pm 0.6$$

$$\sigma = 23.7 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-640 < p < -600$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 5.94 \pm 0.14 \%$$

$$\mu = 429.2 \pm 0.6$$

$$\sigma = 25.5 \pm 0.6$$

WF

$$\frac{\sigma}{\bar{\mu}} = 5.69 \pm 0.14 \%$$

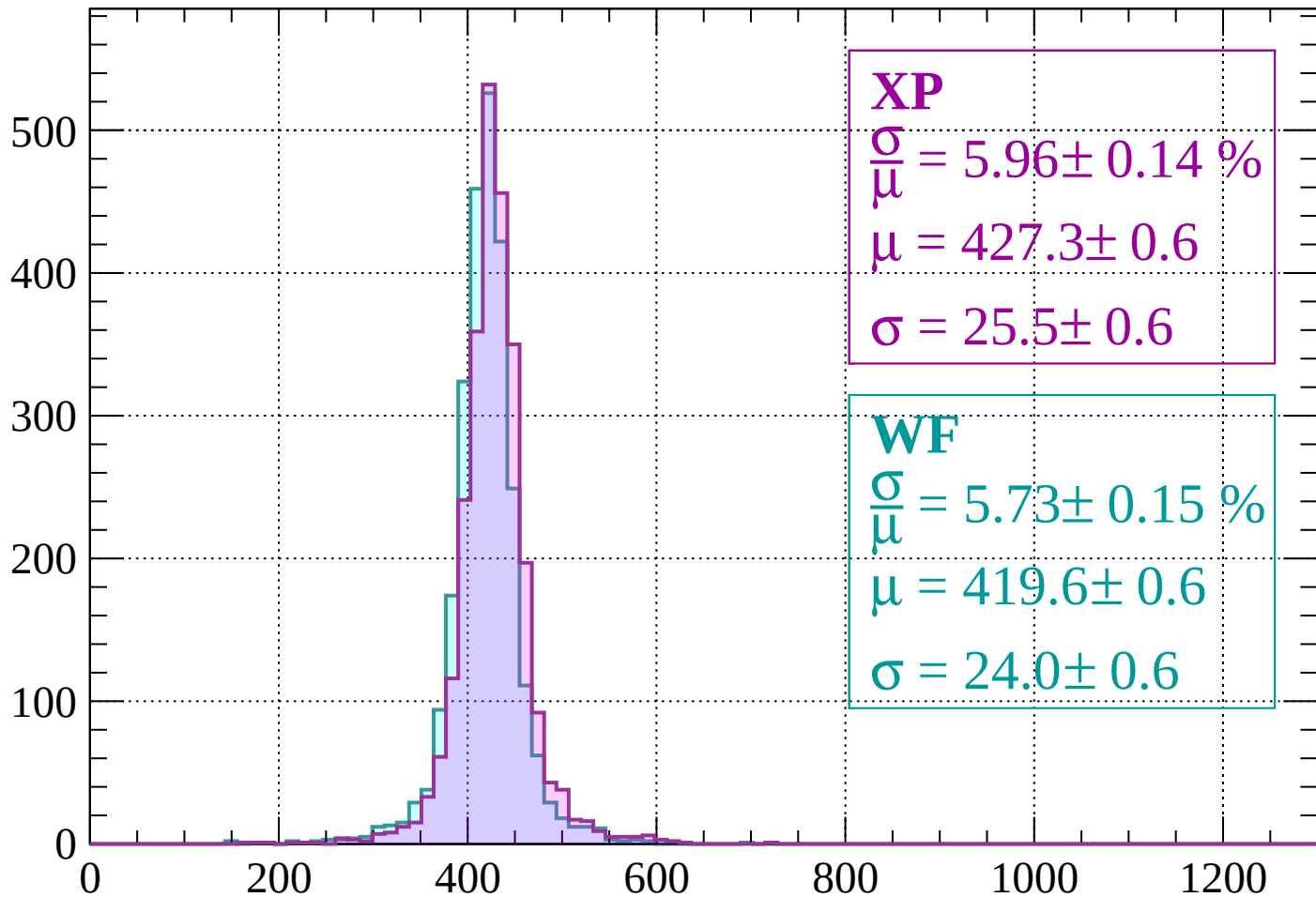
$$\mu = 421.7 \pm 0.6$$

$$\sigma = 24.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-600 < p < -560$

Count



XP

$$\frac{\sigma}{\mu} = 5.96 \pm 0.14 \%$$

$$\mu = 427.3 \pm 0.6$$

$$\sigma = 25.5 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.73 \pm 0.15 \%$$

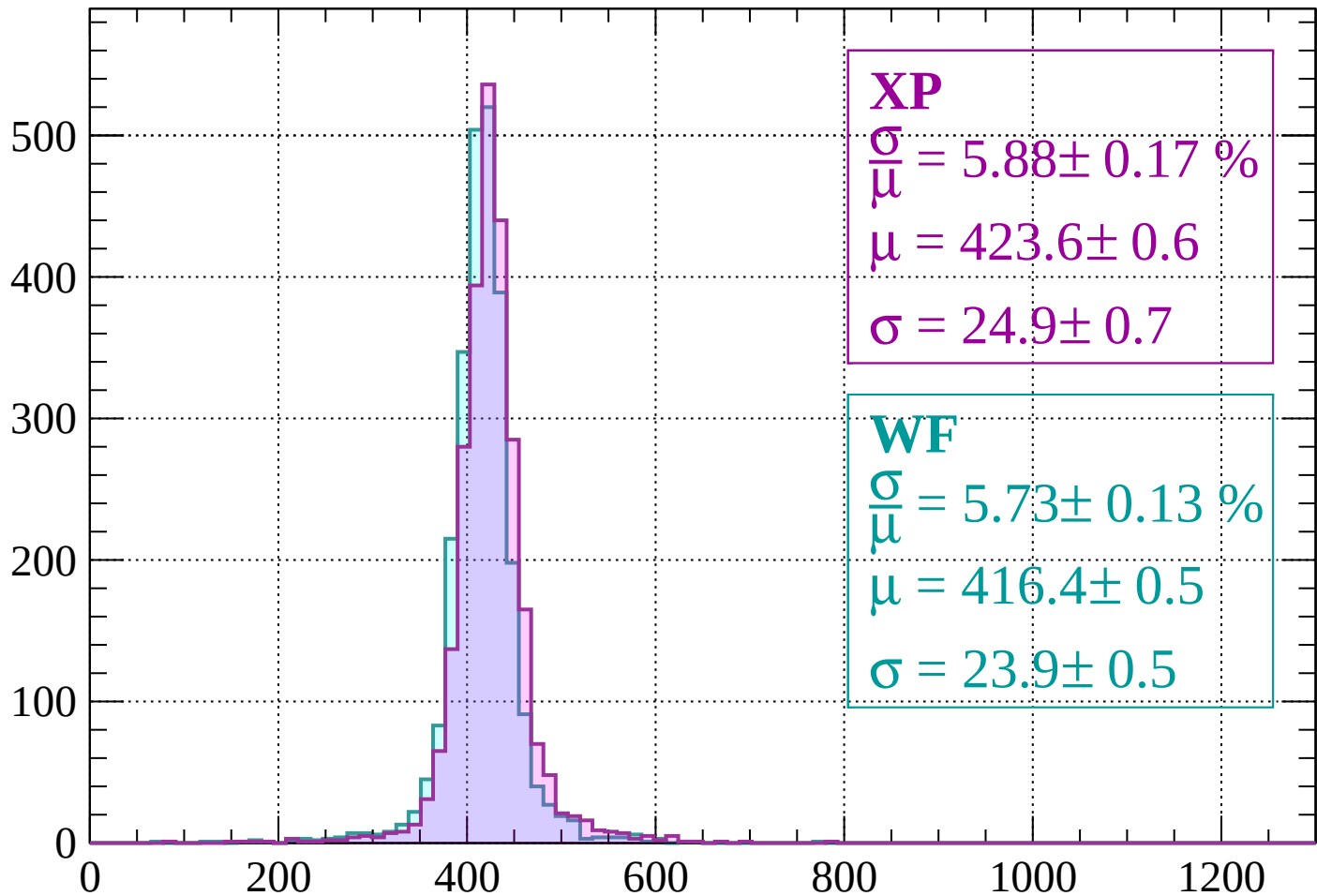
$$\mu = 419.6 \pm 0.6$$

$$\sigma = 24.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $-560 < p < -520$

Count



XP

$$\frac{\sigma}{\mu} = 5.88 \pm 0.17 \%$$

$$\mu = 423.6 \pm 0.6$$

$$\sigma = 24.9 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 5.73 \pm 0.13 \%$$

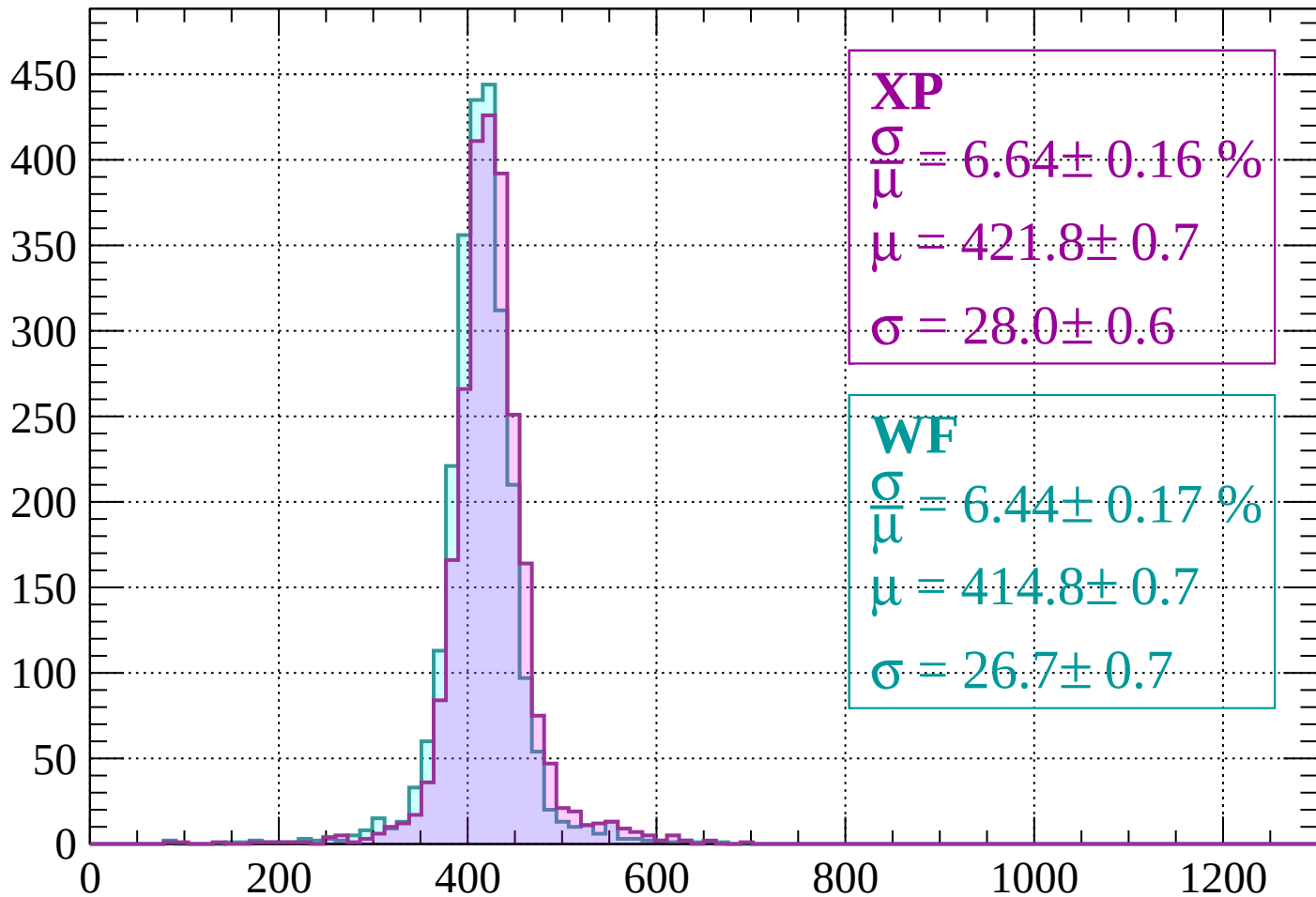
$$\mu = 416.4 \pm 0.5$$

$$\sigma = 23.9 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $-520 < p < -480$

Count



XP

$$\frac{\sigma}{\mu} = 6.64 \pm 0.16 \%$$

$$\mu = 421.8 \pm 0.7$$

$$\sigma = 28.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 6.44 \pm 0.17 \%$$

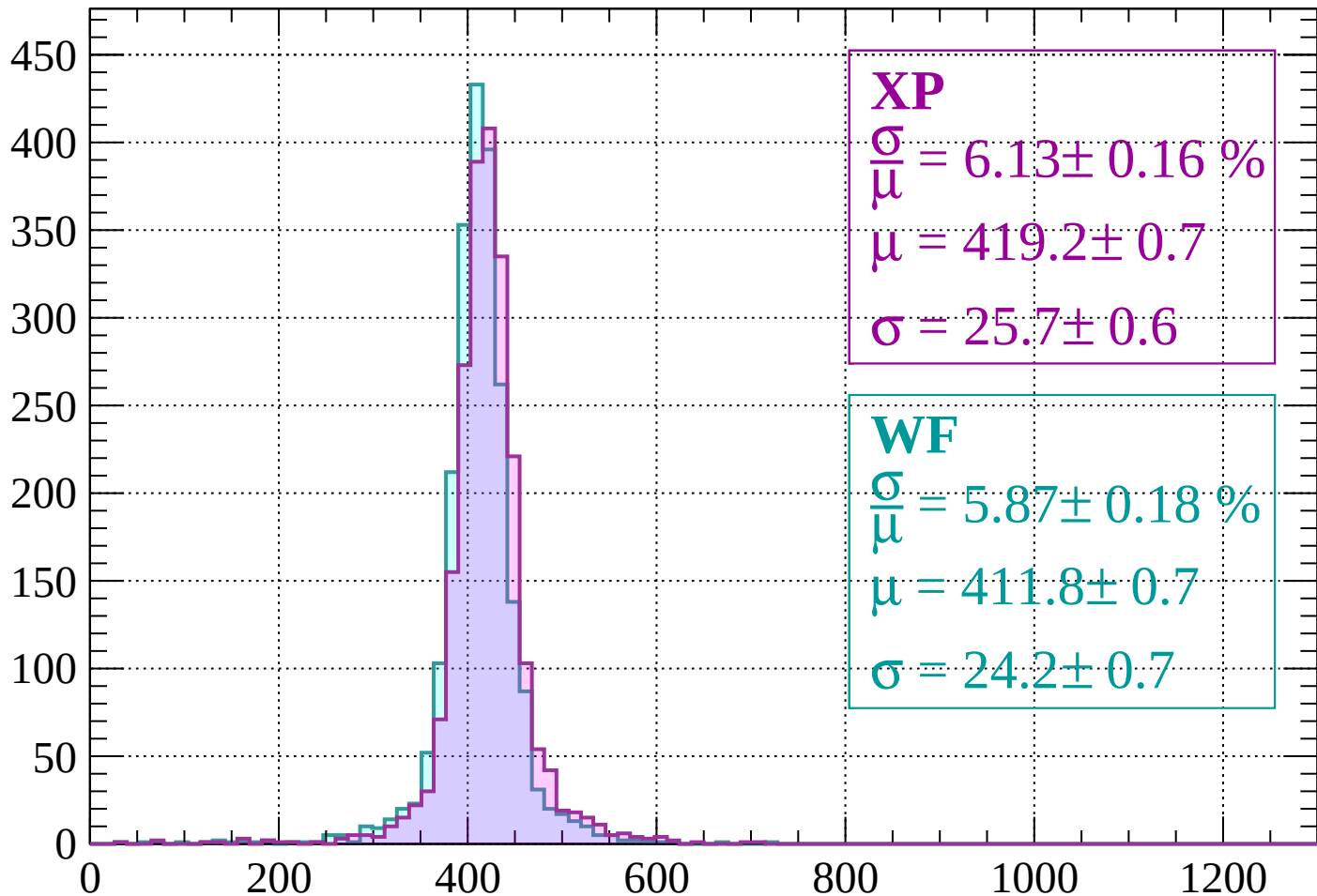
$$\mu = 414.8 \pm 0.7$$

$$\sigma = 26.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-480 < p < -440$

Count



XP

$$\frac{\sigma}{\mu} = 6.13 \pm 0.16 \%$$

$$\mu = 419.2 \pm 0.7$$

$$\sigma = 25.7 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 5.87 \pm 0.18 \%$$

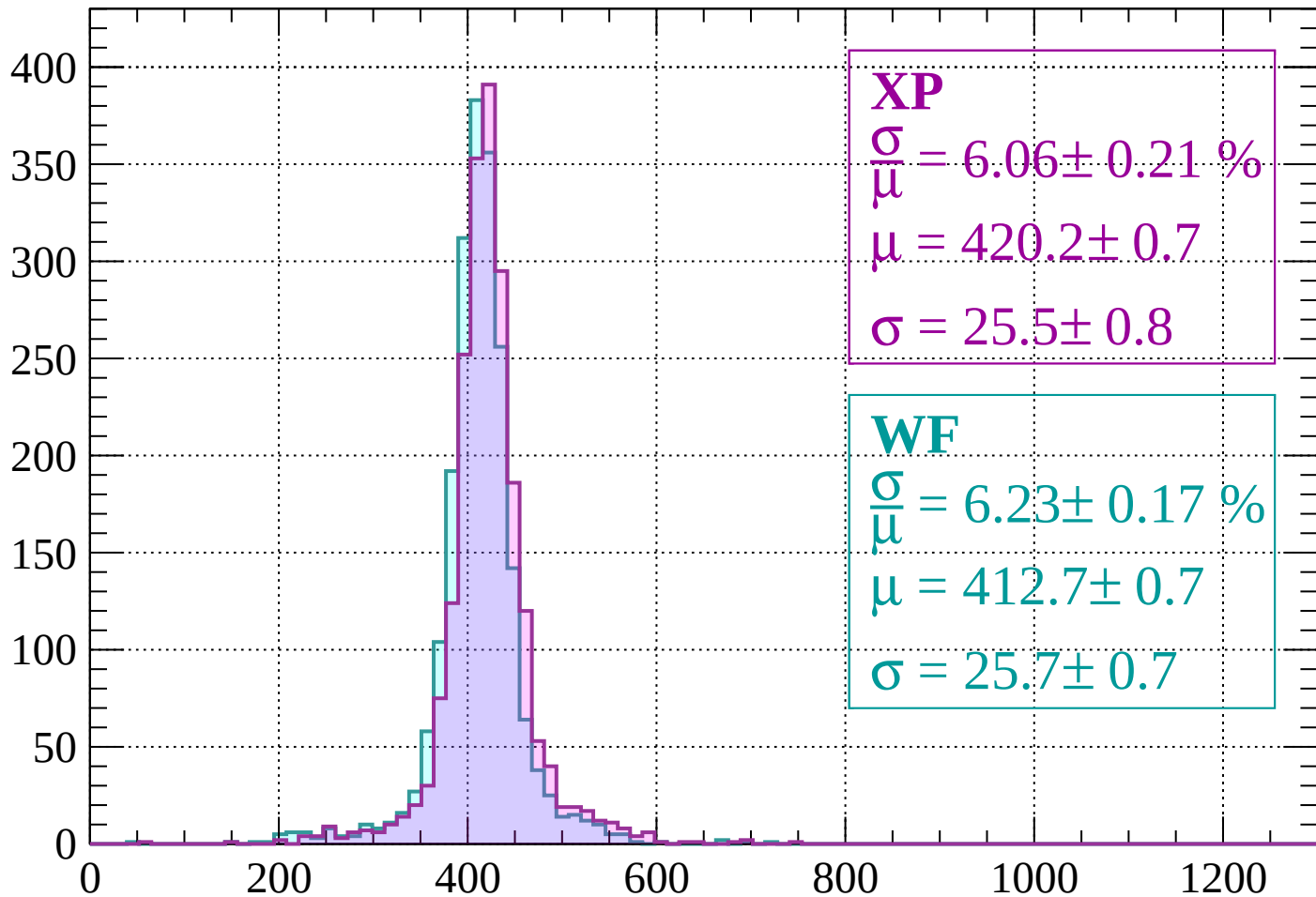
$$\mu = 411.8 \pm 0.7$$

$$\sigma = 24.2 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-440 < p < -400$

Count



XP

$$\frac{\sigma}{\mu} = 6.06 \pm 0.21 \%$$

$$\mu = 420.2 \pm 0.7$$

$$\sigma = 25.5 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.23 \pm 0.17 \%$$

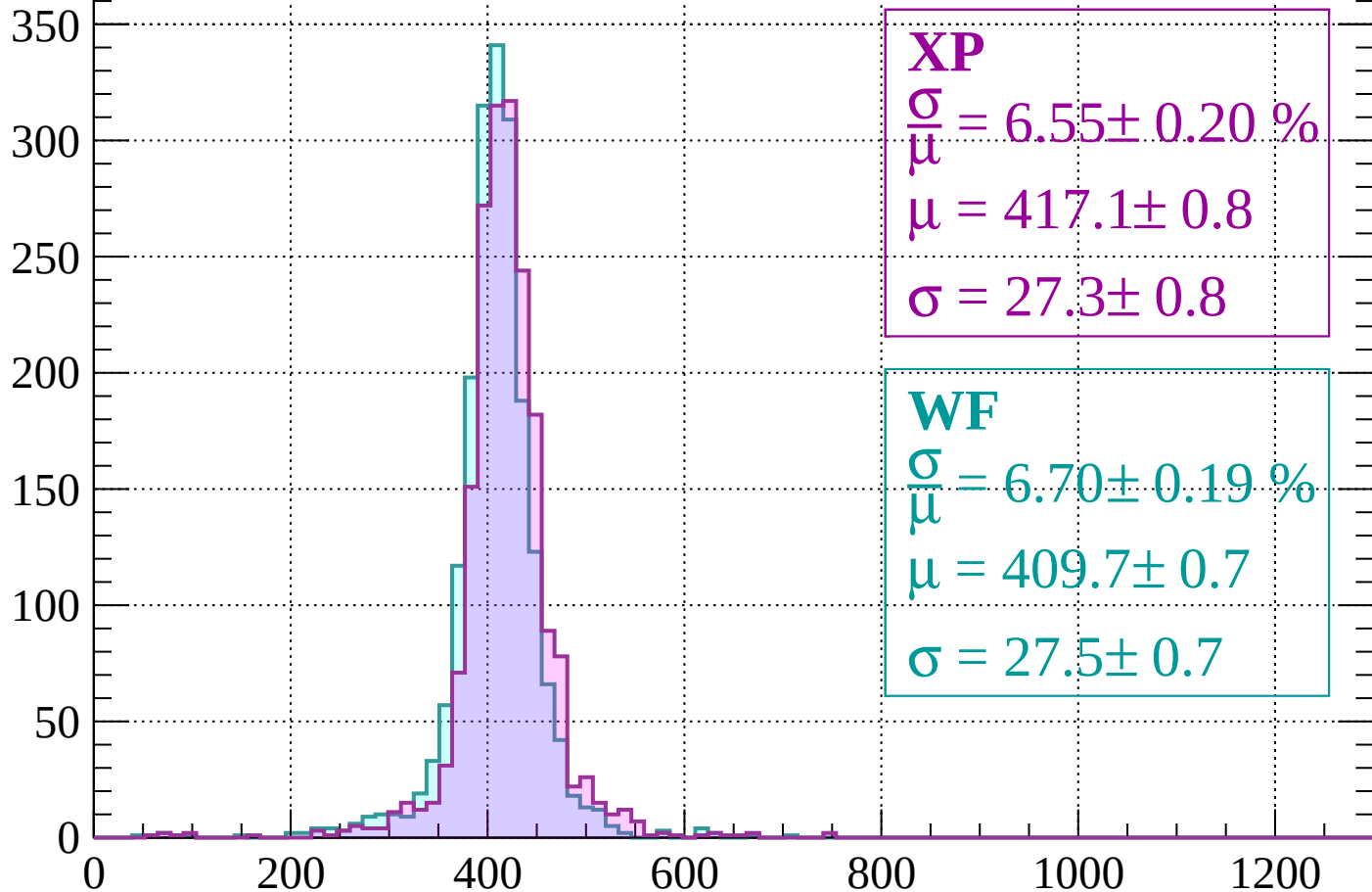
$$\mu = 412.7 \pm 0.7$$

$$\sigma = 25.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-400 < p < -360$

Count



XP

$$\frac{\sigma}{\mu} = 6.55 \pm 0.20 \%$$

$$\mu = 417.1 \pm 0.8$$

$$\sigma = 27.3 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.70 \pm 0.19 \%$$

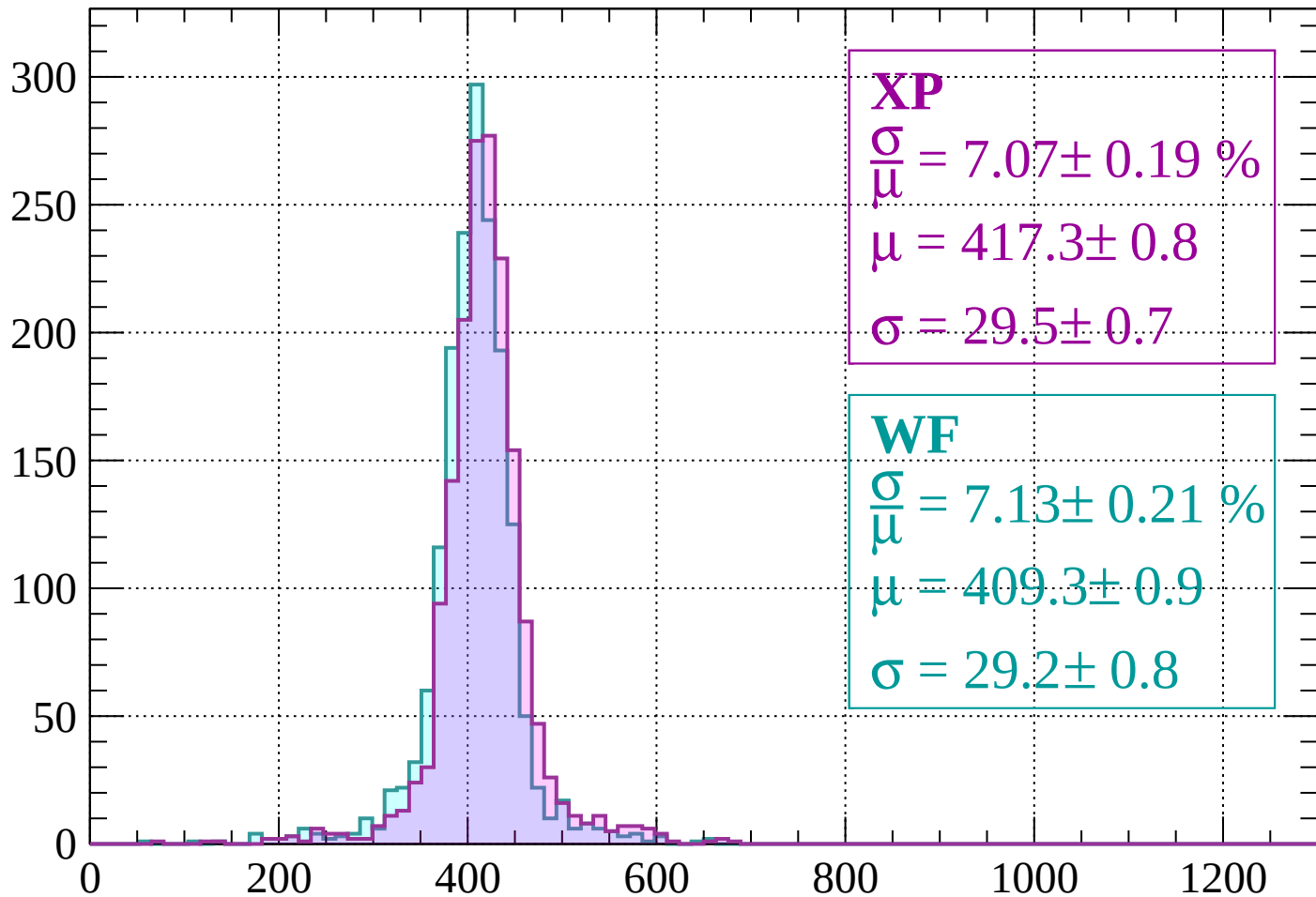
$$\mu = 409.7 \pm 0.7$$

$$\sigma = 27.5 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $-360 < p < -320$

Count



XP

$$\frac{\sigma}{\mu} = 7.07 \pm 0.19 \%$$

$$\mu = 417.3 \pm 0.8$$

$$\sigma = 29.5 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 7.13 \pm 0.21 \%$$

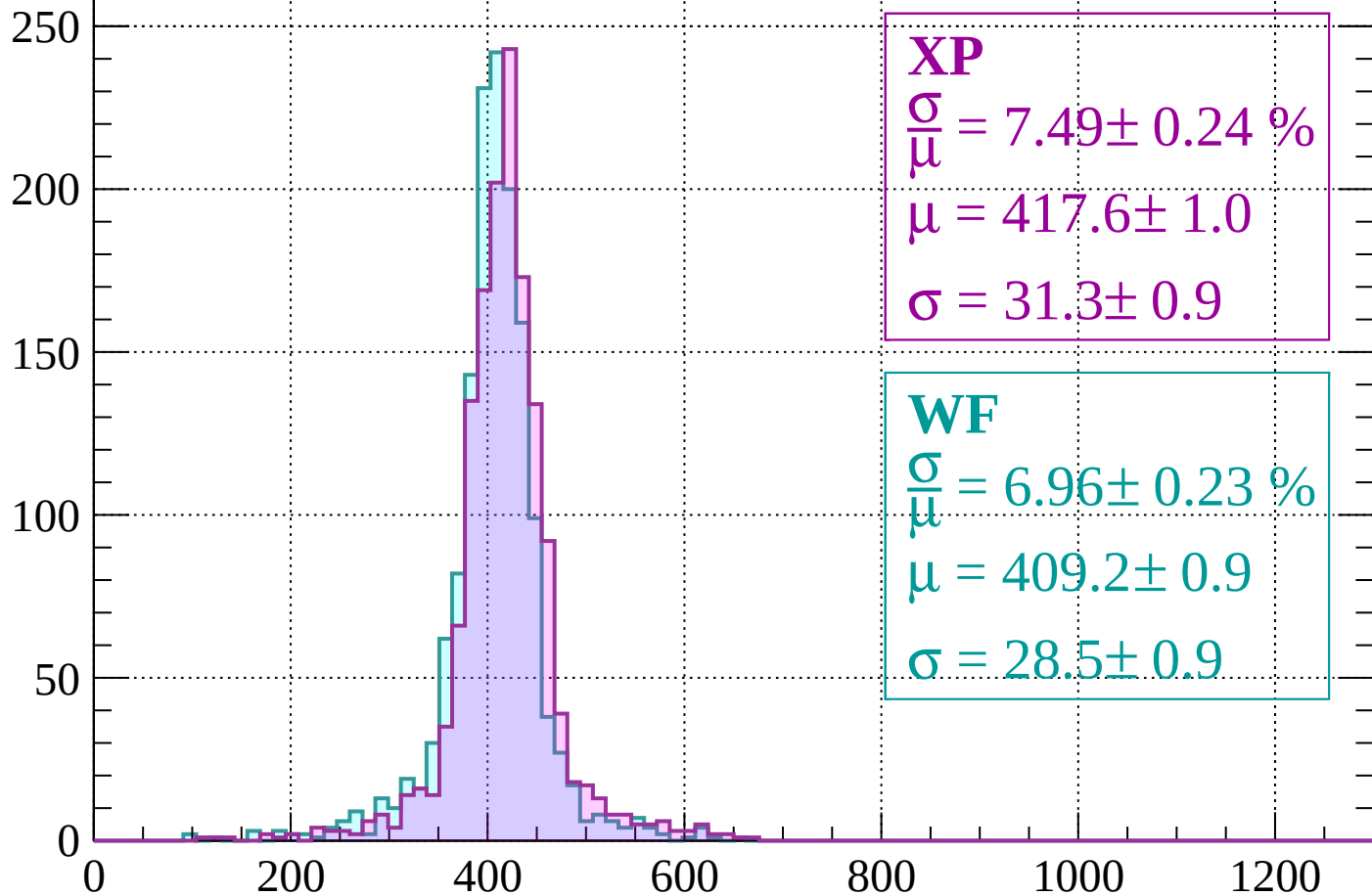
$$\mu = 409.3 \pm 0.9$$

$$\sigma = 29.2 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $-320 < p < -280$

Count



XP

$$\frac{\sigma}{\mu} = 7.49 \pm 0.24 \%$$

$$\mu = 417.6 \pm 1.0$$

$$\sigma = 31.3 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.96 \pm 0.23 \%$$

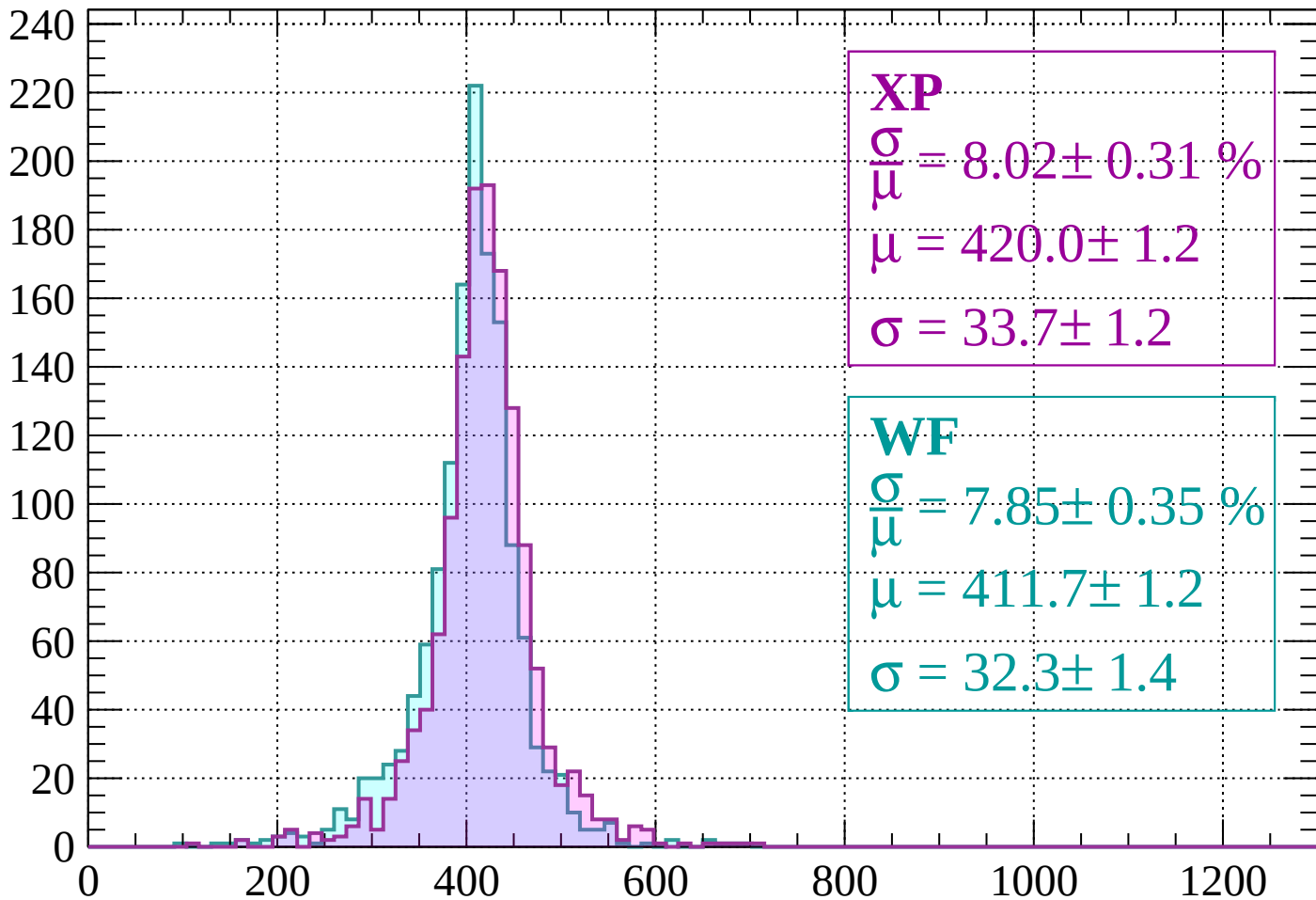
$$\mu = 409.2 \pm 0.9$$

$$\sigma = 28.5 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $-280 < p < -240$

Count



XP

$$\frac{\sigma}{\mu} = 8.02 \pm 0.31 \%$$

$$\mu = 420.0 \pm 1.2$$

$$\sigma = 33.7 \pm 1.2$$

WF

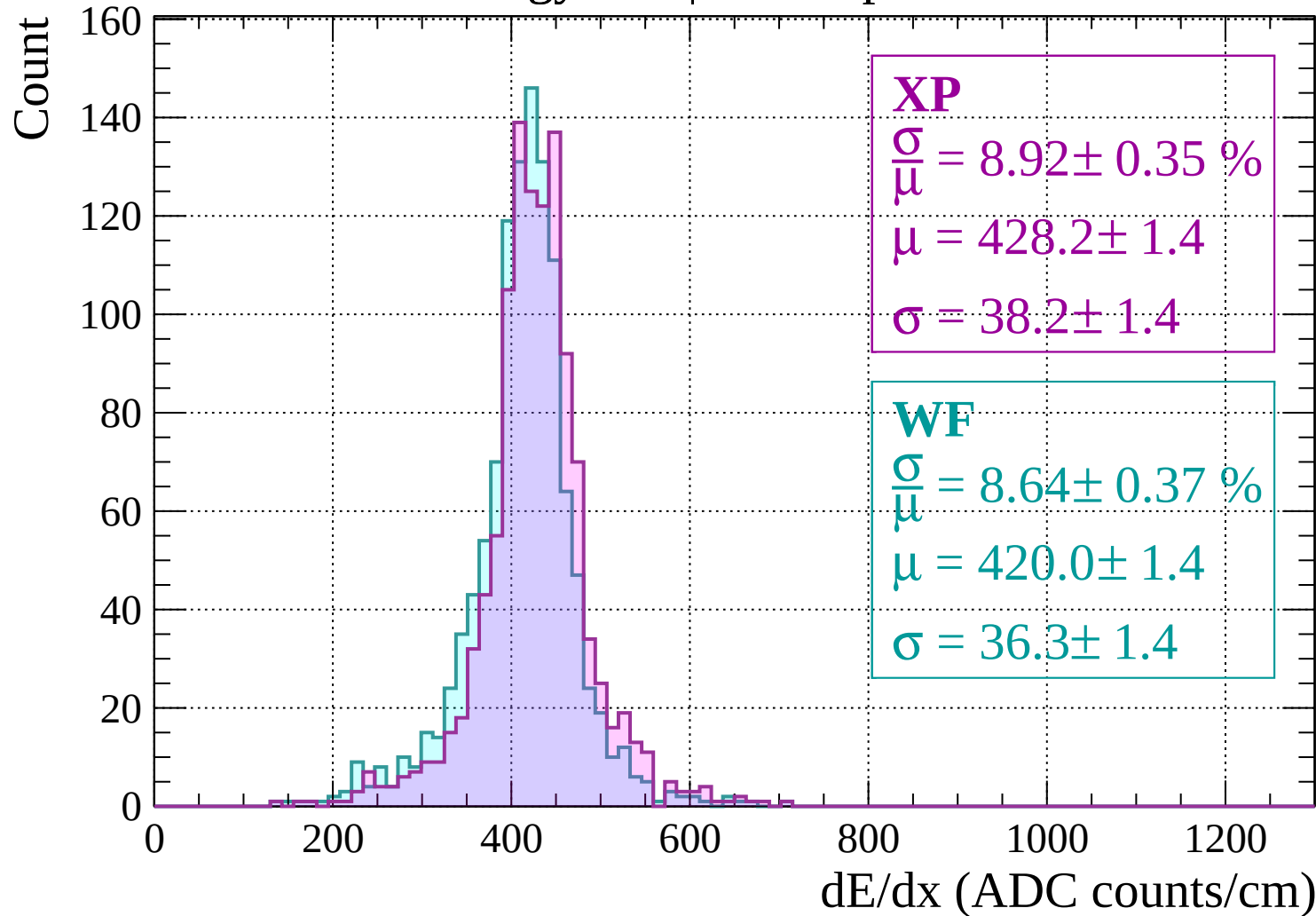
$$\frac{\sigma}{\mu} = 7.85 \pm 0.35 \%$$

$$\mu = 411.7 \pm 1.2$$

$$\sigma = 32.3 \pm 1.4$$

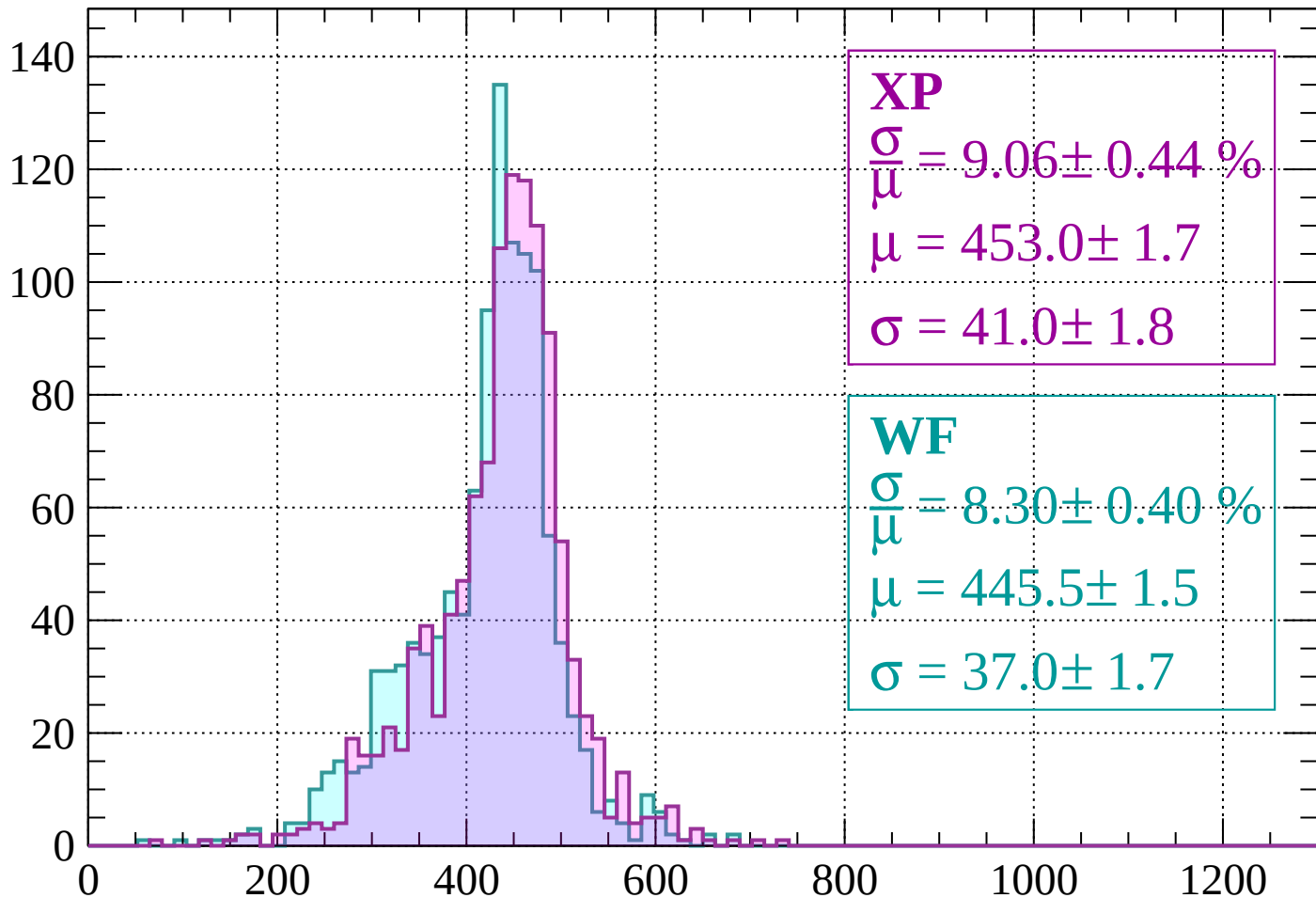
dE/dx (ADC counts/cm)

Energy loss | $-240 < p < -200$



Energy loss | $-200 < p < -160$

Count



XP

$$\frac{\sigma}{\mu} = 9.06 \pm 0.44 \%$$

$$\mu = 453.0 \pm 1.7$$

$$\sigma = 41.0 \pm 1.8$$

WF

$$\frac{\sigma}{\mu} = 8.30 \pm 0.40 \%$$

$$\mu = 445.5 \pm 1.5$$

$$\sigma = 37.0 \pm 1.7$$

dE/dx (ADC counts/cm)

Energy loss | $-160 < p < -120$

Count

XP

$$\frac{\sigma}{\mu} = 16.08 \pm 0.82 \%$$

$$\mu = 478.2 \pm 3.5$$

$$\sigma = 76.9 \pm 3.4$$

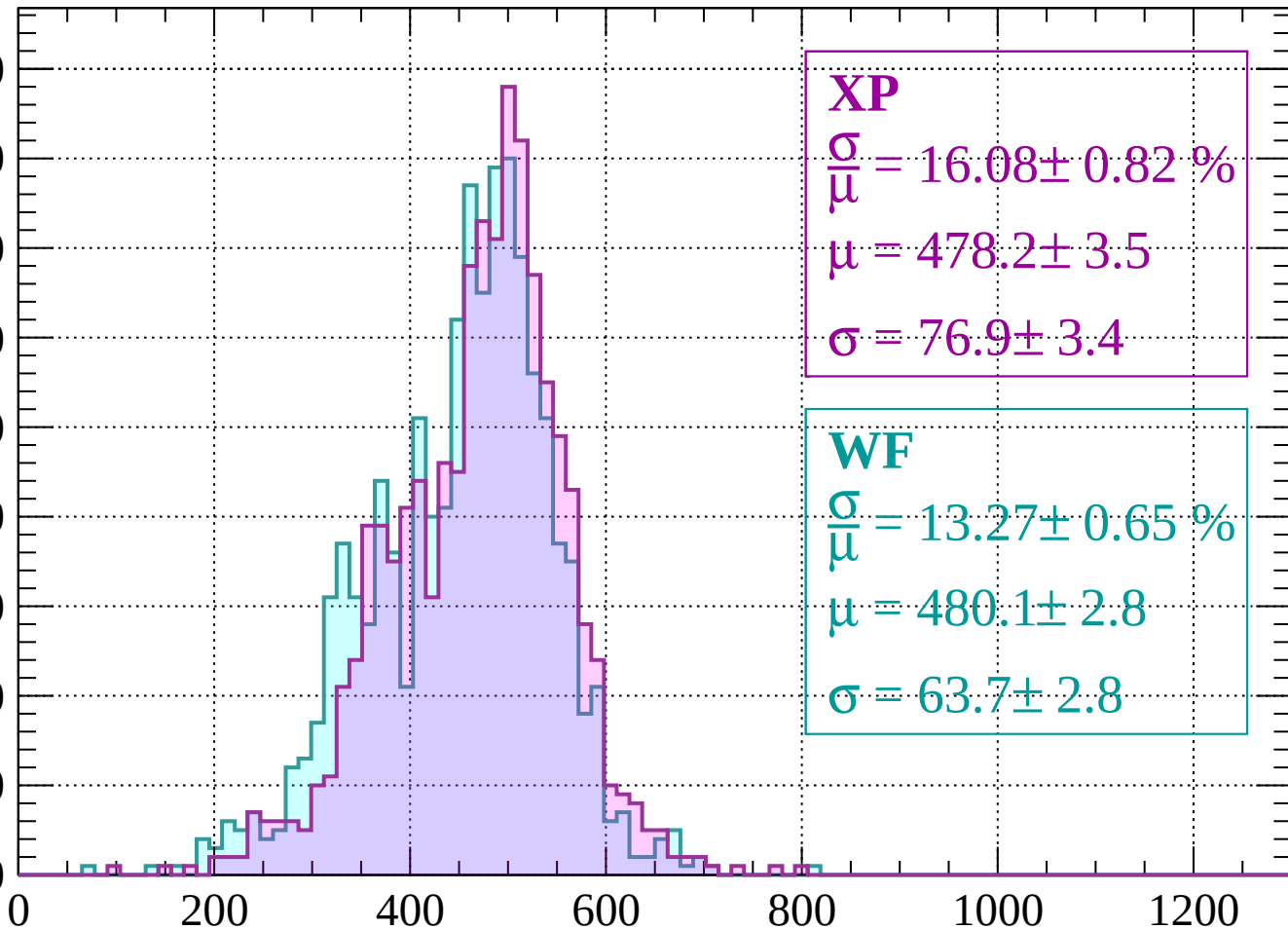
WF

$$\frac{\sigma}{\mu} = 13.27 \pm 0.65 \%$$

$$\mu = 480.1 \pm 2.8$$

$$\sigma = 63.7 \pm 2.8$$

90
80
70
60
50
40
30
20
10
0



dE/dx (ADC counts/cm)

Energy loss | $-120 < p < -80$

Count

XP

$$\frac{\sigma}{\mu} = 12.82 \pm 0.75 \%$$

$$\mu = 590.7 \pm 3.5$$

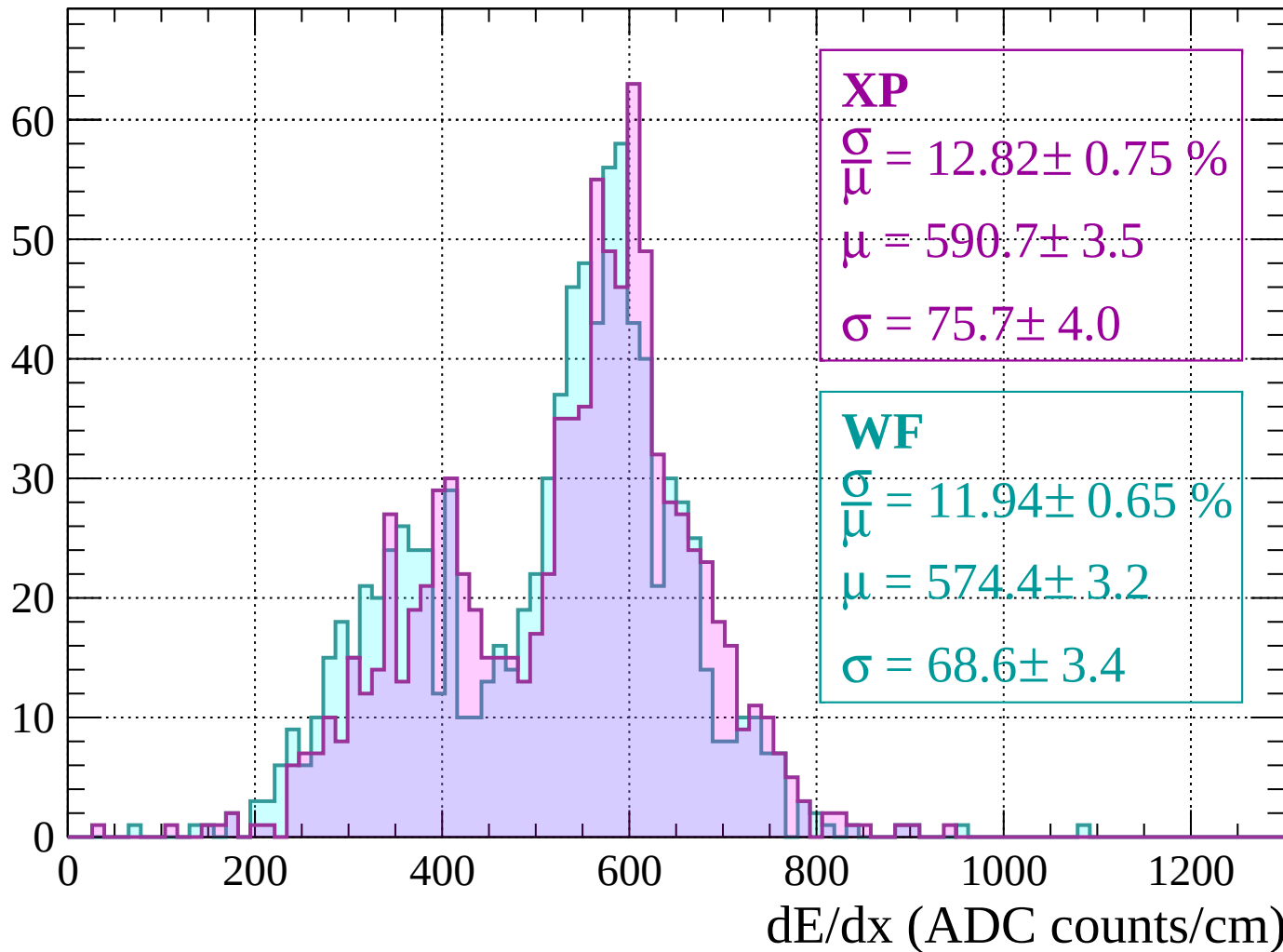
$$\sigma = 75.7 \pm 4.0$$

WF

$$\frac{\sigma}{\mu} = 11.94 \pm 0.65 \%$$

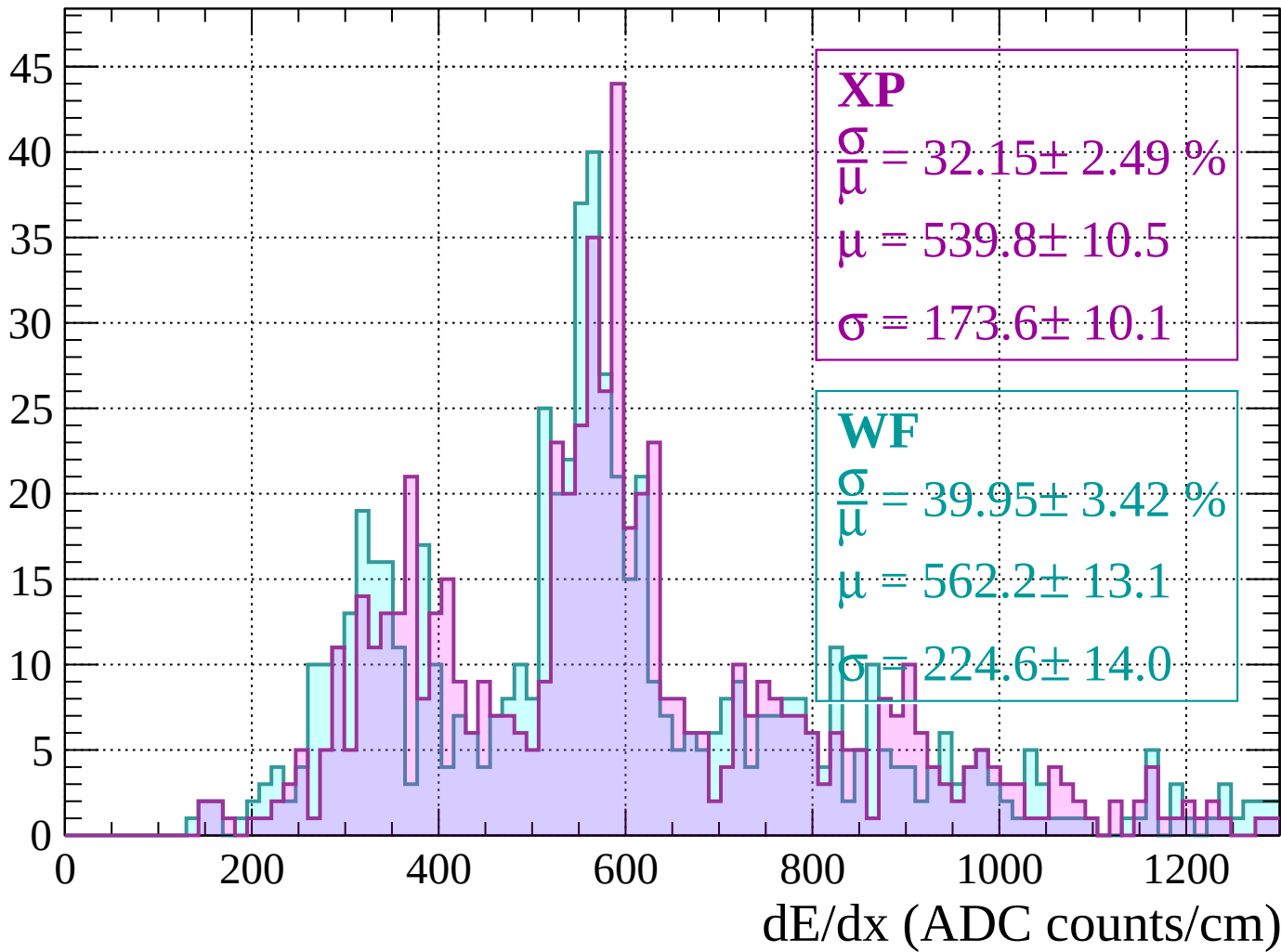
$$\mu = 574.4 \pm 3.2$$

$$\sigma = 68.6 \pm 3.4$$



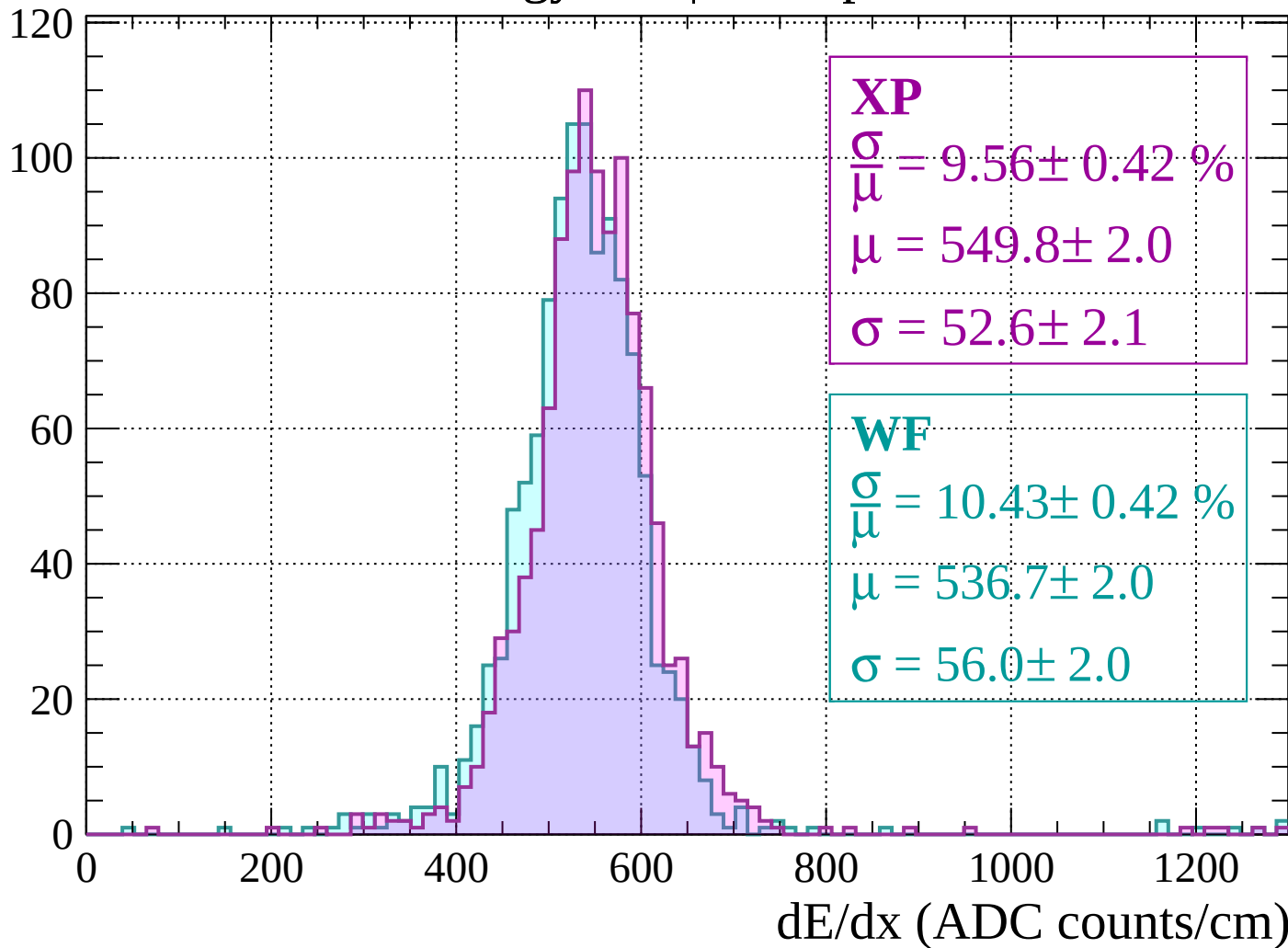
Energy loss | $-80 < p < -40$

Count



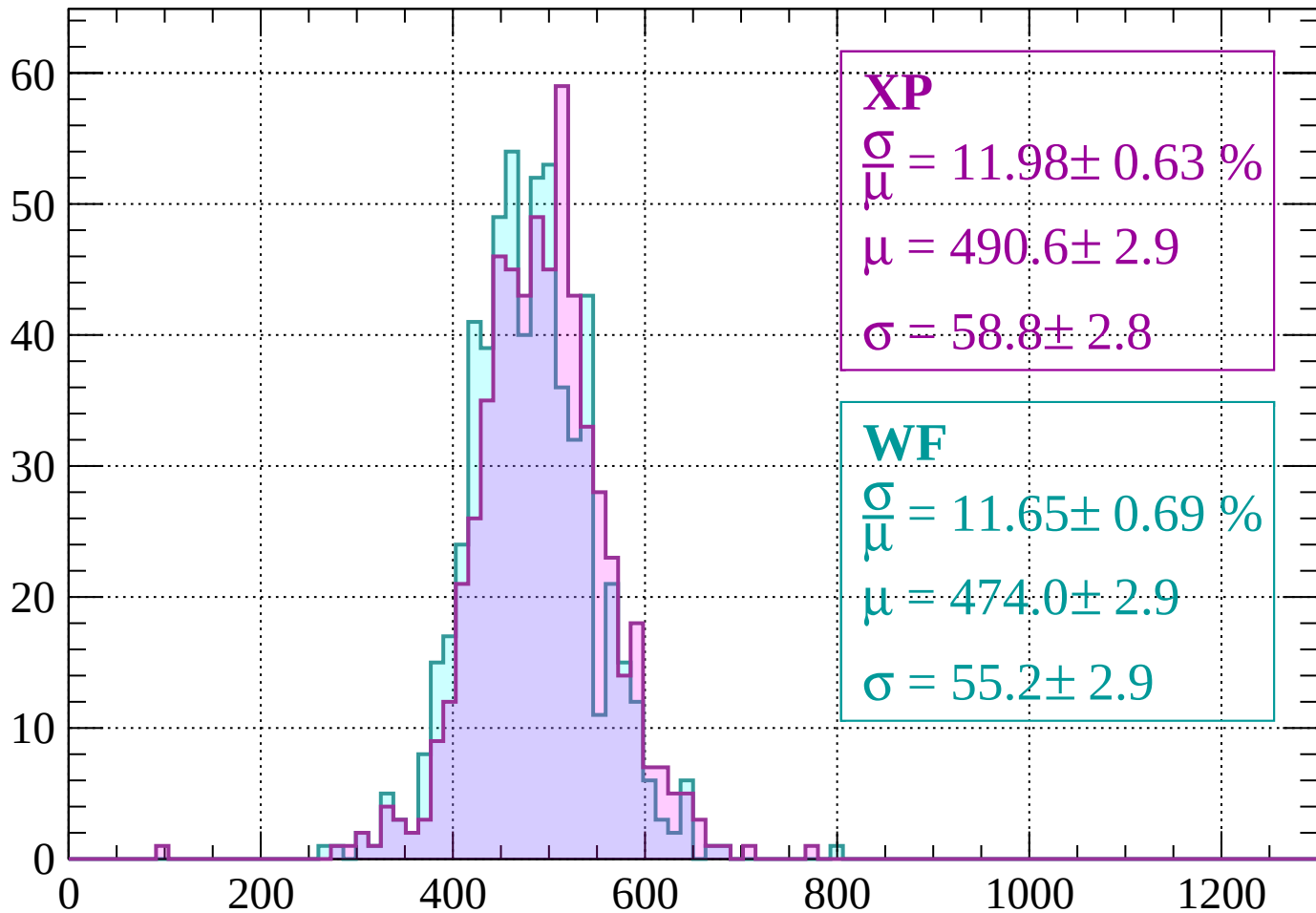
Energy loss | $-40 < p < 0$

Count



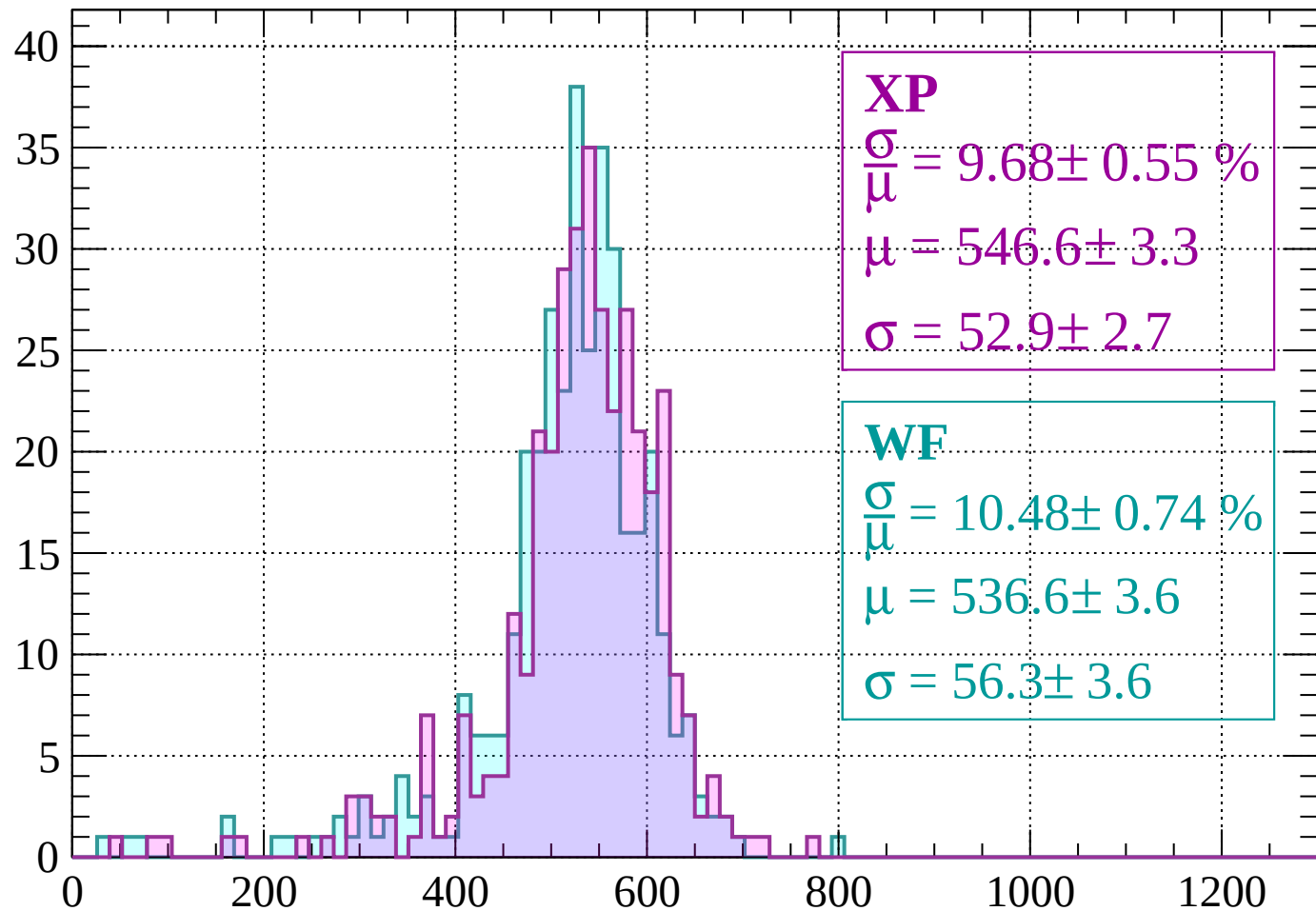
Energy loss | $0 < p < 40$

Count



Energy loss | $40 < p < 80$

Count



XP

$$\frac{\sigma}{\mu} = 9.68 \pm 0.55 \%$$

$$\mu = 546.6 \pm 3.3$$

$$\sigma = 52.9 \pm 2.7$$

WF

$$\frac{\sigma}{\mu} = 10.48 \pm 0.74 \%$$

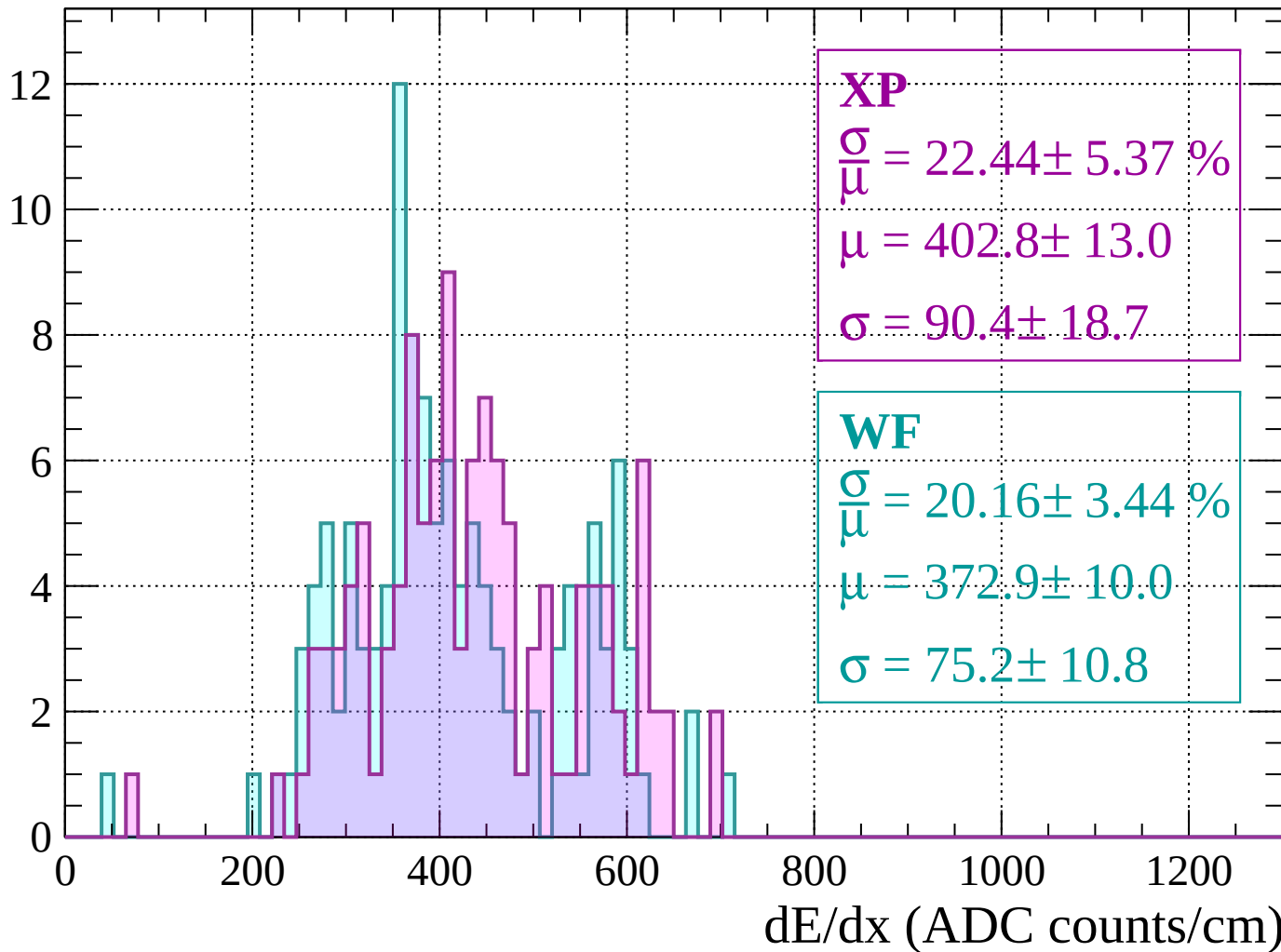
$$\mu = 536.6 \pm 3.6$$

$$\sigma = 56.3 \pm 3.6$$

dE/dx (ADC counts/cm)

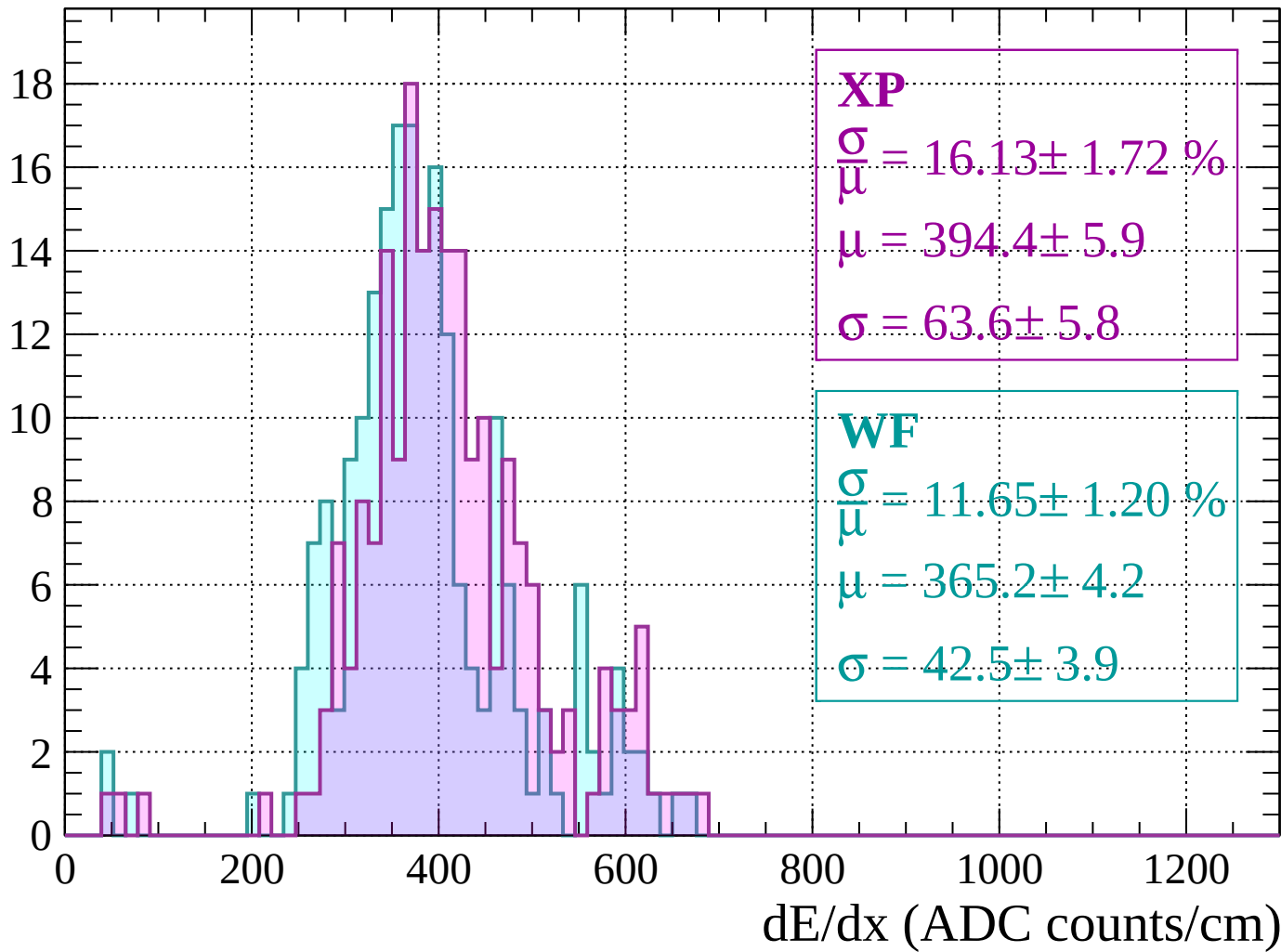
Energy loss | $80 < p < 120$

Count



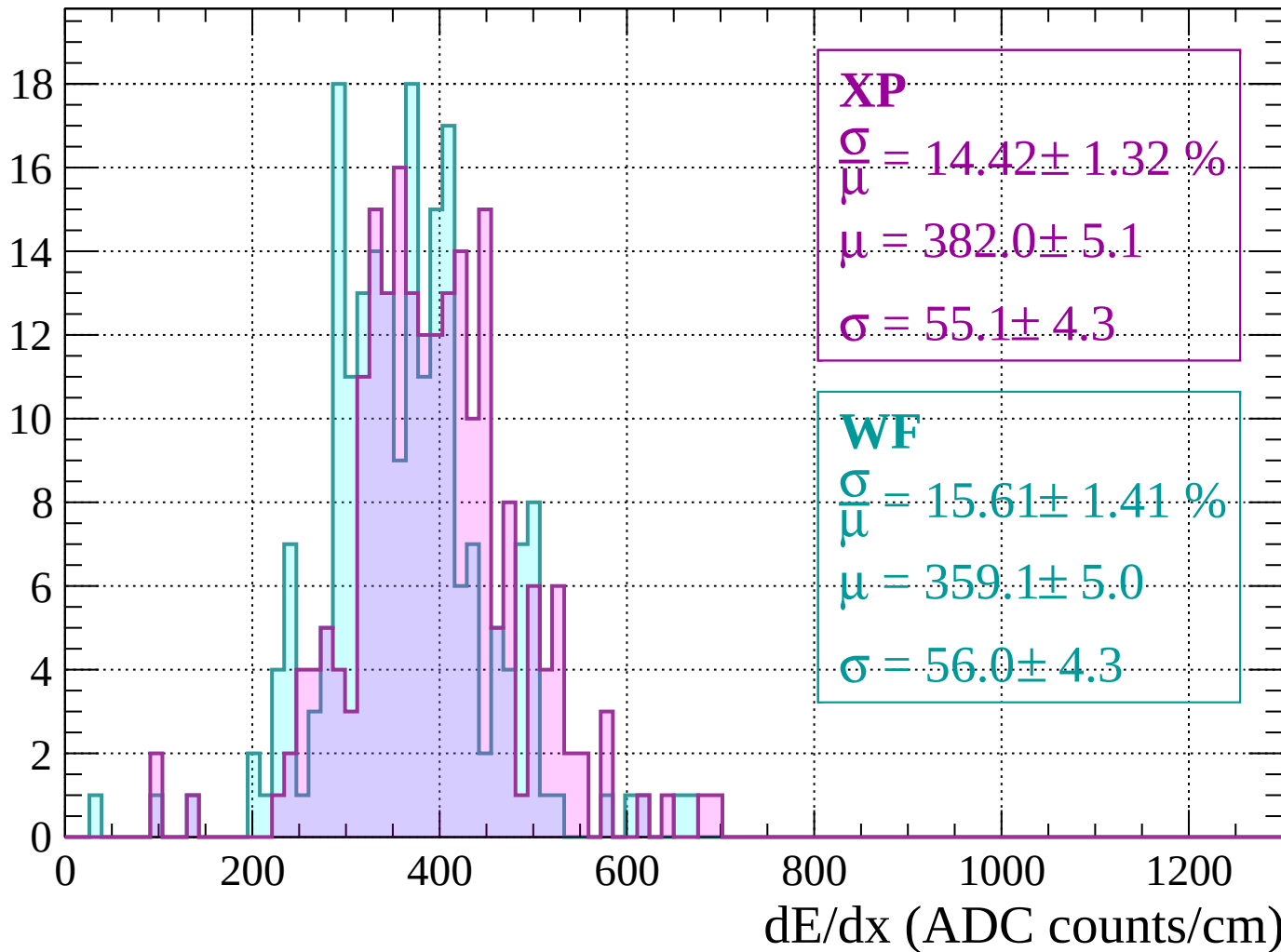
Energy loss | $120 < p < 160$

Count



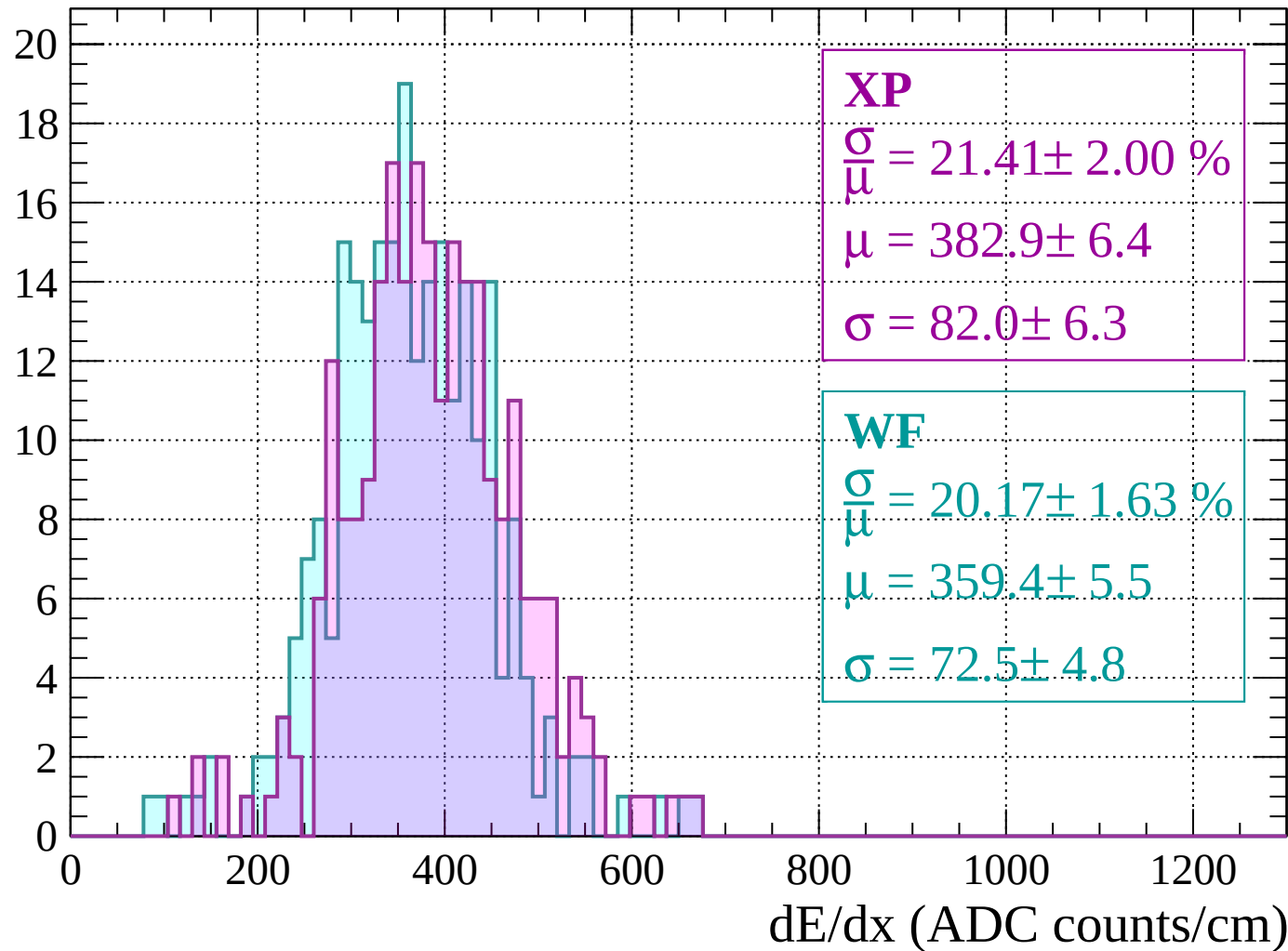
Energy loss | $160 < p < 200$

Count



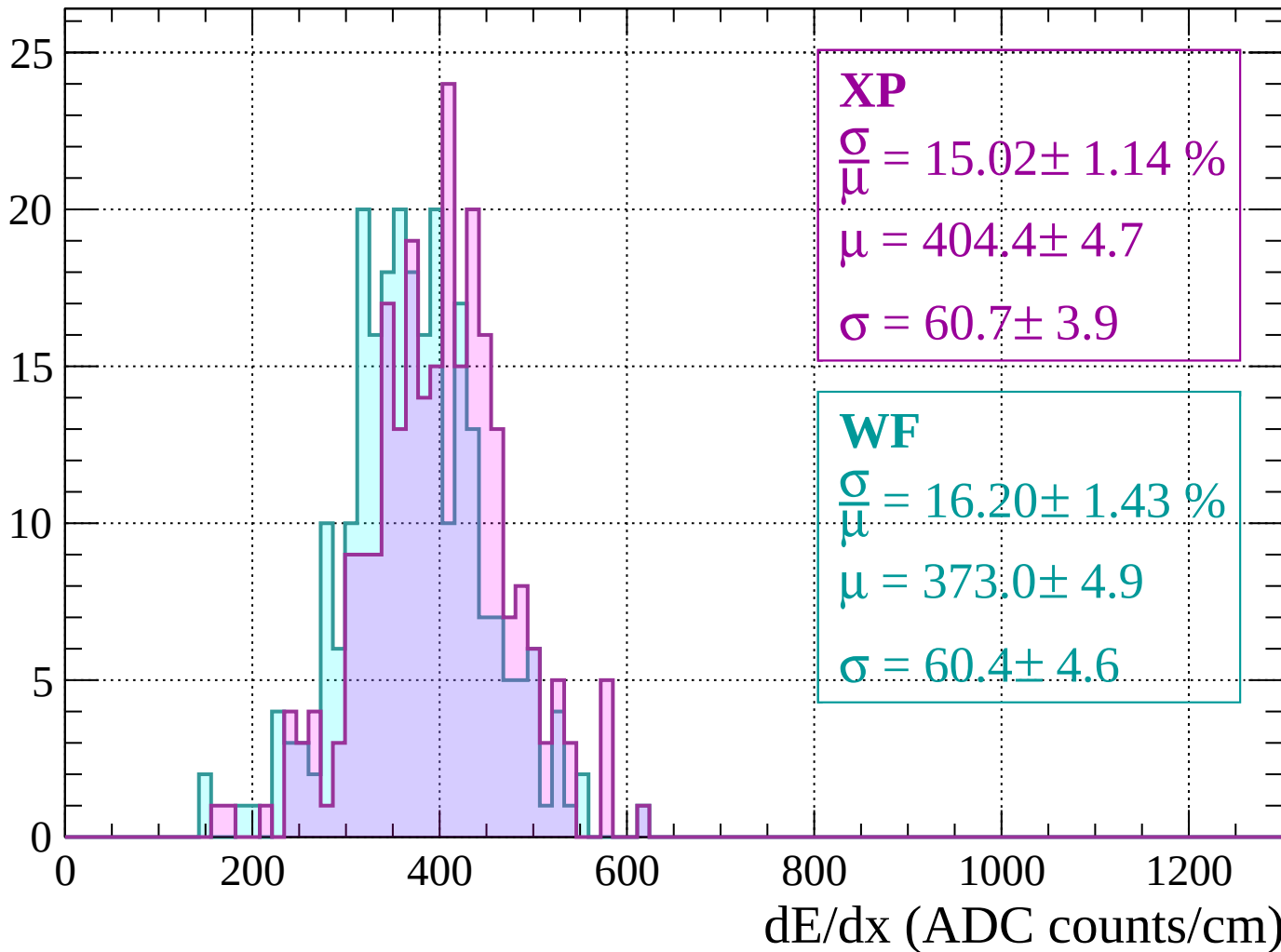
Energy loss | $200 < p < 240$

Count



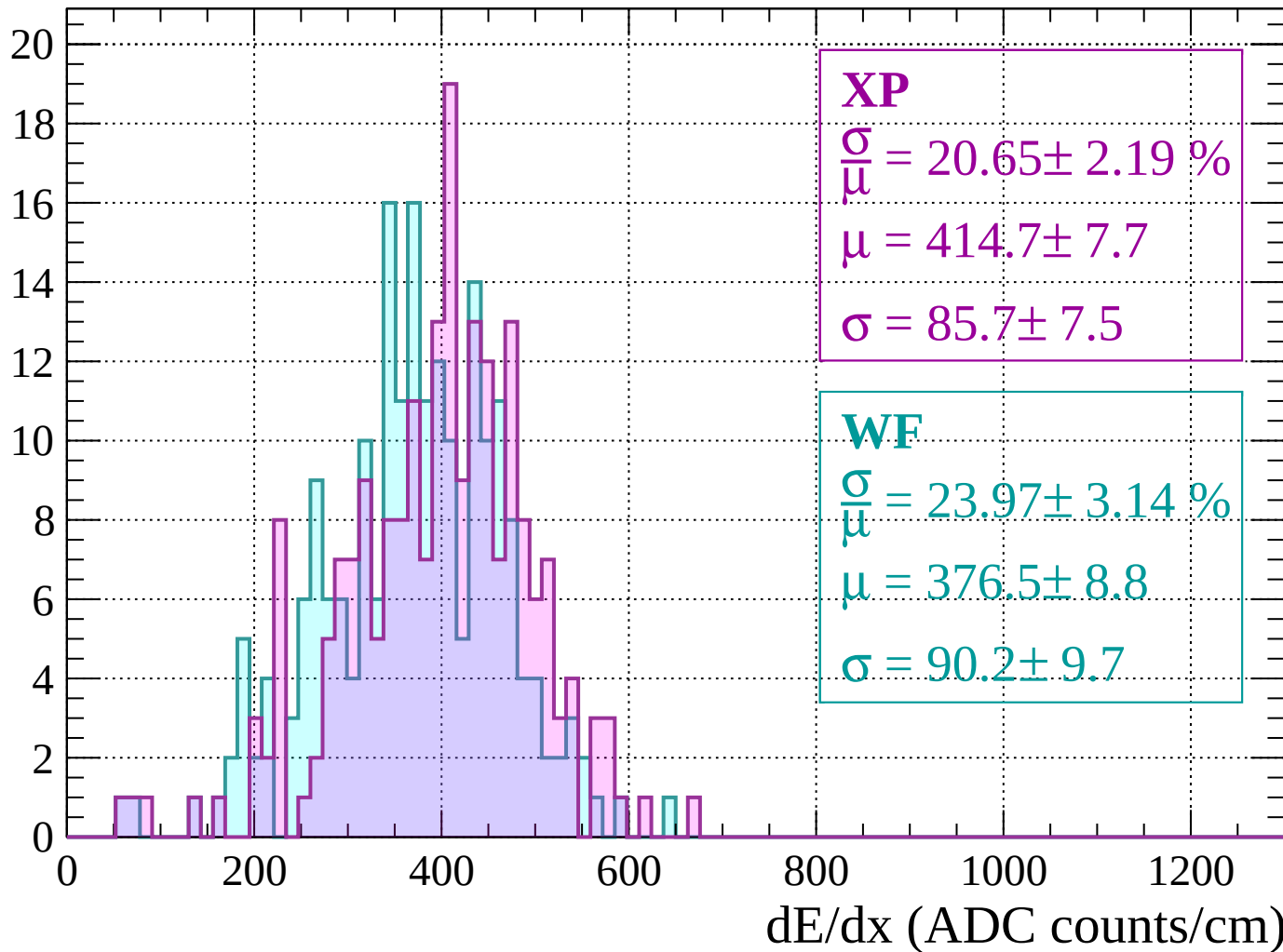
Energy loss | $240 < p < 280$

Count



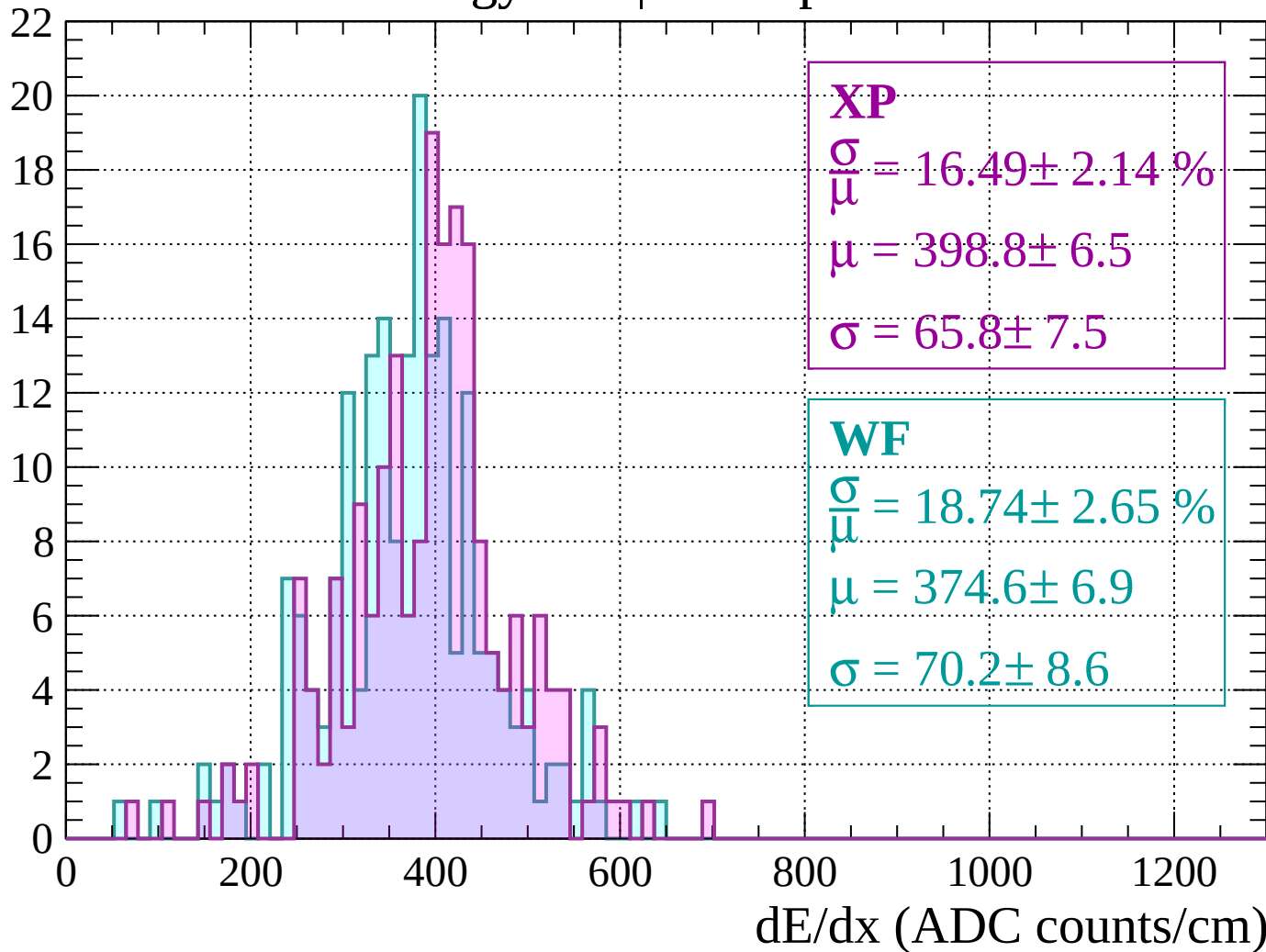
Energy loss | $280 < p < 320$

Count



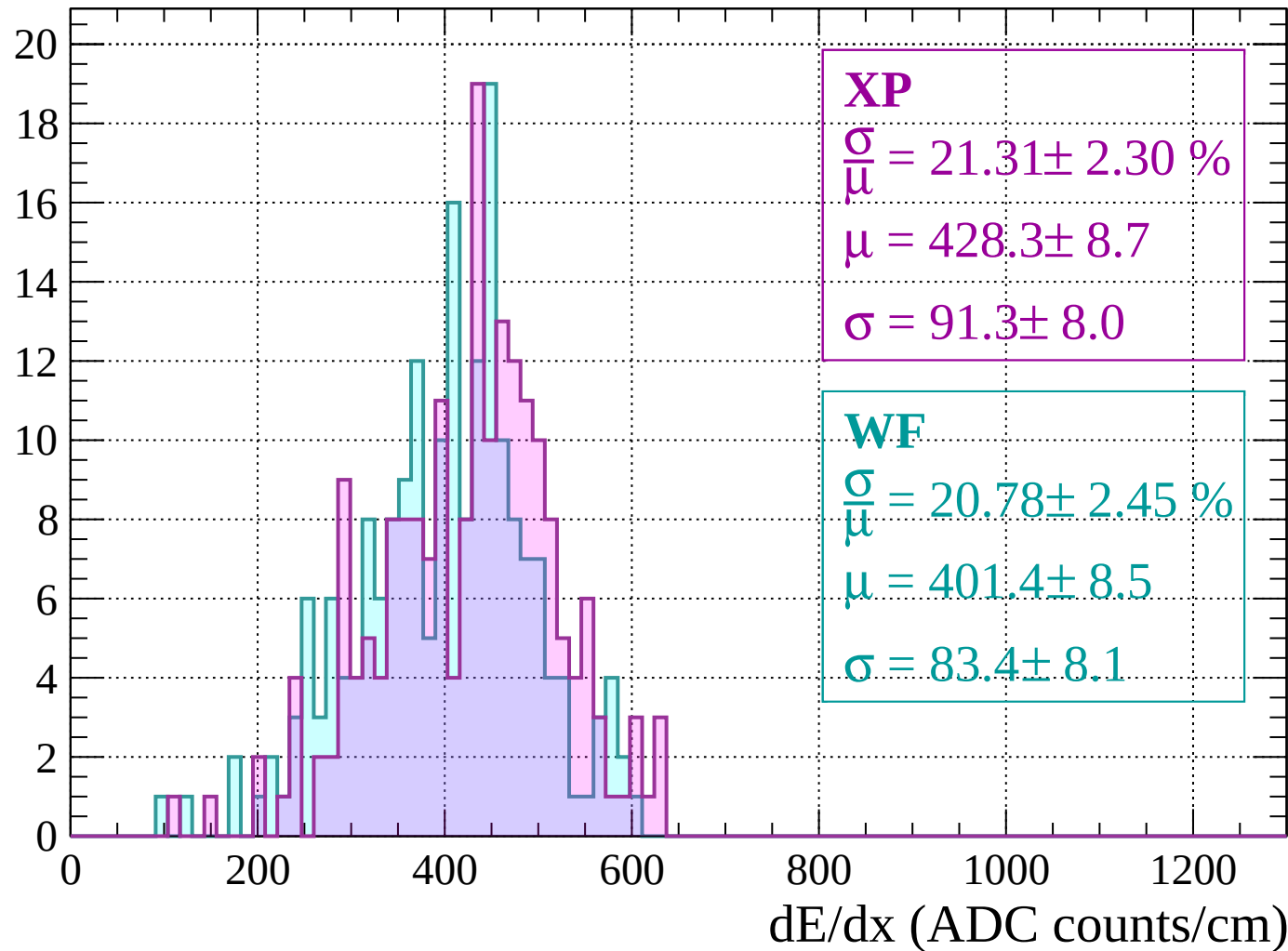
Energy loss | $320 < p < 360$

Count



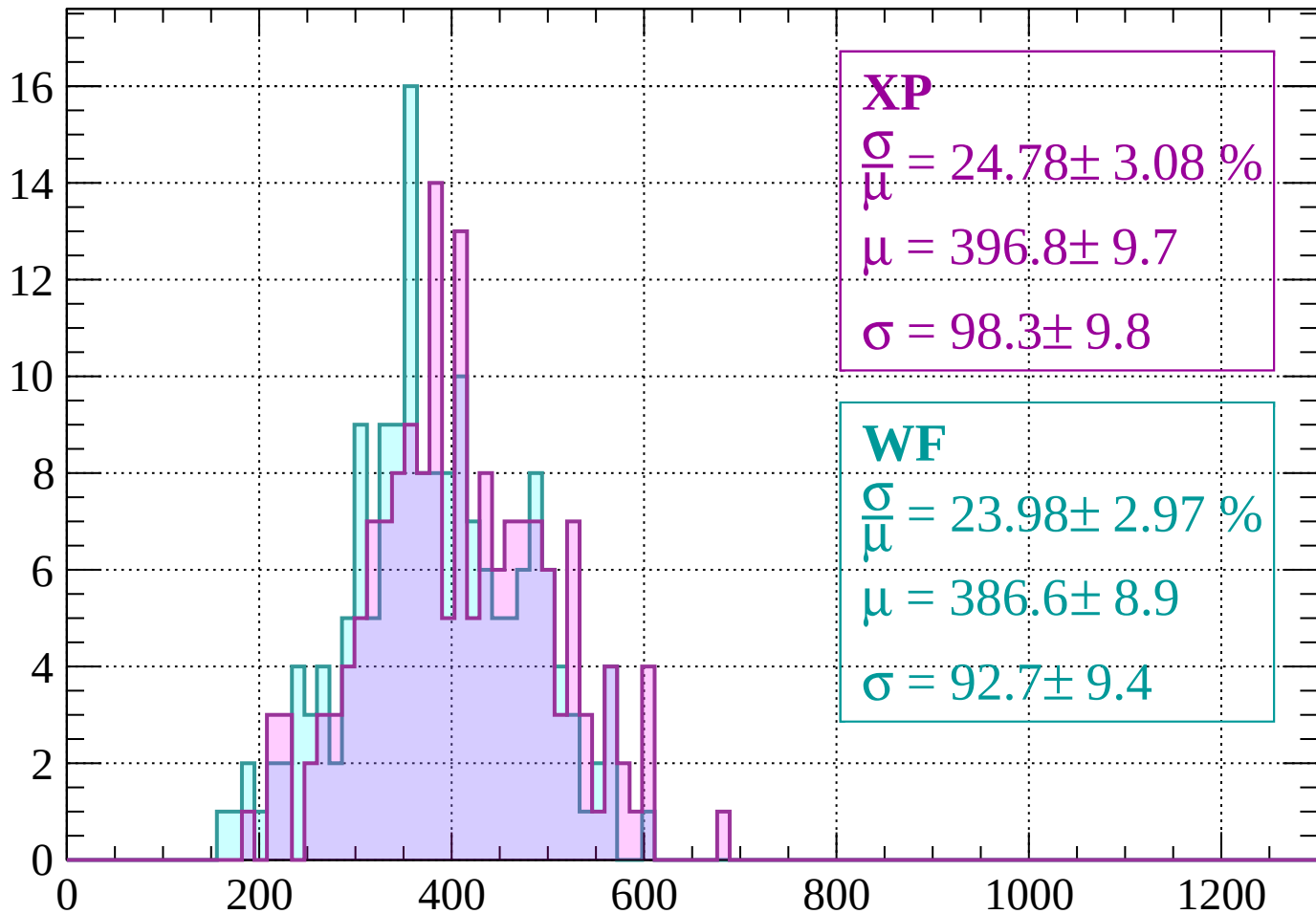
Energy loss | $360 < p < 400$

Count



Energy loss | $400 < p < 440$

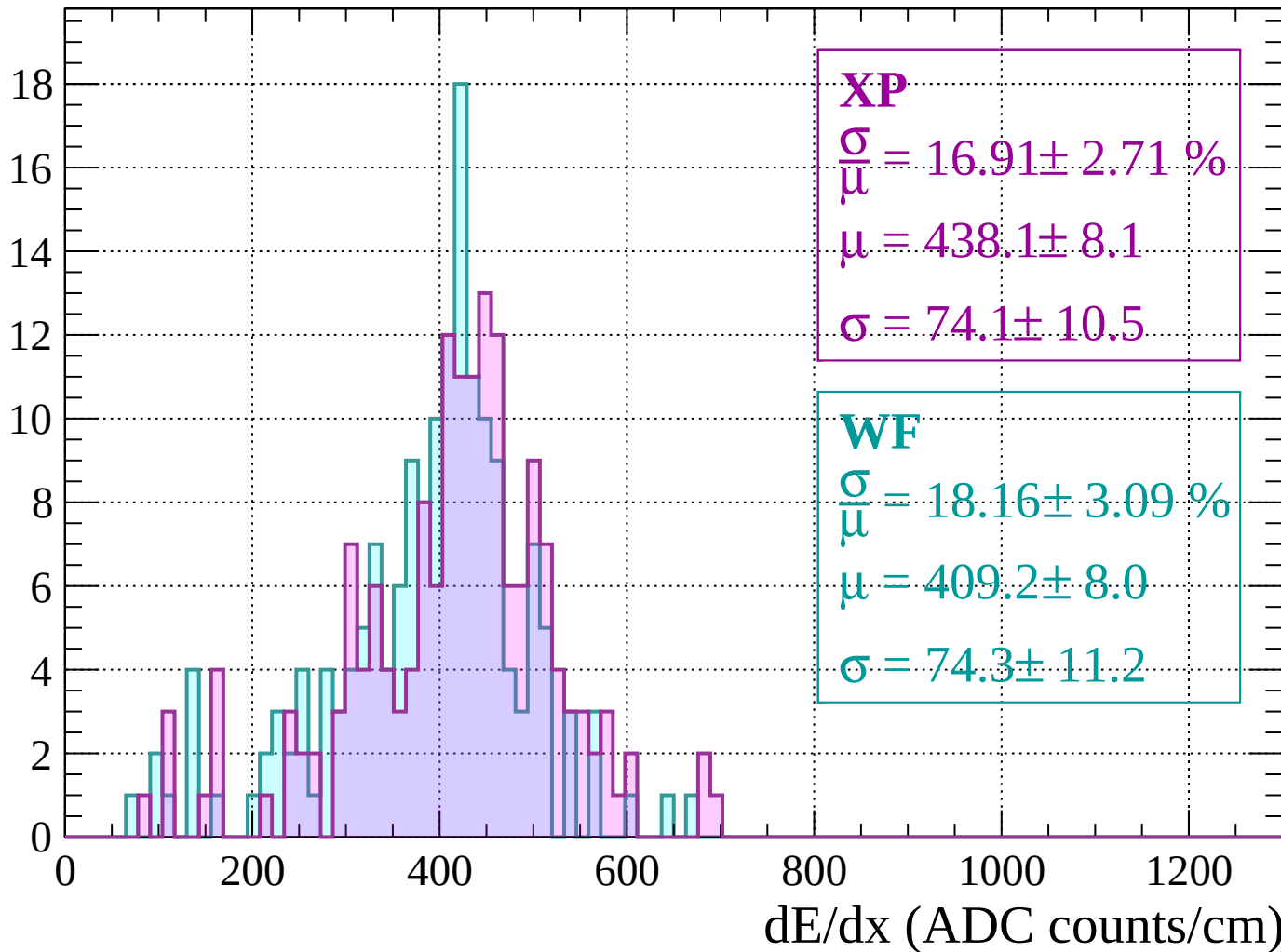
Count



dE/dx (ADC counts/cm)

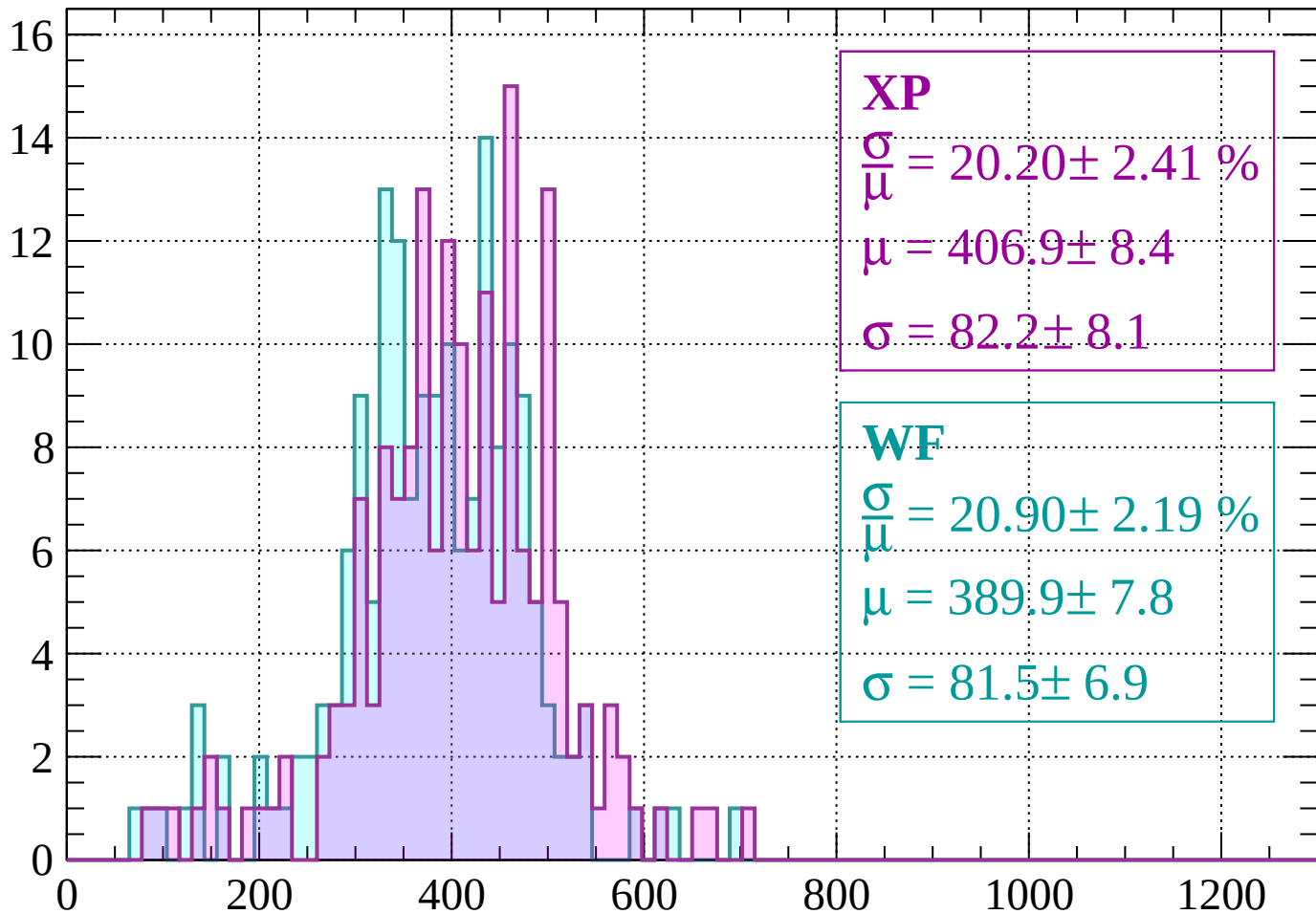
Energy loss | $440 < p < 480$

Count



Energy loss | $480 < p < 520$

Count



XP

$$\frac{\sigma}{\mu} = 20.20 \pm 2.41 \%$$

$$\mu = 406.9 \pm 8.4$$

$$\sigma = 82.2 \pm 8.1$$

WF

$$\frac{\sigma}{\mu} = 20.90 \pm 2.19 \%$$

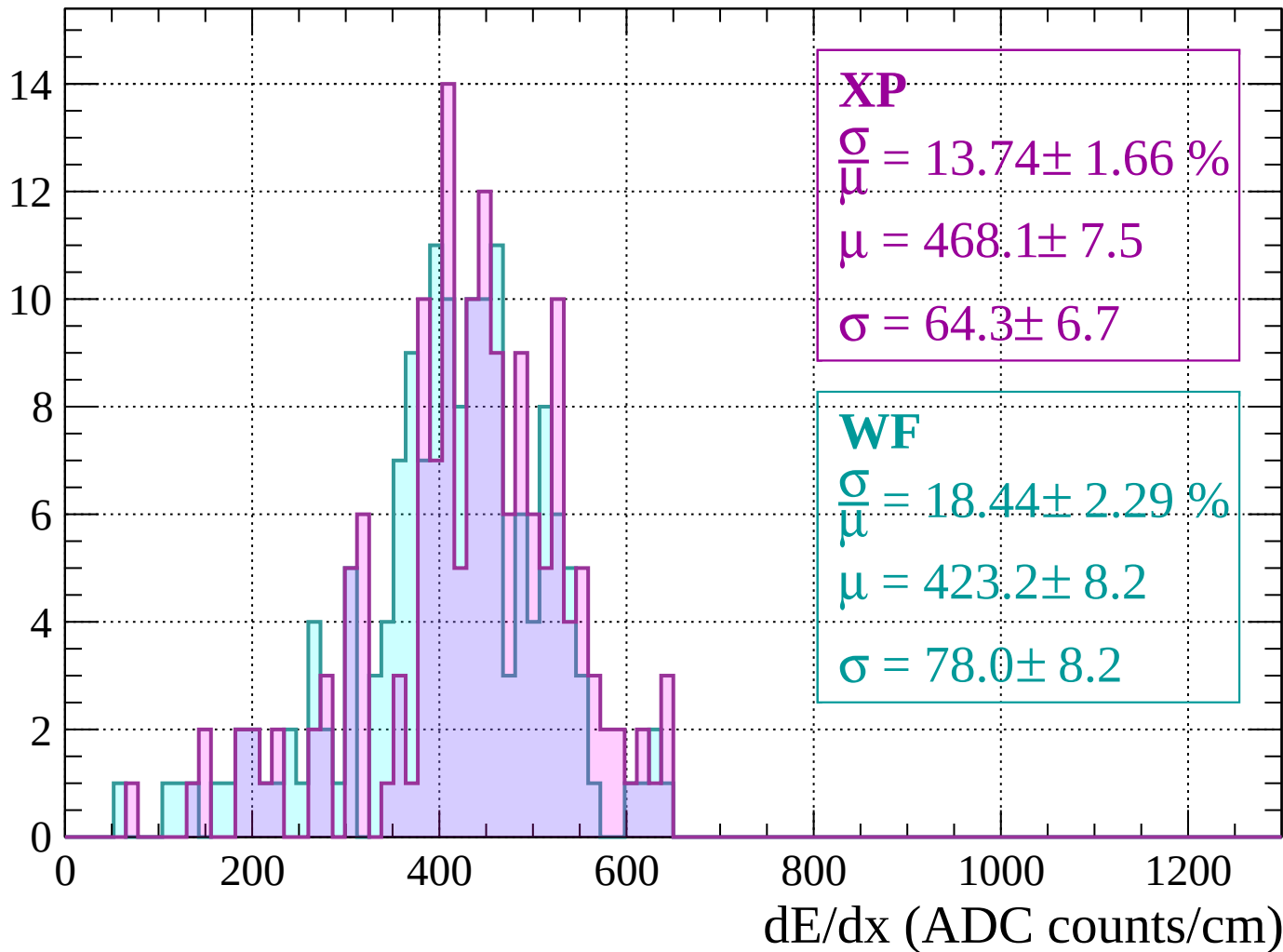
$$\mu = 389.9 \pm 7.8$$

$$\sigma = 81.5 \pm 6.9$$

dE/dx (ADC counts/cm)

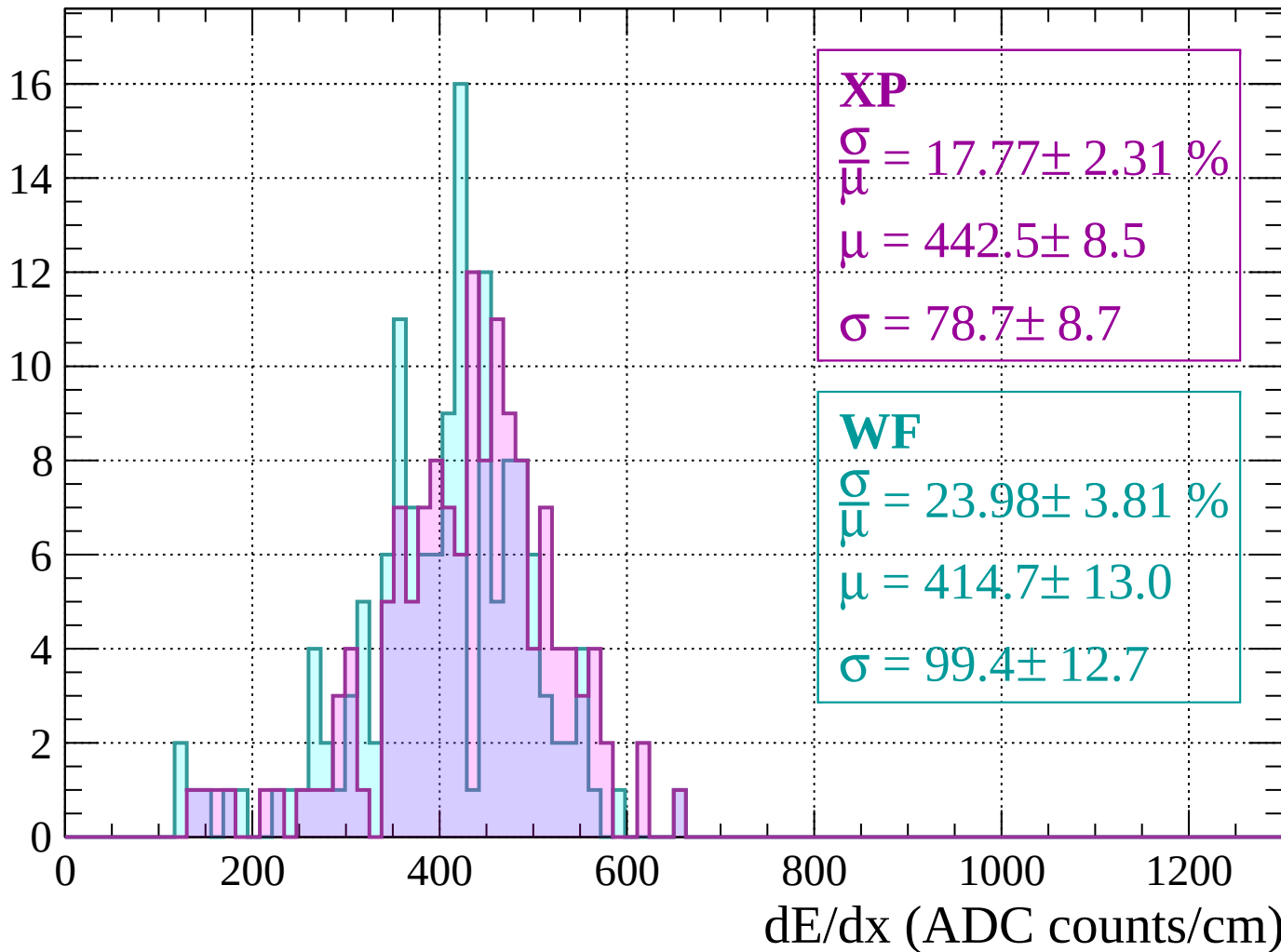
Energy loss | $520 < p < 560$

Count



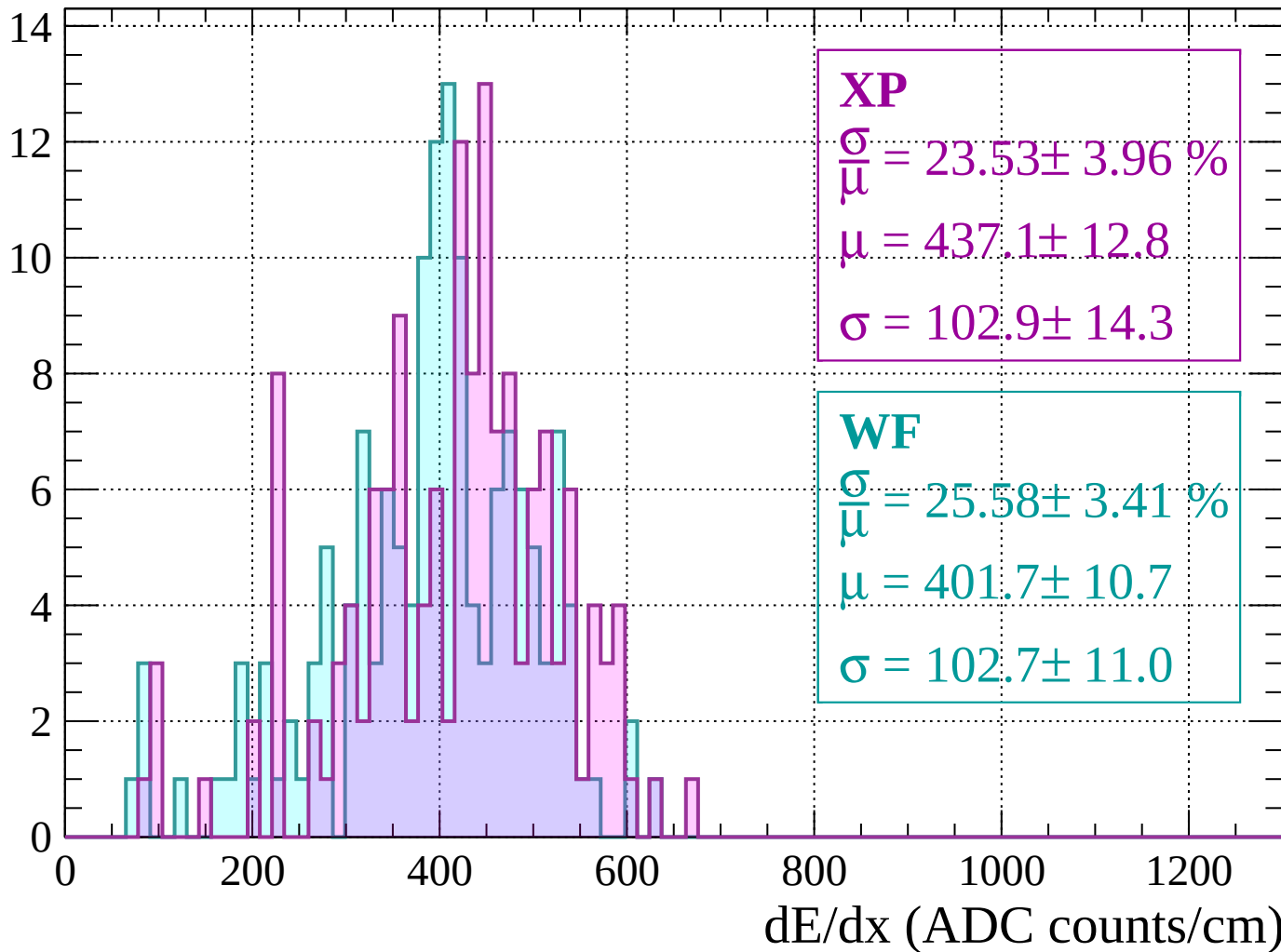
Energy loss | $560 < p < 600$

Count



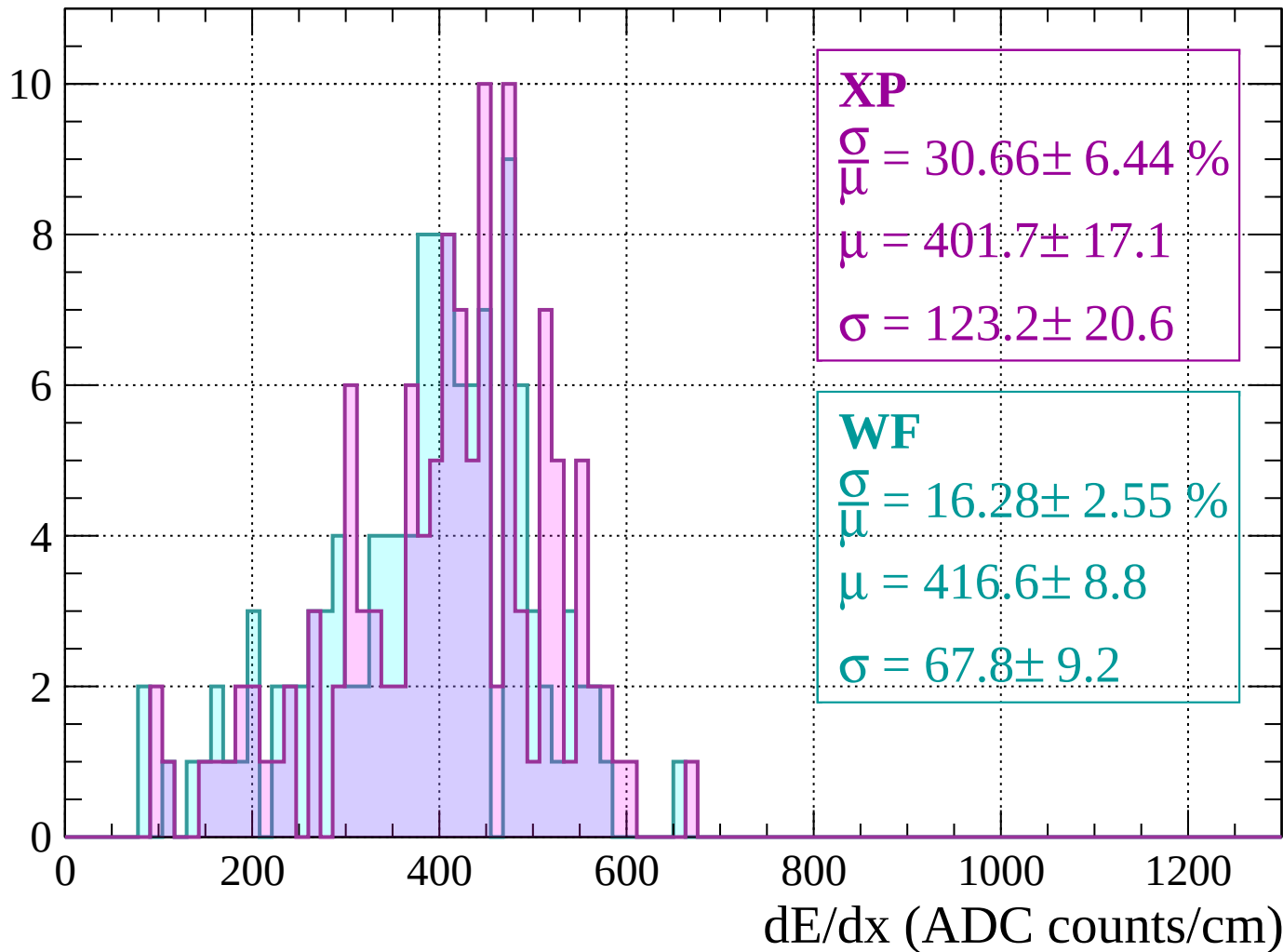
Energy loss | $600 < p < 640$

Count



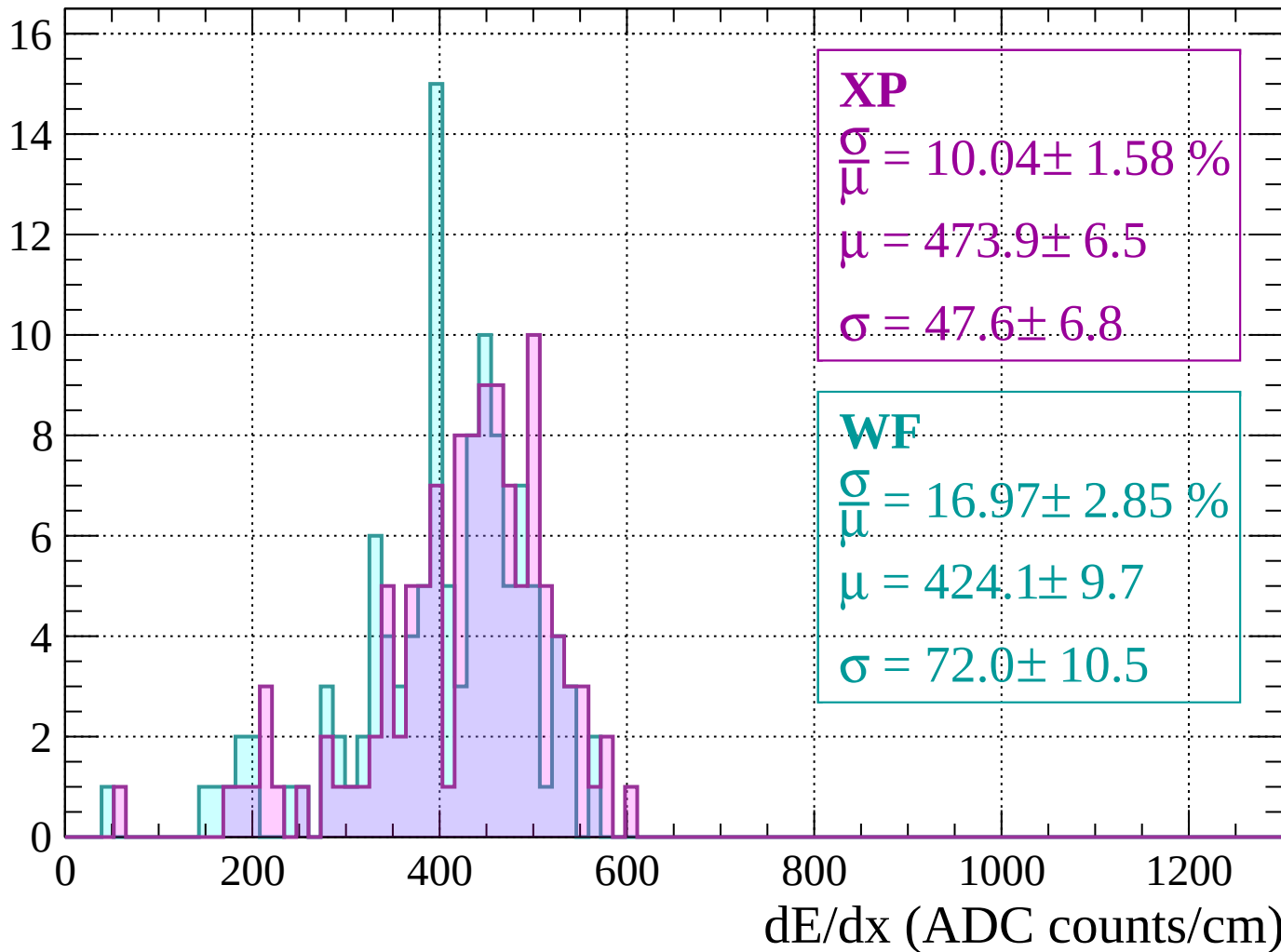
Energy loss | $640 < p < 680$

Count



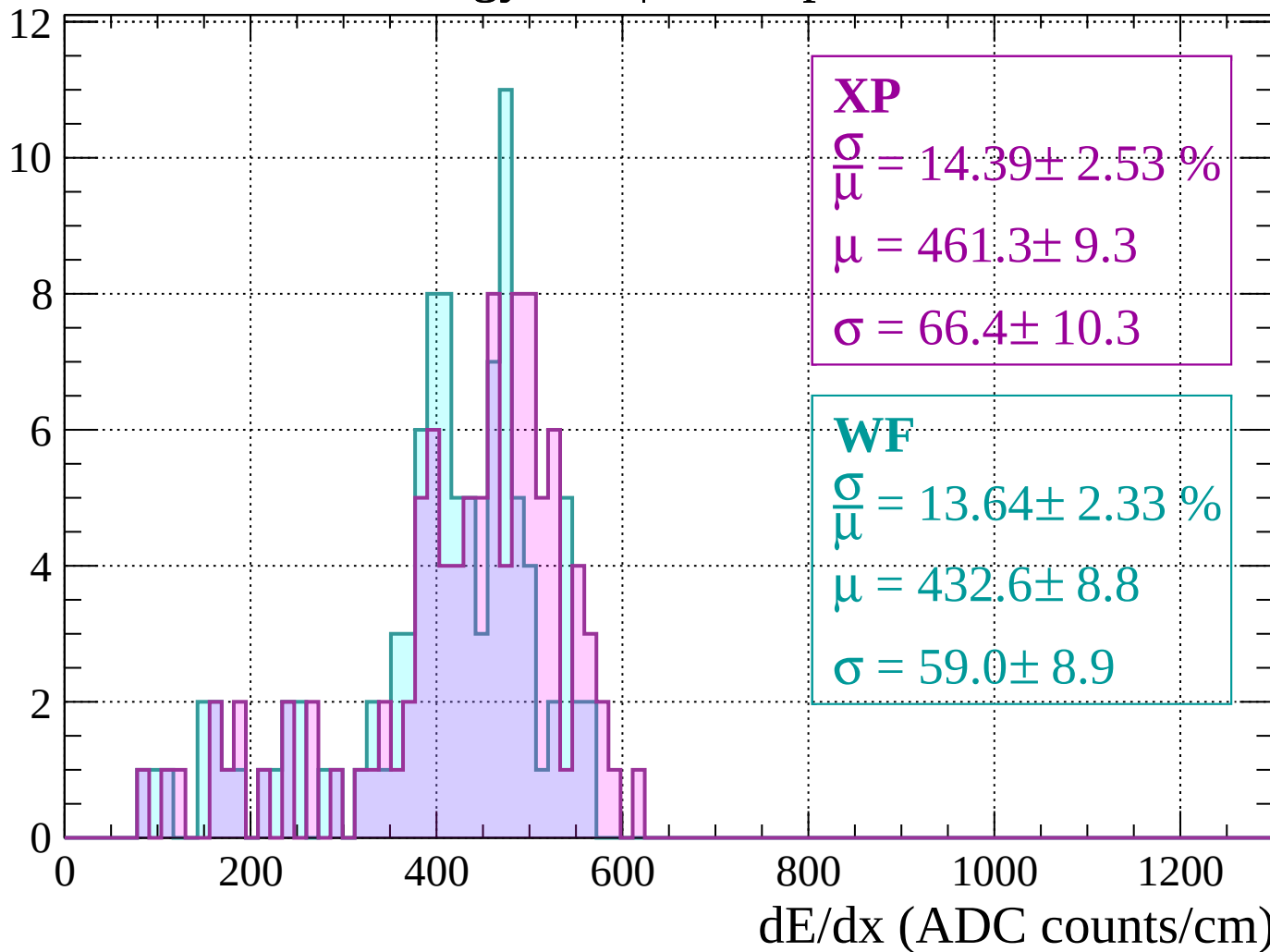
Energy loss | $680 < p < 720$

Count



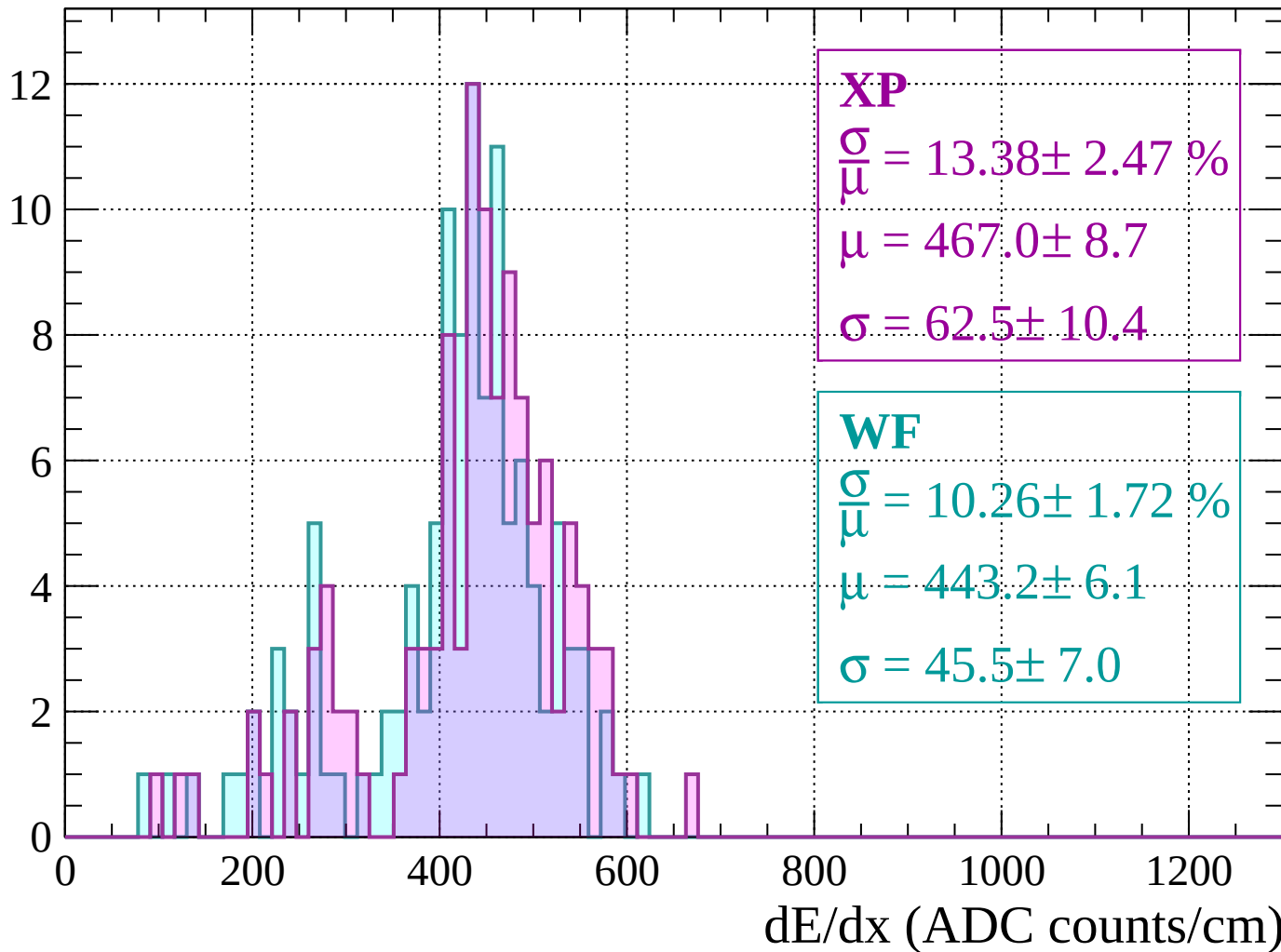
Energy loss | $720 < p < 760$

Count



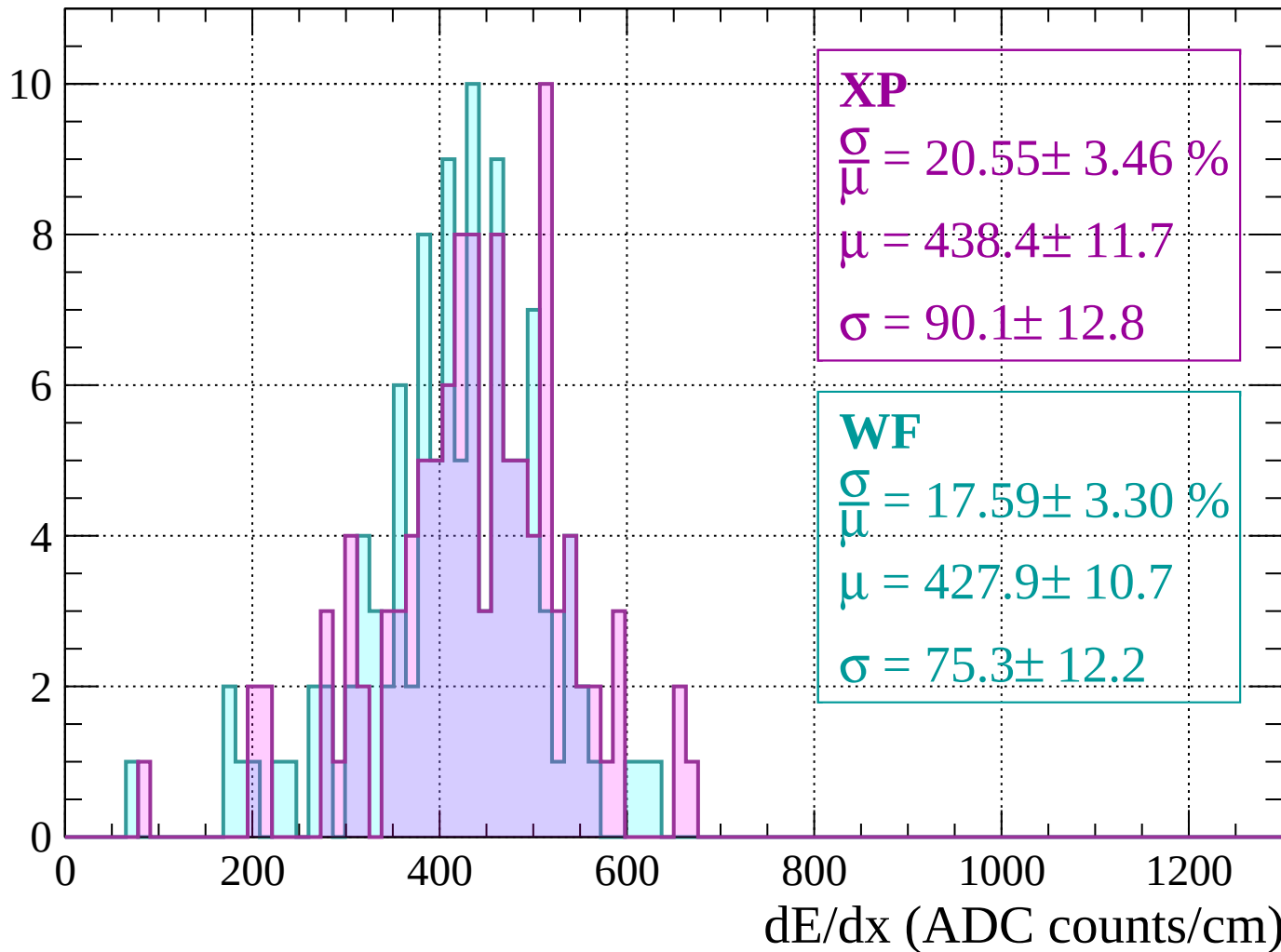
Energy loss | $760 < p < 800$

Count



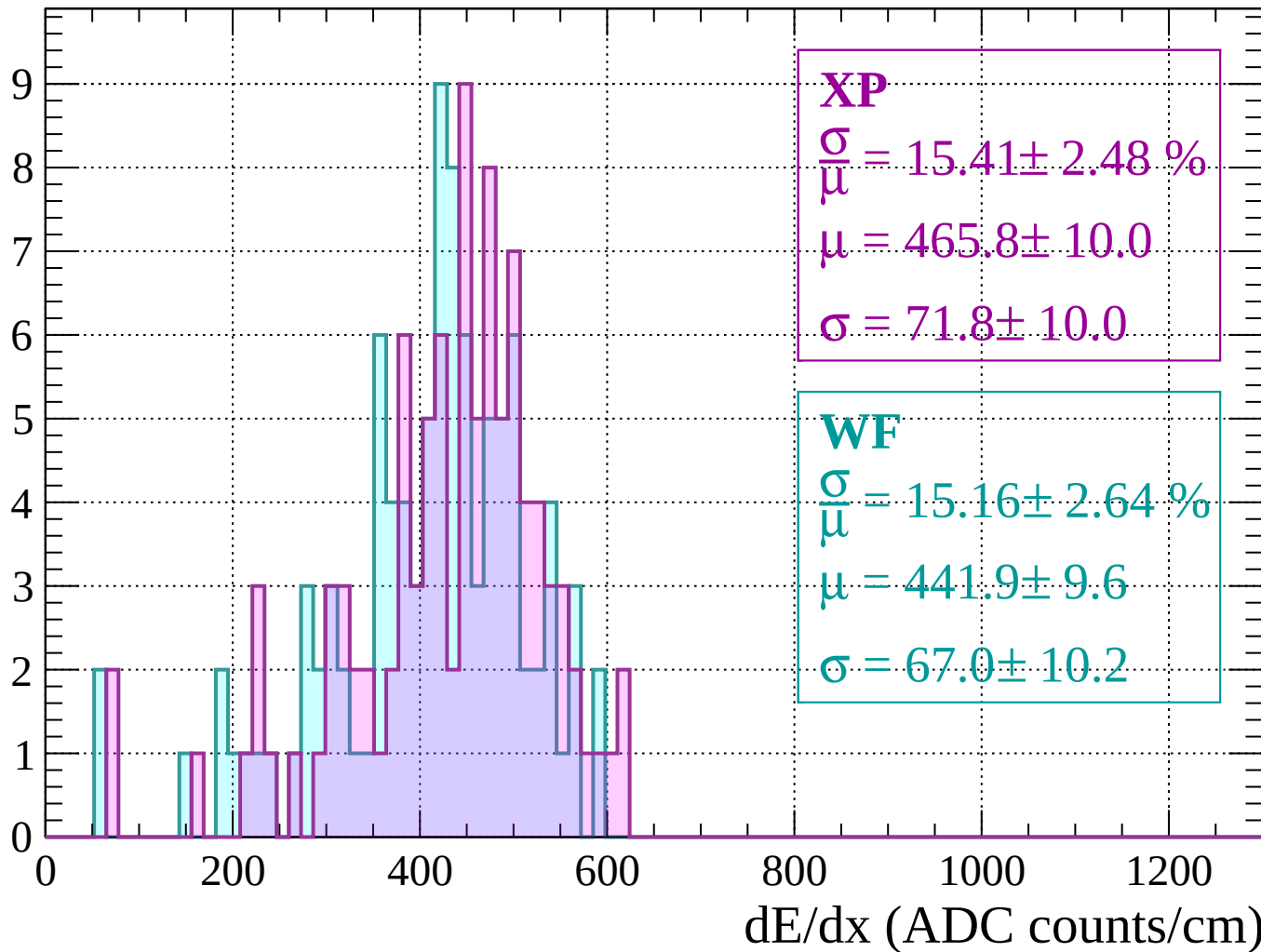
Energy loss | $800 < p < 840$

Count



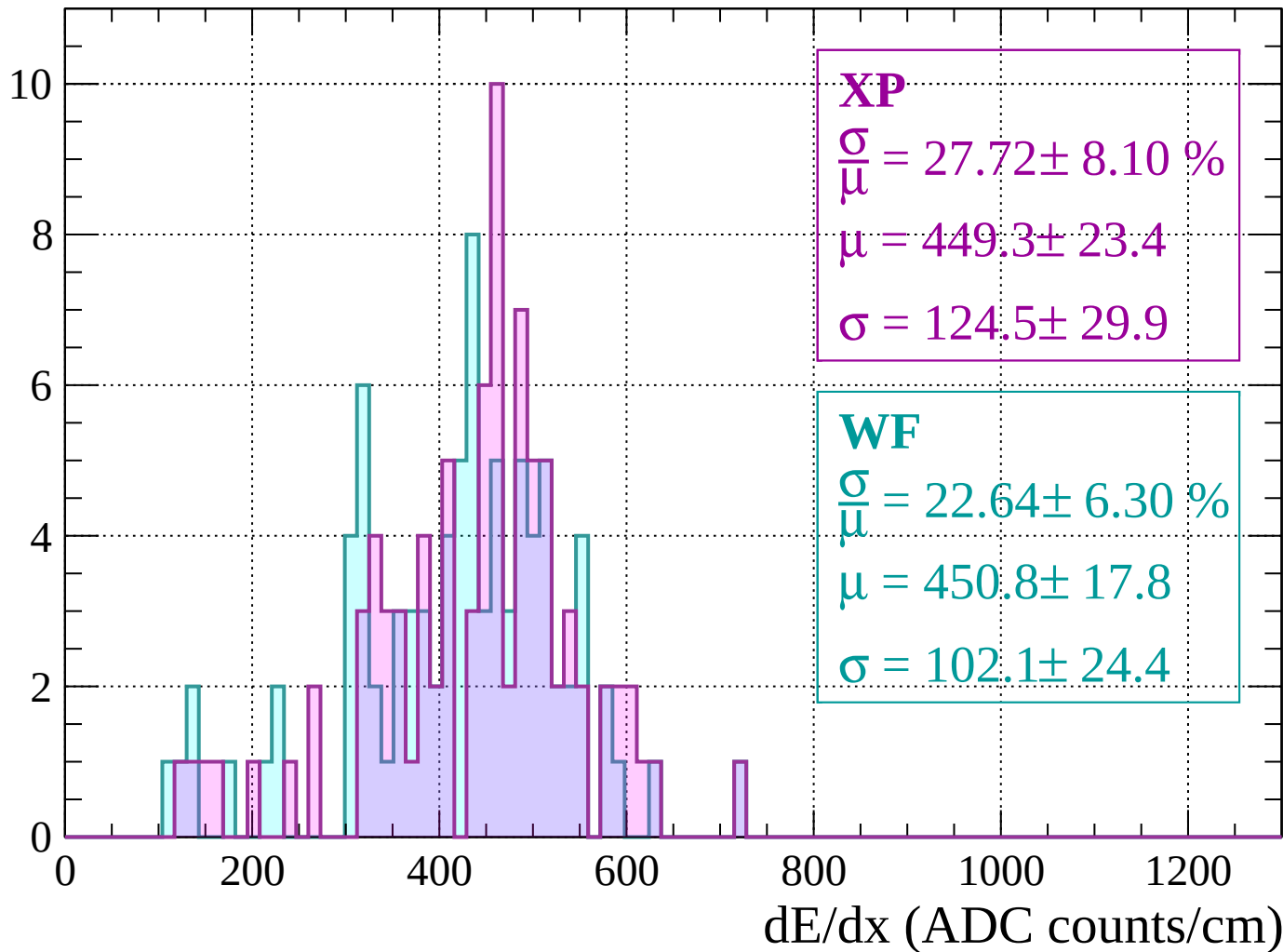
Energy loss | $840 < p < 880$

Count



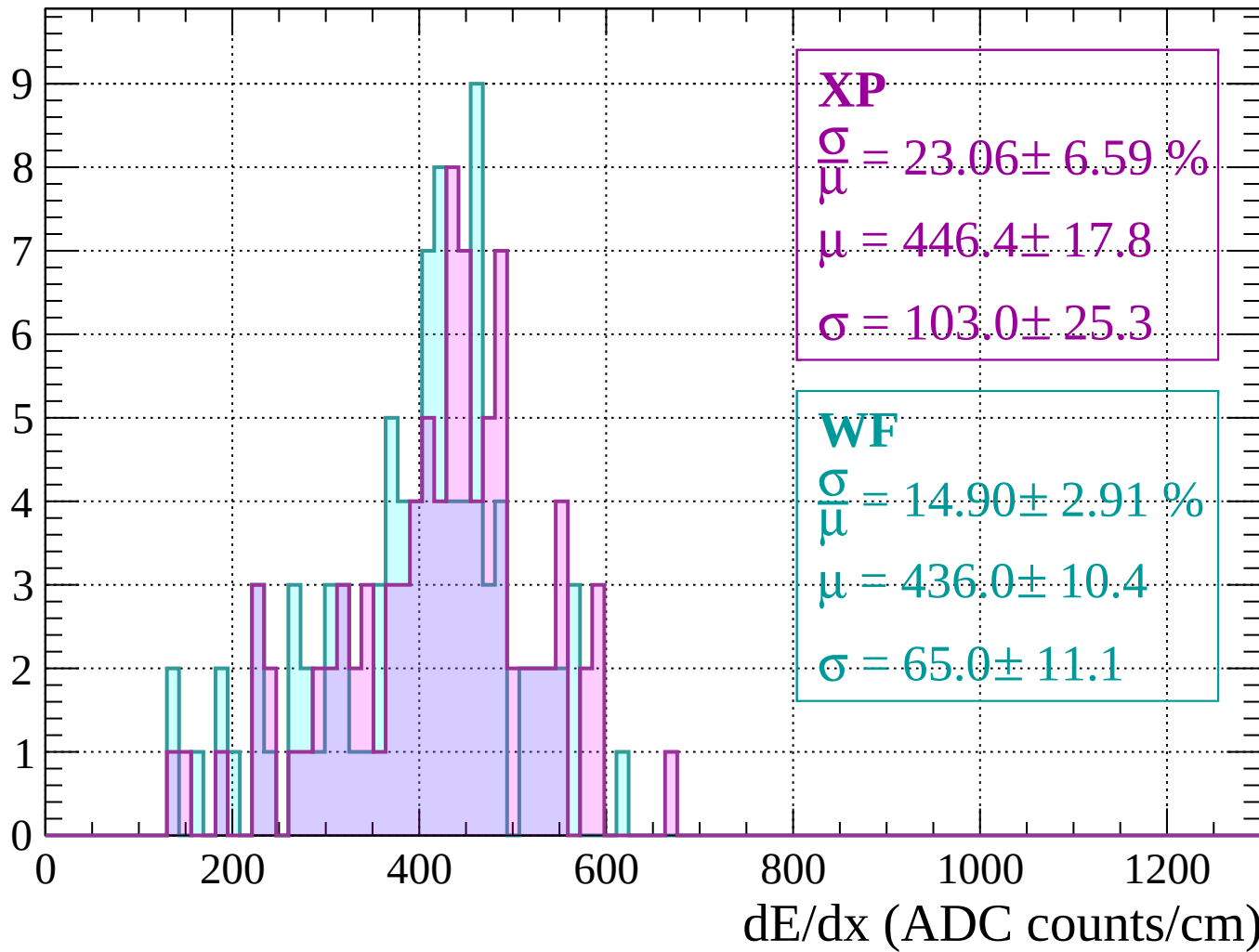
Energy loss | $880 < p < 920$

Count



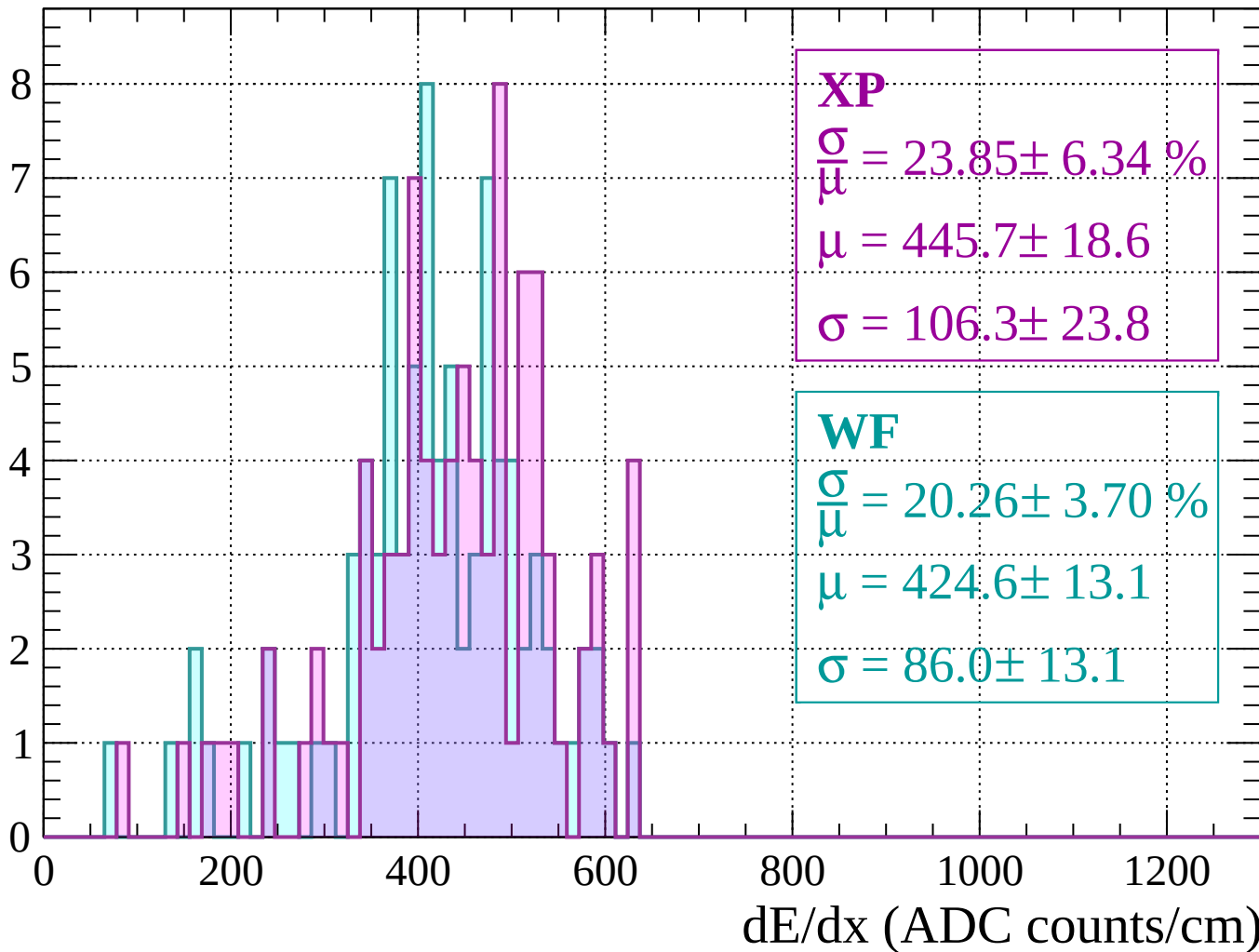
Energy loss | $920 < p < 960$

Count



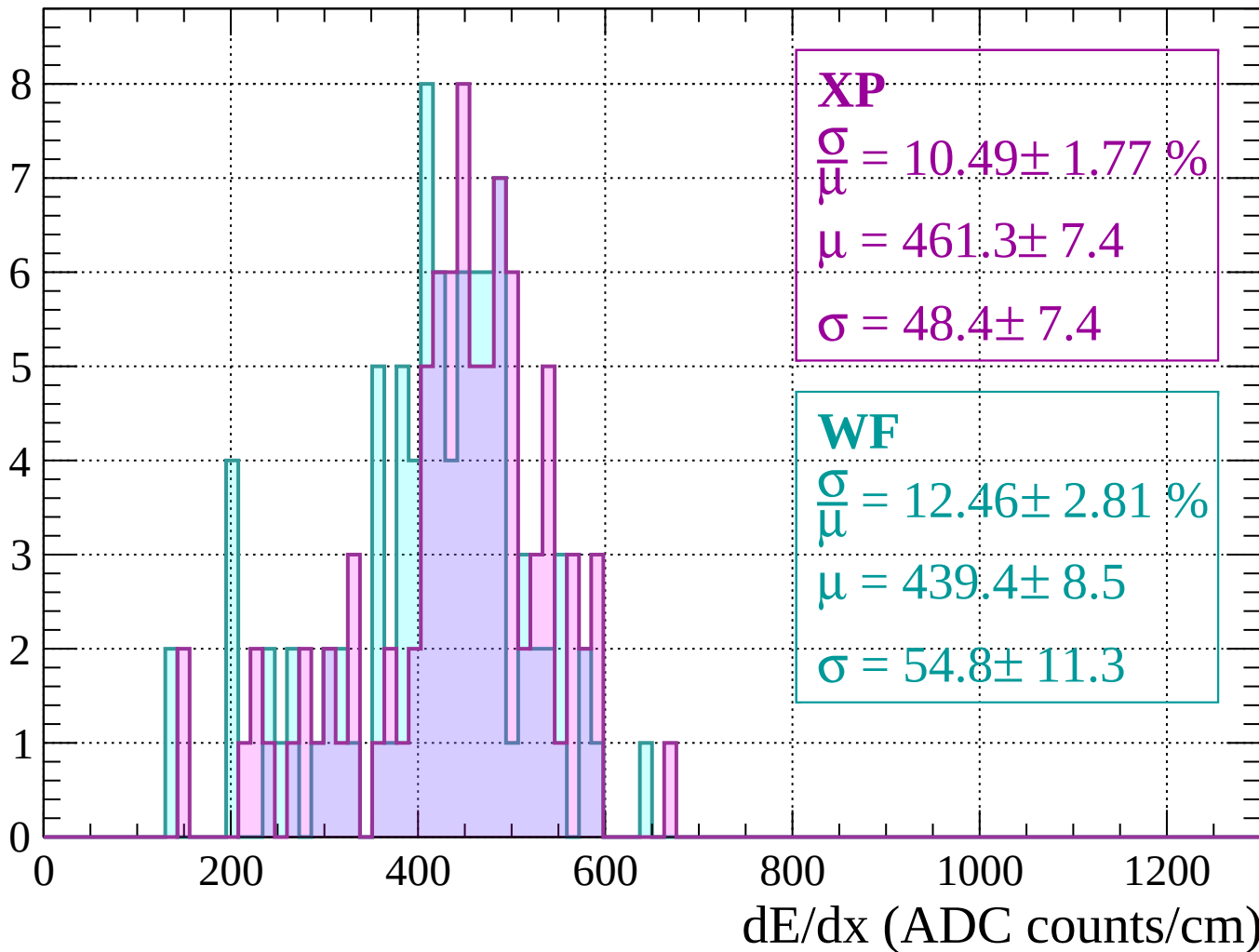
Energy loss | $960 < p < 1000$

Count



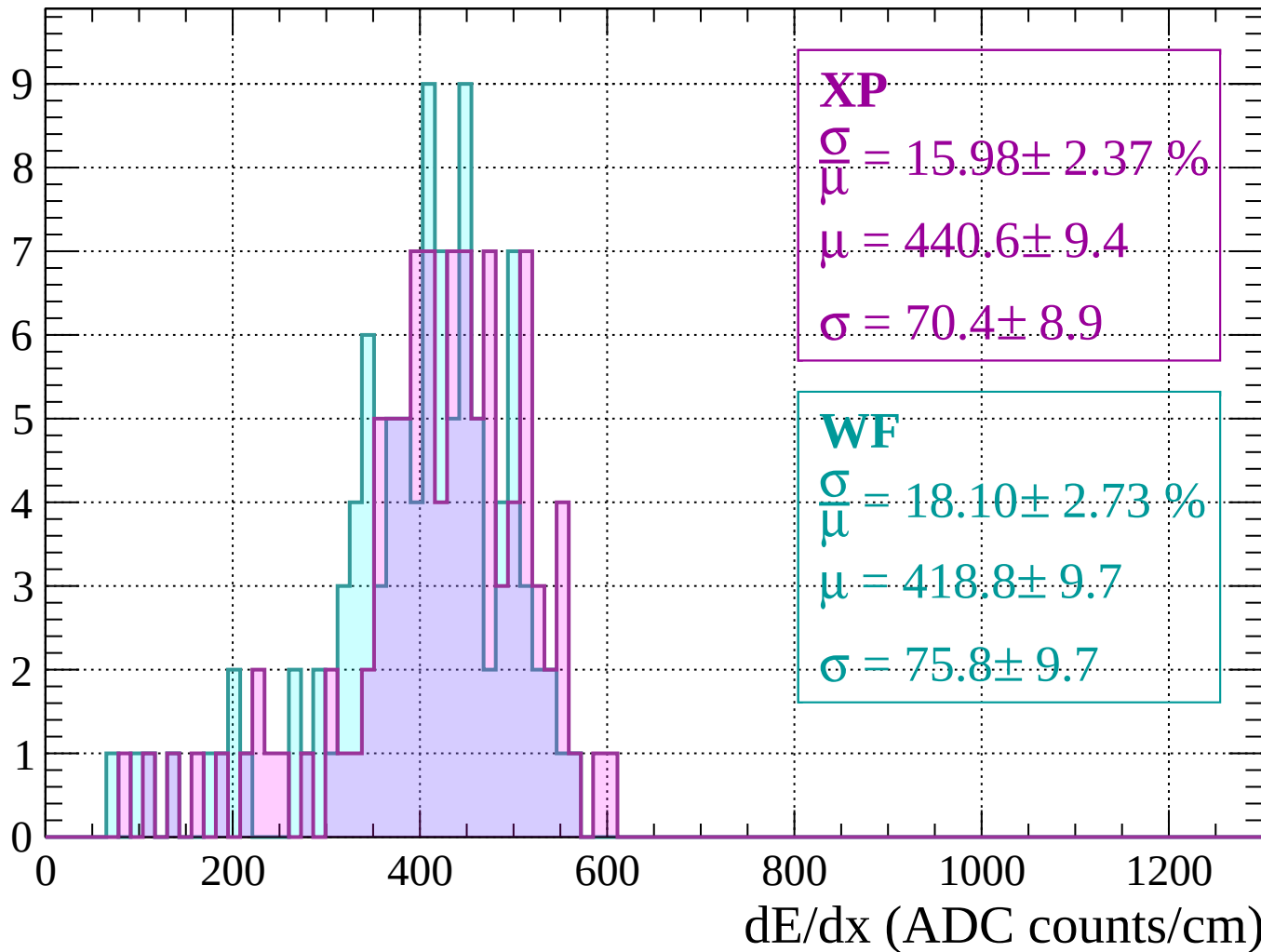
Energy loss | $1000 < p < 1040$

Count



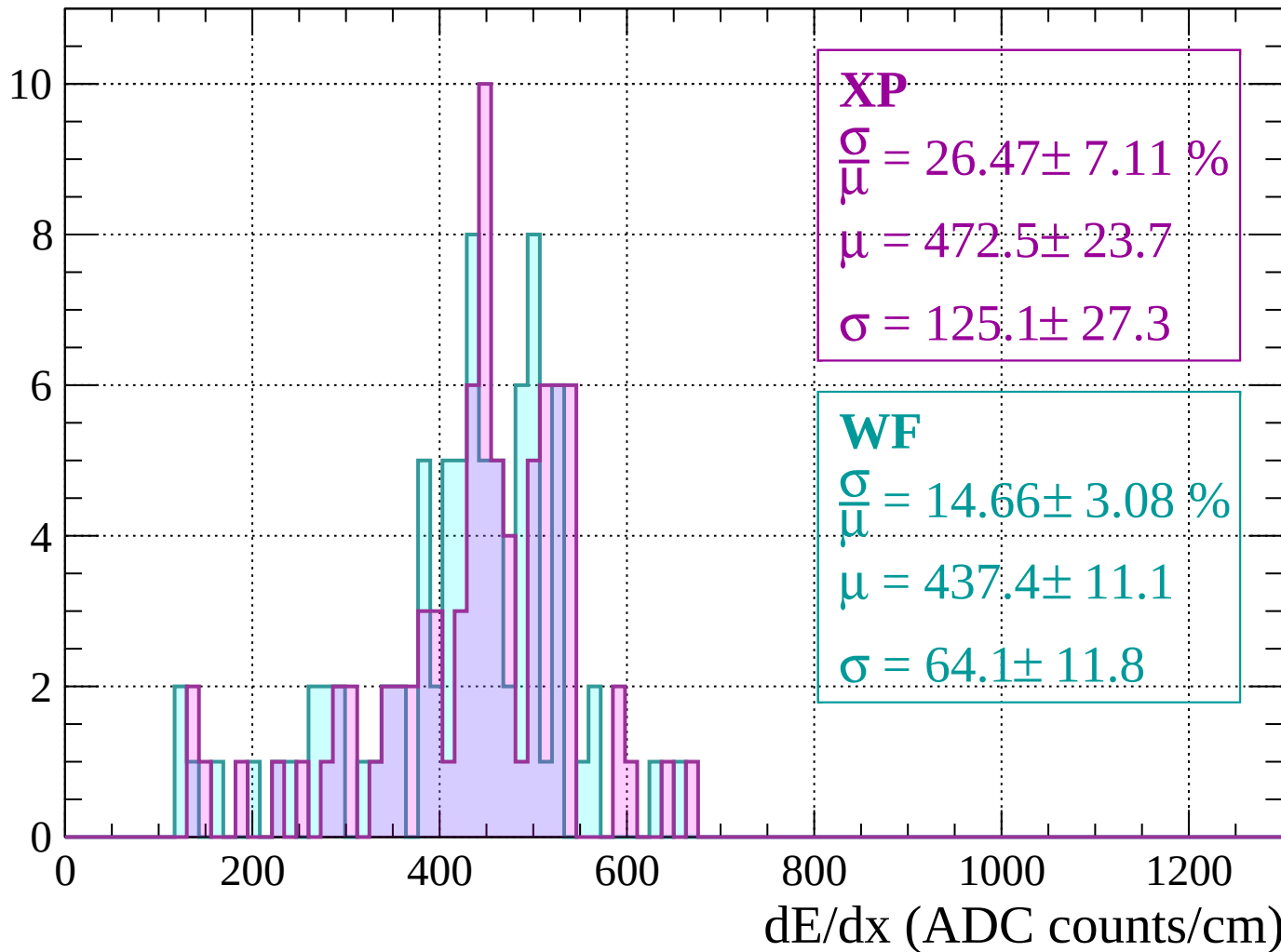
Energy loss | $1040 < p < 1080$

Count



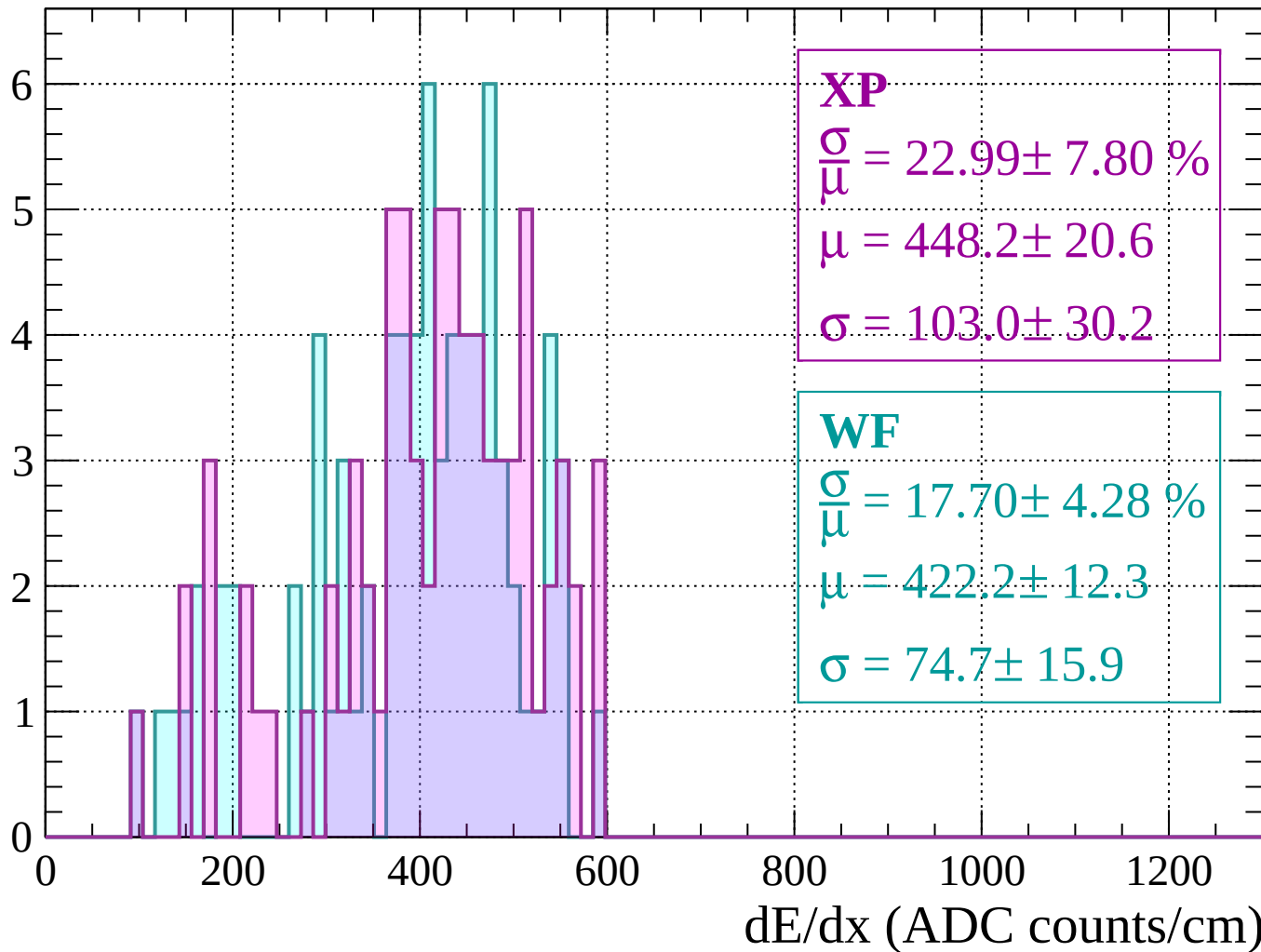
Energy loss | $1080 < p < 1120$

Count



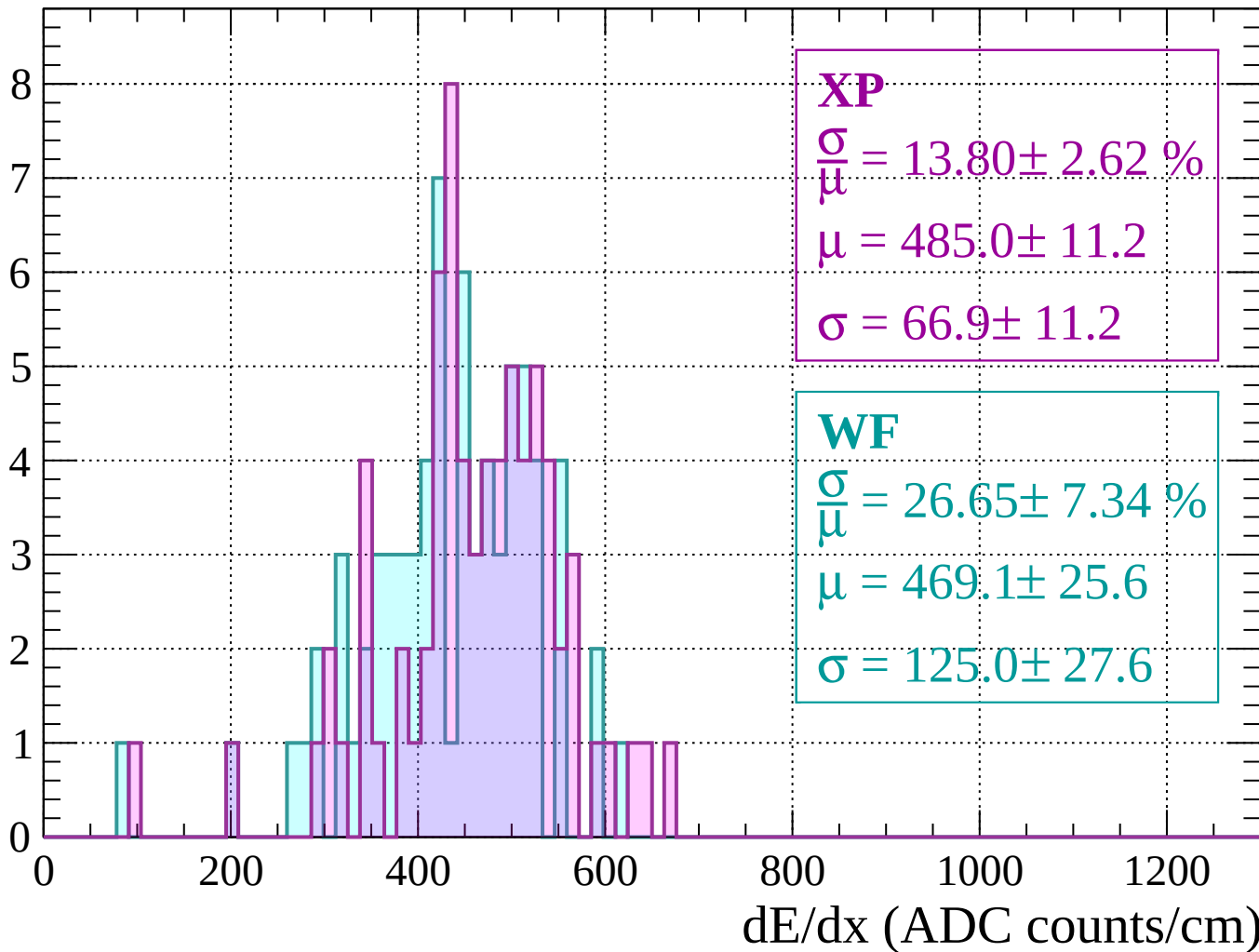
Energy loss | $1120 < p < 1160$

Count



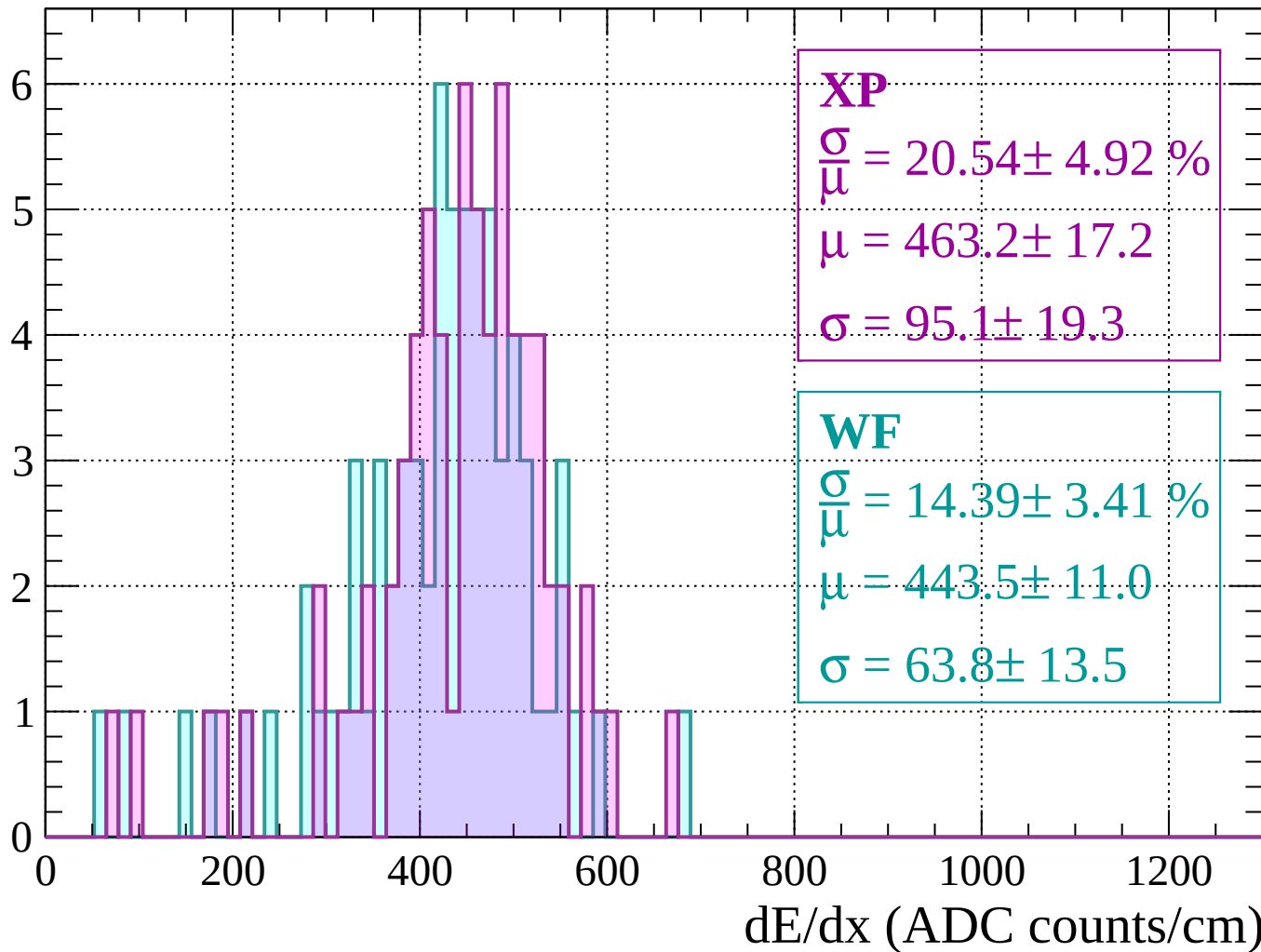
Energy loss | $1160 < p < 1200$

Count



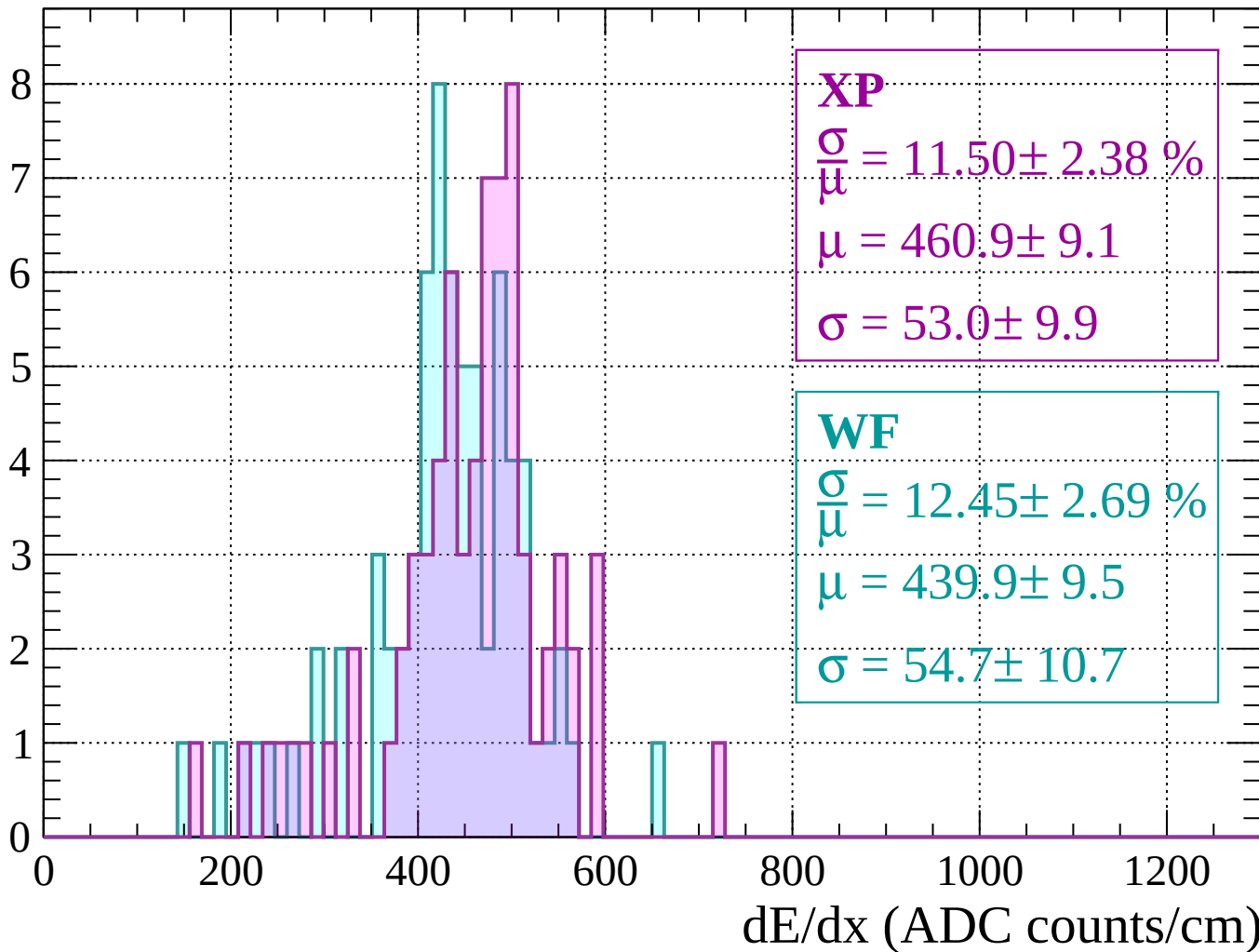
Energy loss | $1200 < p < 1240$

Count



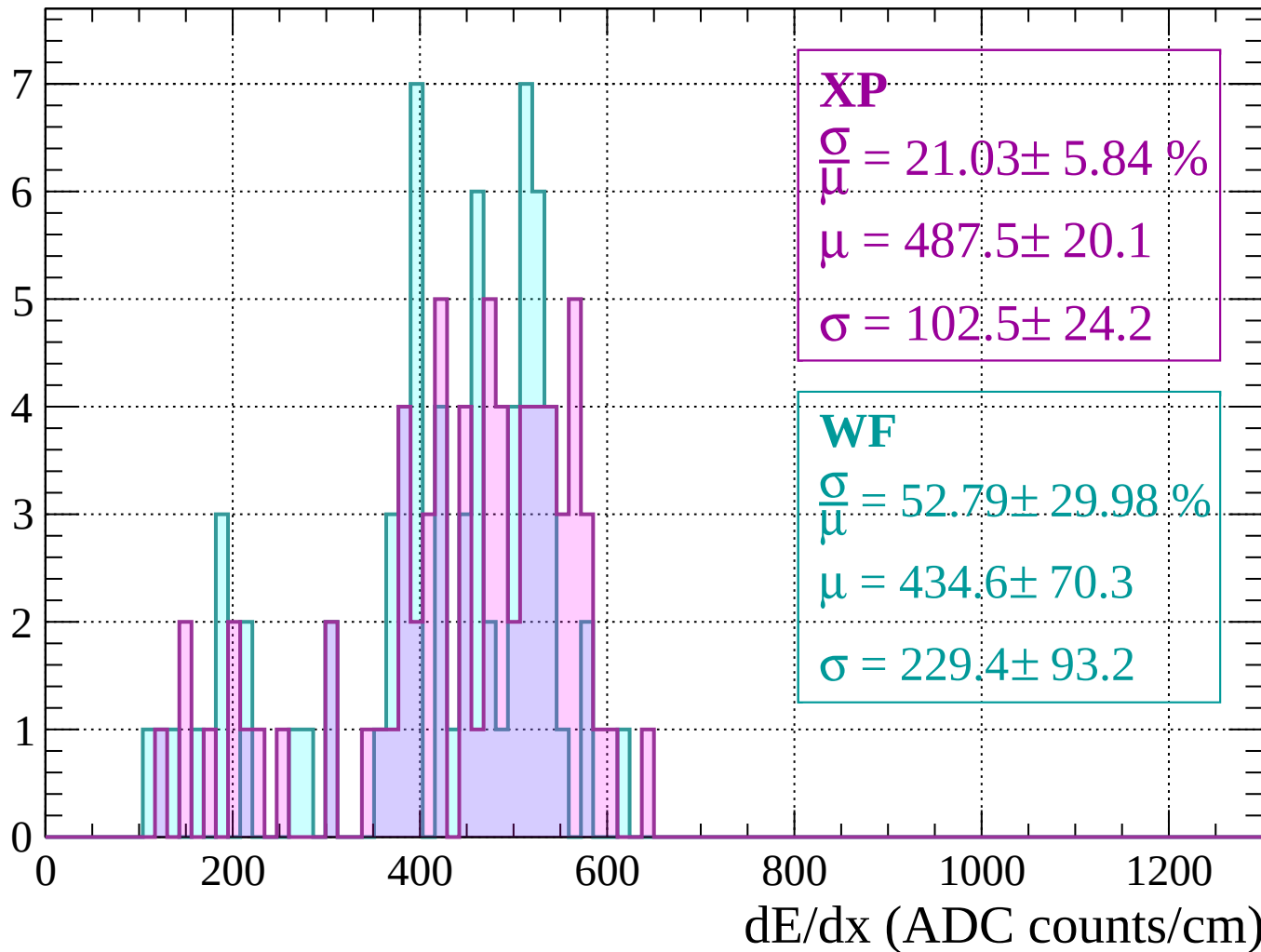
Energy loss | $1240 < p < 1280$

Count



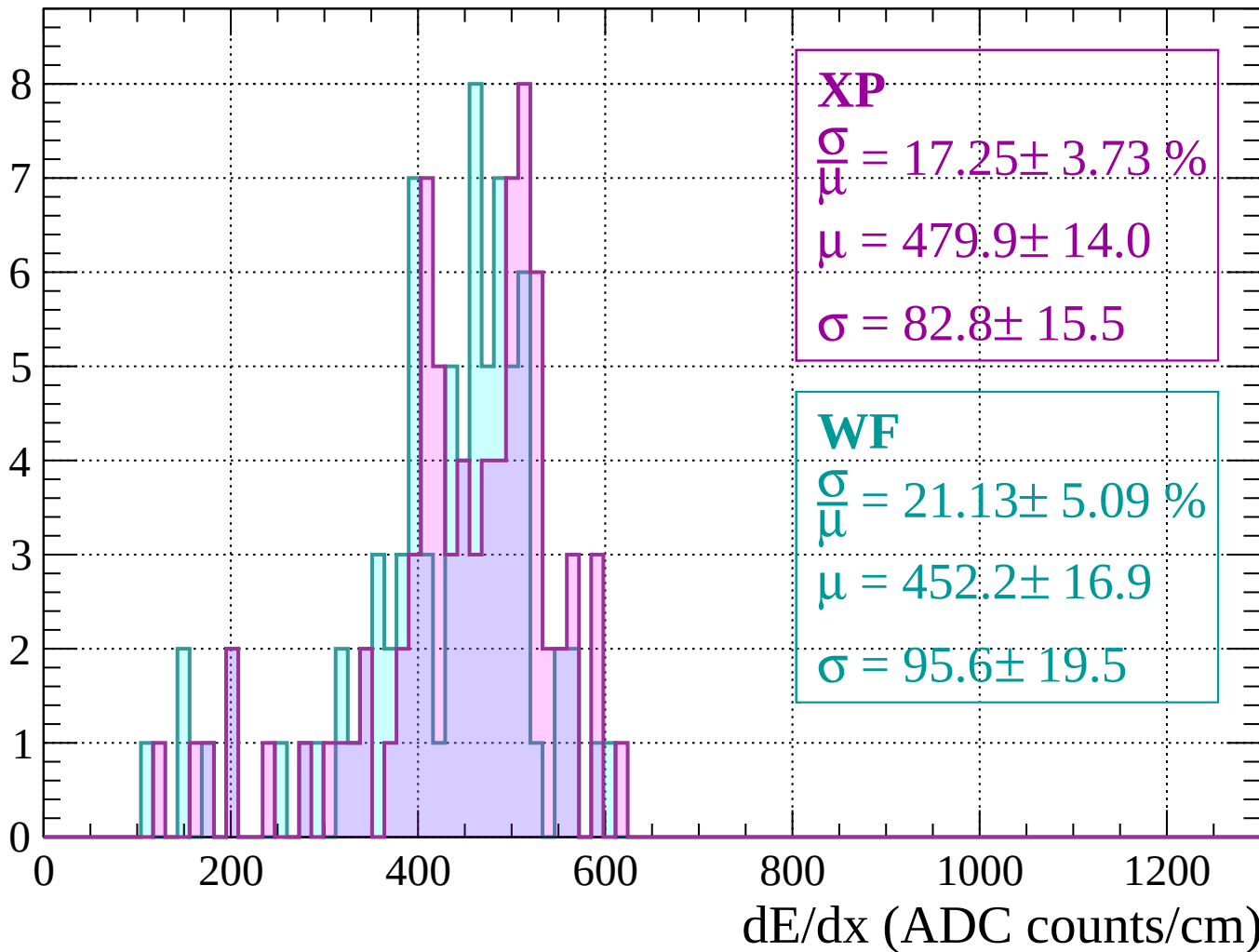
Energy loss | $1280 < p < 1320$

Count



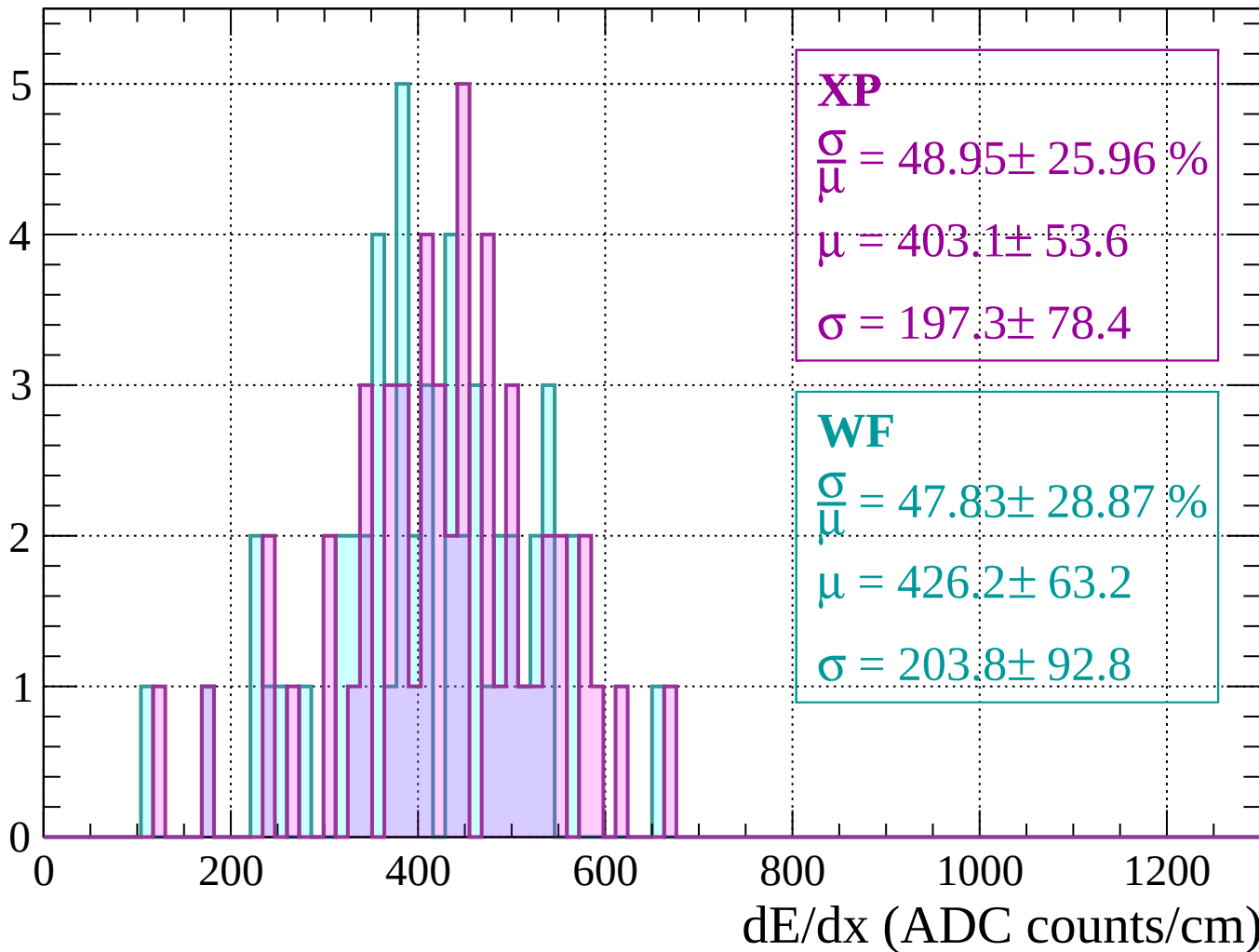
Energy loss | $1320 < p < 1360$

Count



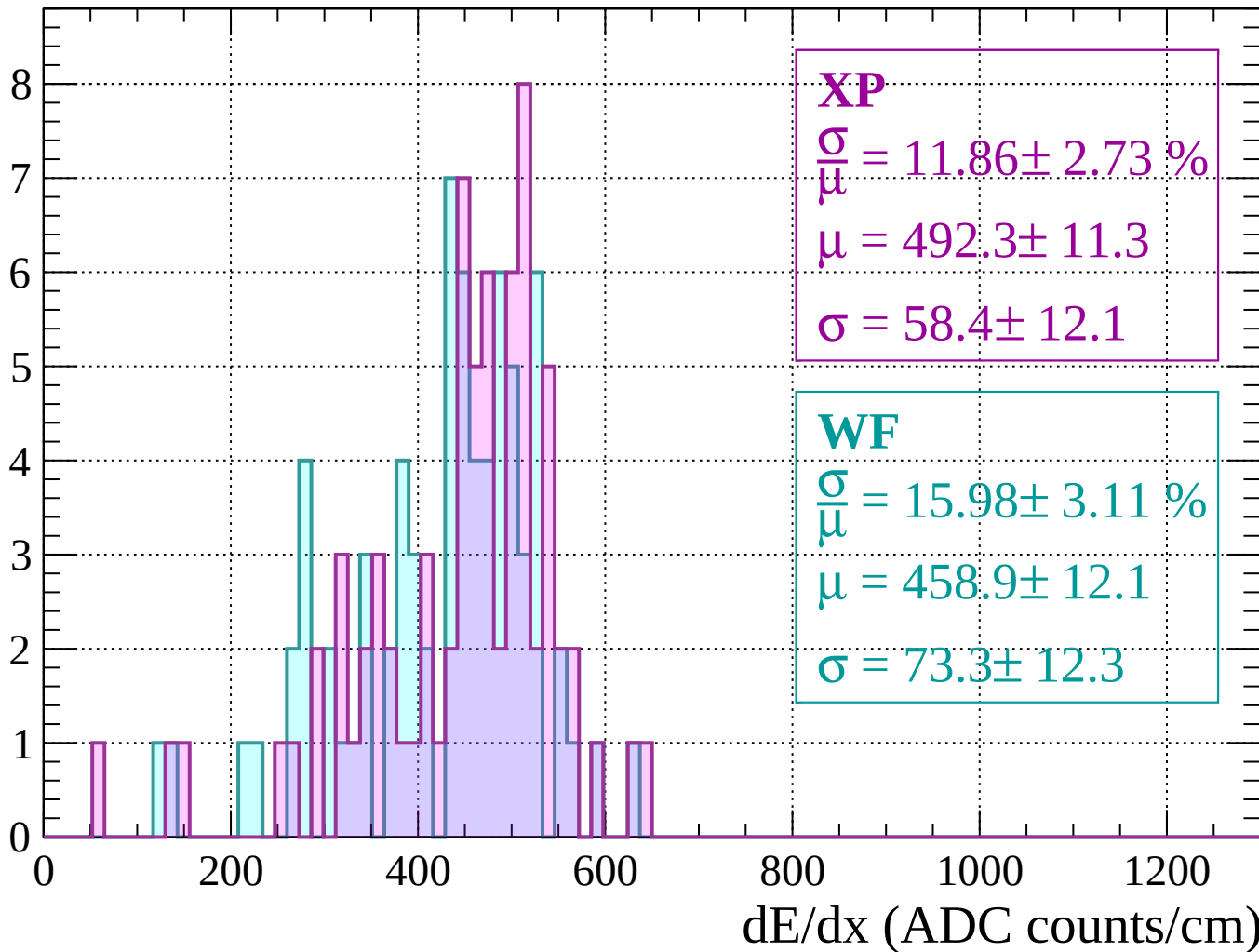
Energy loss | $1360 < p < 1400$

Count



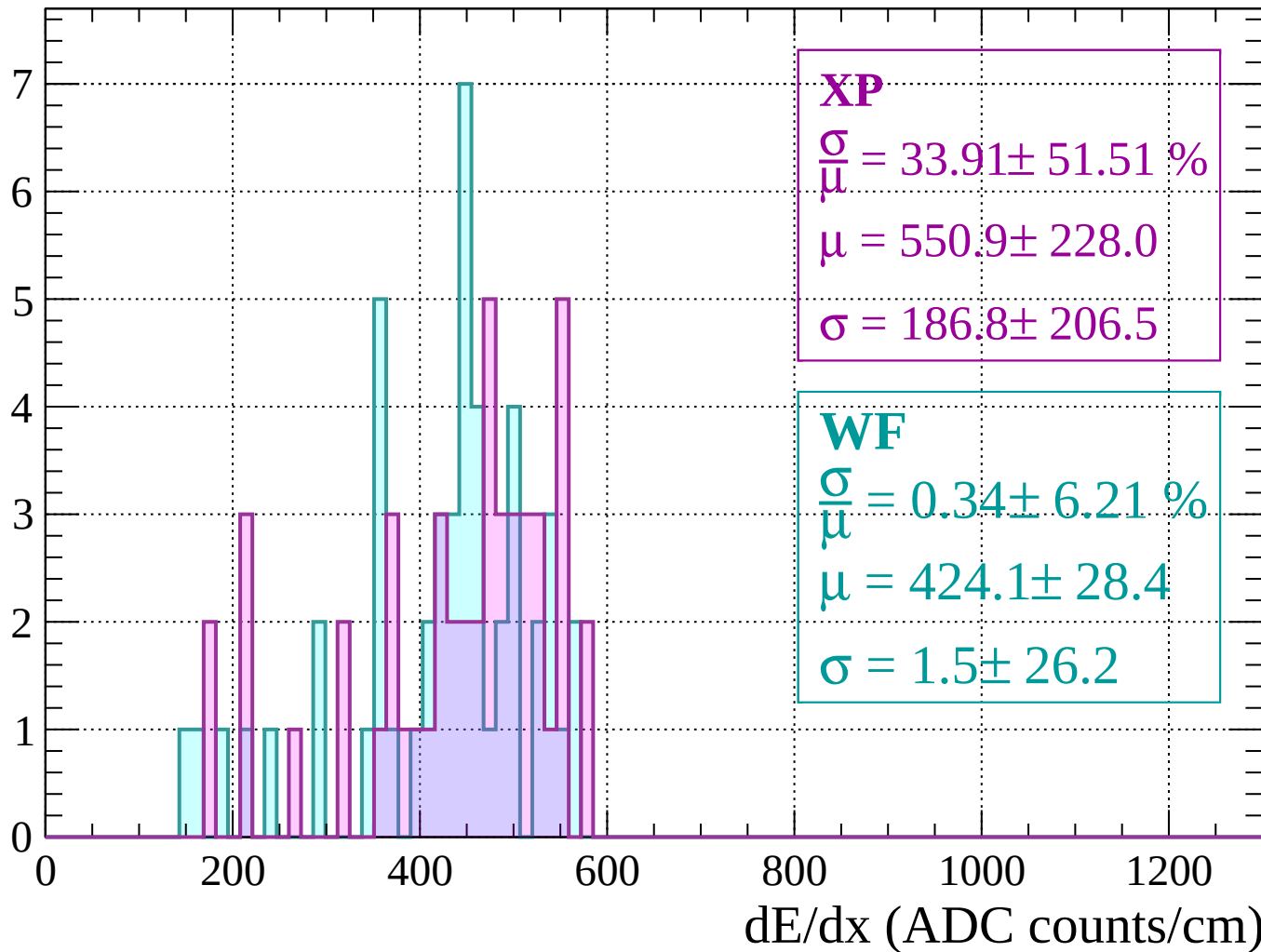
Energy loss | $1400 < p < 1440$

Count



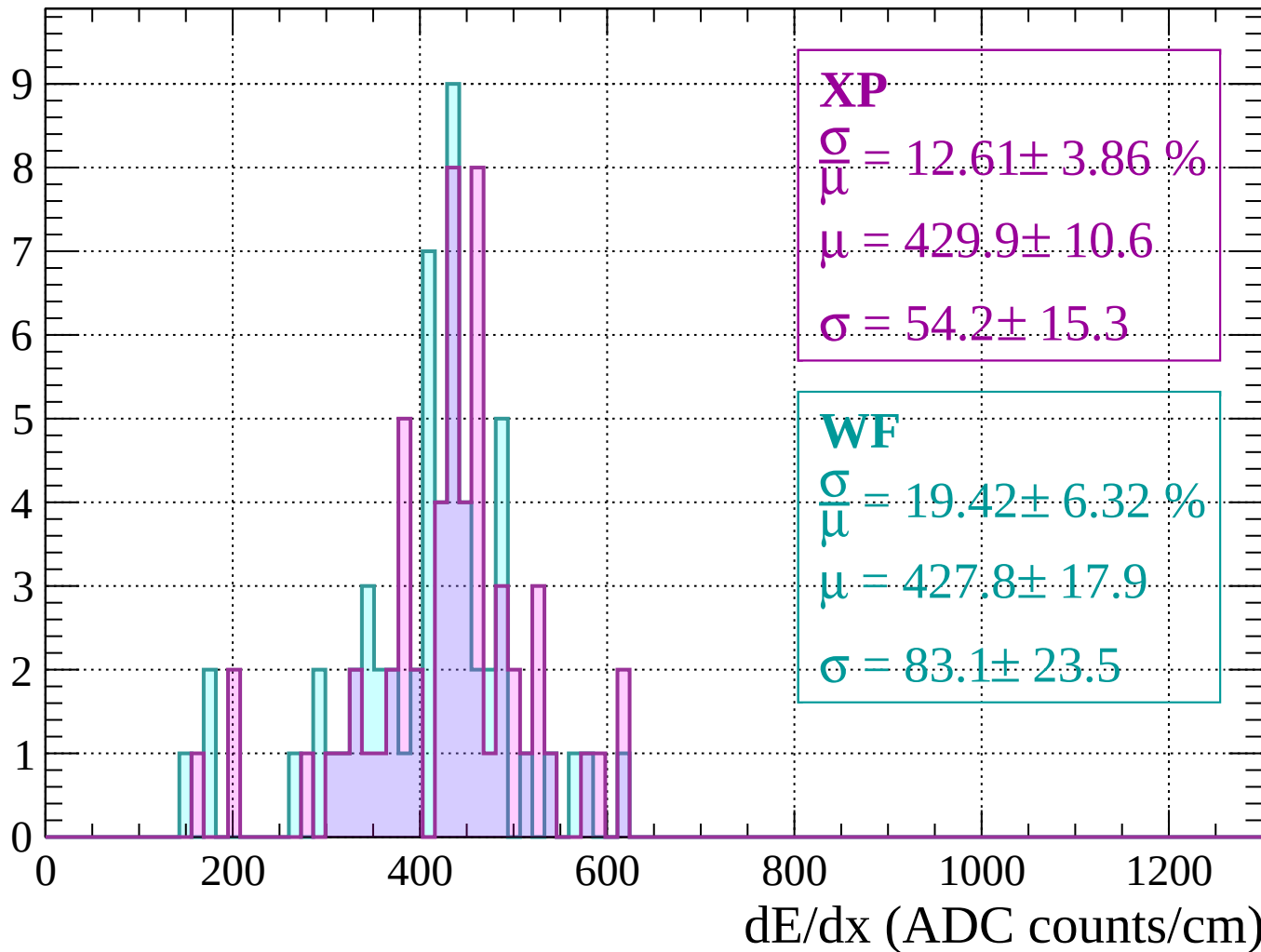
Energy loss | $1440 < p < 1480$

Count



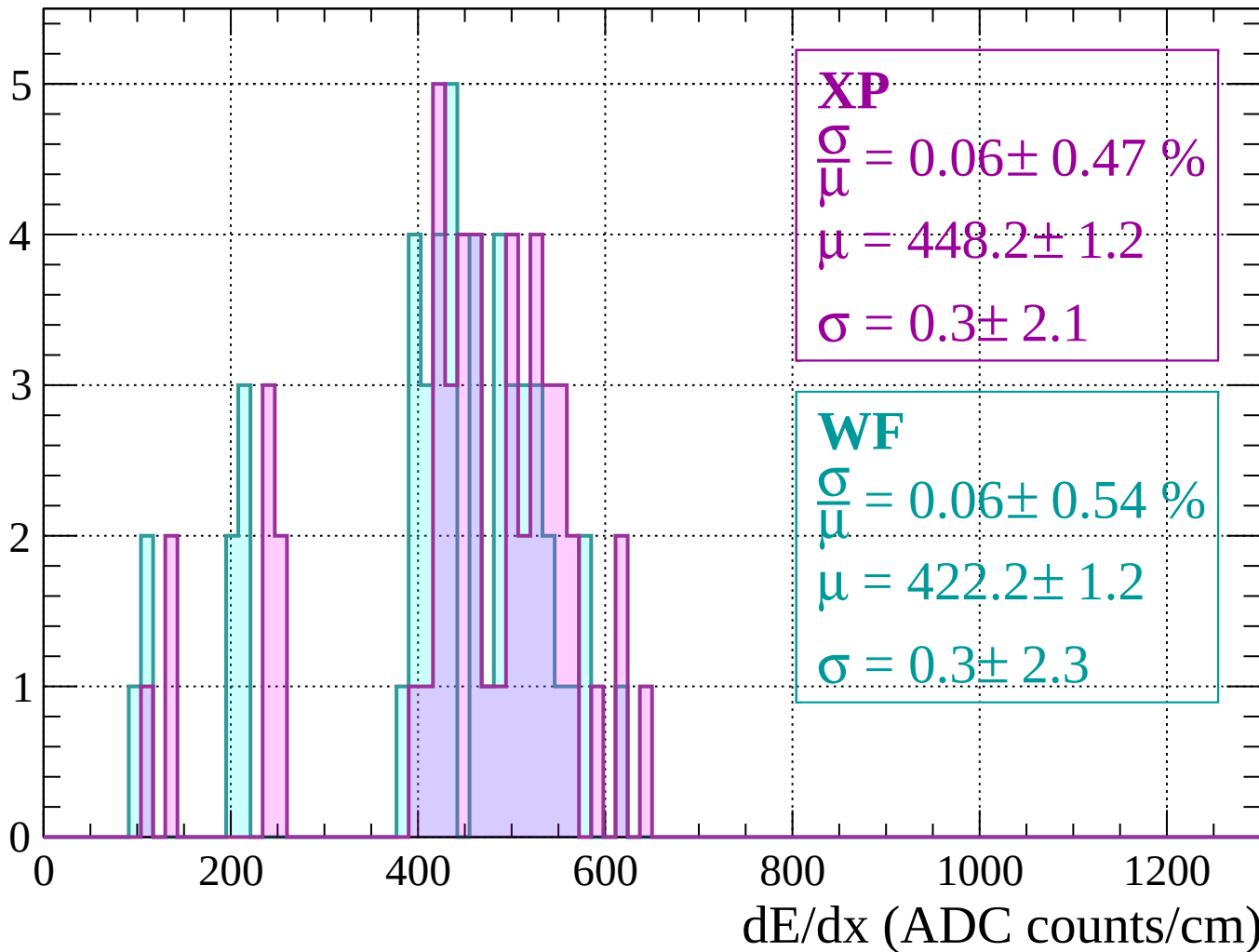
Energy loss | $1480 < p < 1520$

Count



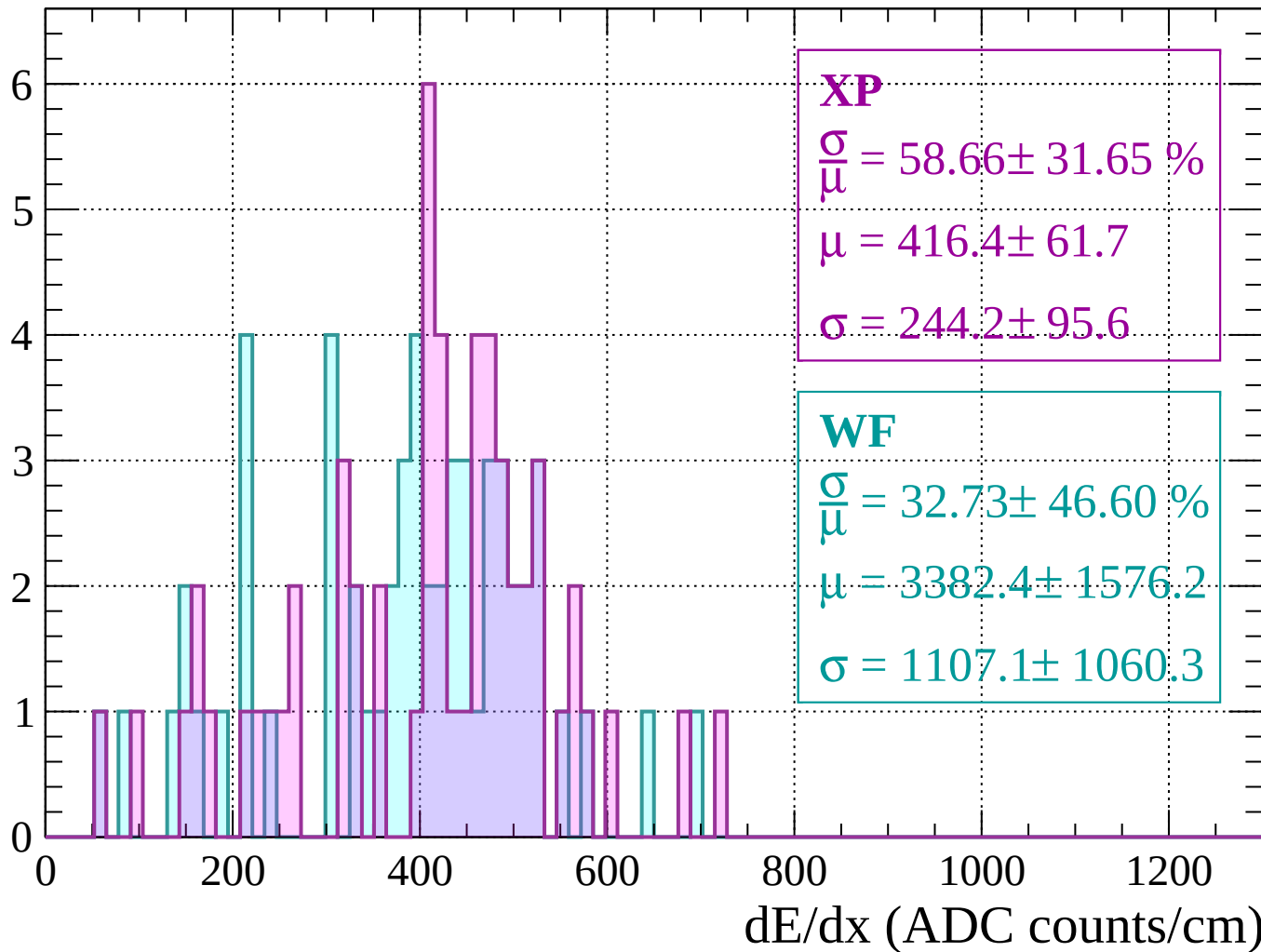
Energy loss | $1520 < p < 1560$

Count



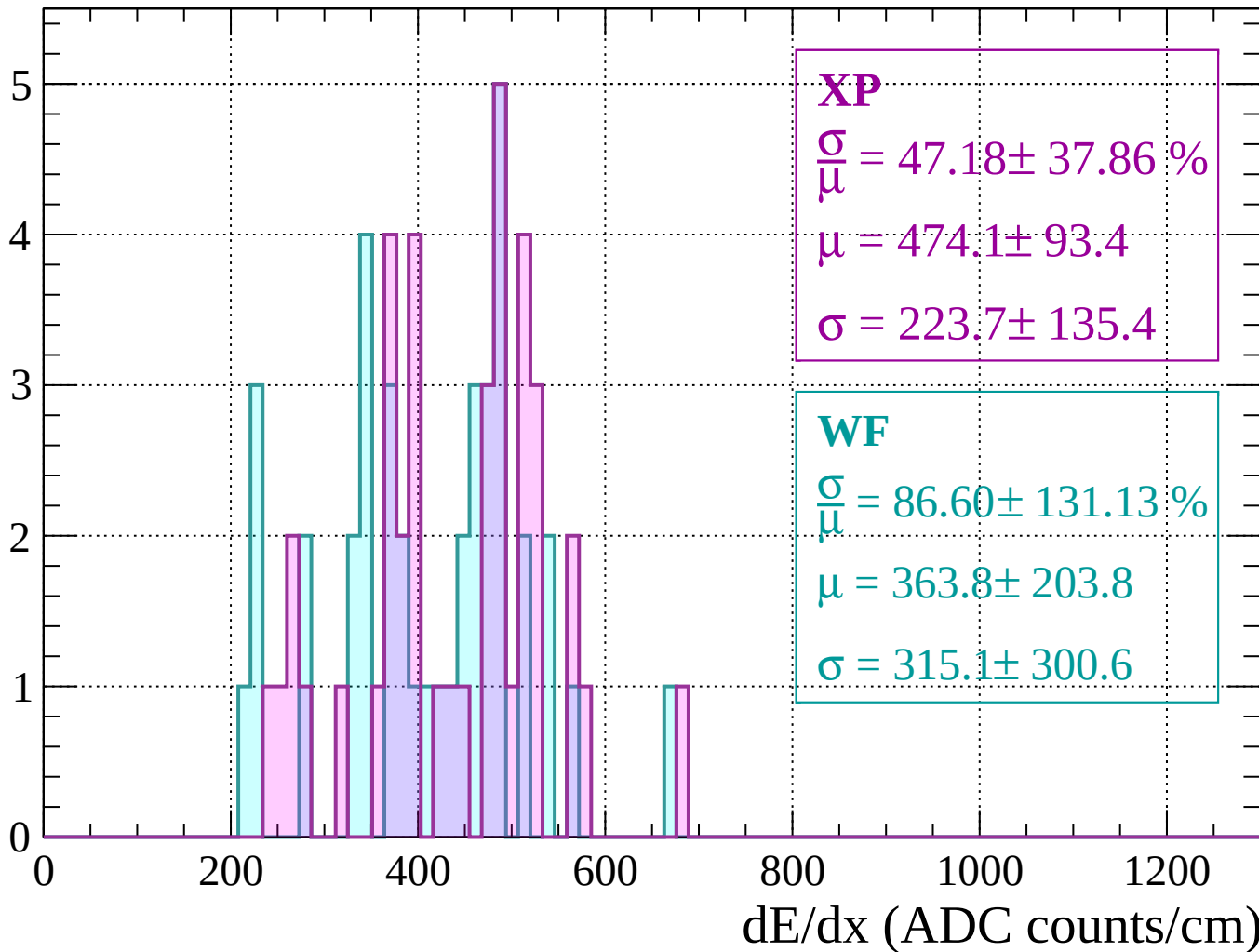
Energy loss | $1560 < p < 1600$

Count



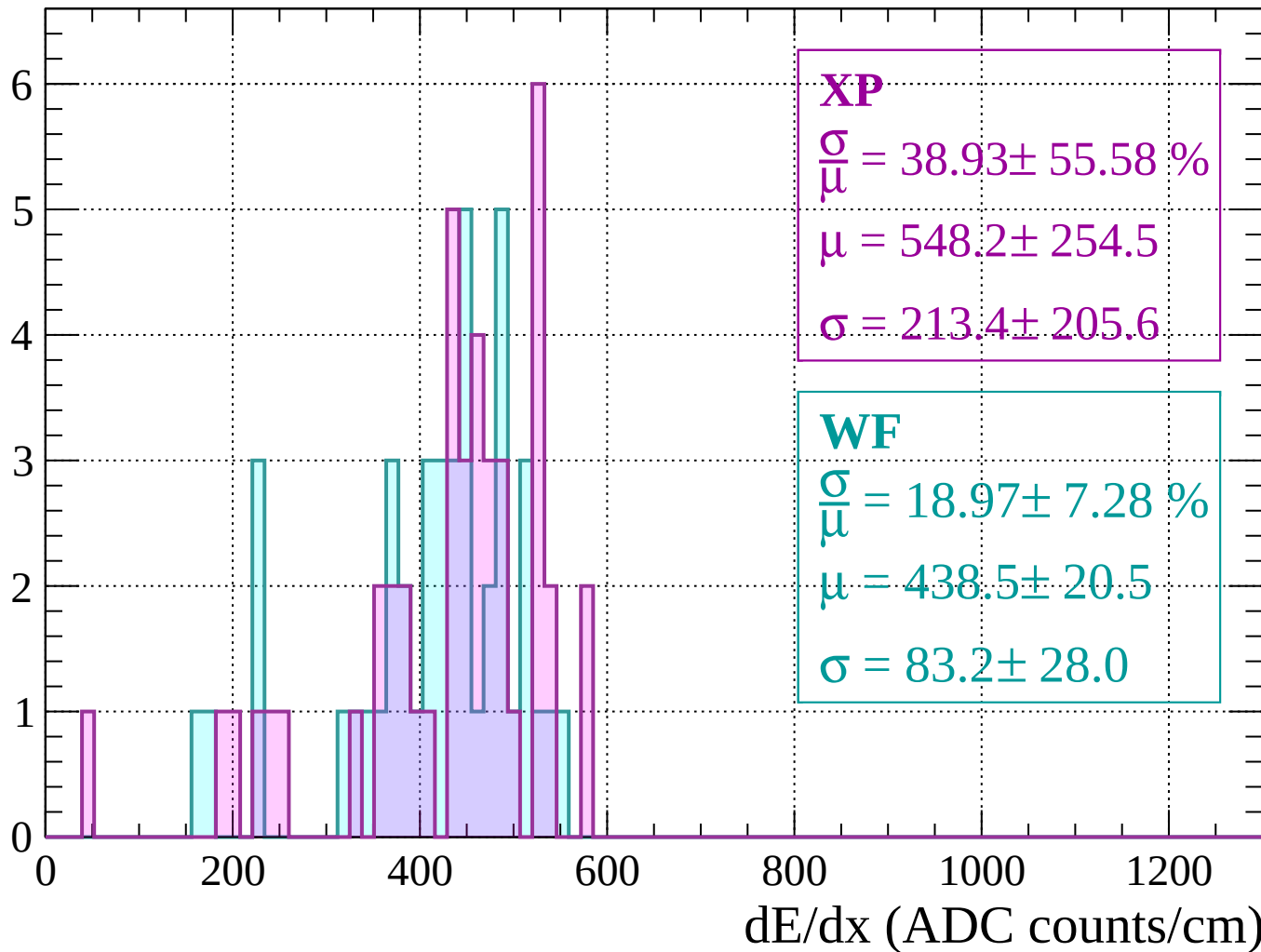
Energy loss | $1600 < p < 1640$

Count



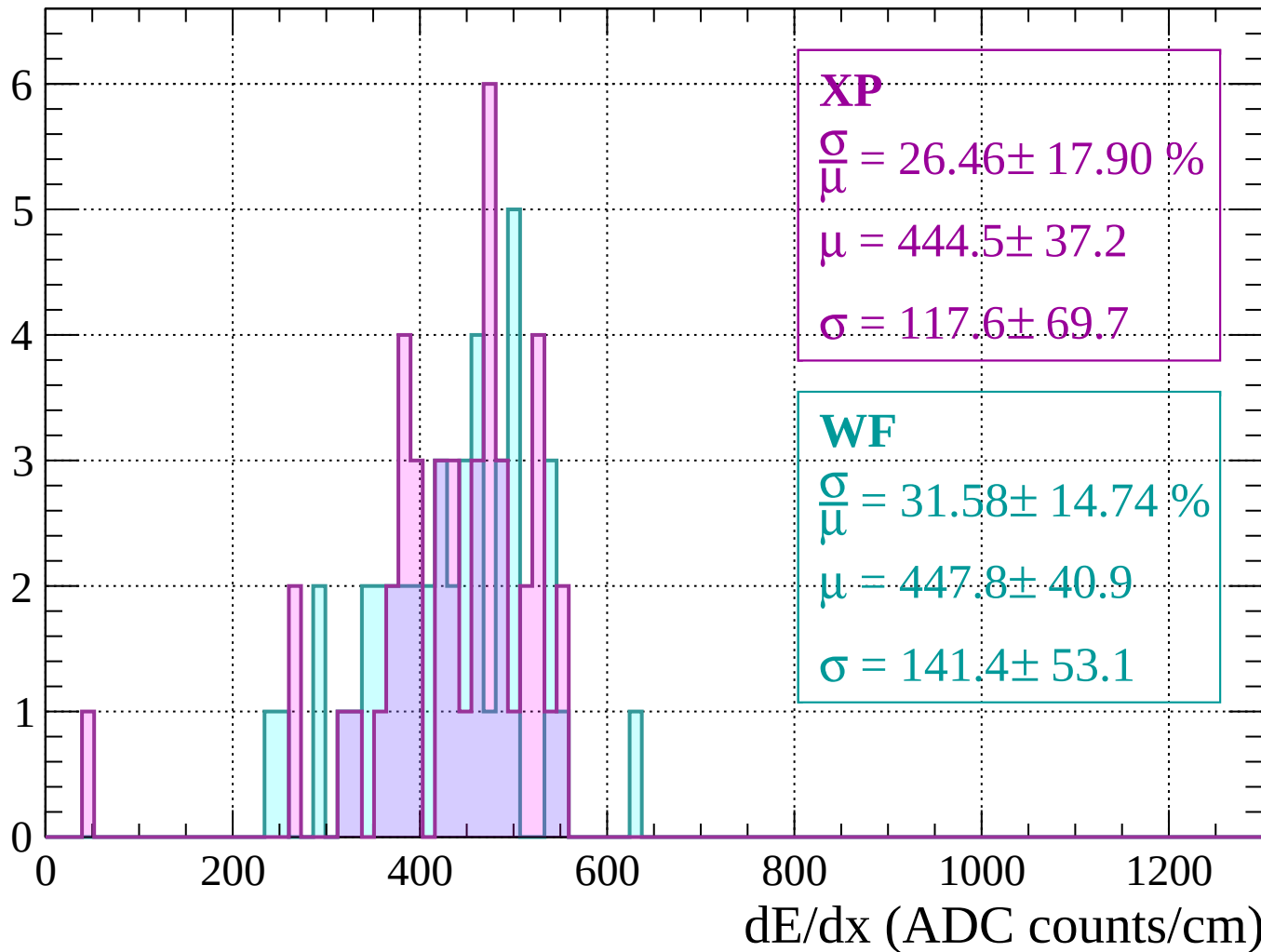
Energy loss | $1640 < p < 1680$

Count



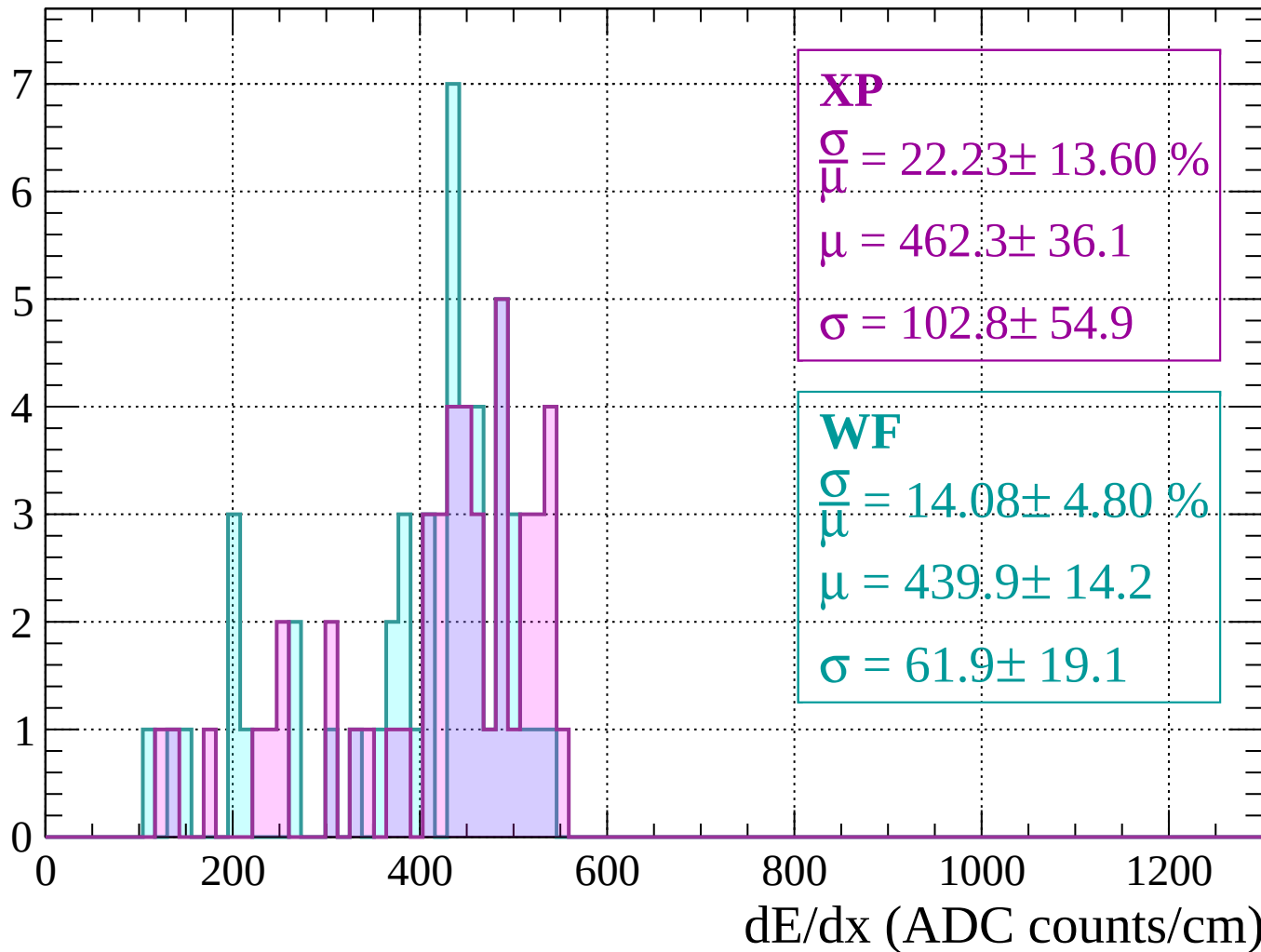
Energy loss | $1680 < p < 1720$

Count



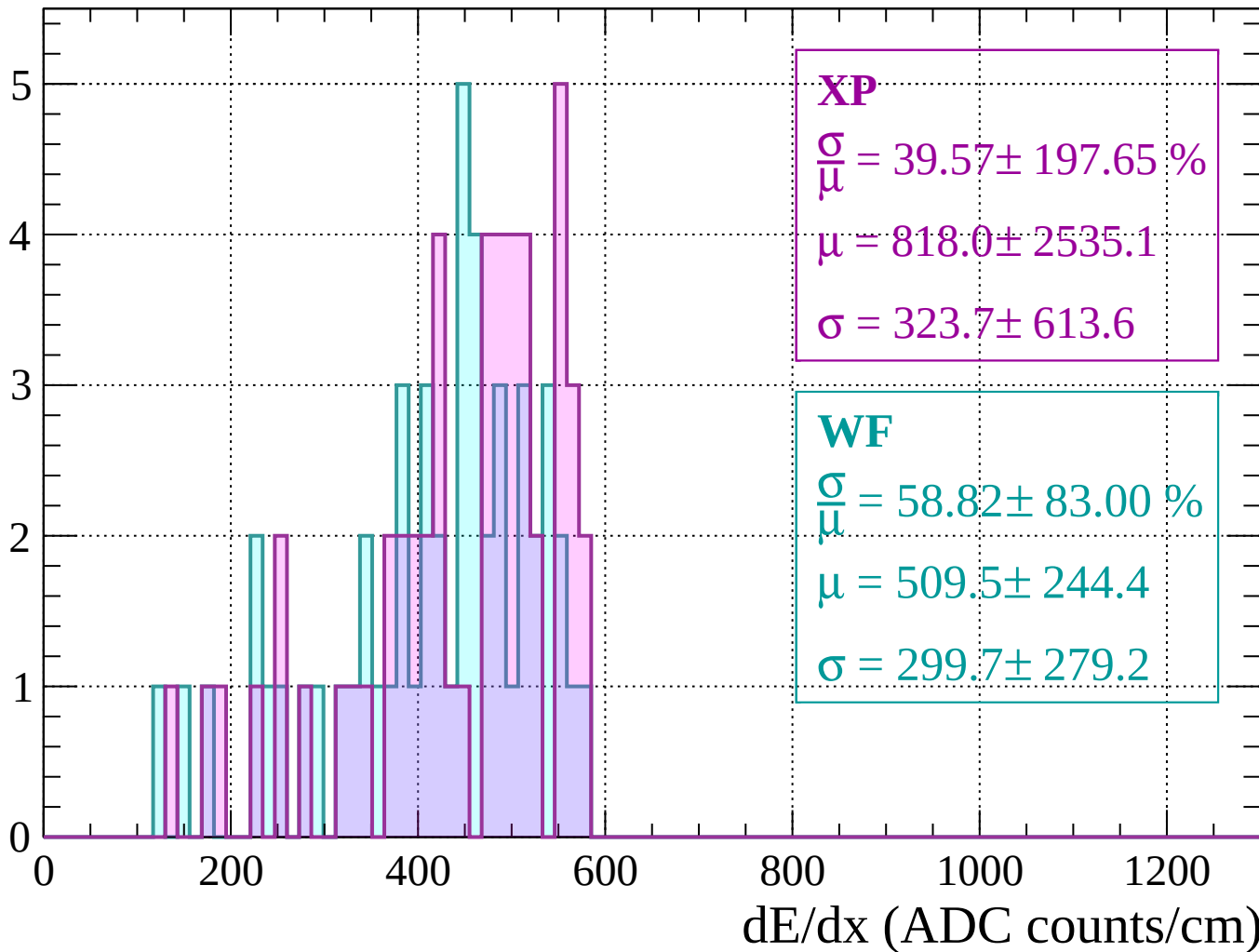
Energy loss | $1720 < p < 1760$

Count



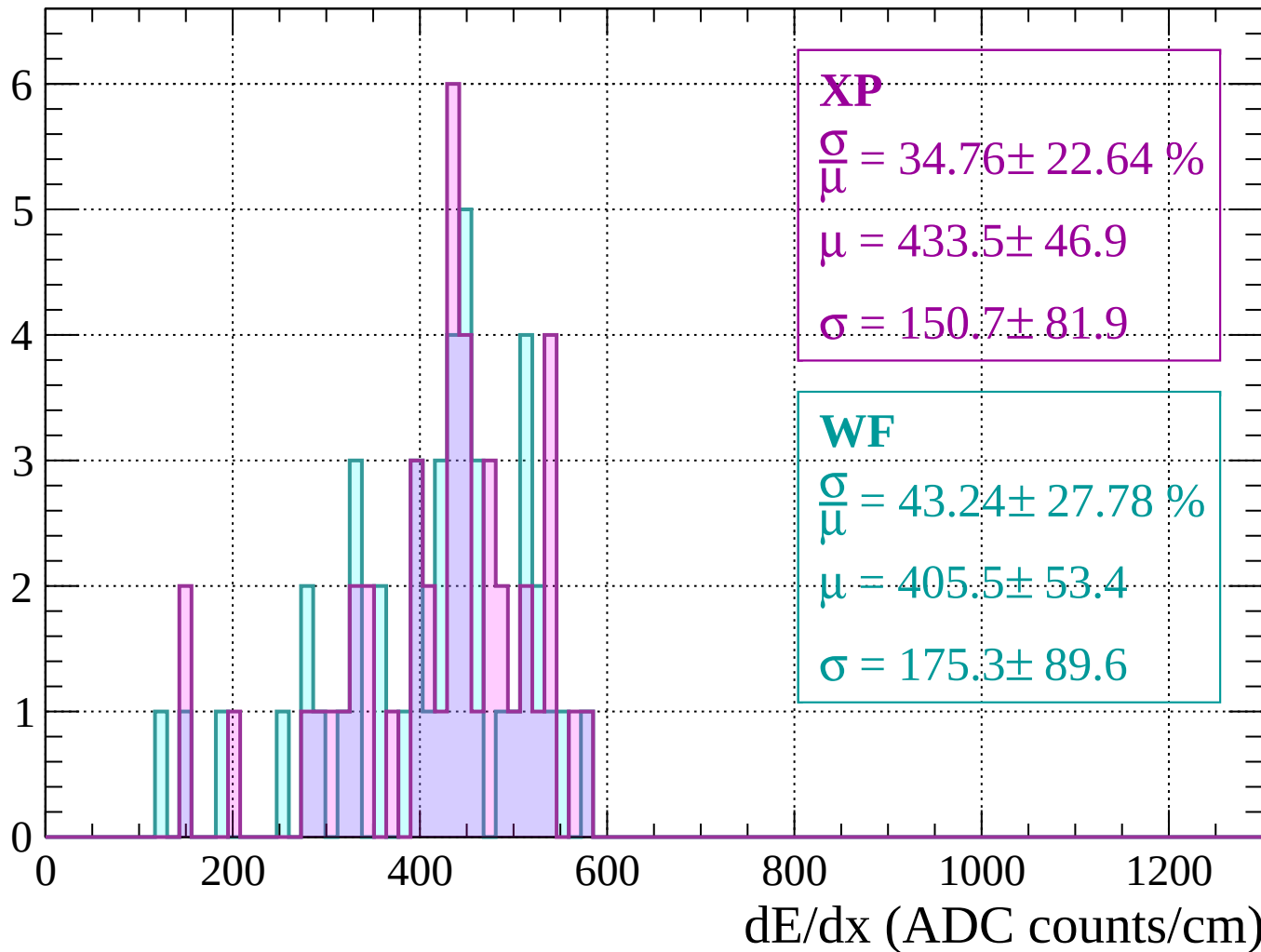
Energy loss | $1760 < p < 1800$

Count



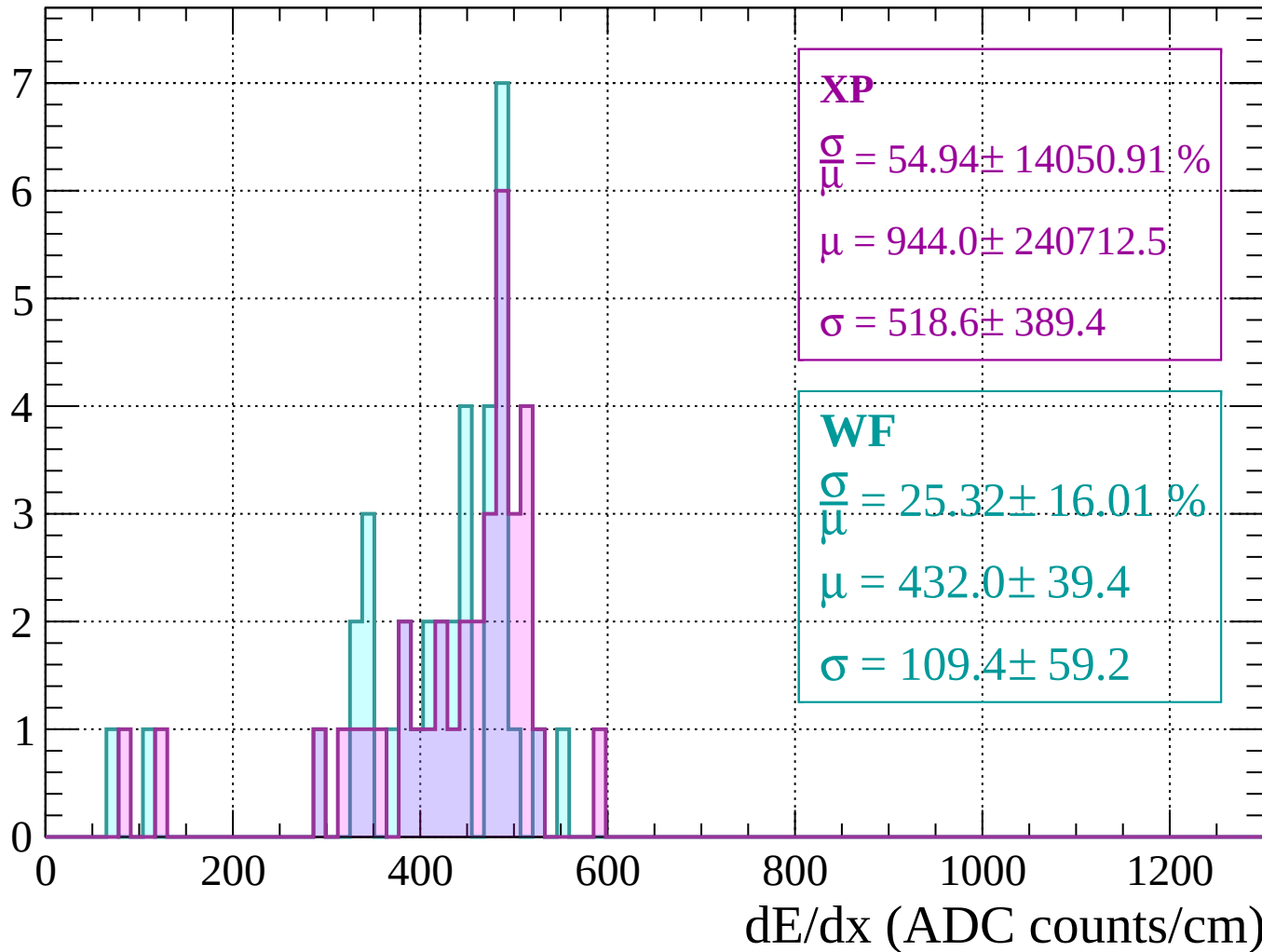
Energy loss | $1800 < p < 1840$

Count



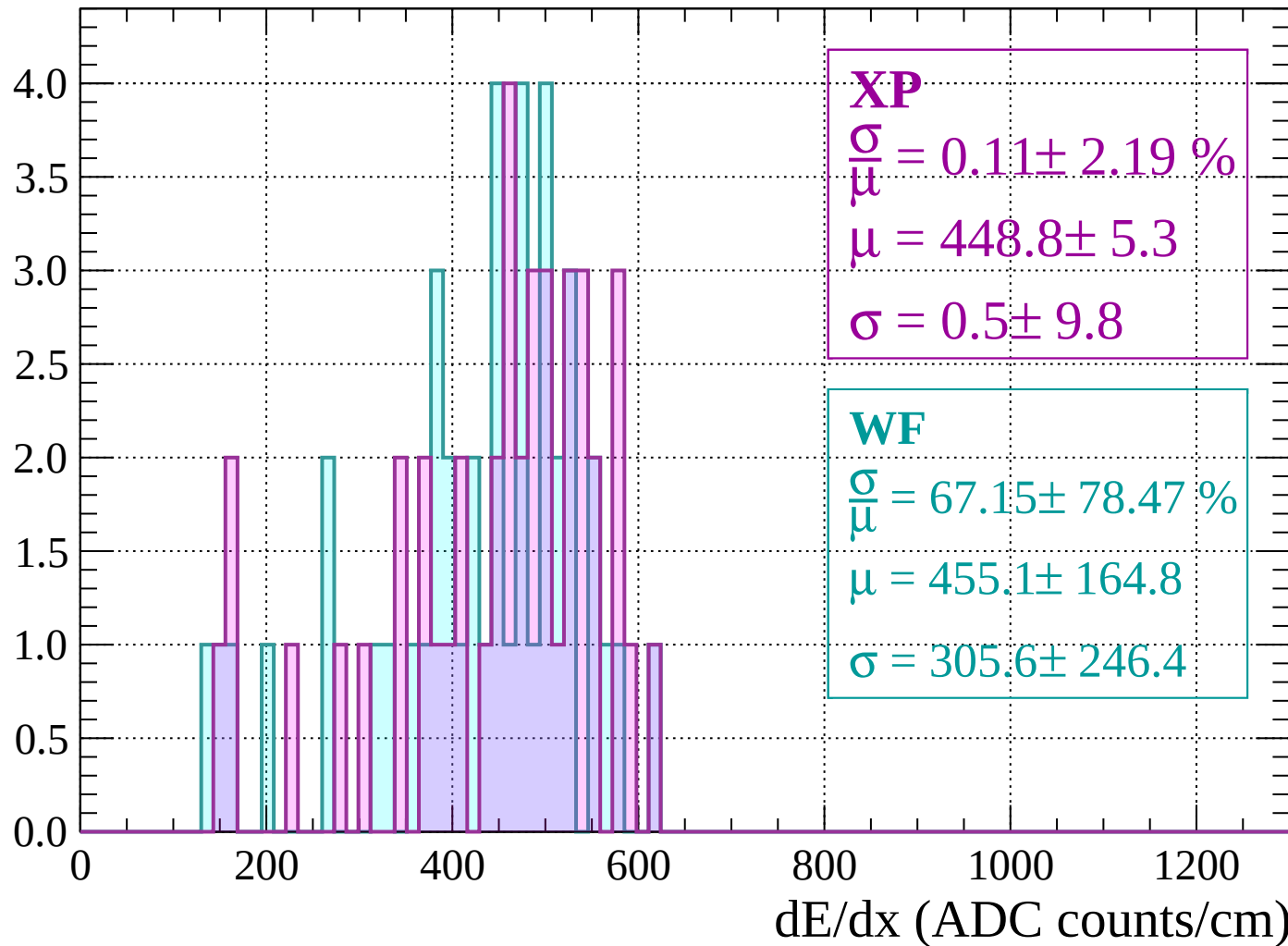
Energy loss | $1840 < p < 1880$

Count



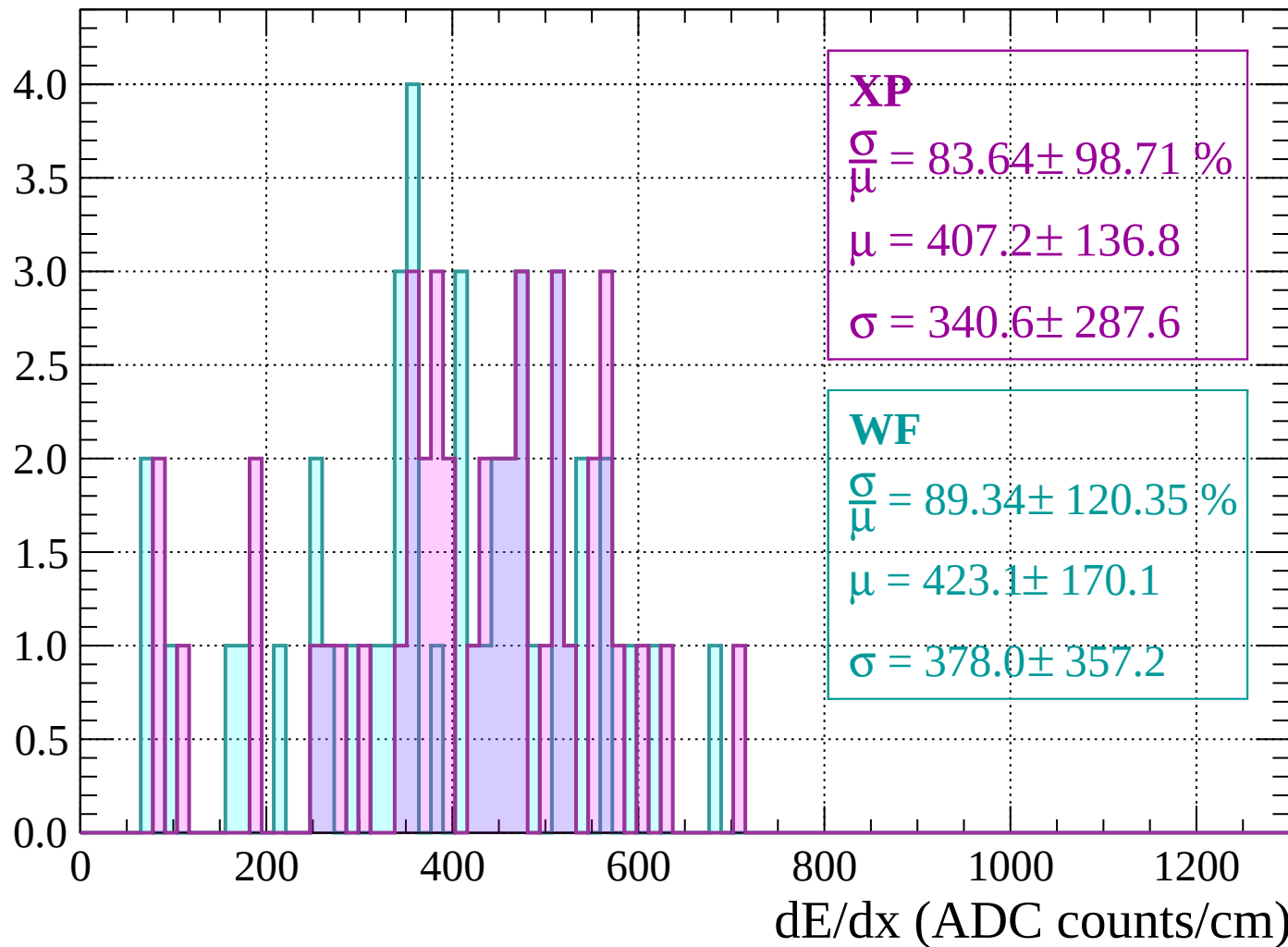
Energy loss | $1880 < p < 1920$

Count



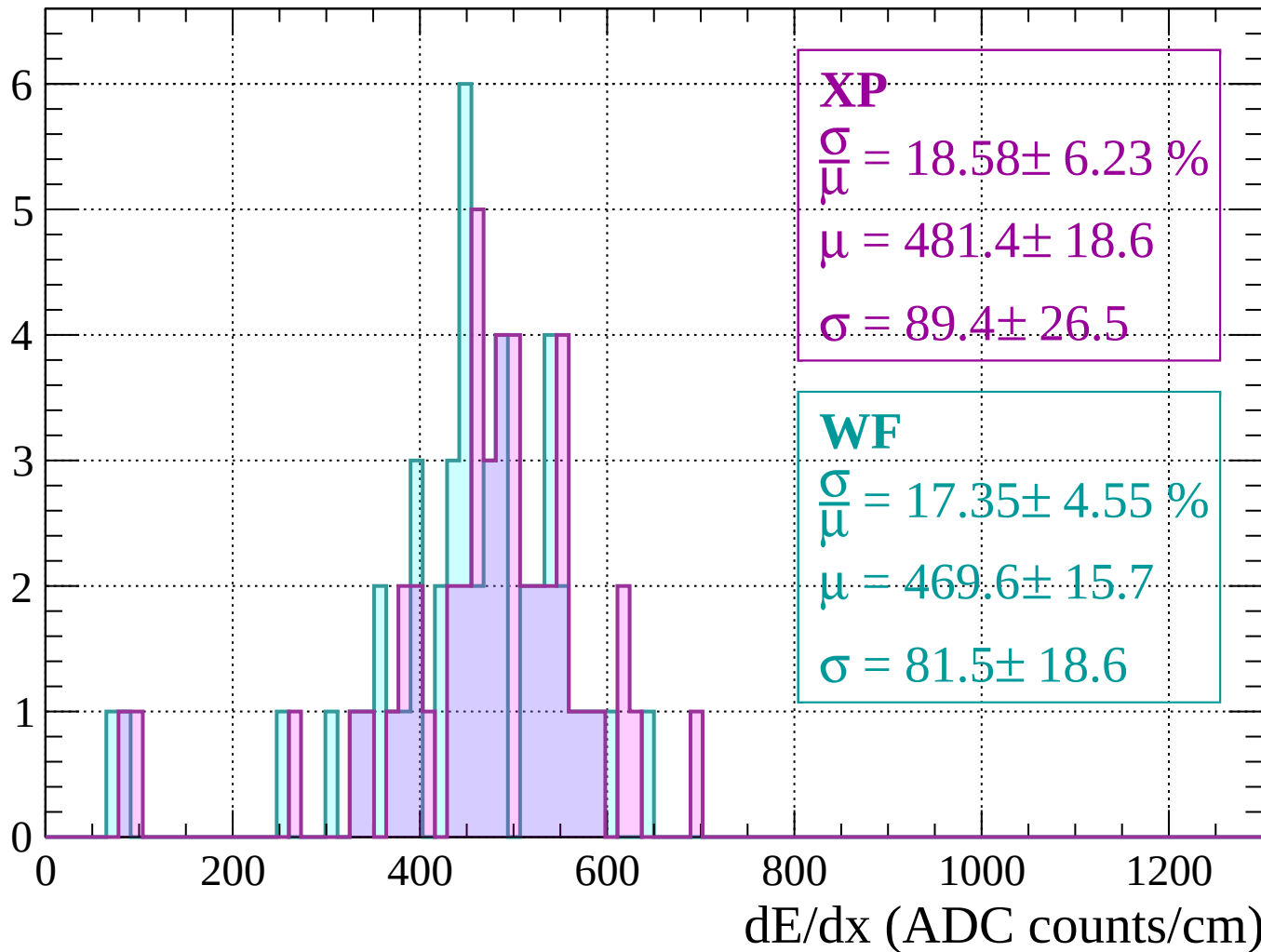
Energy loss | $1920 < p < 1960$

Count



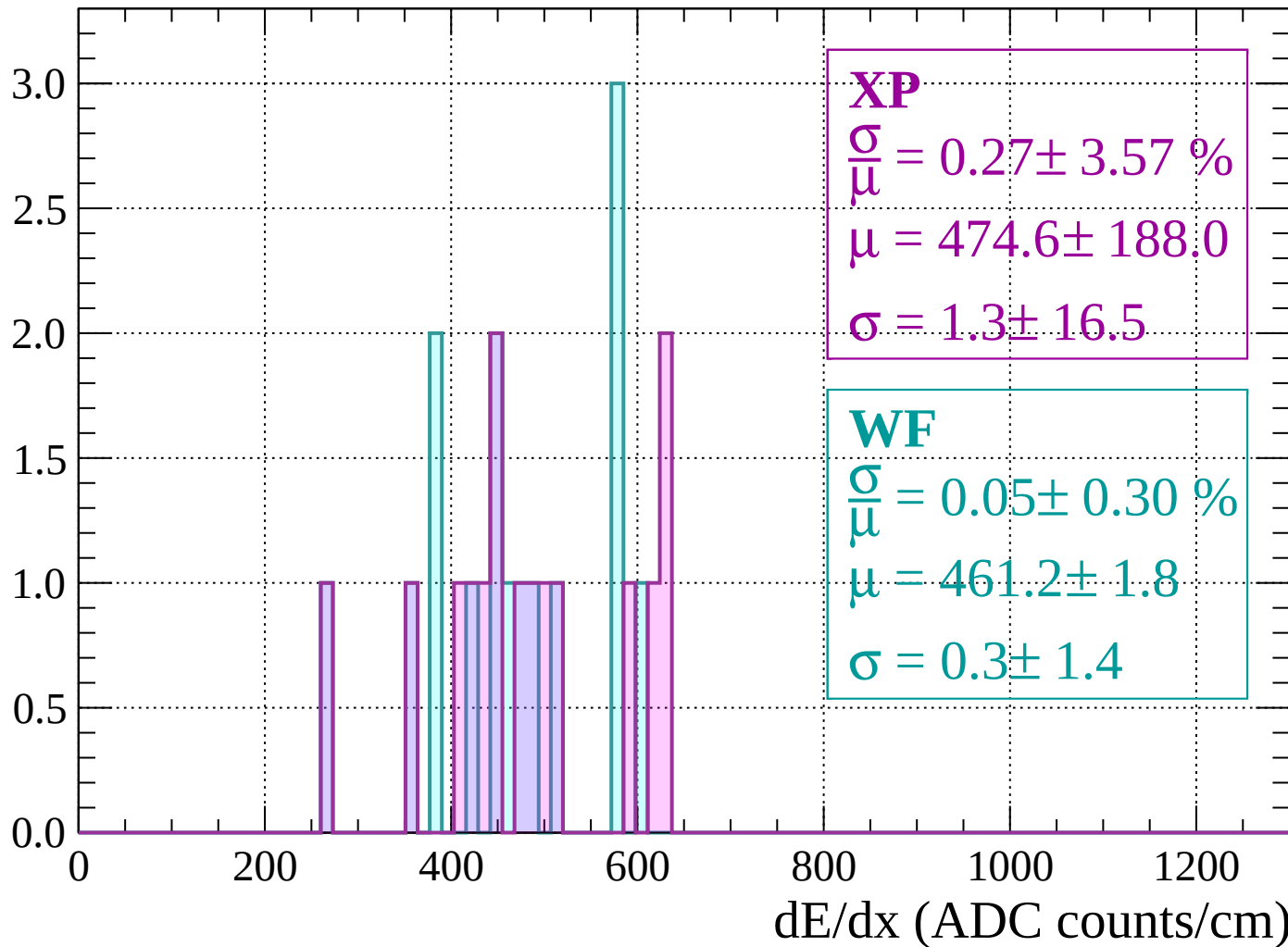
Energy loss | 1960 < p < 2000

Count



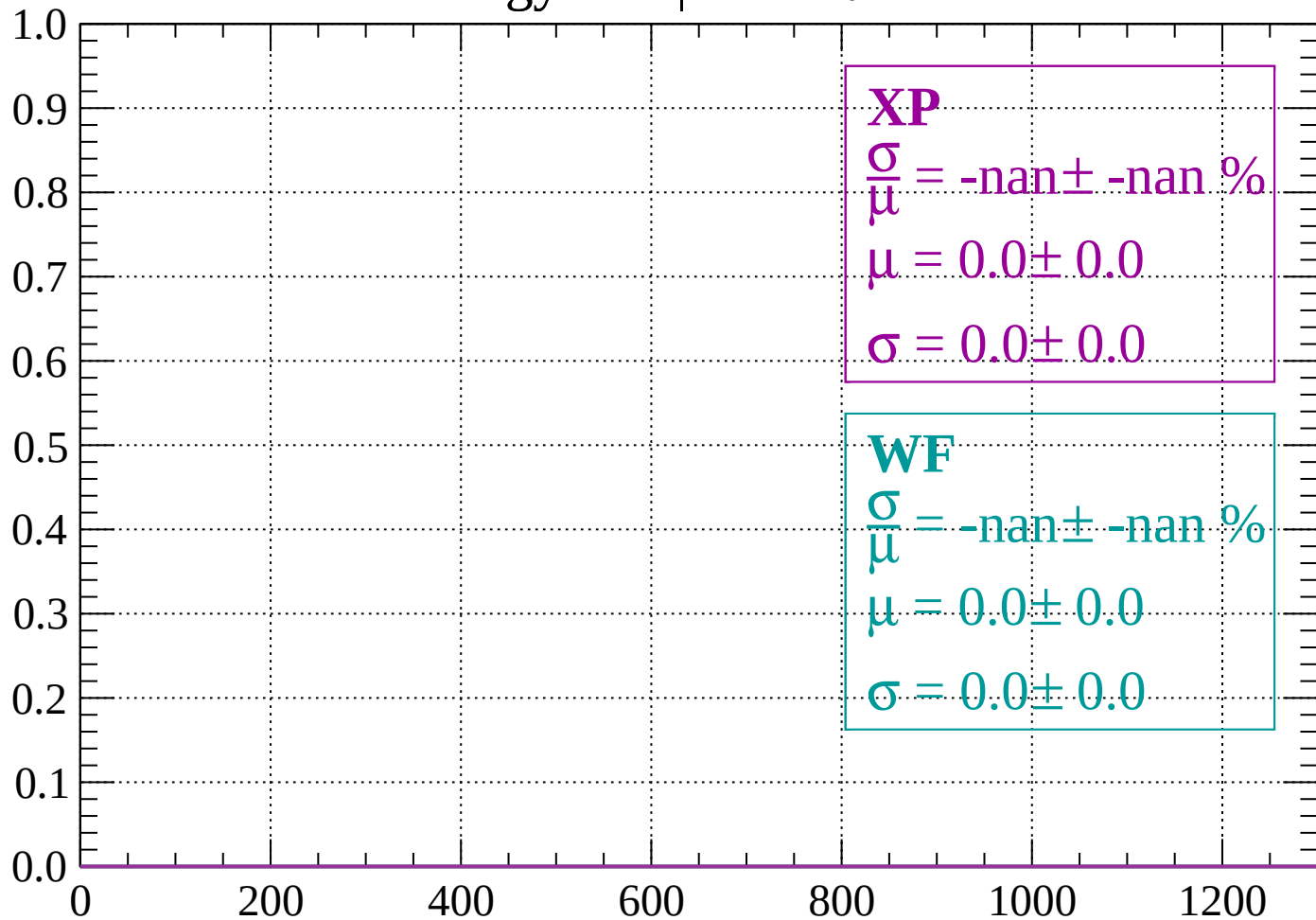
Energy loss | $2000 < p < 2040$

Count



Energy loss | $-90 < \theta < -84$

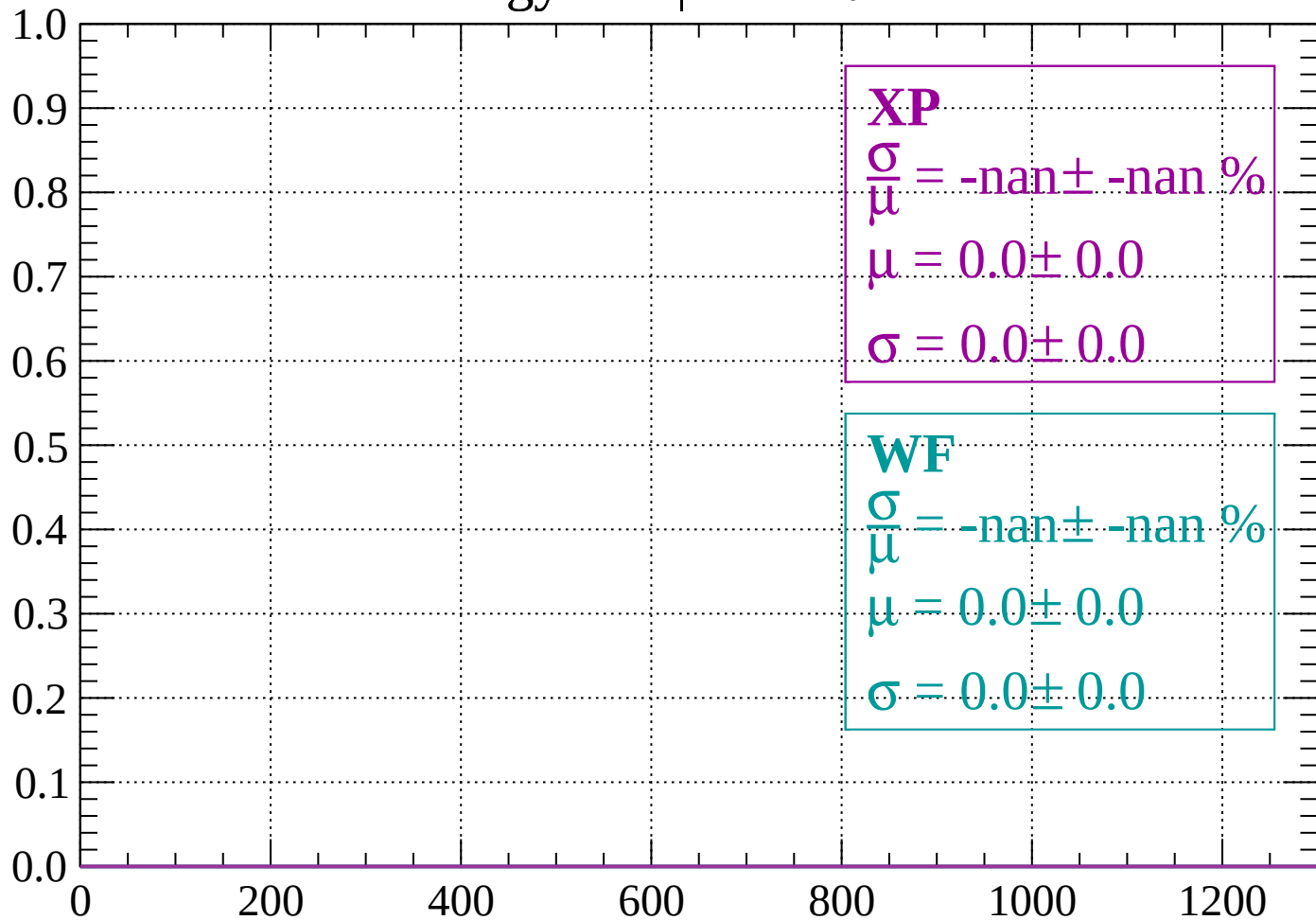
Count



dE/dx (ADC counts/cm)

Energy loss | $-84 < \theta < -78$

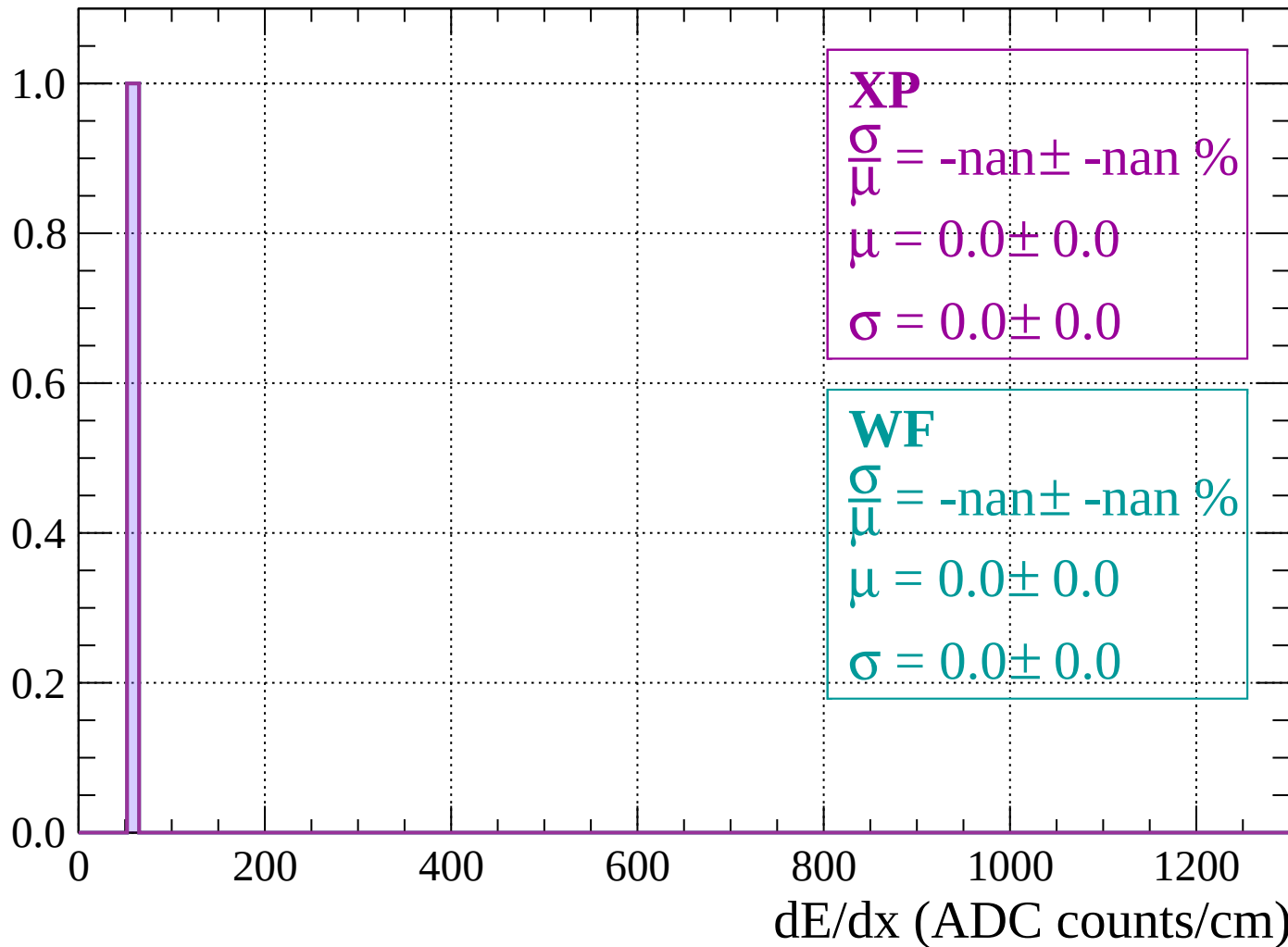
Count



dE/dx (ADC counts/cm)

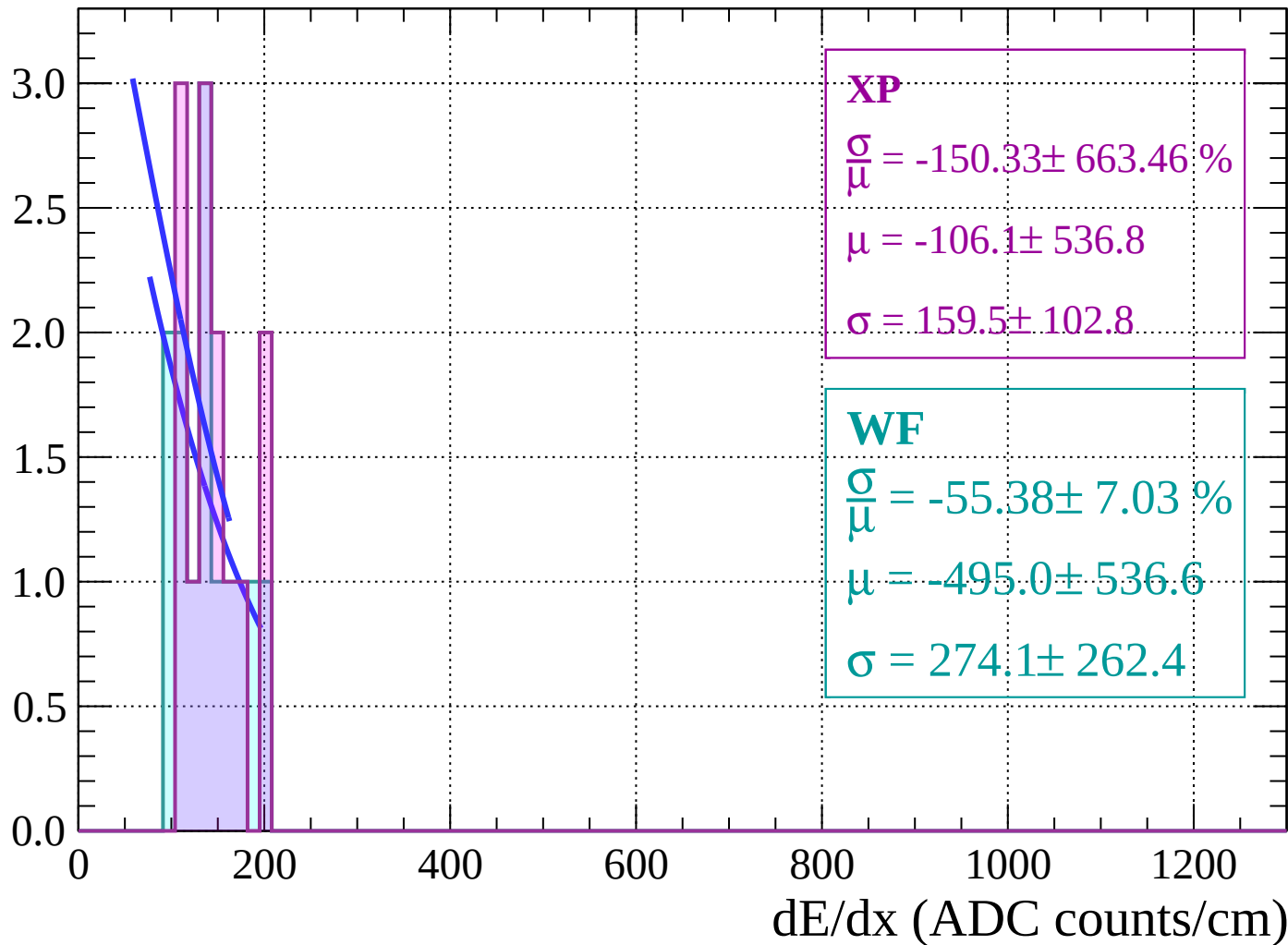
Energy loss | $-78 < \theta < -72$

Count



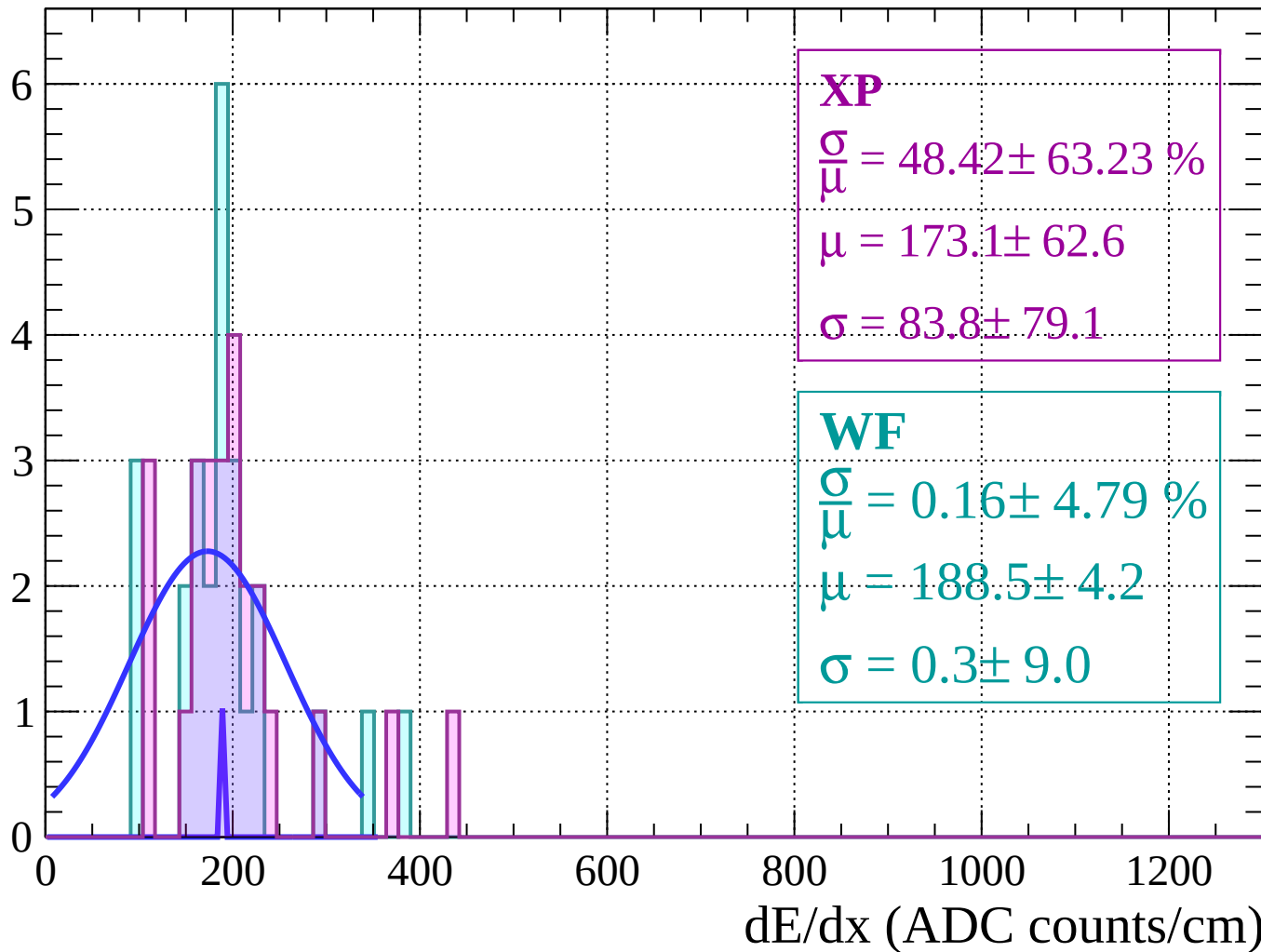
Energy loss | $-72 < \theta < -66$

Count



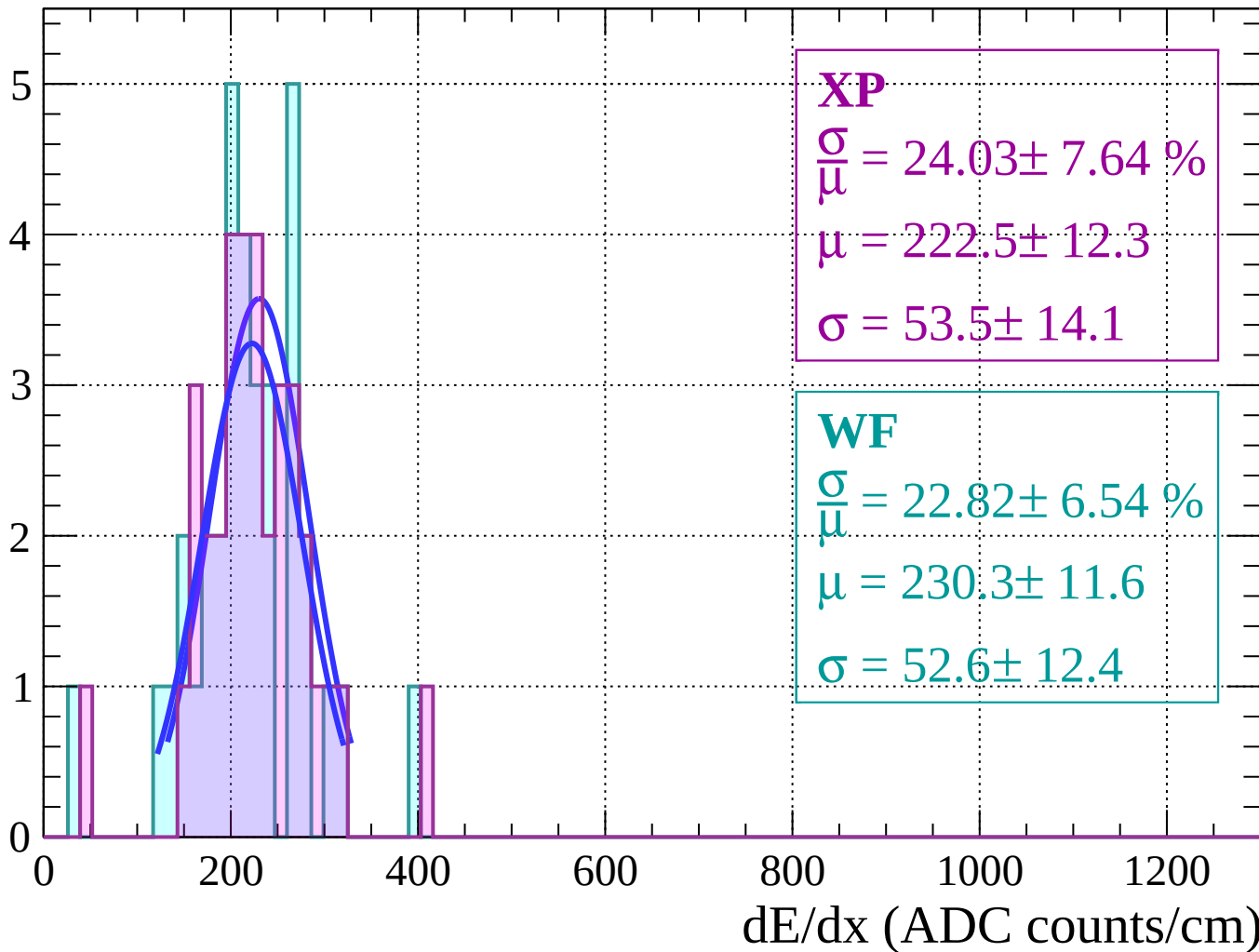
Energy loss | $-66 < \theta < -60$

Count



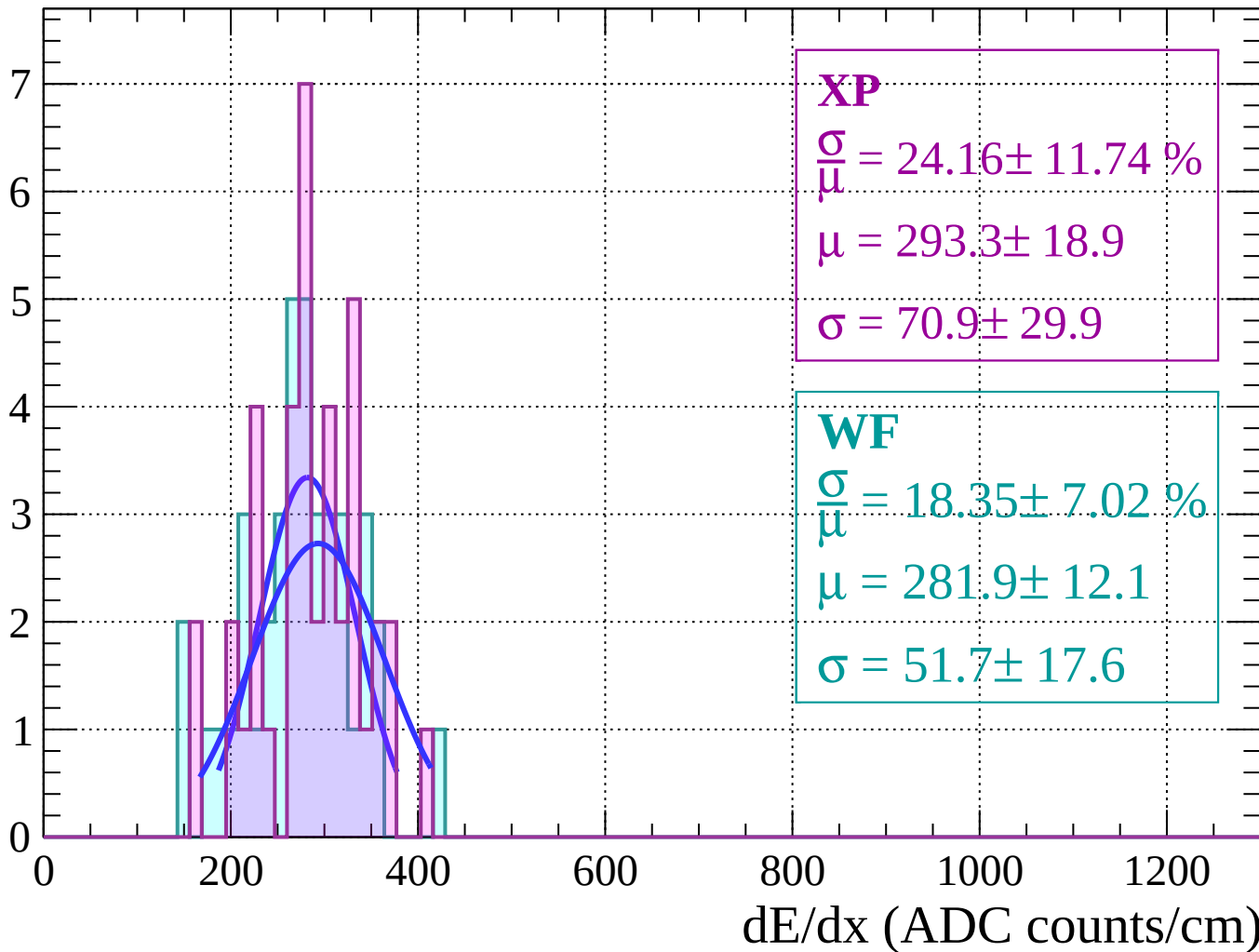
Energy loss | $-60 < \theta < -54$

Count



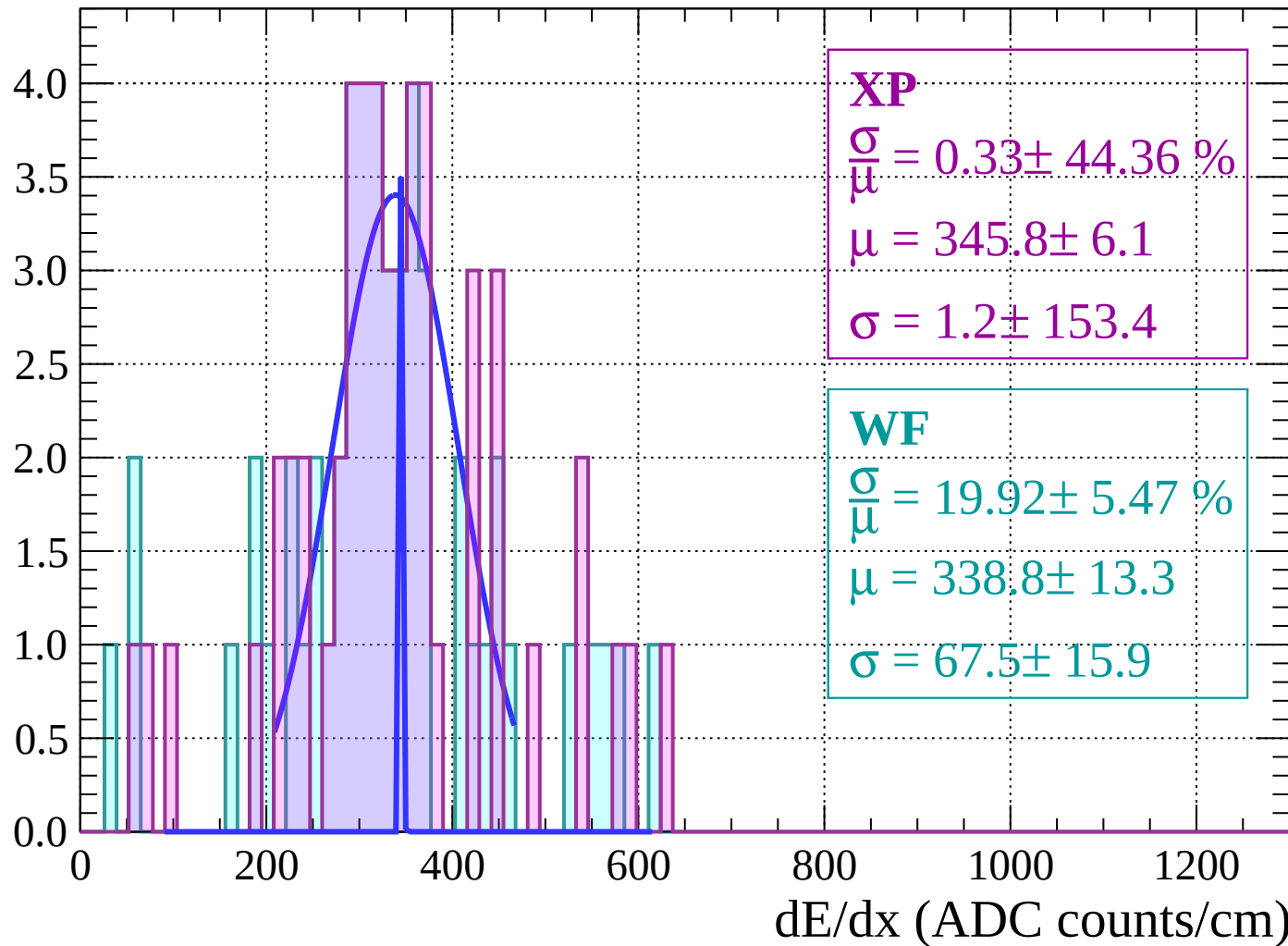
Energy loss | $-54 < \theta < -48$

Count



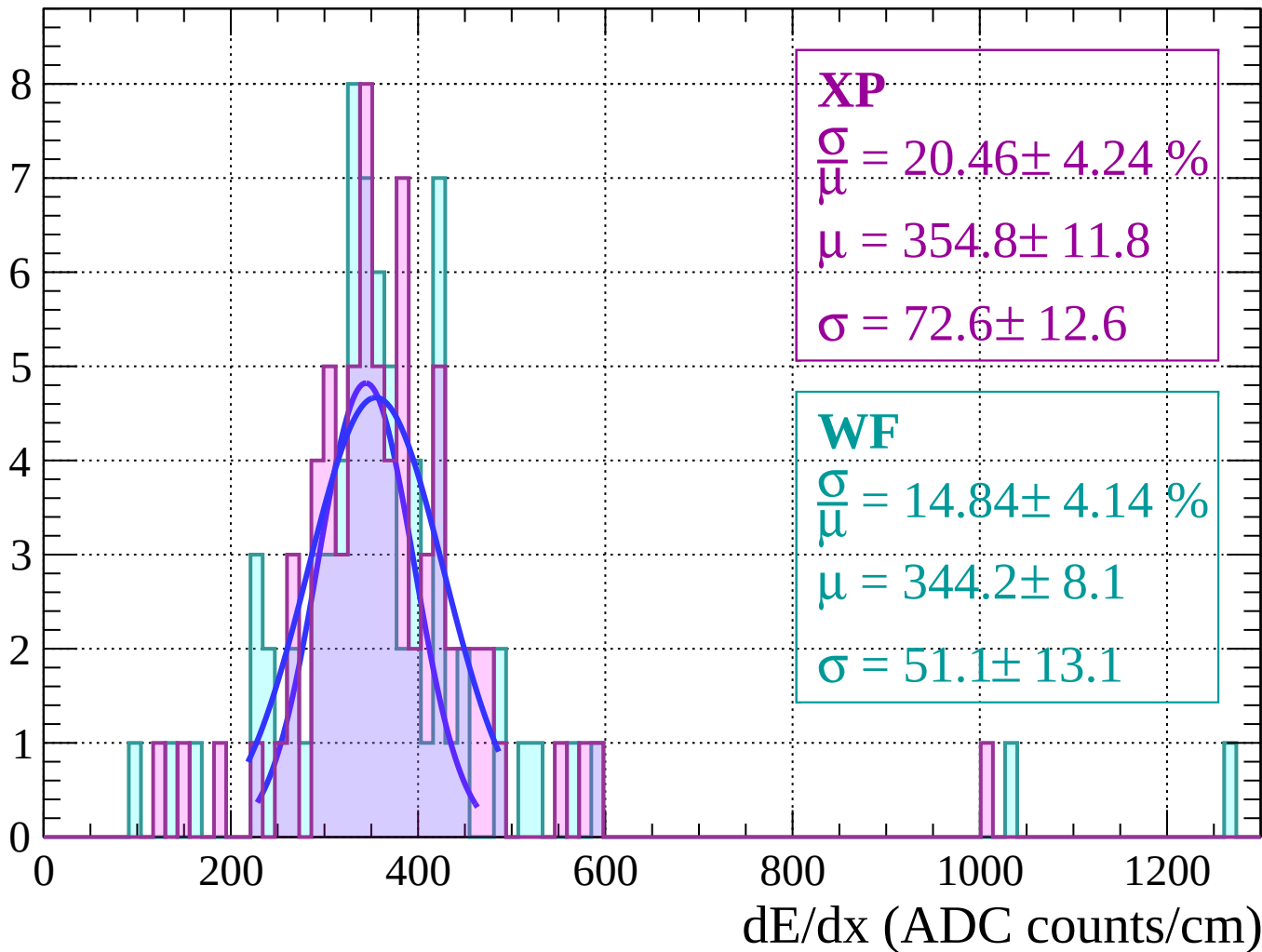
Energy loss | $-48 < \theta < -42$

Count



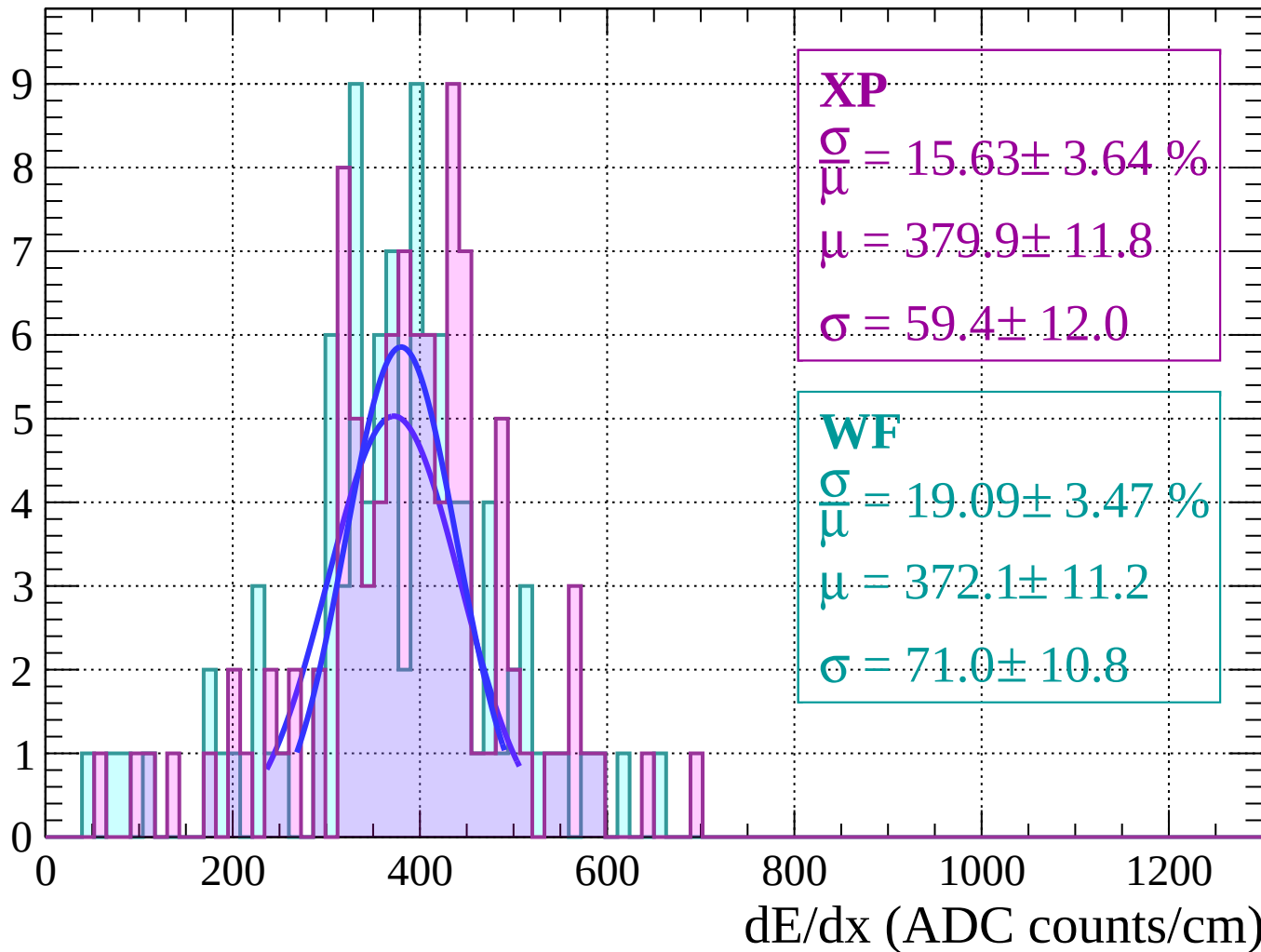
Energy loss | $-42 < \theta < -36$

Count



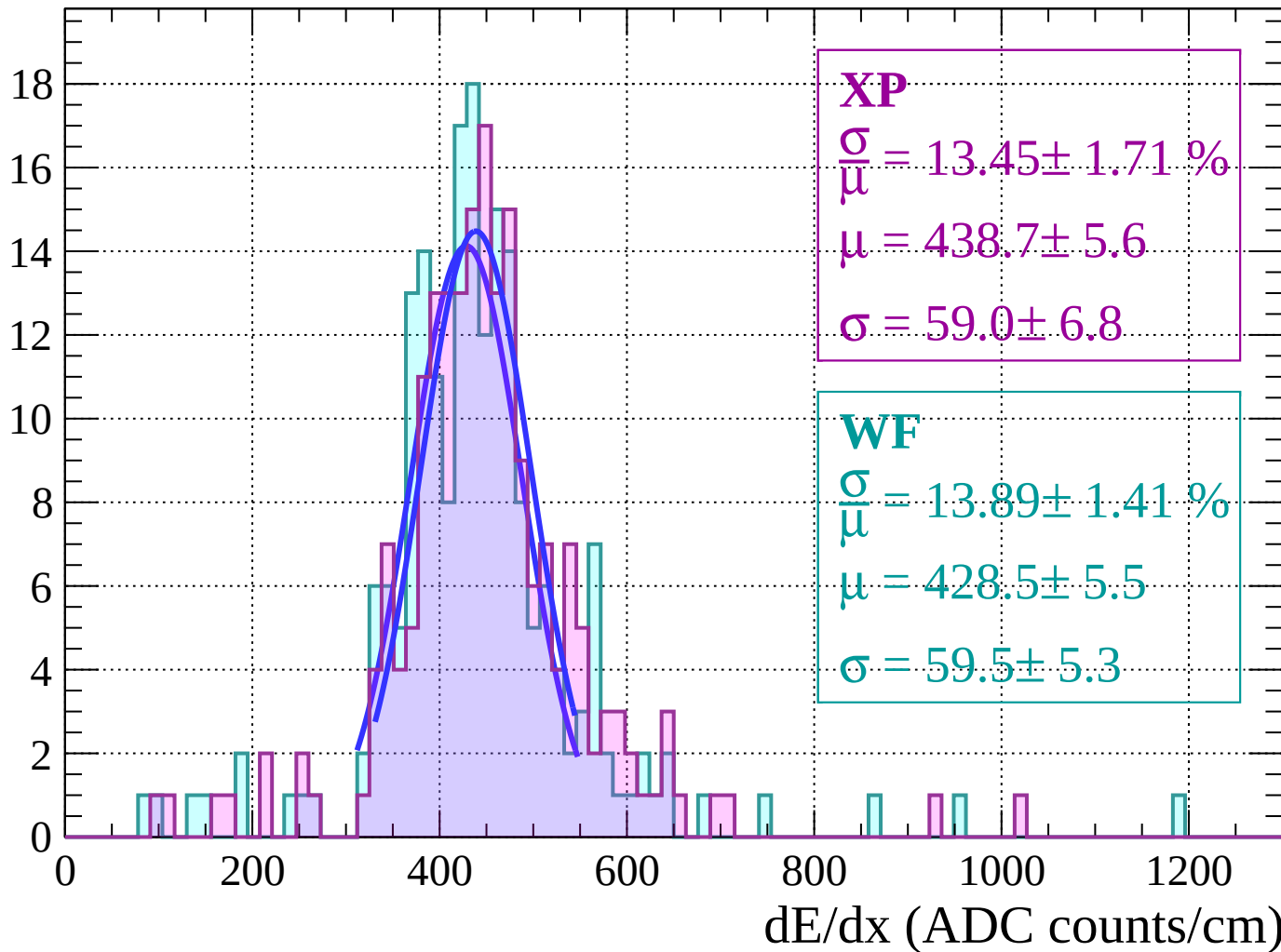
Energy loss | $-36 < \theta < -30$

Count



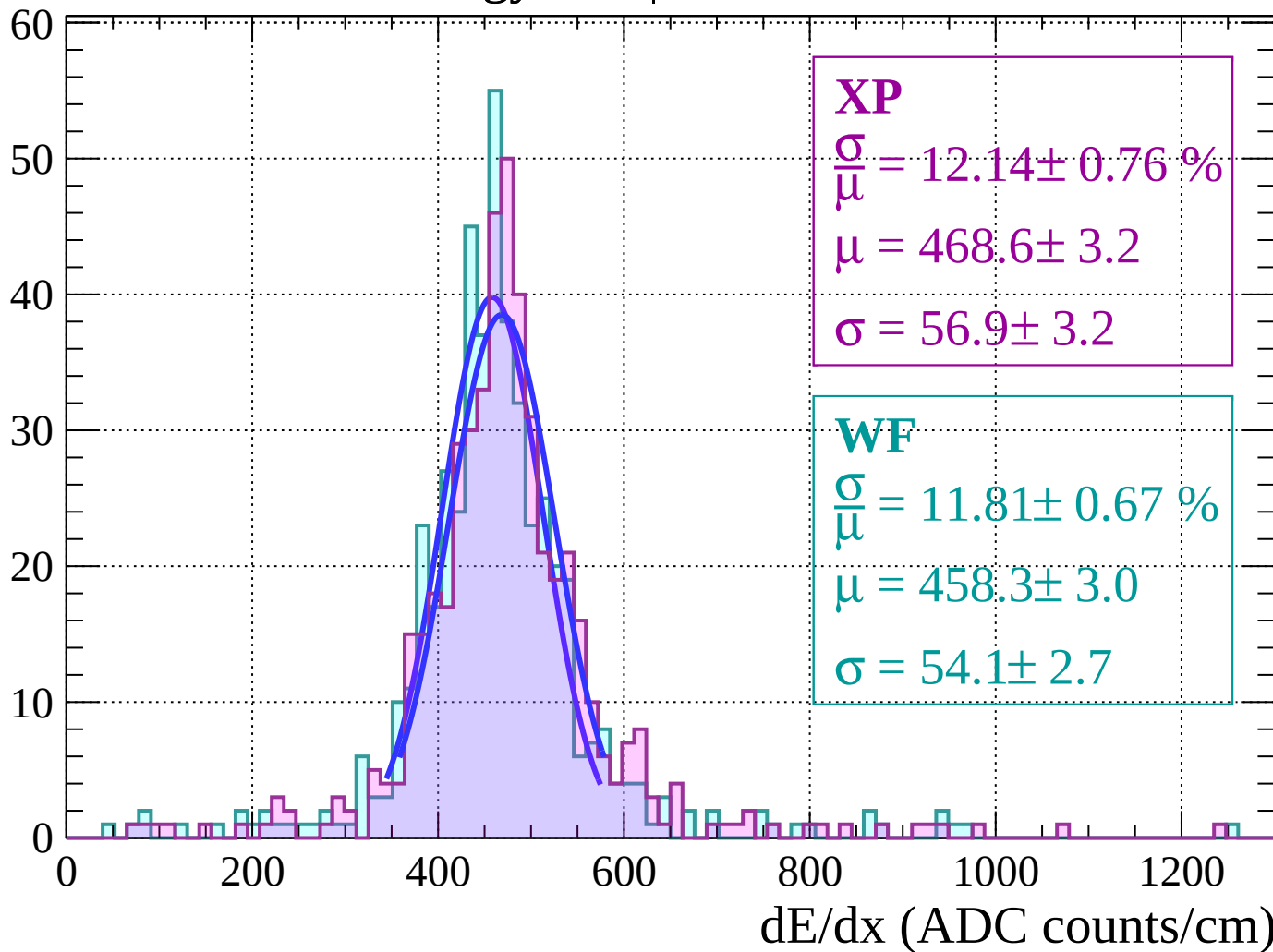
Energy loss $|-30 < \theta < -24$

Count



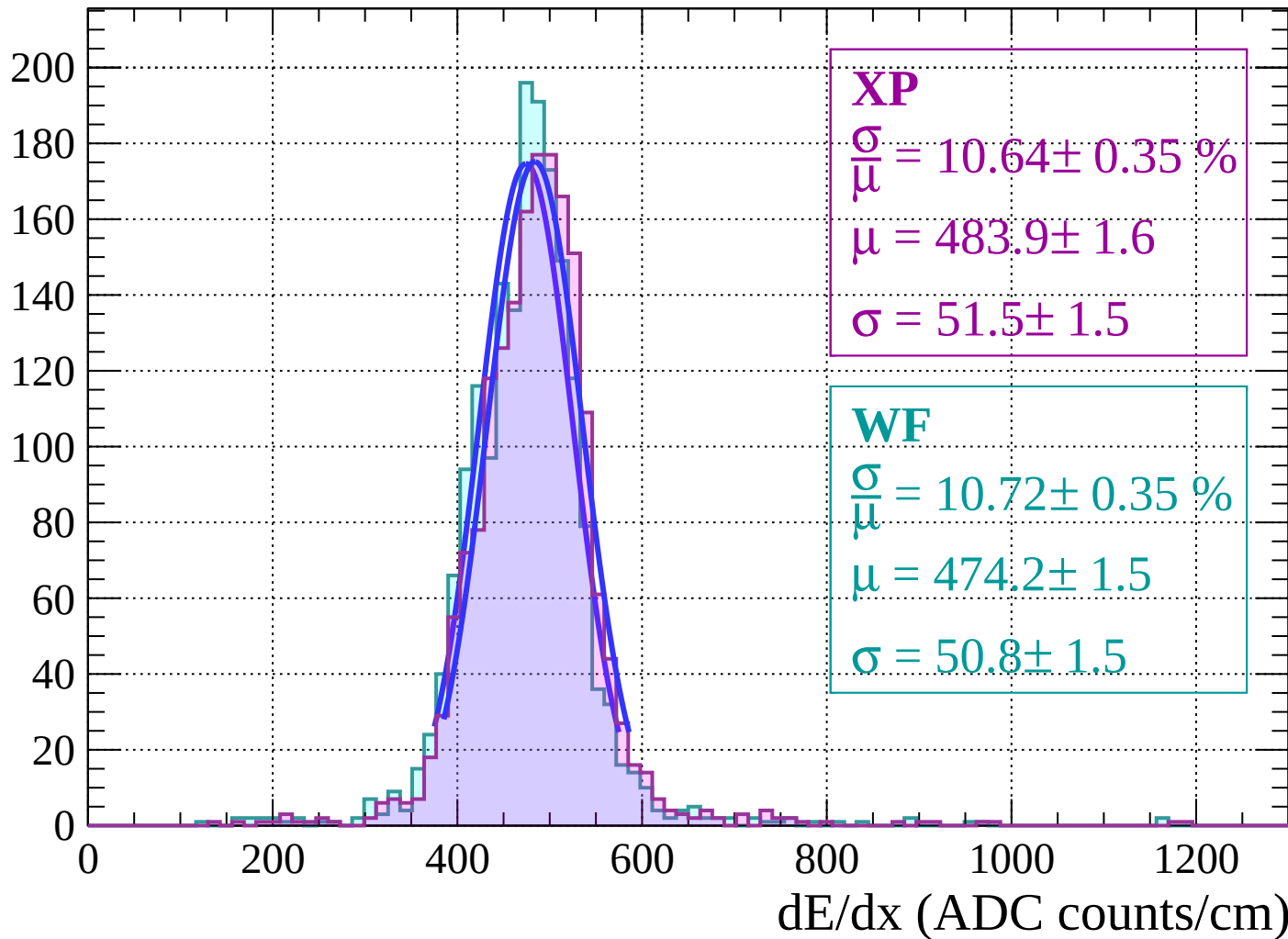
Energy loss | $-24 < \theta < -18$

Count

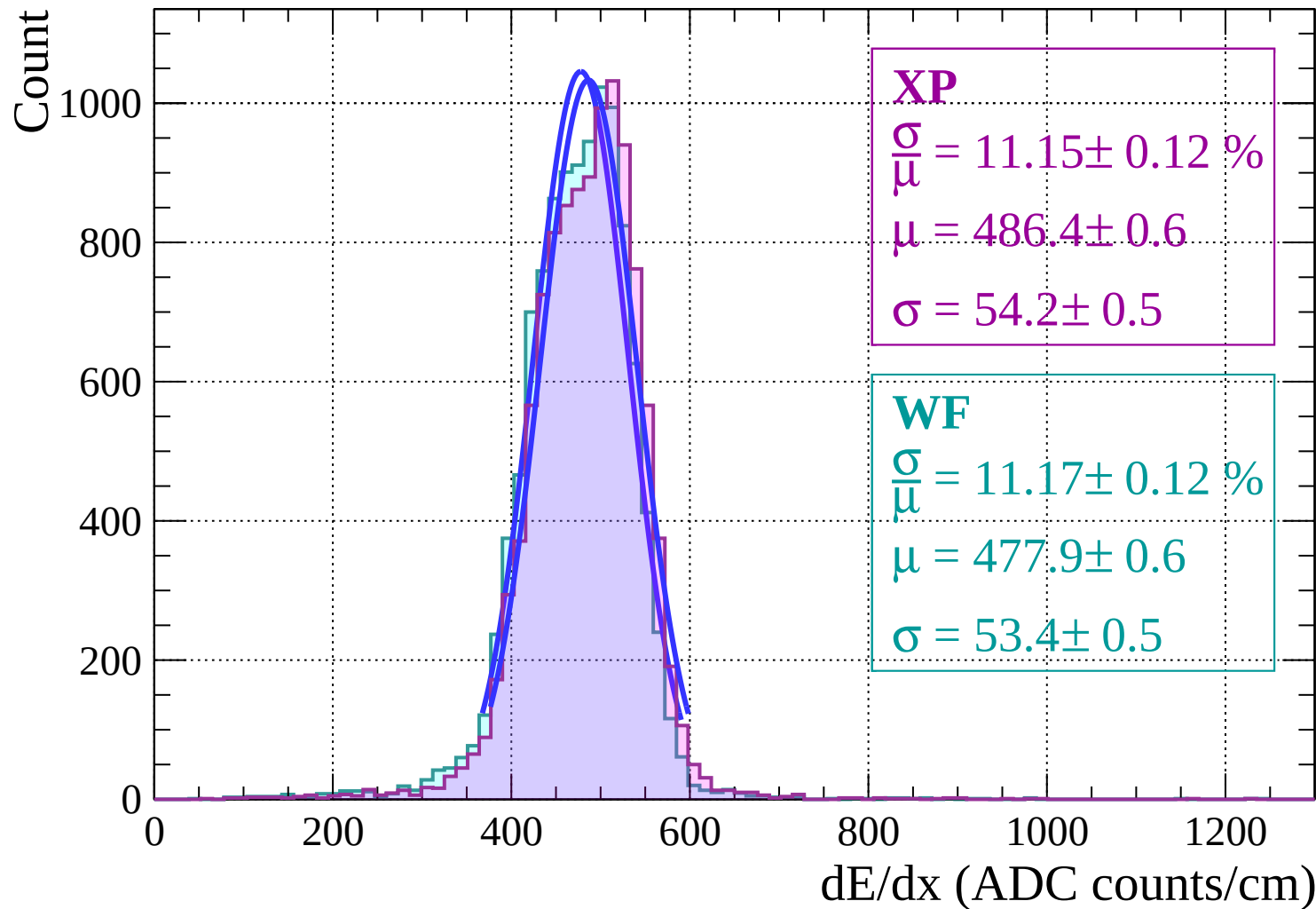


Energy loss | $-18 < \theta < -12$

Count



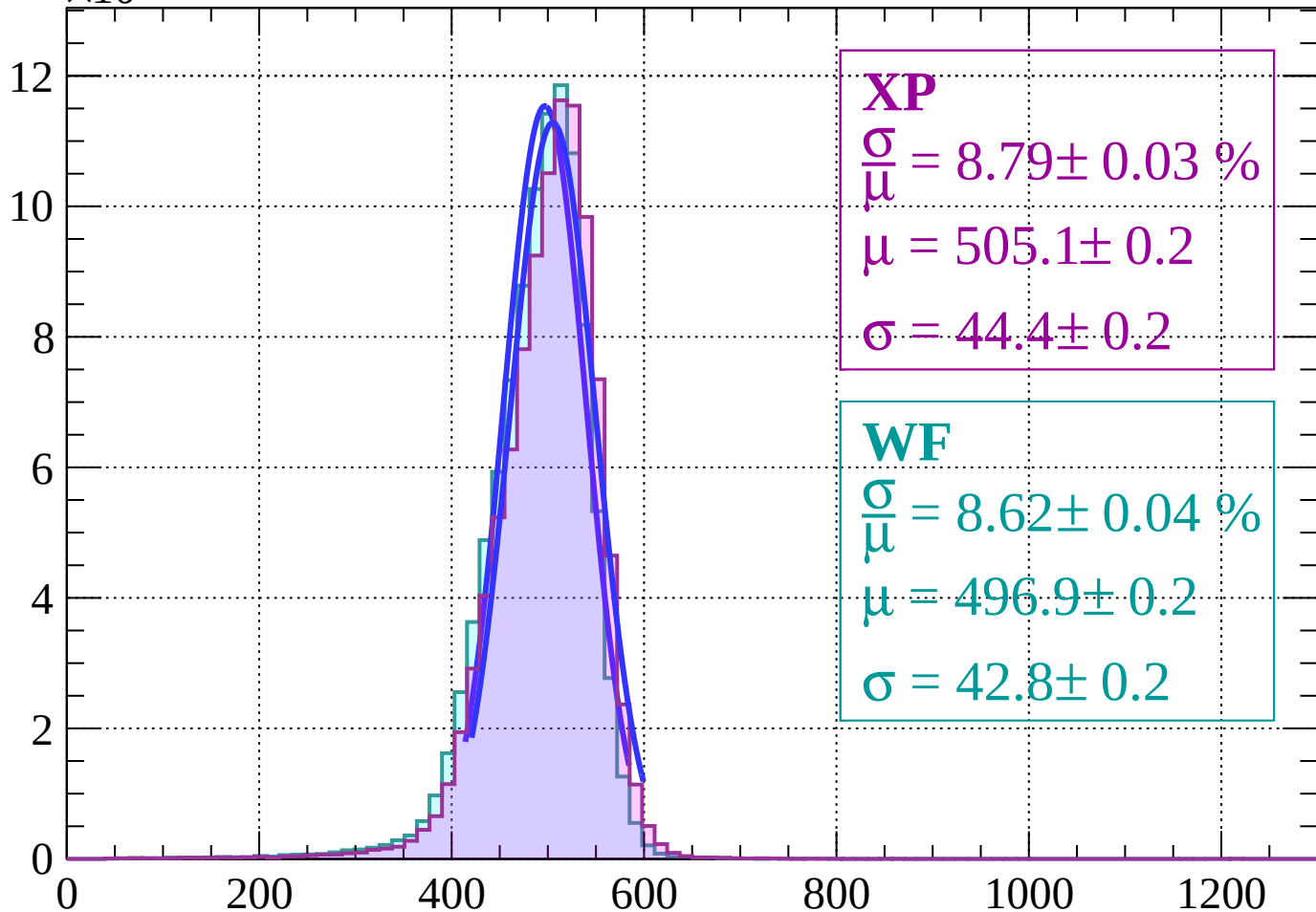
Energy loss | $-12 < \theta < -6$



Energy loss | $-6 < \theta < 0$

Count

$\times 10^3$



XP

$$\frac{\sigma}{\mu} = 8.79 \pm 0.03 \%$$

$$\mu = 505.1 \pm 0.2$$

$$\sigma = 44.4 \pm 0.2$$

WF

$$\frac{\sigma}{\mu} = 8.62 \pm 0.04 \%$$

$$\mu = 496.9 \pm 0.2$$

$$\sigma = 42.8 \pm 0.2$$

dE/dx (ADC counts/cm)

Energy loss | $0 < \theta < 6$

Count

$\times 10^3$

XP

$$\frac{\sigma}{\mu} = 8.78 \pm 0.03 \%$$

$$\mu = 504.8 \pm 0.1$$

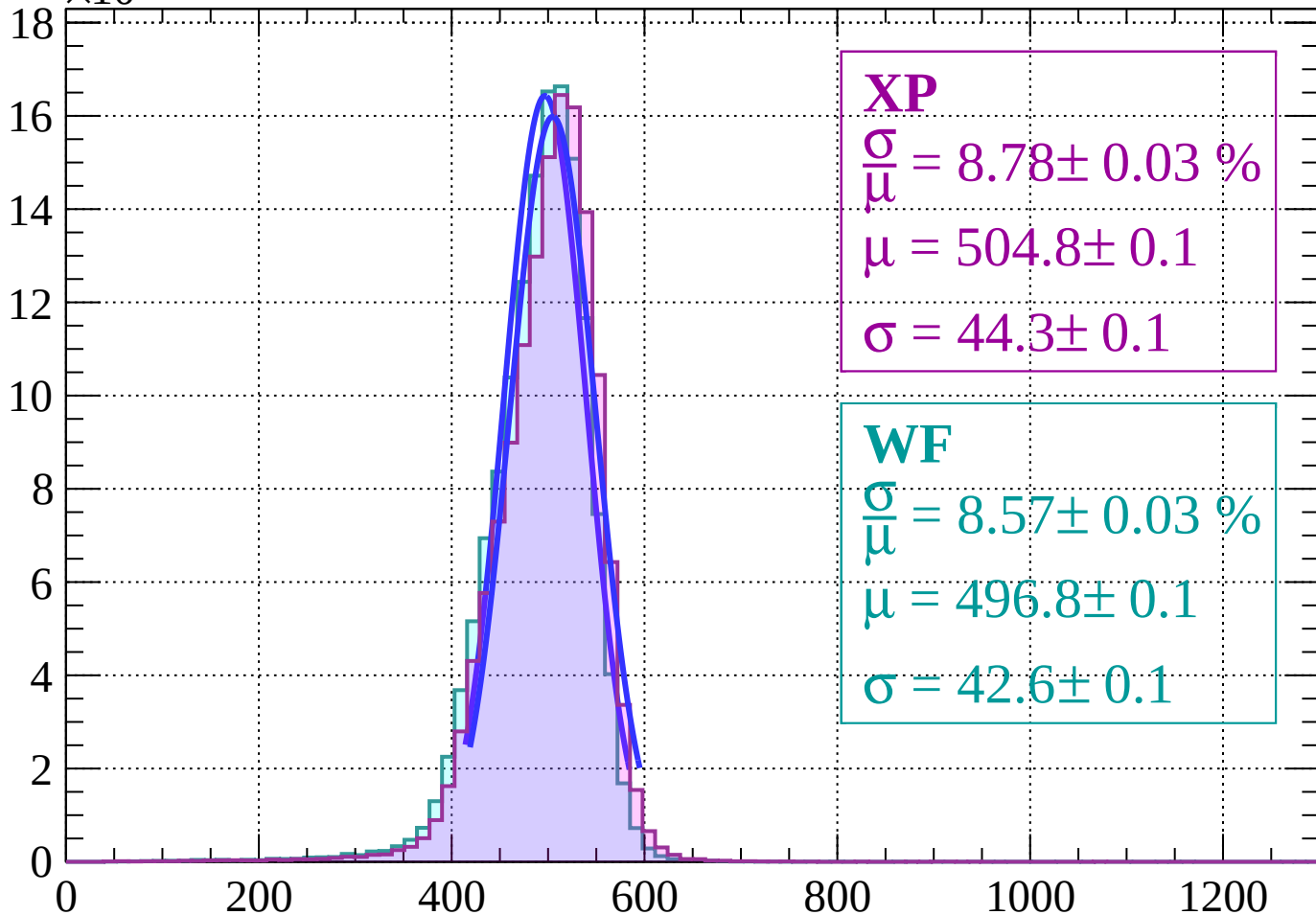
$$\sigma = 44.3 \pm 0.1$$

WF

$$\frac{\sigma}{\mu} = 8.57 \pm 0.03 \%$$

$$\mu = 496.8 \pm 0.1$$

$$\sigma = 42.6 \pm 0.1$$



dE/dx (ADC counts/cm)

Energy loss | $6 < \theta < 12$

Count

$\times 10^3$

XP

$$\frac{\sigma}{\mu} = 8.77 \pm 0.03 \%$$

$$\mu = 505.4 \pm 0.2$$

$$\sigma = 44.3 \pm 0.2$$

WF

$$\frac{\sigma}{\mu} = 8.63 \pm 0.04 \%$$

$$\mu = 497.1 \pm 0.2$$

$$\sigma = 42.9 \pm 0.2$$

12

10

8

6

4

2

0

0

200

400

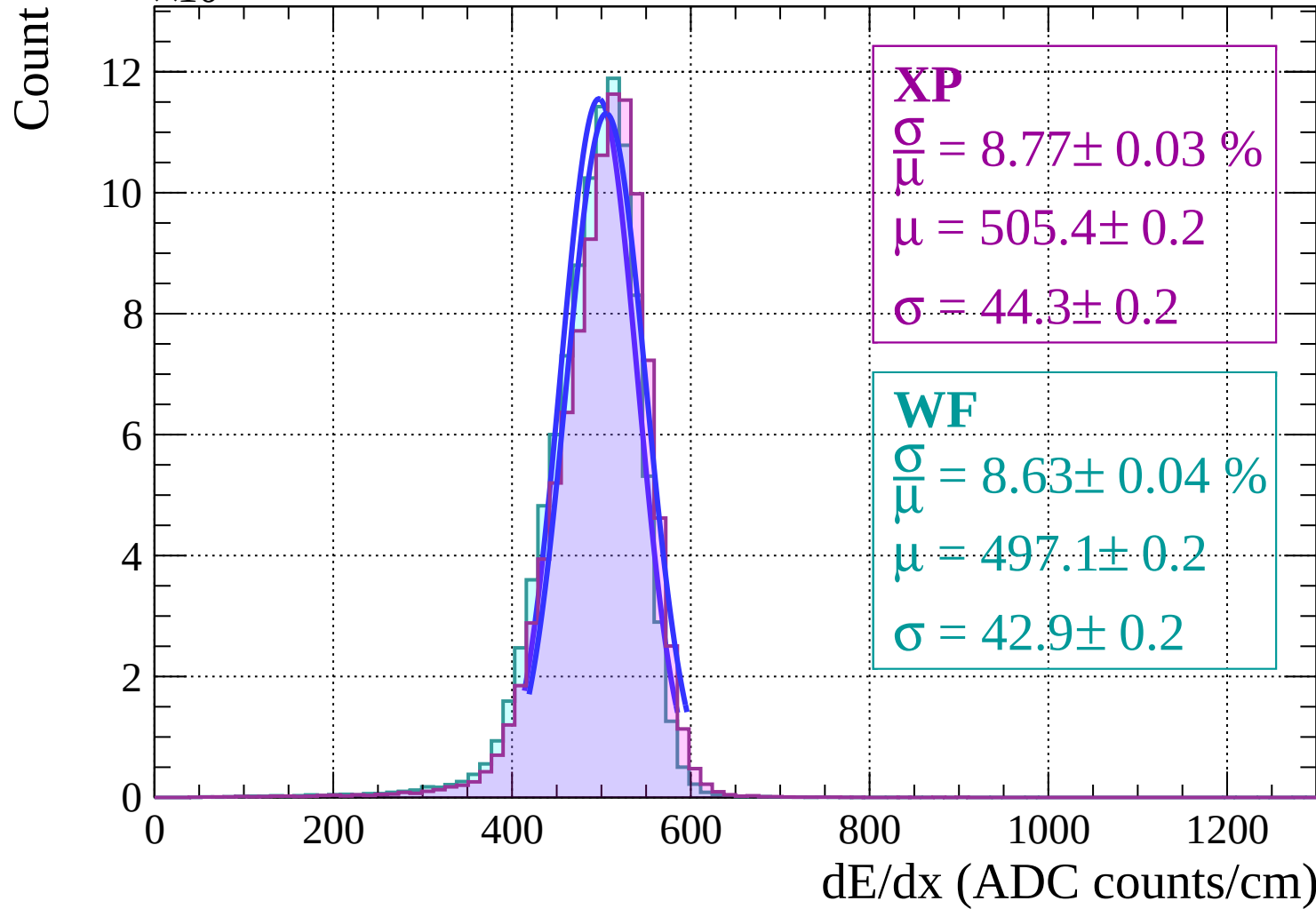
600

800

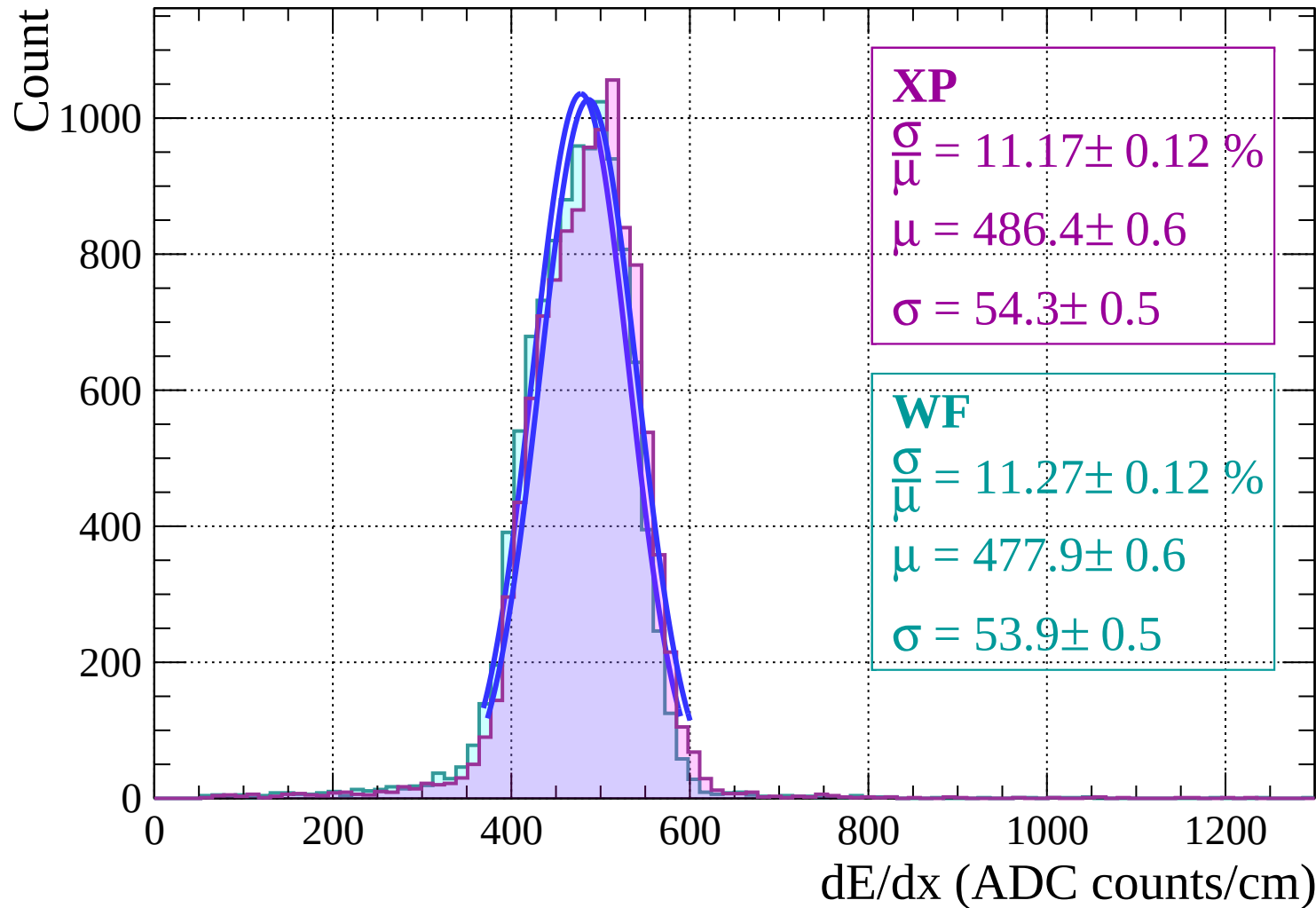
1000

1200

dE/dx (ADC counts/cm)

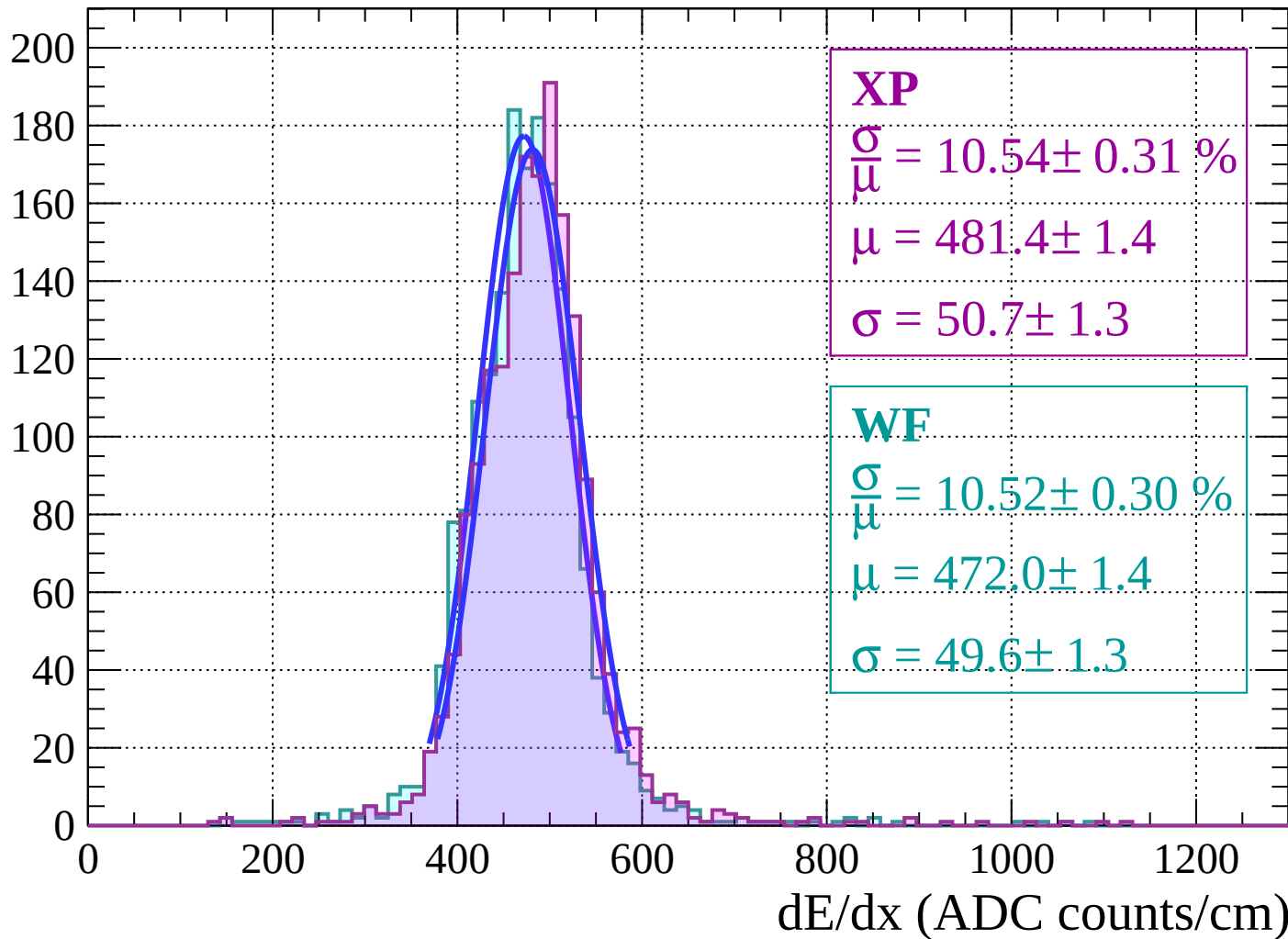


Energy loss | $12 < \theta < 18$



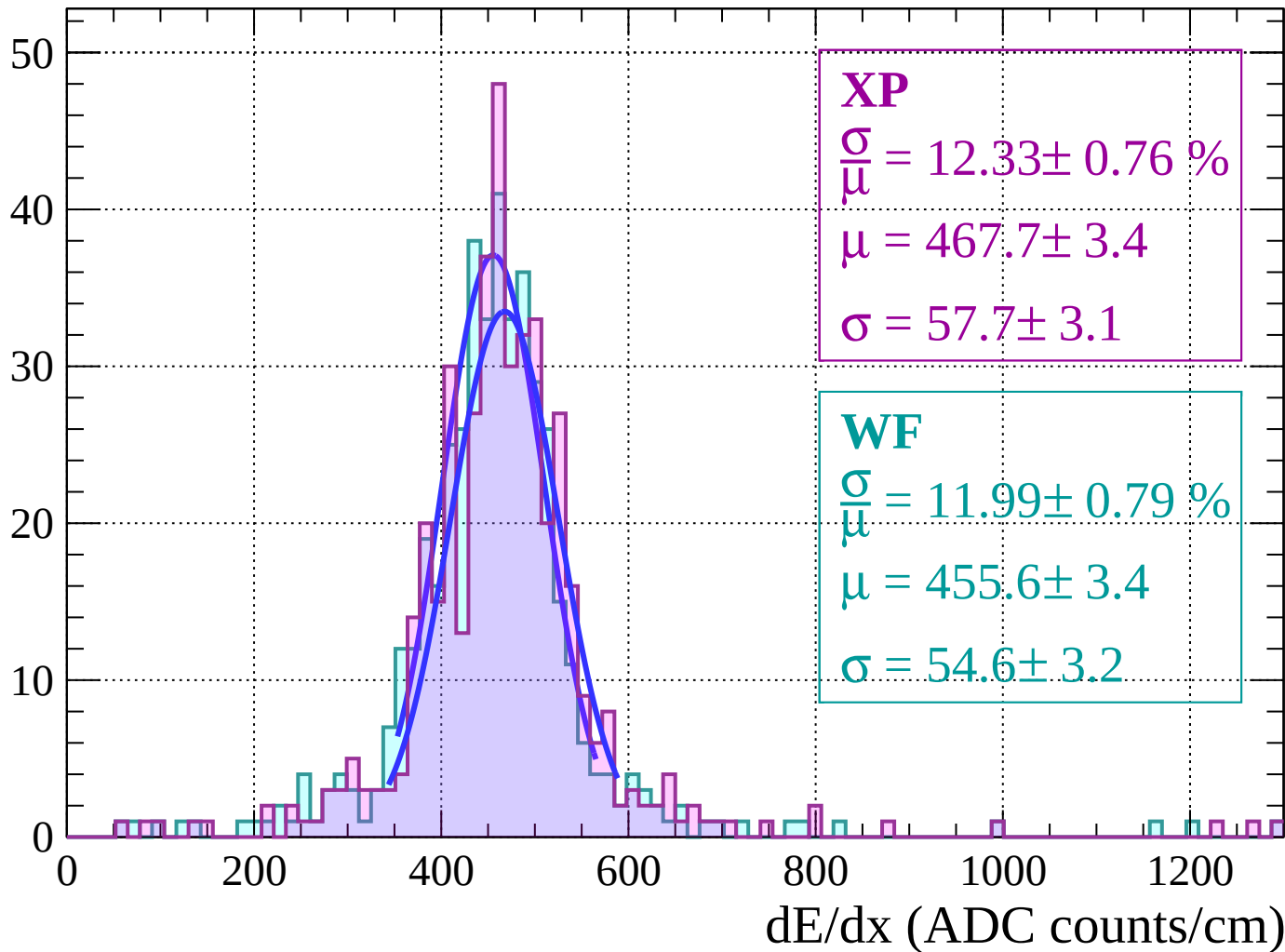
Energy loss | $18 < \theta < 24$

Count



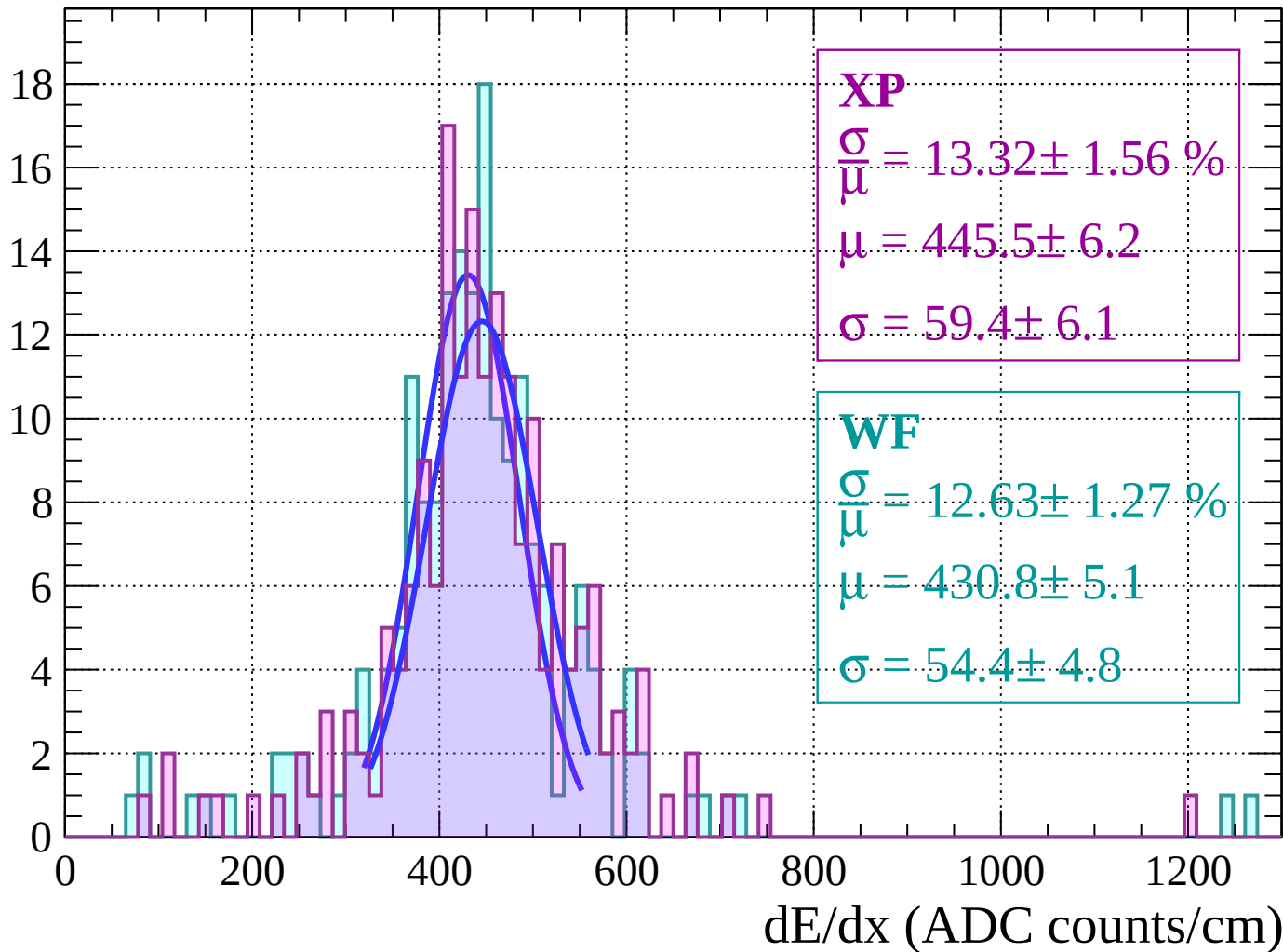
Energy loss | $24 < \theta < 30$

Count



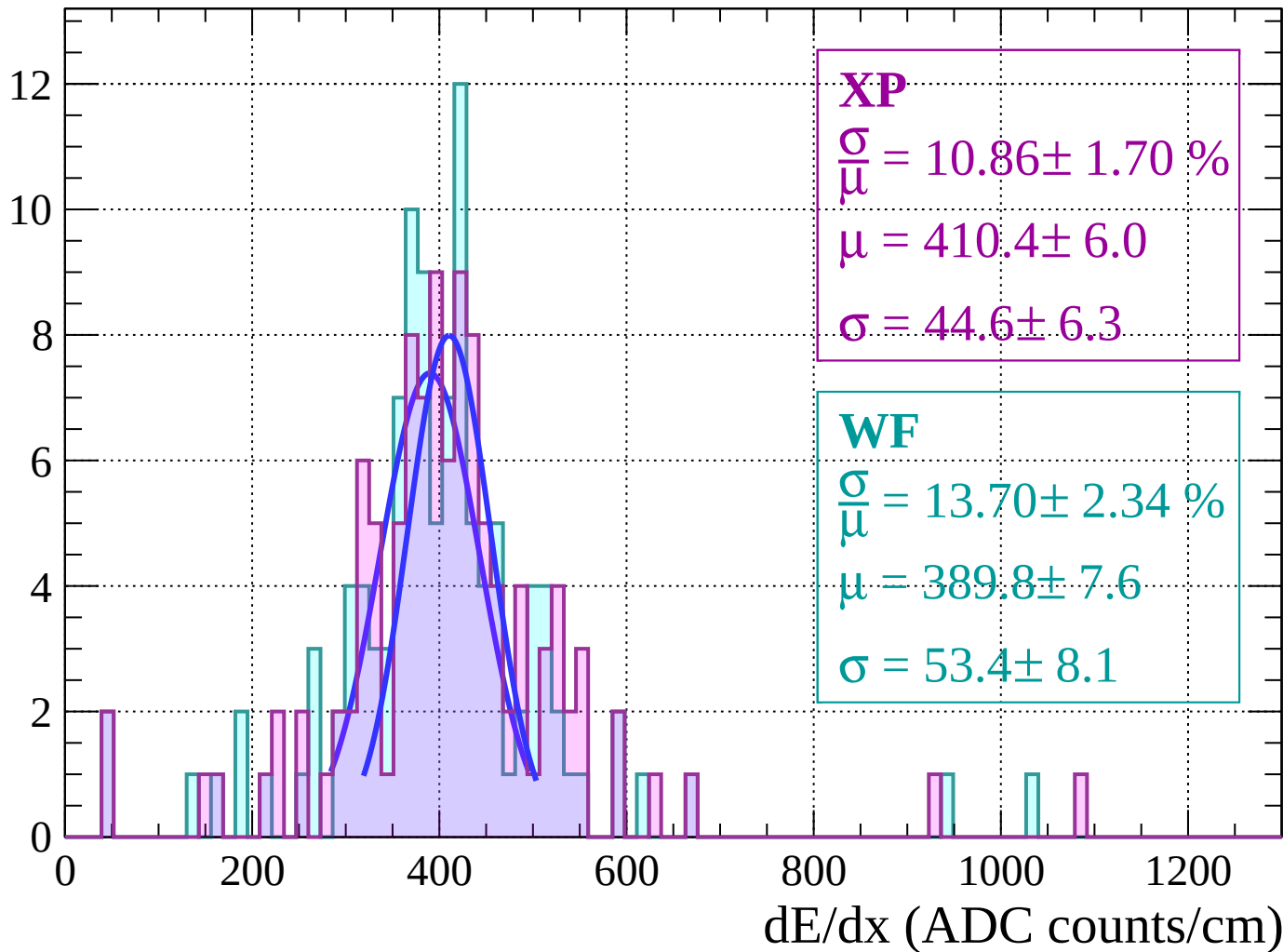
Energy loss | $30 < \theta < 36$

Count



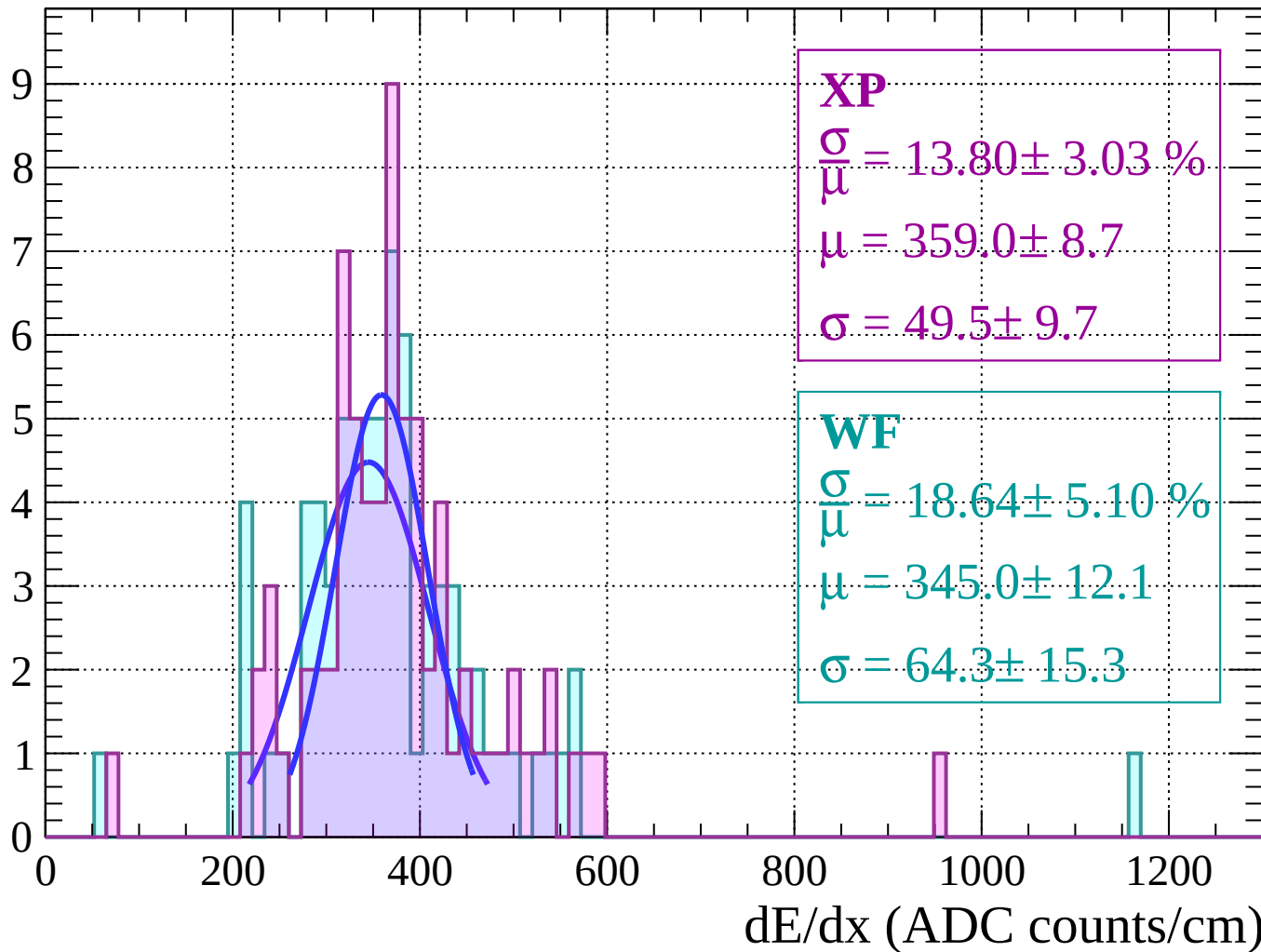
Energy loss | $36 < \theta < 42$

Count



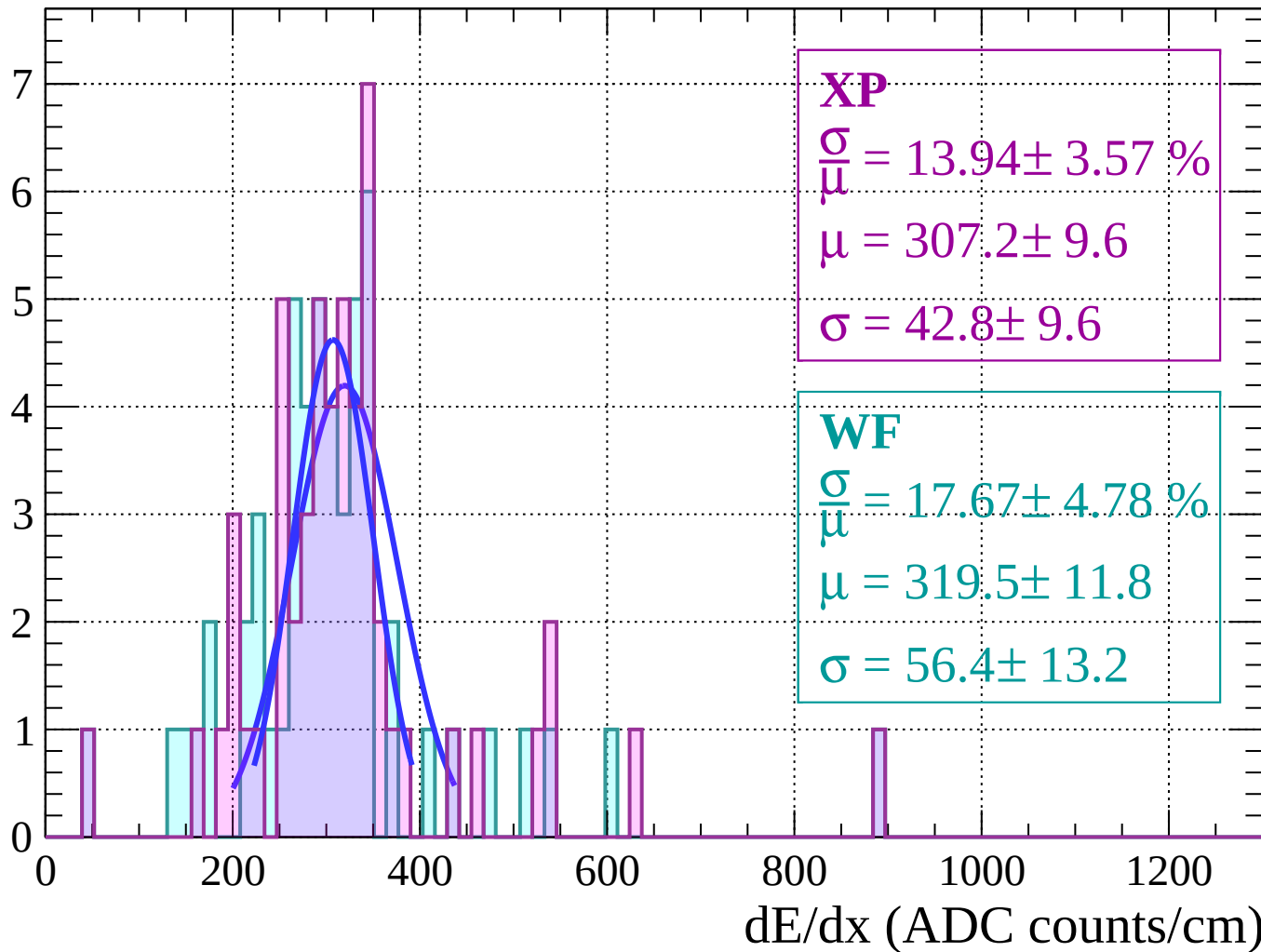
Energy loss | $42 < \theta < 48$

Count



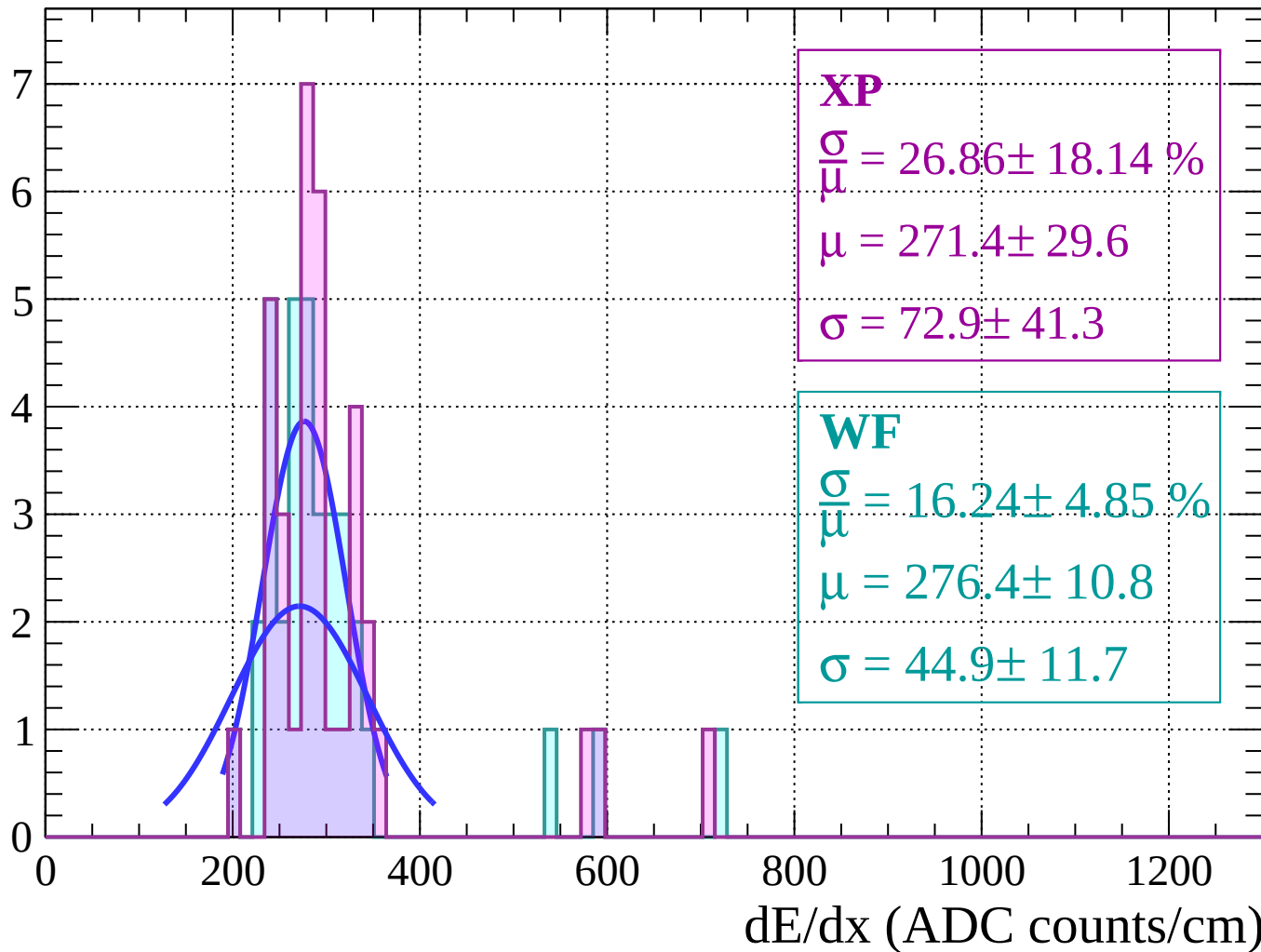
Energy loss | $48 < \theta < 54$

Count



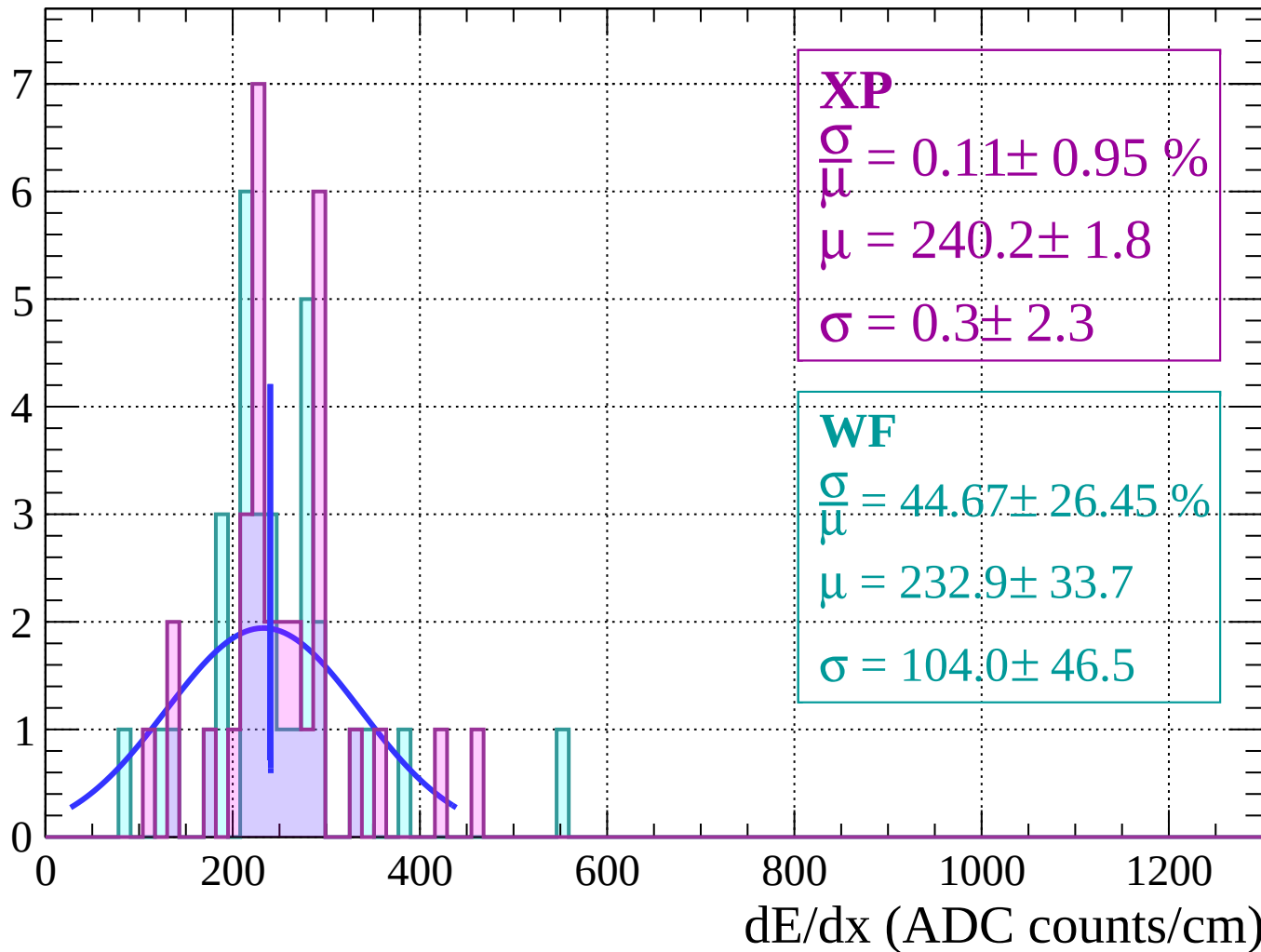
Energy loss | $54 < \theta < 60$

Count



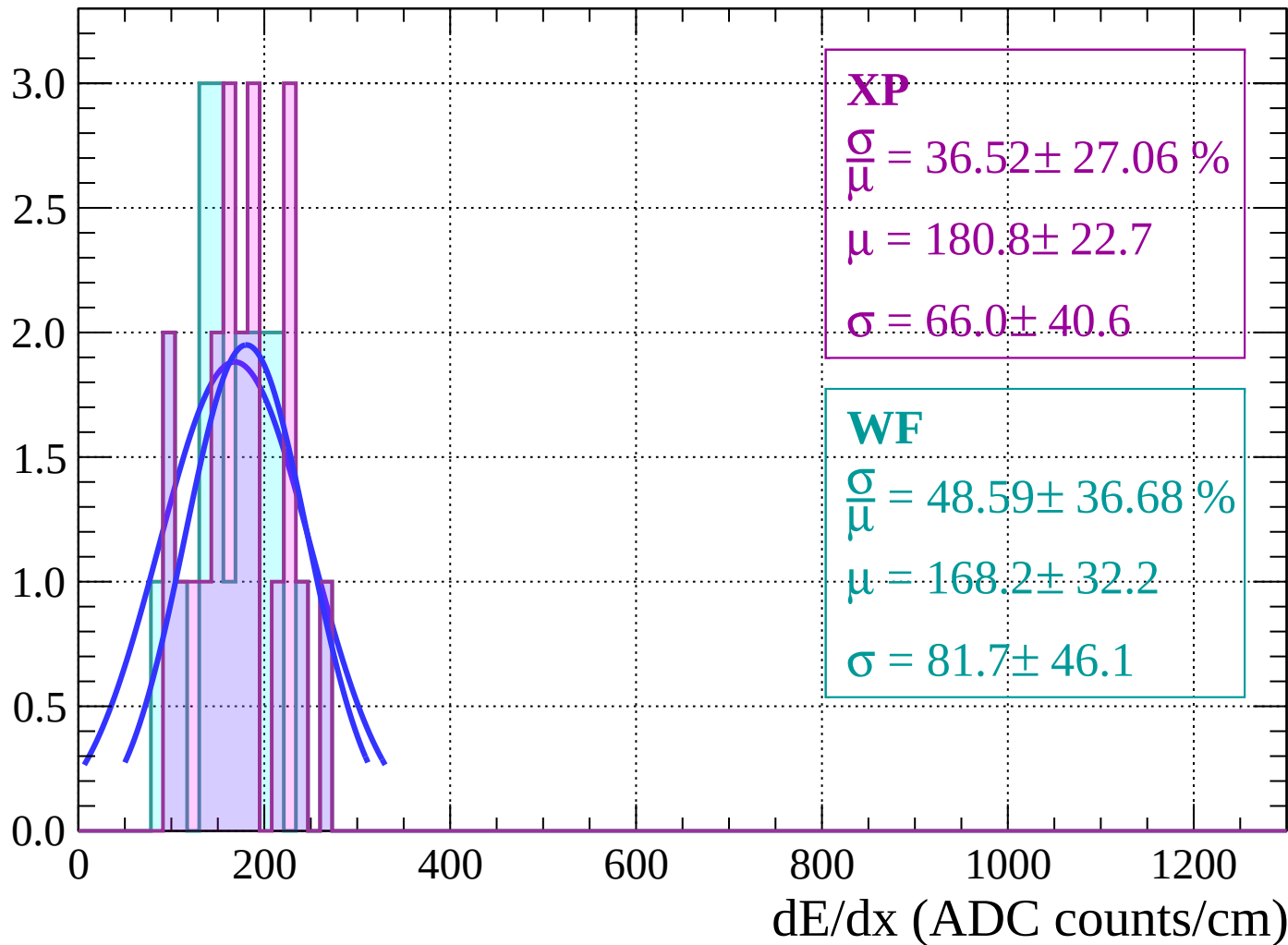
Energy loss | $60 < \theta < 66$

Count



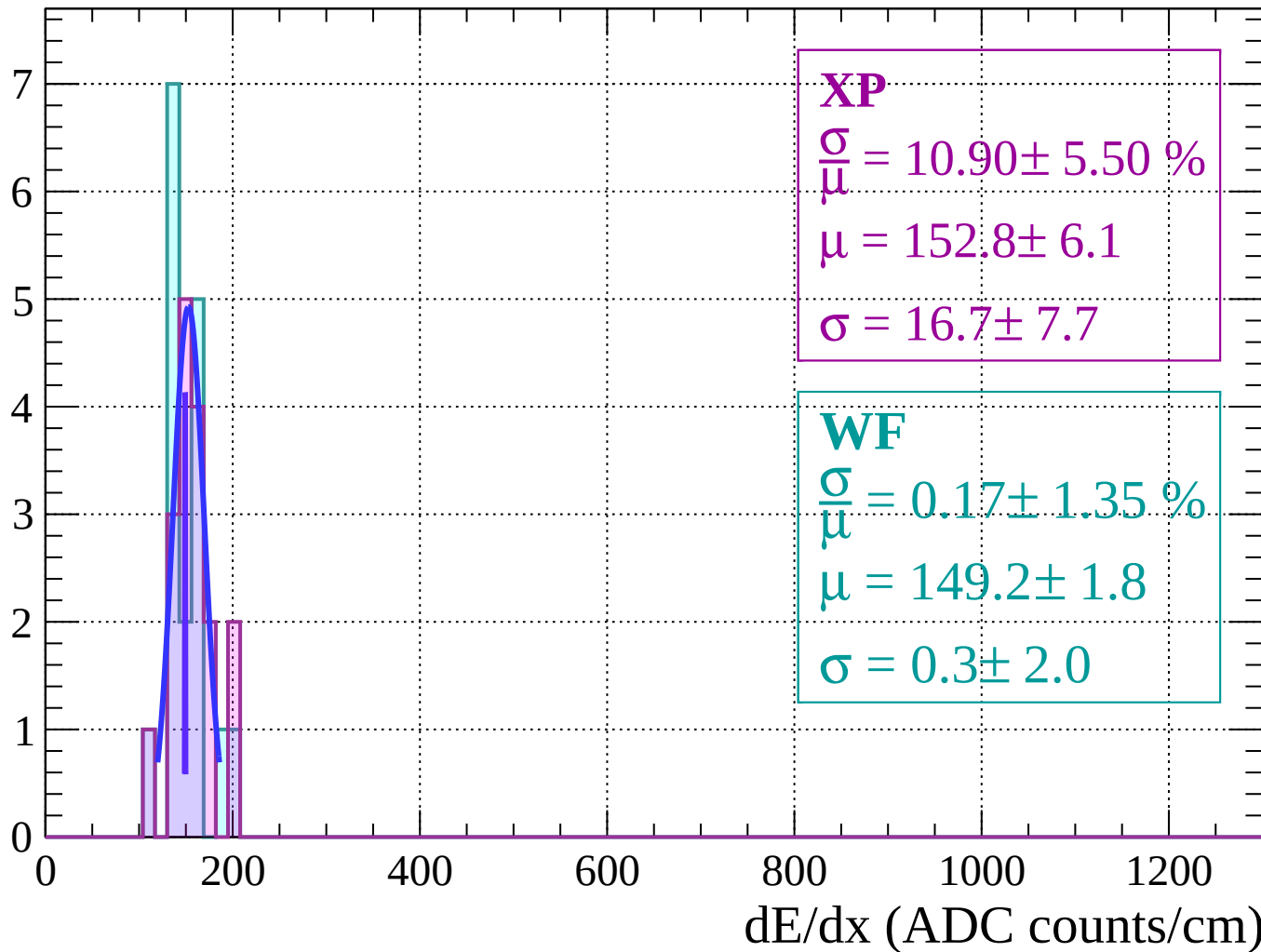
Energy loss | $66 < \theta < 72$

Count



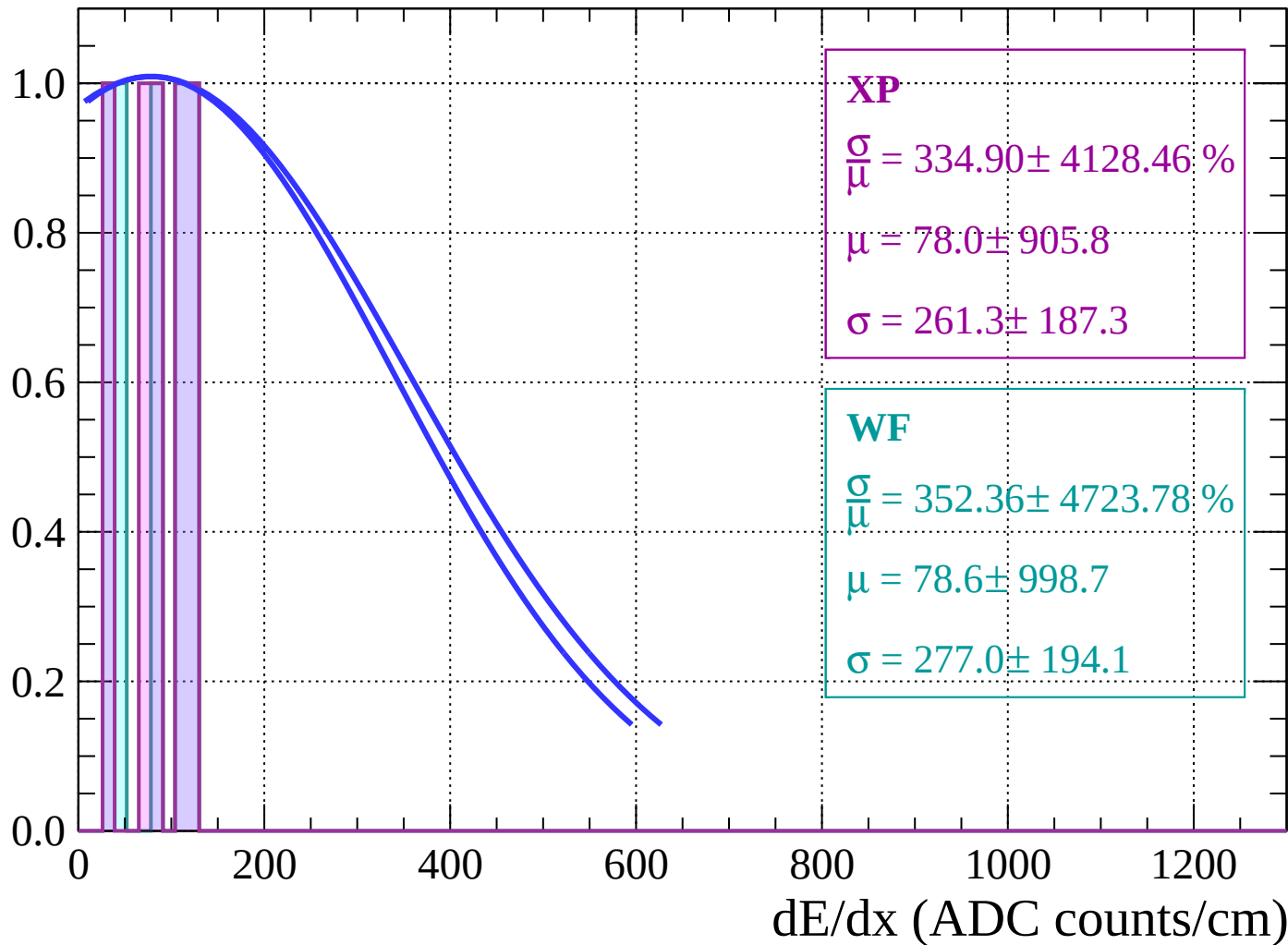
Energy loss | $72 < \theta < 78$

Count



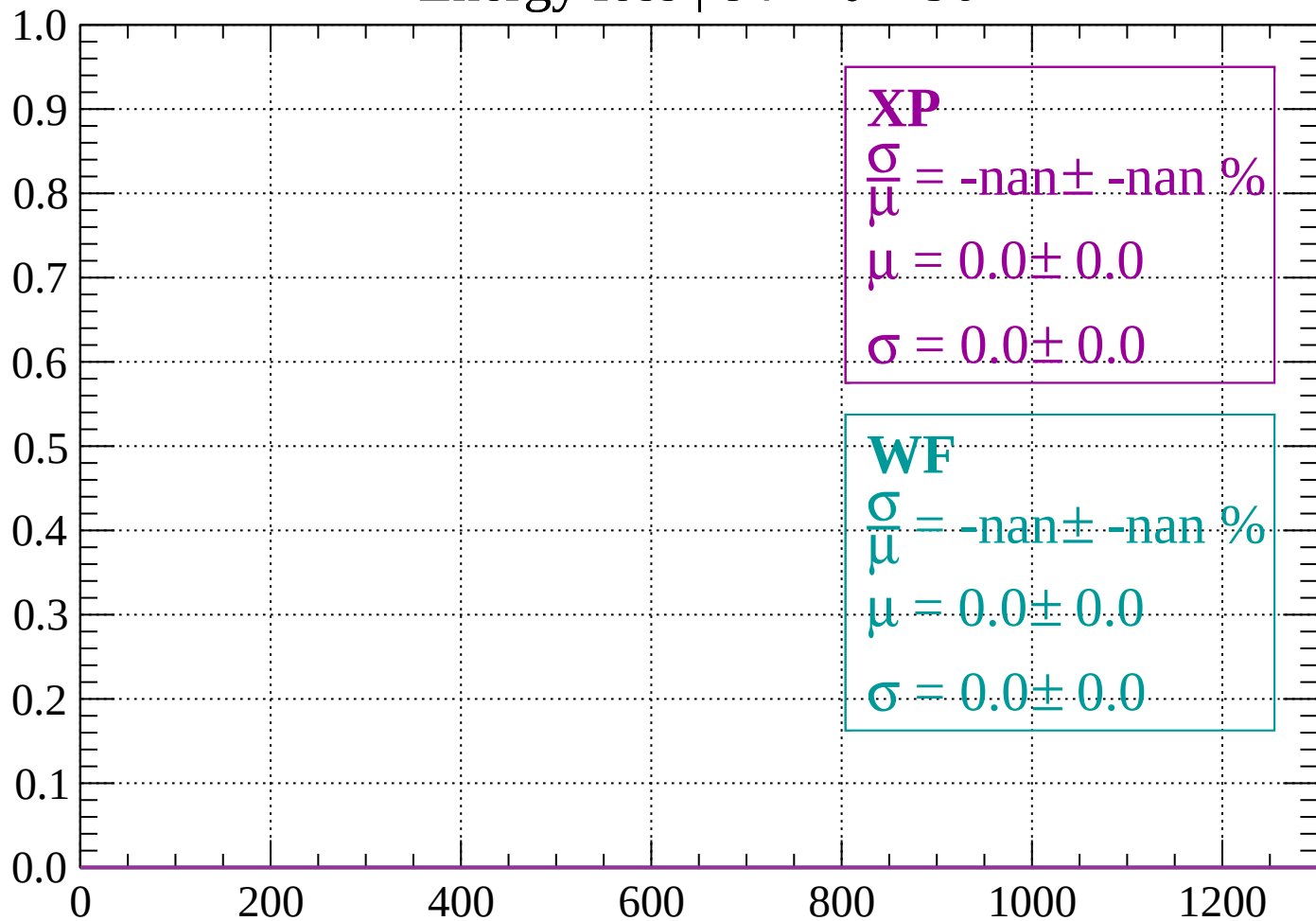
Energy loss | $78 < \theta < 84$

Count



Energy loss | $84 < \theta < 90$

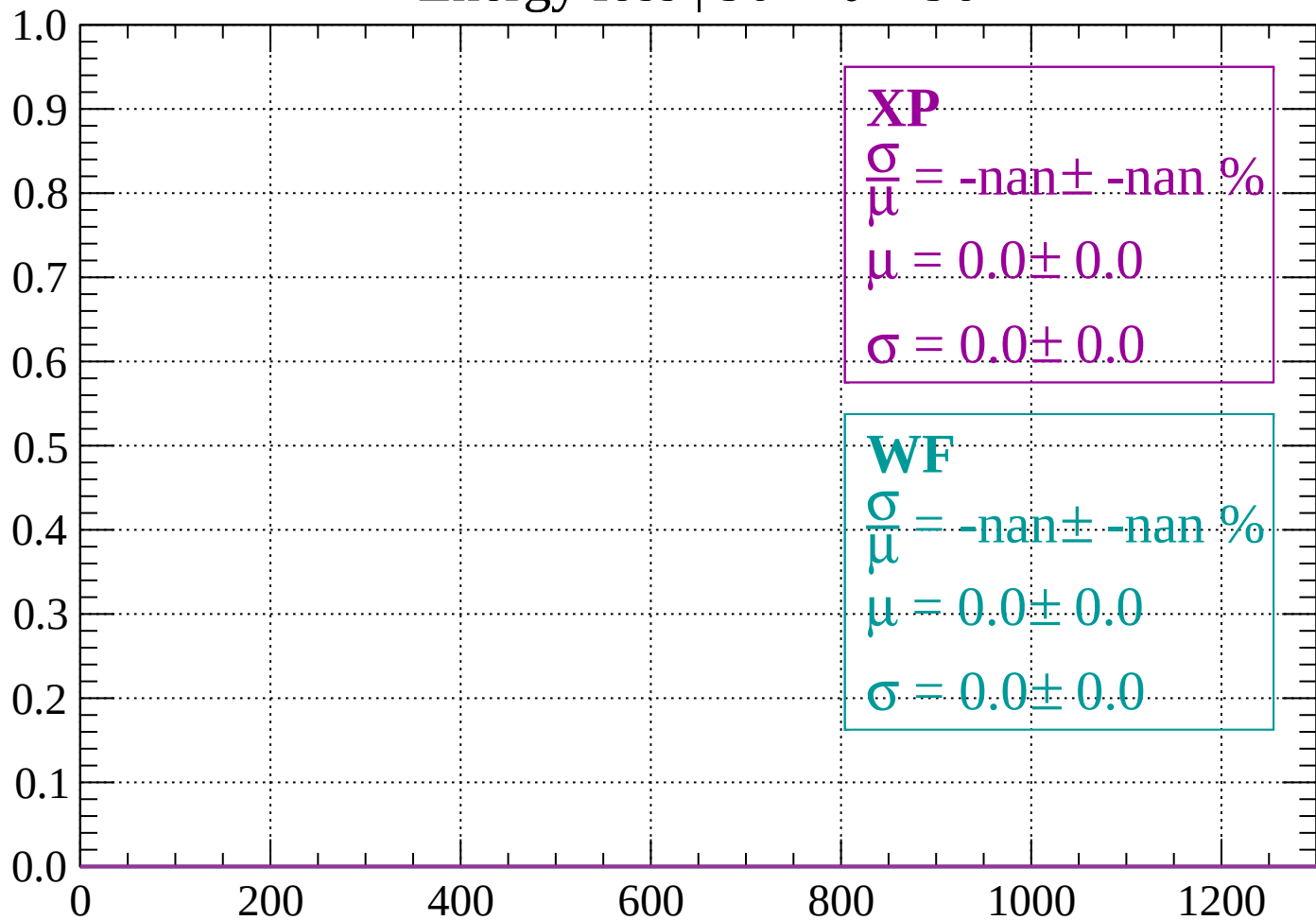
Count



dE/dx (ADC counts/cm)

Energy loss | $90 < \theta < 96$

Count



dE/dx (ADC counts/cm)

